SPACE X · NOTEBOOK · v1

The whole course, in one notebook

Everything from the course in one file, in the order of your paper notebook. 169 pages. The page number is in the top right corner of every page.

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How it is laid out

- **Front:** the whole course on one page, the plan, and the call sheet.
- **Middle:** the seven sections in course order. Each starts with a divider page: what to say, the cue card for that section's talk, and the page of every lecture.
- **Each lecture:** page 1 is the spread you copy (drawing left, cues and notes right). Then the teaching page ("Like you're 5", official names, how it works), then what the instructor got wrong, then every word.
- **Back:** the Sections 1–3 teach-back kit, concept cards, word banks, glossary and Who's Who.

Before the call, read only this: pages 2 to 5, then the seven divider pages. That is 12 pages.

What changed along the way

Sections 4–5 lectures also have copy-into-notes and concept-card pages (C17–C31). Sections 6–7 were made during the deadline push and stop at the vocabulary page. Older versions of any page were left out.

SPACE X · WHOLE COURSE · v1

All seven sections on one page

One sentence to say, one number to know, and one thing the instructor said that you should not repeat. If you can say the middle column from memory, you can hold the conversation.

Section	Say this	Number to know	Don't repeat
1 · Mission design	An orbit is falling and missing. Low is fast and sharp; one exact height hangs still over the equator.	LEO 400 km: 92 min, 7.7 km/s · GEO 35,786 km	"36,000 km from the centre" (that is the altitude)
2 · Space environment	The air did five jobs for you: sunscreen, pressure, damper, fan, ground wire. Space does none.	atomic oxygen rules 200–600 km	"atomic oxygen appears at 320 km"
3 · Orbital mechanics	Six numbers describe any orbit. The lazy half-ellipse is cheapest; a flyby steals speed from a planet.	Mars: 259 days, window every 26 months	"Lambert takes the Δv and gives the time" (reversed)
4 · Propulsion	Every engine throws mass backward. Chemical pushes hard and is thirsty; electric is gentle and efficient; nuclear never flew.	$F = \dot{m} \times v_e$ · Isp 450 s / 3,000 s / 900 s	"nuclear is efficient because it is hotter" (lighter exhaust)
5 · Communications	Write the message on a radio wave, aim it with a dish, send it through the radio window, land louder than the noise.	$c = \lambda \times f$ · Cassini: 20 W, 10^{-16} W received	"a dish adds power" · "k is about wavelength"
6 · Attitude control	Sense, compare, twist, forever. Gyros drift and stars fix them. Wheels swap spin; jets and magnets remove it.	star tracker 0.001° · Hubble 0.007 arcsec	"Explorer 1 failed" · "inertia = mass in a direction"
7 · Power	A sunlight sandwich near the Sun, a bucket for the dark, a warm rock where the Sun doesn't reach.	1,361 W/m² at Earth · cells 30% · Pu-238 87.7 years	"three layers = 3× efficiency" · "RTGs last 500 years"

The thread: a spacecraft has to do everything for itself. Get there (1, 3, 4) · survive (2) · talk (5) · point (6) · stay powered (7).

The four formulas worth carrying

$F = \dot{m} \times v_e$ · thrust = mass thrown per second × exhaust speed

$c = \lambda \times f$ · wavelength × frequency = speed of light

P = flux × area × efficiency · solar array power

battery = load × shadow time ÷ depth of discharge

The one idea that repeats three times

Twice as far = a quarter as much. It is why gravity weakens with height (Section 1), why a radio signal fades (Section 5) and why sunlight fades (Section 7). Anything that spreads out over a sphere does this.

SPACE X · PLAN FOR TOMORROW

About three hours, in five blocks

Sections 1–3 you have rehearsed before. Sections 4–7 you have only copied. So the time goes to 4–7, and most of it to saying things out loud.

The blocks

Time	Do this
40 min	**Read.** For each of the 12 lectures in Sections 4–7: the "Like you're 5" box and the "Where the instructor was loose" list. Nothing else. About 3 minutes each.
40 min	**Quiz.** App → Quiz → new questions. Scenes 8 to 19. Every miss, read the explanation out loud.
45 min	**Talk.** App → Teach. Section 4, 5, 6 and 7 talks, once each, timer running, cue card hidden after the first minute. Send me any one of them.
20 min	**Whole course.** App → Teach → Whole course, twice. The second time, no cue card.
20 min	**Quiz 5** in Udemy (designing a solar array), using the recipe on the right.

If you only have one hour: the page before this one, then the whole-course talk twice, then the Section 4–7 "loose" lists.
If it is a test, not a talk: swap the talk block for a second quiz block and the Numbers tab.

Recipe for Quiz 5: size a solar array

1. Sunlight per m² = 1,361 ÷ (distance in AU)².
2. Usable per m² = that × efficiency × any loss factor they give.
3. Area = power needed ÷ usable per m².
4. If there is a battery to refill, add its recharge power to "power needed" first.
Example: 500 W at Mars, 30% cells, no other losses. 1,361 ÷ 1.52² = 590. 590 × 0.30 = 177 W per m². 500 ÷ 177 = **2.8 m²**.

What is still open

You have not explained anything back to me since Section 3. That is 16 lectures of copying with no check. One talk sent tomorrow is worth more than re-reading everything.

Paused during the push, yours to ask for afterwards: concept-card pages for Sections 6–7, the copy-into-notes pages, and the Who's Who names for Sections 6–7.

SPACE X · CALL SHEET · 1 OF 2

What he will ask, and what you say

Your friend knows the course, so expect why and how questions. These are the three most likely per section, with an answer short enough to say in one breath.

How to answer anything

1. **Picture first.** Start with the one-line picture: "it's falling and missing", "it's a flashlight for radio".
2. **Then the official name.** "That's the Seebeck effect." It shows you know the real term.
3. **Then one number.** 7.7 km/s. 35,786 km. 30%. One is enough.
4. **If you blank,** say what you do know and reason out loud, or sketch it. Reasoning counts for more than recall.
5. **"I don't know that one"** is a fine answer. Guessing with confidence is not.

He learned from the same instructor

So he may say these, because the course did. Don't argue. Say "the course says that, but I checked", and give the reason.

The course says	What is true
Nuclear rockets are efficient because they run hotter	They run cooler. The exhaust is light hydrogen.
Project Orion was NASA's nuclear thermal rocket, 1986	A 1958–65 study of riding nuclear bombs. Not thermal, not NASA's.
Electric propulsion is new, from the 1980s	First flown in 1964.
A directional antenna gives you more power	It aims the same power into a beam.
A triple-junction cell triples the efficiency	About 20% becomes about 30%.
Explorer 1 was a failure	It tumbled, and still discovered the radiation belts.
RTGs last 500 years	The heat halves every 87.7 years.
A low orbit takes two hours	About 92 minutes: 16 a day.

Sections 1 to 3

Sec	He asks	You say
1	**Why doesn't a satellite fall down?**	It is falling. It also moves sideways at 7.7 km/s, so the ground curves away as fast as it drops: 4.3 m every second. Gravity is the string; no engine is needed.
1	**Why is geostationary exactly 35,786 km, and why over the equator?**	It is the one height where a lap takes one Earth rotation, 23 h 56 min. Only over the equator does it stay above one spot; tilted, it draws a figure-8.
1	**Flyby, orbiter, lander: what changes?**	Each step needs more slowing down, so more fuel. A flyby needs no braking burn, an orbiter must brake into orbit, a lander must also get down.
2	**The thermosphere is 1,000 °C. Why doesn't the ISS cook?**	Temperature is how fast the molecules move; heat is how much energy they deliver. The air is so thin that almost nothing hits you.
2	**What is actually dangerous about a vacuum?**	The air did five jobs: blocked UV, held pressure, damped vibration, carried heat away, bled off static. In space you build all five yourself.
2	**What is atomic oxygen?**	Single O atoms made when UV splits O_2: the main gas at 200–600 km. A satellite rams them at 7.5 km/s and they eat plastics such as Kapton.
3	**What are the six orbital elements?**	The Classical (Keplerian) Orbital Elements: a size, e squash, i tilt, Ω swivel, ω where perigee points, ν where you are now. Five fix the orbit, one moves.
3	**Why can't we leave for Mars any time?**	The cheap route is a Hohmann half-ellipse, 259 days. Mars must be about 44° ahead at launch so it arrives when you do. That lines up every 26 months.
3	**Where does a gravity assist's speed come from?**	From the planet. In the planet's view you leave as fast as you came, only bent. In the Sun's view you picked up some of its orbital motion. It slows by an unmeasurable amount.

SPACE X · CALL SHEET · 2 OF 2

Sections 4 to 7: the ones you have only copied

You rehearsed Sections 1–3 out loud. These four you have not. Say each answer aloud once before the call, with this page covered.

Propulsion and communications

Sec	He asks	You say
4	**No air in space: what does a rocket push on?**	Its own exhaust. Newton's third law. Thrust = mass thrown per second × exhaust speed.
4	**What is specific impulse?**	Exhaust speed ÷ 9.81, in seconds: the engine's mileage. Chemical about 450 s at best, ion about 3,000 s.
4	**Why not ion engines for everything?**	Thrust is limited by electric power: 0.09 N from 2.3 kW. Perfect for years of cruising, useless for launch, landing or a capture burn.
5	**How does 20 W reach Earth from Saturn?**	A 4 m dish squeezes it into a 0.6° beam, a gain of about 50,000, and a 70 m dish listens. The dish adds no power. It aims it.
5	**Why radio and not something else?**	The air is a wall with two windows: visible light and radio (about 1 cm to 10 m). UV and X-rays are absorbed; long radio bounces off the ionosphere.
5	**Why is deep-space data so slow?**	The signal spreads: twice as far, a quarter the power. What counts is signal ÷ noise at the receiver, and that sets bits per second. Voyager: 160.

Attitude control and power

Sec	He asks	You say
6	**How does it know where it is pointing?**	Gyros feel turning but drift. Sun sensor, star tracker, horizon sensor and magnetometer each see one outside reference. One direction gives only two angles, so they are combined.
6	**How does it turn with nothing to push on?**	Spin a reaction wheel one way and the craft turns the other: angular momentum is conserved. Wheels fill up (saturate); thrusters or magnetic torquers empty them.
6	**Why did Explorer 1 tumble?**	A pencil spun about its long axis, its smallest moment of inertia. Flexing antennas leaked energy, so it ended up spinning about its largest: end over end.
7	**How does a solar cell work?**	Two layers, n and p, with a built-in field at the junction. A photon frees an electron, the field pushes it to one side, it returns through the wire. Triple-junction cells reach about 30%.
7	**What is depth of discharge?**	How much of the battery you drain each cycle. Shallow drains last far longer but need a bigger battery. Low orbit means about 16 cycles a day.
7	**How is Voyager still powered? Is that a reactor?**	RTGs: plutonium-238 decays and makes heat; thermocouples turn hot-against-cold into electricity, about 6%. Not a reactor: no chain reaction.

One idea ties three sections together: twice as far = a quarter as much. Gravity (Section 1), a radio signal (Section 5) and sunlight (Section 7) all thin out the same way, because each spreads over a sphere.

SPACE X · SECTION 1 OF 7

Section 1 · Mission design

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

An orbit is falling and missing. Low is fast and sharp; one exact height hangs still over the equator.

The lectures

#	Lecture	Page	Length
1	**Satellite types and orbits**	7	3 pages
2	**Types of interplanetary spacecraft**	10	3 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Number to know
LEO 400 km: 92 min, 7.7 km/s · GEO 35,786 km above the ground

Don't repeat
"36,000 km from the centre" (that figure is the altitude; from the centre it is 42,164 km)

Talk card · 3¼ min

He opens with: "So what even is a satellite, and why are there different kinds?"

- 1 · FALL & MISS — ball off a mountain
- 2 · 4 JOBS — look out / look down / relay / live
- 3 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
- 4 · LADDER — flyby → orbiter → probe → lander → rover → penetrator
- LAND — "every rung costs more and gives more"

Land on: Every spacecraft is a different answer to one question: how close do you dare get, and what will it cost?

In the app: scenes 1, 2 in Learn, and the "Section 1" talk in Teach.

SEC 1 · SATELLITE TYPES & ORBITS

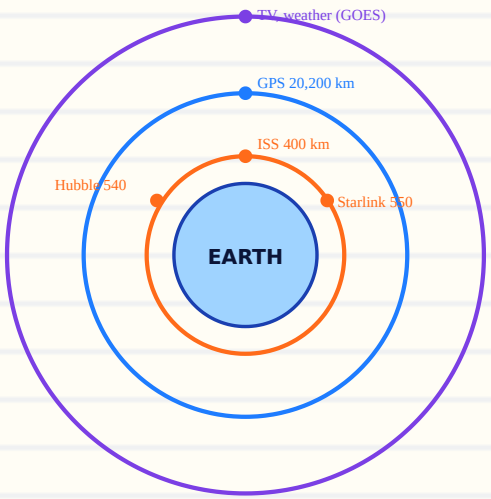
Your spread, upgraded

Your right page was already the correct shape. This is the same content with a drawn left page, harder cues, grouped lists, and the summary box. Copy what helps.

LEFT PAGE · DRAW

SEC 1 — SATELLITE TYPES & ORBITS

date · pg __



GEO = 35,786 km
1 orbit = 1 day → hangs over one spot
(geoSTATIONARY if also over the equator)

MEO = 2,000 – 35,786 km
GPS: 12-hour orbit

LEO = 180 – 2,000 km
~90-minute orbit, 7.7 km/s
drag + atomic oxygen live here

rule: higher = slower = longer day
LEO 90 min → MEO 12 h → GEO 24 h

not to scale! GEO is ~6 Earth-radii out →

⭐ 3 TAKEAWAYS

1. Higher orbit = slower = longer trip around (90 min → 12 h → 24 h)
2. GEO is special only because its day matches Earth's day
3. Group the 8 satellite types by JOB: look out / look down / relay / live

RIGHT PAGE · CORNELL

CUES = questions | **NOTES = keywords**

Which satellite types **look out**, which **look down**, which **relay**?

Where does the ISS live? GPS? TV satellites?

Why is GEO exactly 35,786 km?

Geosynchronous vs geostationary?

Higher orbit → faster or slower?

? what orbit do spy sats use, and why

LOOK OUT — astronomical (Hubble, JWST)
LOOK DOWN — Earth obs (Landsat), weather (GOES), recon (spy)
RELAY — comms (Starlink, DirecTV), navigation (GPS)
LIVE — biosatellites (Laika!), space stations (ISS)

LEO 180–2000 km · ISS 400 · ~90 min
MEO 2000–35,786 · GPS 20,200 · 12 h
GEO 35,786 · 1 orbit = 1 Earth day → stays put
geoSYNC = 1-day orbit, any tilt. geoSTATIONARY = 1-day + over equator
higher = slower (bigger circle, weaker gravity)

job table:

type	job (5 words)	example
navigation	tells you where you are	GPS
comms	bounces signals across Earth	Starlink
weather	watches clouds from GEO	GOES
…	…	…

🧠 SUMMARY — video closed, from memory (5 lines)

1. _____
2. _____
3. _____
4. _____
5. _____

FEEDBACK · YOUR PAGE

What was right, what to sharpen

Honest read of the photo you sent. The structure was correct on the first try, which is the hard part.

✔ Correct

- **Cues are questions.** "What are the types of satellites?" not "Satellite types." That's the thing most people never do.
- **Keywords, not sentences.** Nothing to fix there.
- **All three altitude ranges are right.** LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km.
- **You started a third cue mid-lecture** ("job of each satellite"). That's exactly how the column is supposed to grow.

▲ Sharpen

- **Geosynchronous ≠ geostationary.** Geosynchronous = the orbit takes one day, at any tilt. Geostationary = one day AND over the equator, so it hangs over one spot. TV and weather sats are geostationary. Write both words. Classic quiz trap.
- **Add the one "why."** GEO is 35,786 km because at that height one orbit takes exactly one Earth rotation (23 h 56 min). Lower = faster, higher = slower. Every other orbit number you'll just memorize; this one you can explain.
- **Cues are too easy.** "What are the orbit types?" you can answer with half a glance. Use cues you might fail: "Where does GPS live?", "Why 35,786?", "Higher = faster or slower?"
- **Eight-item list → four groups.** Flat lists of 8 don't retrieve. "Look out / look down / relay / live" does.
- **Job table = type · job · one real example.** The example is what sticks.

🖍 Left page was blank

This lecture is a picture: Earth, three rings, residents on each ring. Page 1 shows it. Draw yours badly first, then fix it. Don't copy page 1 neatly — the drawing is the memory, the copying isn't.

🧠 Missing: header + summary

- Top of page: section / lecture / date, so the index can point here.
- Bottom: five lines from memory, video closed. Can't fill five? That's your first gap. Good — now you know where it is.

📒 For the Question bank (back pages)

- Why is GEO exactly 35,786 km?
- Geosynchronous vs geostationary: which one hangs over one spot?
- Name the orbit and altitude of: ISS, GPS, a TV satellite.
- Group the 8 satellite types by job.

⚡ **Next step: finish the lecture, write the 5-line summary, send me a photo. I'll check it and write the next cards.**

SEC 1 · VOCABULARY

Satellite types & orbits, in plain English

Every word used on the previous pages, including the ones I introduced in the notes column. Read once; on the +1 day review, explain three of them out loud without looking.

Satellite — anything that orbits something bigger. The Moon is Earth's natural satellite; everything we launch is an artificial one. *A ball on a string, swung so fast the string is gravity.*

Altitude — height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

Period — how long one full trip around takes. LEO ~90 min, GPS 12 h, GEO 24 h. Higher = longer. *Bigger circle, slower runner.*

Geostationary — geosynchronous AND directly over the equator with no tilt, so it appears frozen in one spot. TV dishes point at these and never move. *Sync = same time every day. Stationary = never leaves.*

Sidereal day — one true rotation of Earth relative to the stars: 23 h 56 min, not 24 h. That's the real period a GEO satellite matches. *The 4-minute difference is Earth also moving around the Sun.*

Earth observation (EO) — cameras and radar pointed down at Earth for mapping, farming, disasters (Landsat). *Looks down.*

Reconnaissance satellite — a military spy satellite. Same idea as Earth observation, different customer and classified detail.

Communication satellite — a radio relay in the sky: receives a signal from one place on Earth and sends it back down somewhere else (Starlink, DirecTV). *A mirror for radio.*

Space station — a satellite people live in (ISS, China's Tiangong).

GPS — Global Positioning System, the US navigation constellation at ~20,200 km in MEO.

Starlink — SpaceX's internet constellation: thousands of small comms satellites in LEO at ~550 km.

Eclipse (orbital) — when Earth blocks the Sun from the satellite. LEO sats get ~15 a day; batteries have to cover them.

Orbit — the closed path a satellite follows around a body. It's falling toward Earth and missing, forever. *Throw a ball sideways fast enough and the ground curves away as fast as it falls.*

LEO / MEO / GEO — Low, Medium, Geosynchronous Earth Orbit: the three height bands. LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km. *Downstairs, middle floor, penthouse.*

Geosynchronous — an orbit whose period is exactly one day, at any tilt. The satellite returns to the same spot in the sky once a day but may drift north-south in between.

Equator — the imaginary line around Earth's middle. A geostationary satellite has to sit above it; a satellite above Texas can't hover there because Earth's spin would carry Texas away underneath.

Astronomical satellite — a telescope in space pointed away from Earth (Hubble, JWST). *Looks out.*

Weather satellite — looks down at clouds and storms, usually from GEO so it watches the same region nonstop (GOES).

Navigation satellite — broadcasts a precise time signal; your phone hears four of them and works out where it is (GPS, Galileo). *Tells you where you are.*

Biosatellite — carries living things (plants, animals, cells) to study how they cope with space. Sputnik 2 carried Laika the dog in 1957.

ISS — the International Space Station, in LEO at ~400 km, orbiting every ~90 minutes.

Hubble / JWST — two famous space telescopes. Hubble orbits Earth at ~540 km; JWST sits 1.5 million km away at a balance point called L2, which is why it isn't in any of the three bands.

Landsat / GOES — Landsat = US Earth-observation series since 1972 (LEO). GOES = US weather satellites (GEO).

Atomic oxygen / drag — two LEO-only problems: single oxygen atoms that chew plastics, and thin air that slowly slows the satellite down. Both vanish above ~1,000 km. *Covered in the Space Environment section.*

SEC 1 · LEC 3 · TYPES OF INTERPLANETARY SPACECRAFT

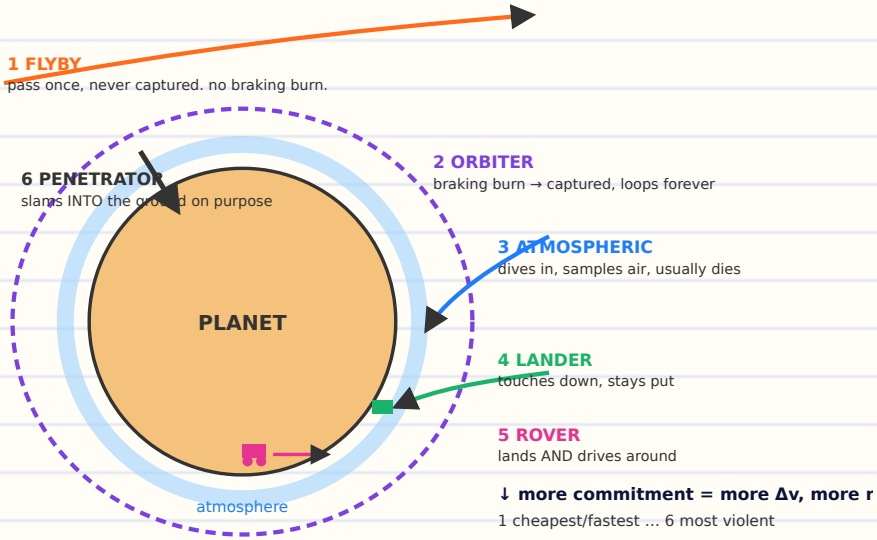
The slide is a ladder. Draw the ladder.

Eight bullet points on the slide. Six of them are the same idea at increasing commitment: how close do you get to the planet? The other two aren't about a planet at all.

🖉 LEFT PAGE · DRAW

🚀 SEC 1 / LEC 3 — TYPES OF SPACECRAFT

date · pg __



🔭 THE OTHER TWO aren't about a planet

7. OBSERVATORY — a telescope parked in space (Hubble, JWST). The target is the sky, not a body.
8. COMMUNICATIONS — a relay so the ones above can phone home (Mars orbiters relay for rovers; China's Queqiao relays for the Moon's far side).

❓ RIGHT PAGE · CORNELL

CUES = questions | **NOTES = keywords**

Rank the 6 "go to a planet" types from least to most committed.

Which type needs NO braking burn? Why is that a big deal?

Why ever fly by if you could orbit?

Atmospheric probe vs lander: what's the difference?

Which 2 types aren't aimed at a planet at all?

? did any penetrator mission actually work

FLYBY — pass once · no capture burn · can chain planets (Voyager grand tour, New Horizons → Pluto)
ORBITER — brake into orbit · long stay · maps (MRO, Cassini, Juno)
ATMOSPHERIC — parachute/balloon into the air · short life (Galileo probe → Jupiter, Huygens → Titan)
LANDER — survives touchdown · fixed spot (Philae, Viking, InSight)
ROVER — lander + wheels (Sojourner, Spirit/Opportunity, Curiosity, Perseverance)
PENETRATOR — impact on purpose · buries sensors or kicks up material (Deep Impact → comet, LCROSS → Moon)
OBSERVATORY — telescope platform (Hubble, JWST)
COMMS — relay for the others

ladder: closer = more Δv to slow down = heavier = riskier = more data
flyby wins on: cost, speed, multi-target. loses on: time at target (hours)
one craft can be 2 types: Cassini (orbiter) carried Huygens (atmospheric + lander)

🧠 SUMMARY — video closed, from memory (5 lines)

1. _____
2. _____
3. _____
4. _____
5. _____

WHY THIS SLIDE MATTERS

It's not a vocabulary list. It's a cost menu.

The instructor is setting up the whole course here. Every later section (propulsion, comms, power) is "what does it cost to climb one more rung of this ladder?"

🪜 The ladder, in plain English

- **Flyby:** you drive past the house at highway speed and take photos out the window. Cheap, fast, and you can hit several houses on one trip. But you get a few hours, once.
- **Orbiter:** you pull into the driveway and stay. That means **braking**, and braking costs fuel you had to carry all the way there. In return: months or years of looking.
- **Atmospheric:** you throw a sensor out the window as you pass. It parachutes down, reports until it's crushed or out of battery. Short life, but it's the only way to taste the air.
- **Lander:** you actually get out. Surviving the landing is the hard part (entry, descent, landing — "seven minutes of terror" on Mars).
- **Rover:** a lander that can walk around. Everything a lander needs, plus wheels, plus way more power and comms.
- **Penetrator:** you throw a dart into the lawn to see what's under the grass. Violent, cheap-ish, high failure rate.

⚡ The one sentence to remember: every rung down the ladder costs more fuel to slow down, more mass, more risk — and buys more science.

🔎 Fact checks on the slide

- The caption said a flyby goes "around" the planet. Sharpen that: it goes **past** the planet, bending its path but never getting captured. "Around" is what an orbiter does.
- Philae (the lander on the slide) is a good example of why landers are hard: it bounced twice on the comet and ended up in a shadow with no solar power. Right type, rough landing.
- The rover pictured is actually a Mars Exploration Rover (Spirit/Opportunity design), not Curiosity. Minor, but notice it: slides have errors too. Verify, don't just absorb.

📒 Question bank cards

- Rank: flyby, orbiter, atmospheric, lander, rover, penetrator — by commitment. What rises with each step?
- Which spacecraft type needs no braking burn at the target?
- Give one real mission for each of the 8 types.
- Why did Voyager fly by four planets instead of orbiting one?
- Observatory and communications spacecraft: what's their target?

✏️ In your notebook, right now

Pause. On your left page, draw a circle (planet) and six lines showing each type's path, numbered 1–6. Then write the ladder sentence under it. That's the whole lecture in one drawing. Then press play.

SEC 1 · LEC 3 · VOCABULARY

Types of spacecraft, in plain English

Every word used on the previous pages, including the ones I introduced in the notes column. Read once; on the +1 day review, explain three of them out loud without looking.

Interplanetary — between planets. An interplanetary spacecraft leaves Earth's neighborhood entirely, as opposed to a satellite that stays in Earth orbit.

Orbiter — a spacecraft that fires its engine to slow down at the planet so gravity captures it into orbit. *Pulling into the driveway and staying.*

Lander — a craft that survives touchdown and sits in one place doing science (Viking, InSight, Philae).

Penetrator — a craft designed to slam into the surface on purpose, either to bury sensors or to kick up material for another craft to analyze (Deep Impact hit a comet; LCROSS hit the Moon).

Communications spacecraft — a relay whose job is passing messages between other spacecraft and Earth. Mars orbiters relay for the rovers; China's Queqiao relays for the far side of the Moon.

Braking burn / capture burn — firing the engine against your direction of travel when you arrive, to slow down enough for the planet's gravity to hold you. A flyby skips this; an orbiter can't.

Gravity assist — swinging close past a planet to steal a bit of its orbital speed, for free. New Horizons used Jupiter this way. *Grabbing a moving merry-go-round to fling yourself faster.*

Commitment ladder — my phrase, not the instructor's: flyby → orbiter → atmospheric → lander → rover → penetrator, ordered by how much fuel, mass, and risk each needs, and how much science it returns.

Kuiper Belt — the ring of icy bodies beyond Neptune. Pluto is the biggest known member. New Horizons kept flying into it after Pluto.

Voyager — two 1977 NASA flyby craft that visited Jupiter, Saturn, Uranus and Neptune on one trip, then left the solar system. The classic example of why a flyby can do more targets.

Cassini / Huygens — Cassini = NASA Saturn orbiter (2004–2017). Huygens = the European probe it carried, which parachuted onto Titan. One mission, two types.

InSight / Philae / Rosetta — InSight = NASA Mars lander (2018) that listened for marsquakes. Rosetta = European comet orbiter; Philae = its lander, which bounced into a shadow and lost power.

Queqiao — China's relay satellite parked beyond the Moon so the Chang'e-4 lander on the far side could talk to Earth.

Flyby — a spacecraft that passes a planet once at high speed and keeps going. It never slows down enough to be caught. *Driving past a house and photographing it out the window.*

Atmospheric probe — a small craft that dives into a planet's air on purpose, usually on a parachute or balloon, and reports until it's crushed or runs out of power (Galileo probe at Jupiter, Huygens at Titan).

Rover — a lander with wheels: it can drive to new spots (Sojourner, Spirit, Opportunity, Curiosity, Perseverance).

Observatory spacecraft — a telescope platform whose target is the sky, not a planet (Hubble, JWST, Kepler).

Δv (delta-v) — "change in velocity": how much speeding up or slowing down a mission needs, in km/s. It's the real currency of spaceflight, because every bit of Δv costs fuel you had to carry. *Read it as "fuel budget."*

Prograde / retrograde — prograde = firing so you speed up along your direction of motion; retrograde = firing to slow down. *Gas pedal vs brake.*

EDL — Entry, Descent, Landing: the sequence a lander or rover goes through from hitting the top of the atmosphere to touching down. On Mars it's ~7 minutes and fully automatic. *"Seven minutes of terror."*

C3 (launch energy) — a number for how hard you have to throw a spacecraft to leave Earth and reach a target, in km²/s². Bigger C3 = bigger rocket. Earth escape is C3 = 0; Mars needs ~10–15; Pluto direct needed far more.

Comet — a chunk of ice and dust that grows a tail when it nears the Sun. Rosetta orbited one and dropped the Philae lander onto it.

New Horizons — the 2006 NASA flyby craft that reached Pluto in 2015 (9.5 years). Launched too fast to ever stop there.

Juno / MRO — Juno = NASA Jupiter orbiter. MRO = Mars Reconnaissance Orbiter, which also relays data for rovers on the ground.

Deep Impact / LCROSS — Deep Impact (2005) fired an impactor into comet Tempel 1. LCROSS (2009) crashed a rocket stage into a lunar crater to look for water in the spray. Both are penetrator-style missions.

SPACE X · SECTION 2 OF 7

Section 2 · Space environment

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

The air did five jobs for you: sunscreen, pressure, damper, fan, ground wire. Space does none.

The lectures

#	Lecture	Page	Length
1	**Space vacuum**	14	3 pages
2	**Earth's atmosphere**	17	3 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Number to know
atomic oxygen is the main gas from 200 to 600 km

Don't repeat
"atomic oxygen appears at 320–340 km" (it dominates from about 200 km)

Talk card · 3 min

He opens with: "What's actually so dangerous about space? There's nothing there."

- 1 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
- 2 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
- 3 · WHAT'S IN IT — N_2/O_2 → atomic O (200–600) → He → H · ozone · ionosphere
- LAND — "space does nothing for you; build all five"

Land on: Space isn't dangerous because something's there. It's dangerous because nothing is, and you have to bring all five of the air's jobs with you.

In the app: scenes 4, 5 in Learn, and the "Section 2" talk in Teach.

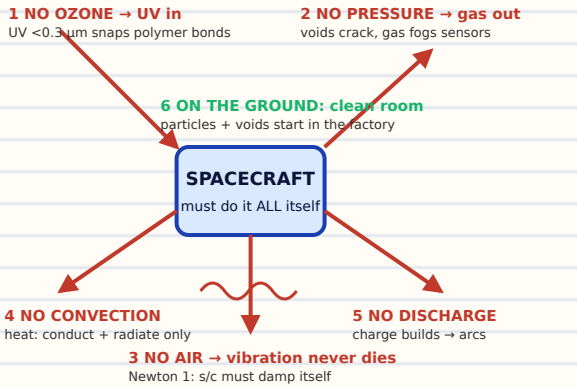
SEC 2 · LEC 4 · SPACE VACUUM CONDITIONS

Vacuum = six things Earth does for free

Six slides, one idea: on Earth the air quietly does jobs for you. In vacuum the spacecraft has to do every one of them itself. Drawn at your small-page budget.

🖊 LEFT · DRAW

S2 / L4 — Space vacuum **11/7 · p__**



⭐ **3 TAKEAWAYS**

1. Vacuum removes 5 free services: ozone, pressure, air damping, convection, discharge
2. UV photon energy > bond energy ∴ polymers break
3. Fix most of it before launch: materials, bake-out, clean rooms

🧠 **SUMMARY · video closed · 3 lines**

1. _____
2. _____
3. _____

❓ RIGHT · CORNELL

CUES	NOTES
What does ozone block, and in what unit?	no ozone → UV < 0.3 µm (= 300 nm) reaches s/c
Why does UV eat polymers but not metal?	UV photon energy > C–C bond ∴ polymers go brittle/yellow (not metals)
Outgassing: cause → 2 harms → 2 fixes?	outgassing: trapped gas in voids, 1 atm in / 0 out → cracks escaped gas lands on cold lenses/sensors → keep outgassers away from sensors fix: no voids in mfg · bake-out · low-outgas materials
Why does a vibration die on Earth but not in space?	clean room + constant testing (JUNO arrays) — 1 particle can kill a sensor Newton 1: on Earth air absorbs vibration; in space nothing does → s/c must damp itself
Only 2 ways heat moves in vacuum?	no convection → heat only by conduction + radiation → hot spots stay hot
Where does static go in space?	no discharge path → charge builds up → arcs; need grounding/coatings/active bleed
? what IS "spacecraft charging"	∴ Earth's air = free ozone, damper, fan, ground wire. Space: build all 4.

How to use this

- **Draw the center box and 5 red arrows yourself first.** Label each arrow with what's missing. Mine is for checking, not copying.
- Notes column is at its 10-line limit. Yours can be shorter.
- The ∴ line at the bottom is the one to green-highlight on your +1 day review if you can say it from memory.
- Page 2 → fact checks: two things the lecture got loose, one it got wrong.

⚡ Page 3 is the vocabulary. Read it BEFORE you copy anything: a word you can't explain is a word you can't remember.

SEC 2 · LEC 4 · TEACH + FACT CHECK

The air is doing four jobs for you right now

Plain-English version of the lecture, then where the instructor was loose or wrong. Read this on screen; it does not go in the notebook.

🌬 What Earth's air does for free

- **Sunscreen (ozone).** The ozone layer swallows the nastiest ultraviolet, everything shorter than about 0.3 µm (300 nanometers). In space that light hits the spacecraft raw. A UV photon at 300 nm carries about 4 eV, and the carbon-carbon bonds that hold plastics together are about 3.6 eV. The photon wins. Chains snap, the plastic goes brittle and yellow. Metals don't care; their bonds aren't the same kind.
- **Pressure (outgassing).** Every material made in air has tiny pockets of air inside it at one atmosphere. In vacuum the outside is zero, so that trapped gas pushes out. Two harms: the material can crack, and the escaping gas drifts off and condenses on the coldest nearby surface, which is usually a lens or sensor. That's the real reason the slide says "keep outgassing materials away from sensors." Fixes: manufacture without voids, pick low-outgassing materials, and bake the parts in a vacuum oven before launch so they outgas on the ground instead.
- **A damper (Newton's 1st law).** Flick a ruler on your desk and it stops buzzing because the air steals its energy. In space nothing steals it. A vibration, a wobble, a spin: it keeps going until the spacecraft itself kills it with dampers or reaction wheels. That's why the ADCS section exists.
- **A fan (no convection).** On Earth hot air rises and carries heat away. In vacuum there is no air to rise. Heat can only move by conduction through solid parts or by radiating into space. A hot chip stays hot unless you give it a metal path to a radiator.
- **A ground wire (no discharge).** Static on you bleeds off into the air and the floor. In space it has nowhere to go, so charge builds up on the spacecraft until it jumps as a spark between two surfaces. Fixes: conductive coatings, grounding straps between every part, sometimes an active device that spits charge away.

🔎 Fact checks

- **Wrong (in the captions):** "wavelengths smaller than 0.3 *millimeters*." The slide is right: 0.3 **micrometers** (µm), a thousand times smaller. 0.3 mm would be infrared and harmless. Write µm.
- **Loose:** "corrosion" of polymers. Corrosion is a chemical attack, usually with oxygen. UV damage is photons breaking bonds. Different mechanism, same result (brittle, cracked). In low Earth orbit the thing that really chews polymers is atomic oxygen, which *is* corrosion. Keep the two separate in your head.
- **Loose:** "no dissipation in space." Materials still have a little internal damping, so a vibration does fade eventually. It just fades far too slowly to rely on. The point stands; the word "no" is an exaggeration.
- **Simplified:** outgassing presented mainly as cracking. In practice the contamination of optics is the bigger headache; the slide's own last bullet hints at it.

📒 Question bank cards

- Ozone absorbs UV below what wavelength? (unit!)
- Why does UV damage plastics but not aluminum?
- Name two harms of outgassing and two fixes.
- A reaction wheel starts vibrating. Why doesn't it settle on its own in orbit?
- List the two heat paths that still work in vacuum.
- What's the Earth-side equivalent of a grounding strap on a spacecraft?
- Five "free services" the air provides on Earth that vacuum removes?

⚡ Next: write the 3-line summary with the video closed, then send me a photo of the spread. Then play the "atmospheric conditions" lecture.

SEC 2 · LEC 4 · VOCABULARY

Every word in this lecture, in plain English

The instructor uses these as if you already know them. You don't need to copy this page. Read it once, then on the +1 day review try to explain three of them out loud without looking.

Vacuum — space with (almost) no air in it. Not "nothing"; just no gas pushing on things. *Like the inside of a sealed jar after you pump all the air out.*

Ultraviolet (UV) — light that's a bit "bluer than blue," too high-energy for your eyes to see. It's what sunburns you. *The invisible part of sunlight that does damage.*

µm (micrometer / micron) — one millionth of a meter. 0.3 µm = 300 nm. A human hair is ~70 µm. *The caption said "millimeters": wrong by 1,000×.*

Organic chemical bond — the "glue" between carbon atoms that holds those chains together. Takes a certain amount of energy to break. *UV has more than that amount, so it snaps the glue.*

Corrosion — chemical attack on a material, like rust on iron. The instructor used it loosely for UV damage; strictly, UV damage is bond-breaking, not corrosion. *Rust = corrosion. Sun-rotted plastic = degradation.*

Void — a tiny bubble or pocket inside a material left over from manufacturing, usually full of air. *A bubble in a cake.*

Clean room — a sealed workspace with filtered air, bunny suits, and strict rules, so dust and skin flakes don't land on the spacecraft. *An operating theater for hardware.*

Dissipation — energy leaking away into the surroundings until the motion stops. *The reason a plucked guitar string goes quiet.*

Vibration / damping — a shaking back and forth / anything that makes it fade out. *Shock absorbers on a car are dampers.*

Conduction — heat moving through a solid by touch, atom to atom. *A metal spoon getting hot in soup.*

Equilibrium / "heat inequilibrium" — equilibrium = everything settled at the same temperature. Inequilibrium = a hot spot that hasn't spread out yet. *In vacuum, hot spots stay hot spots.*

Spacecraft charging — the spacecraft slowly collecting electric charge from space (sunlight knocks electrons off, charged particles stick). With no air to bleed it away, it builds until it sparks. *You on a carpet, forever, with no doorknob.*

Ozone layer — a thin band of a special oxygen (O_3) high in Earth's air that soaks up harsh sunlight before it reaches you. *Earth's sunscreen.*

Wavelength — the size of one ripple of a light wave. Shorter wavelength = more energy = more damage. UV is shorter than visible light. *Small ripples hit harder.*

Polymer — any plastic, rubber, or similar material made of long repeating chains of molecules. Kapton, Teflon, epoxies, cable insulation. *Long molecular spaghetti.*

Degradation / decomposition — the material slowly falling apart: going brittle, yellowing, cracking, flaking. *Like a garden hose left in the sun for 5 years.*

Outgassing — gas slowly leaking out of a solid material once it's in vacuum. *Like a new car's "new car smell": plastics releasing trapped stuff, but in space it lands on your camera lens.*

Pressure difference — one side pushing harder than the other. Air trapped in a void pushes out at 1 atmosphere; vacuum pushes back with nothing. *A balloon taken up a mountain swells for the same reason.*

Perturbation — any small unwanted push, nudge, or wobble. A thruster puff, a solar panel unfolding, a wheel spinning up. *Someone bumping your desk.*

Newton's 1st law (inertia) — a thing keeps doing what it's doing (moving or sitting still) unless something pushes on it. *On Earth, air and friction are the "something." In space, there isn't one.*

Convection (natural) — heat carried away by moving air or liquid: hot air rises, cool air replaces it. *Why a radiator heats a whole room.*

Radiation (heat) — heat glowing away as invisible light, needing no air at all. *Feeling a campfire on your face from a distance.* In vacuum, conduction + radiation are the only two left.

Electrical discharge — built-up charge suddenly jumping somewhere: a spark. *The zap from a doorknob after walking on carpet.*

Passive vs active system — passive = works with no power or decisions (a coating, a strap, a radiator). Active = does something on command using power (a heater, a reaction wheel, a charge-bleeding device). *A window shade vs. air conditioning.*

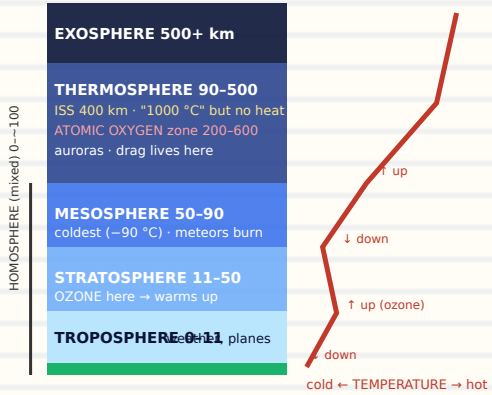
SEC 2 · LEC 5 · EARTH ATMOSPHERE

Five layers, one zigzag, one hungry atom

Three slides: the layers, what the air is made of at each height, and atomic oxygen. One drawing covers all three: the layer stack with the temperature zigzag and who lives where.

🖊 LEFT · DRAW

S2 / L5 — Earth atmosphere **11/8 · p__**



⭐ **3 TAKEAWAYS**

1. Temp zigzags: down, up (ozone eats UV → heat), down, way up
2. Thermosphere is "hot" but too thin to burn you: temp ≠ heat
3. Atomic O = UV-split oxygen; dominant 200–600 km; s/c hits it at 7.5 km/s

🧠 **SUMMARY · video closed · 3 lines**

1. _____
2. _____
3. _____

❓ RIGHT · CORNELL

CUES	NOTES
5 layers, bottom to top, with km?	tropo 0–11 · strato 11–50 · meso 50–90 · thermo 90–500 · exo 500+
Why does temp go UP in the stratosphere?	temp: ↓ ↑ ↓ ↑↑ · strato ↑ b/c ozone absorbs UV → heat (L4 callback)
Thermosphere is 1000 °C. Would the ISS melt? Why not?	thermo "500–1500 °C" but air ~nothing ∴ no heat to transfer (no convection, L4)
What's the air made of low down vs at 300 km?	homosphere 0–~100 km = gases mixed (N_2 78%, O_2 21%, Ar) above ~100: gases sort by weight · chart is log scale (each line = 100,000×)
Atomic O: what, where from, where does it live?	atomic O (single atom, not O_2) = UV splits O_2 → dominant gas ~200–600 km = LEO
Why does 7.5 km/s matter if the atoms barely move?	very reactive → oxidation eats polymers (Kapton), metals slower atoms ~still; s/c slams into them at 7.5 km/s → ~5 eV per hit > bond energy
Metals vs polymers under atomic O?	fix: coatings (SiO_2, Al), AO-resistant materials, shield ram-facing side
? what is "anomalous O"	corrosion IS the right word here (oxidation), unlike UV in L4

How to use this

- **Draw the stack first.** Five bands, heights on the left, the red temperature zigzag on the right. Then put the ISS and the atomic-oxygen zone in the thermosphere band.
- Notes column is at the limit again. Cut what you already knew.
- Two callbacks to last lecture: ozone + UV, and no convection. Green-highlight those on review if you can explain the link.
- Page 2: the captions mangled 8 words. The list is there so you don't copy a wrong one.

⚡ Slide 1 (the title) wasn't sent. Fine. The three you sent are the whole content.

SEC 2 · LEC 5 · TEACH + FACT CHECK

The air gets thinner, weirder, and hungrier as you go up

Plain-English version, then the caption errors and the two places the instructor was loose. Read on screen; don't copy.

🪜 The stack, floor by floor

- **Troposphere, 0–11 km.** Everything you've ever experienced: weather, clouds, airliners at 10 km. Gets colder as you climb, about −50 °C at the top.
- **Stratosphere, 11–50 km.** Temperature goes *up* with height, which feels backwards. Reason: the ozone layer lives here and it soaks up UV. Absorbed light becomes heat. Last lecture's "sunscreen" is also a heater.
- **Mesosphere, 50–90 km.** Coldest place on the whole stack, about −90 °C. Meteors burn up here.
- **Thermosphere, 90–500 km.** Temperature shoots up to 500–1500 °C. But the air is so thin there's almost nothing to carry that heat into you. A thermometer reads hot; your hand would feel nothing and the ISS, which orbits right in the middle of this layer at 400 km, does not melt. Temperature is how fast the few molecules move; heat is how much energy they can hand over. Here, almost none.
- **Exosphere, 500 km and up.** So thin it blends into space. Atoms here can escape Earth entirely.
- **Homosphere** is the instructor's "first region": the bottom ~100 km where the gases are stirred together so the recipe is the same everywhere (78% nitrogen, 21% oxygen, 1% argon). Above that, gases separate out by weight, heavy ones low, hydrogen and helium highest.

🦷 Atomic oxygen, the hungry atom

- Normal oxygen is two atoms holding hands: O_2. Up high, UV light (no ozone to stop it) snaps the pair apart into single O atoms. A single O atom is desperate to bond with anything. That's "highly reactive."
- The chart shows it: the brown line (O) rises above the N_2 line around 200 km and stays on top. From roughly 200 to 600 km, atomic oxygen is the main gas. That is exactly where LEO satellites fly.
- The atoms are nearly stationary. The spacecraft is the thing moving, at 7.5 km/s. So every atom hits the front face like a bullet, with about 5 eV of impact energy. More than enough to break a bond.
- Polymers like Kapton erode visibly in months (the slide photo). Metals form a thin oxide skin and then mostly stop. Fixes: thin glass or aluminum coatings, oxygen-resistant materials, and keeping delicate stuff off the side that faces forward.

🔎 Caption errors (don't copy these)

Captions said	Instructor meant
"the comet"	the **homosphere**
"density and competence"	density and **components**
"the great of a time"	**degrade** over time
"corrosion and explosion"	corrosion and **erosion**
"destroying the bones"	destroying the **bonds**
"polymers are followable"	polymers are **vulnerable**
"increasing the acceleration rate"	increasing the **oxidation** rate
"320–340 comets"	320–340 **km** (see below)

▲ Where the instructor was loose

- **"Atomic oxygen appears around 320–340 km."** Not quite. His own chart shows atomic O present from the left edge (80 km) and dominant from ~200 km. What appears at 320–340 km on that chart is the black "anomalous O" line, a minor extra form. Write: atomic O dominant ~200–600 km.
- **"Mainly only atomic nitrogen increases."** Atomic N does rise, but it's a minor player. The big story on the chart is atomic O taking over from N_2 and O_2. Don't let the nitrogen line steal the lesson.
- **"Corrosion" is correct this time.** Last lecture he used it loosely for UV. Here atomic oxygen really does oxidize materials, which is what corrosion means. Same word, now accurate.
- **Not mentioned at all: drag.** That thin thermosphere air also slows satellites down. The ISS loses ~2 km of altitude a month and has to fire thrusters to climb back. It'll show up later; note it now.

📒 Question bank cards

- Order the 5 layers and give the height where each starts.
- Why is the stratosphere warmer at the top than the bottom?
- The thermosphere is 1000 °C. Why doesn't the ISS cook?
- How is atomic oxygen made, and where is it the main gas?

SEC 2 · LEC 5 · VOCABULARY

Every word used, in plain English

Includes the words on the chart slide, which the instructor didn't explain at all. Copy only the ones that were new to you into the word bank, 3–5 word meaning, with the page number.

Atmosphere — the blanket of gas held around a planet by gravity. Thick at the bottom, fading to nothing a few hundred km up. *Earth's air, all of it.*

Troposphere — bottom floor, 0–11 km. Weather, clouds, airliners. *"Tropo" = turning: the air churns here.*

Mesosphere — 50–90 km. The coldest layer; meteors burn up here. *"Meso" = middle.*

Exosphere — 500 km and beyond. So thin that atoms can drift off into space. *"Exo" = outside.*

Temperature vs heat — temperature = how fast molecules move. Heat = how much energy they can actually give you, which also depends on how many there are. *One sparkler is 1000 °C and won't warm a room.*

Log scale — the chart's vertical axis where each tick (10^{10}, 10^{15}, 10^{20}) is 100,000× the one below. Used when numbers span huge ranges. *Zooming out so a mountain and a pebble fit on one page.*

N_2, O_2, Ar — nitrogen gas (two N atoms), oxygen gas (two O atoms), argon. The three main ingredients of normal air: 78%, 21%, 1%.

Atomic oxygen (AO) — a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

NRLMSISE-00 — just the name of the computer model that produced the chart (a US Navy atmosphere model). Not a thing to learn; it's a citation.

VLEO / LEO — Very Low Earth Orbit (below ~250 km, heavy drag, short life) vs Low Earth Orbit (up to 2,000 km). The green line on the chart marks the split.

Oxidation / corrosion — oxygen bonding to a material and changing it. Rust is iron oxidizing. Atomic oxygen oxidizes spacecraft surfaces; here "corrosion" is the right word. *Rust, but in orbit, and fast.*

Kinetic energy / relative velocity — kinetic energy = energy of motion. Relative velocity = how fast two things approach each other; it doesn't matter which one is "really" moving. The s/c at 7.5 km/s hitting a still atom is the same as the atom hitting a still s/c at 7.5 km/s.

Ram-facing — the side of the spacecraft pointing in the direction of travel, which takes all the atomic-oxygen hits. *The windshield.*

Protected vs exposed — the slide photo's labels: a sample with a coating vs. one bare, after 10 months and 5.8 years in orbit. The bare one is shredded.

Layer / region — a slice of the atmosphere defined by how temperature behaves there. The boundaries are where the temperature trend flips. *Floors of a building.*

Stratosphere — 11–50 km. Calm, layered, holds the ozone layer, warms with height. *"Strato" = layered.*

Thermosphere — 90–500 km. Extremely hot by thermometer, almost no air. The ISS and most LEO satellites fly inside it. *"Thermo" = heat.*

Homosphere / heterosphere — homo = the bottom ~100 km where gases are stirred into one uniform mix. Hetero = above that, where gases separate by weight. *Smoothie vs. a layered cocktail.*

Density — how much stuff is packed into a space. **Numerical density (m^{-3})** on the chart = how many molecules per cubic meter. *People per square meter in a crowd.*

Constituent / component — one of the gases the air is made of. *Ingredients.*

O, N, H, He — single atoms of oxygen, nitrogen, hydrogen, and helium gas. The "atomic" forms. Up high, radiation breaks the paired molecules into these singles. *Loose Lego bricks instead of snapped-together pairs.*

Anomalous O — a tiny extra population of unusually energetic oxygen atoms that the chart's model adds above ~300 km. Minor; not something you need to remember beyond "it's still oxygen."

F10.7 / Ap — two numbers on the chart title describing how active the Sun and Earth's magnetic field were for the model run. The atmosphere puffs up when the Sun is active. Ignore the values.

Reactive — eager to bond chemically with other stuff. *Sticky.*

Erosion — material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

eV (electron-volt) — a tiny unit of energy used for single atoms and bonds. A chemical bond is a few eV; an atomic-oxygen hit at 7.5 km/s carries ~5 eV. *Enough to snap the bond.*

Kapton — a gold-colored polymer film used everywhere on spacecraft for insulation. Famous for being eaten by atomic oxygen unless coated.

S/C — shorthand for spacecraft, used on the slide.

SPACE X · SECTION 3 OF 7

Section 3 · Orbital mechanics

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

Six numbers describe any orbit. The lazy half-ellipse is the cheapest way to go; a flyby steals speed from a planet.

The lectures

#	Lecture	Page	Length
1	**Orbit definition and the six parameters**	21	4 pages
2	**Earth orbits and constellations**	25	4 pages
3	**Interplanetary transfers and flybys**	29	4 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Number to know
Earth to Mars: 259 days, a window every 26 months

Don't repeat
"Lambert's problem takes the Δv and gives the time" (it takes two places and a time, and gives the orbit and its Δv)

Talk card · 4¼ min

He opens with: "How do you even steer something in space? Isn't it just floating?"

- 1 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
- 2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
- 3 · FUEL BILL — lazy throw 8.5 mo · every 26 mo · Lambert = by-Tuesday calculator · fast = $$$
- 4 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
- LAND — "time vs fuel, unless you borrow a planet"

Land on: Going anywhere is a trade between time and fuel, unless you can borrow a planet.

In the app: scenes 3, 6, 7 in Learn, and the "Section 3" talk in Teach.

SEC 3 · LEC 1 · ORBITAL MECHANICS

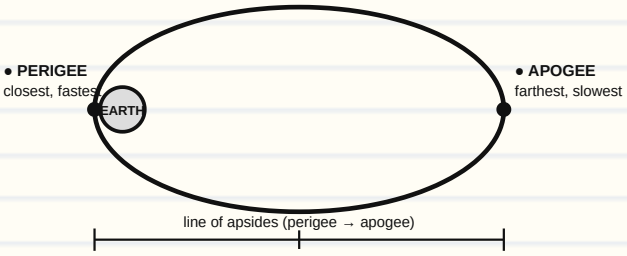
Six numbers pin down any orbit

Left = your left notebook page. Right = your right page. Copy what's here; draw the pictures yourself before you look closely at mine.

LEFT PAGE · DRAW

S3 / L1 — Orbit parameters **date · pg __**

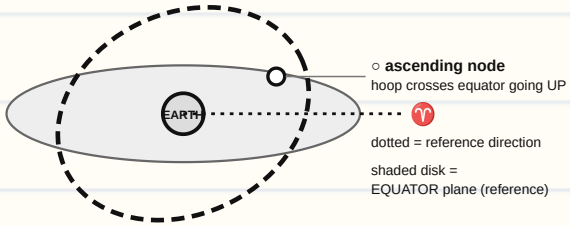
1 · THE ELLIPSE: Earth sits OFF-CENTER (at a focus)



a = (r_perigee + r_apogee) ÷ 2 → SIZE · half of this line

e = how squashed: 0 circle · 0.5 egg · near 1 cigar · over 1 = open path (flyby)
both radii measured from Earth's CENTER, not the surface

2 · THE TILTED HOOP: i, Ω, ω



i = TILT of the hoop vs the equator
Ω = SWIVEL: angle from ♈ around to the ascending node
ω = SPIN in plane: angle from node to perigee · **ν** = where it is NOW

✏️ **PENCIL KEY** · solid = the orbit · dashed = the tilted hoop · shaded disk = equator plane · ● filled dot = perigee / apogee · ○ hollow dot = ascending node · dotted line = ♈ reference direction

⭐ **3 TAKEAWAYS** 3 lines

1. a = size, e = shape, i / Ω / ω = how the hoop is turned, ν = where the bead is
2. Perigee and apogee are measured from Earth's center; a is their average
3. Reference for Earth orbits = equator plane + ♈ direction (NOT ecliptic / Sun)

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Orbital mechanics in one line?	orbital mechanics = the math of where a spacecraft goes, given its position + speed when the rocket cuts off
a = ? from what two points, measured from where?	2 jobs: orbits around Earth · transfers Earth → another planet
e = 0 is what? 0.7? more than 1?	perigee = closest point · apogee = farthest · both from Earth's CENTER
Where are you fastest / slowest? Where do you fire the engine to change the orbit?	perigee = fastest, apogee = slowest · burn at perigee → apogee goes up · burn at apogee → perigee goes up
Which 5 numbers freeze the orbit, which 1 moves?	a = (r_p + r_a) ÷ 2 → SIZE · for a circle, a = r
Hula hoop: which one is tilt, swivel, spin-in-place?	e = squash: 0 circle · 0.5 egg · near 1 cigar · over 1 = open path (flyby)
Reference plane + direction for an Earth orbit?	i = tilt vs equator: 0° = equatorial · 90° = polar · ISS = 51.6°
Same Ω, different ω: what moved?	Ω = swivel: where the orbit crosses the equator going UP (ascending node), measured from ♈
? what e does a flyby have	ω = spin in-plane: angle from node → perigee
	ν = where the s/c is NOW · the only one that changes minute to minute
	ref plane = EQUATOR · ref direction = ♈ (instructor said ecliptic / "toward the Sun": wrong for Earth orbits)

SEC 3 · LEC 1 · TEACH + FACT CHECK

The hula hoop, one parameter at a time

Start with the box. If the box makes sense, everything under it is just adding names to things you already get.

🧒 Like you're 5

You're on a playground and you throw a ball sideways, really hard. It falls, but it also goes forward, so it lands far away. Now throw it so hard that by the time it falls, the ground has curved away underneath it. It never lands. It keeps falling *around* the Earth forever. That's an orbit: **falling and missing.**

Now picture the ball's path as a **hula hoop around a basketball**. To tell a friend exactly how your hoop sits, you'd say six things: How big is it? How squished? How tilted? Which way does the tilt face? Is the squished end pointing up or sideways? And where on the hoop is the ball right now? That's all the six fancy names below are. **Six ways to describe a hoop.**

🏀 Basketball + hula hoop

Earth is the basketball. The orbit is a hoop around it. To describe exactly how the hoop sits, you need six things:

- **a, semi-major axis: how big.** Half the long way across the ellipse. Equivalently the average of the closest distance (perigee) and farthest distance (apogee), both measured from Earth's center. Bigger a = bigger orbit = longer period (Kepler's 3rd law, coming later).
- **e, eccentricity: how squashed.** 0 is a perfect circle. 0.3 is a gentle egg. 0.9 is a cigar with Earth near one tip. Exactly 1 means the path never closes (escape). More than 1 is a flyby: the spacecraft swings past and leaves. Three orbits with the same e but different a look like the same shape at three zoom levels. That's the left picture on the concepts slide.
- **i, inclination: how tilted.** The angle between the hoop and Earth's equator. 0° = hoop lies flat on the equator (GEO sats). 90° = hoop passes over both poles (polar orbit). 51.6° = the ISS.
- **Ω, longitude of the ascending node: swivel.** A tilted hoop crosses the equator at two points. The one where the satellite is heading north is the **ascending node**. Ω says how far around the equator that crossing point is, measured from a fixed starting direction. Keep the tilt, swivel the hoop around Earth's axis: Ω changes, nothing else does. That's the top-right picture: three identical tilted orbits swiveled to different Ω.
- **ω, argument of periapsis: spin in place.** Keep the hoop exactly where it is and rotate it within its own plane, so the squashed end (perigee) points somewhere else. Measured from the ascending node to perigee. The middle and bottom-right pictures: same node, same shape, perigee rotated.
- **ν, true anomaly: where the bead is.** The other five freeze the hoop in space. ν is the angle from perigee to where the spacecraft is right now. It's the only one that changes every second.

🔎 Caption errors (this transcript was rough)

Captions said	Instructor meant
"radius of perjury"	radius of **perigee**
"Perry axis", "paradoxes", "Perry obsessed", "UPS's"	**periapsis** (= perigee, general word)
"the cryptic plane"	the **ecliptic** plane
"your mega"	**omega**, Ω
"lunches do for sending note"	longitude of ascending node
"DMA major axis", "some major accident"	**semi-major** axis
"IC centric"	**eccentric**
"two anomaly"	**true** anomaly

▲ The instructor's real mistake

- **He said the reference plane is "the plane containing Earth and the Sun, the ecliptic," and the reference direction "points from Earth to the Sun."** He hedged with "I believe," and he's wrong for Earth orbits. For a satellite around Earth, the reference plane is **Earth's equator**, and the reference direction is the **vernal equinox** (the ♈ symbol on his own slide): where the Sun sits on the first day of spring, fixed against the stars. "Toward the Sun" moves every day, so it can't be a reference. The ecliptic is the reference for orbits around the *Sun*. Write the equator version.
- **Minor:** "semi-major axis is the distance from one point of the orbit to the other divided by two." Only if those two points are perigee and apogee. The formula on your page is the precise one.
- **Correct and worth noting:** e runs 0 to 1 for closed orbits. He stopped there; the e ≥ 1 cases are escape and flyby, which is

SEC 3 · LEC 1 · VOCABULARY

Every word used, in plain English

Lots of Greek letters this time. They're just labels; the meaning column is what matters. Copy only the new ones into the word bank.

Orbital mechanics — the math of how things move under gravity: given where a spacecraft is and how fast it's going, where will it be later? *Ballistics, but the "ground" is a planet.*

Transfer orbit — an orbit you use to get from one place to another, like Earth to Mars, rather than to stay. *The highway, not the parking lot.*

Semi-major axis (a) — half the longest width of the ellipse. Sets the orbit's size and its period. *"Semi" = half, "major axis" = the long way across.*

Perigee / apogee — closest and farthest points of an orbit around Earth, measured from Earth's center. *"Peri" = near, "apo" = far, "gee" = Earth.*

Apsis (plural apsides) — either of those two extreme points. The line joining them is the long axis of the ellipse.

Equator / equatorial plane — the flat sheet through Earth's middle, perpendicular to its spin axis. The reference plane for Earth orbits.

Plane of reference — the flat sheet you measure tilt from. Equator for Earth orbits, ecliptic for Sun orbits.

Node / ascending node — the two points where a tilted orbit pierces the reference plane. The ascending node is the one where the satellite is crossing from south to north. *Where the hoop pokes up through the table.*

Argument of periapsis (ω, small omega) — the angle, measured in the orbit's own plane, from the ascending node to perigee. The orbit's spin-in-place. *"Argument" is old math-speak for "angle." Nobody is arguing.*

Greek letters — Ω omega (big), ω omega (small), ν nu. Just labels, like x and y. Say them out loud once so they stop looking scary.

Polar orbit / equatorial orbit — i = 90° (passes over both poles, sees the whole Earth as it spins below) / i = 0° (stays over the equator).

Constraints / requirements — what the mission needs (constraints: limits you can't break; requirements: things it must do). The orbit is chosen to satisfy them.

Celestial body — any big natural object in space: planet, moon, Sun, asteroid, comet. The thing you orbit.

Orbital parameters / elements — the six numbers that fully describe an orbit: a, e, i, Ω, ω, ν. *An orbit's address.*

Eccentricity (e) — a number from 0 up that says how squashed the ellipse is. 0 = circle, near 1 = very stretched, 1 or more = doesn't close. *"Eccentric" literally means off-center.*

Periapsis / apoapsis — the same two points, generic word for any body. Perihelion/aphelion = around the Sun. Perilune/apolune = around the Moon. *Same idea, different suffix per planet.*

Inclination (i) — the tilt angle between the orbit's plane and the reference plane (the equator, for Earth). 0° flat, 90° over the poles.

Ecliptic — the flat sheet Earth's orbit around the Sun lies in. The reference plane for orbits around the Sun. *Not the same as the equator; they're tilted 23.4° apart.*

Reference direction / vernal equinox (♈) — a fixed "12 o'clock" direction in space to measure angles from: the direction to the Sun on the first day of spring. The ♈ symbol (Aries) marks it on diagrams. *North on a compass, but for space.*

Longitude of the ascending node (Ω, capital omega) — the angle, measured in the reference plane from ♈, to the ascending node. The orbit's swivel. *Which direction the hoop's "up-crossing" points.*

True anomaly (ν, nu) — the angle from perigee to where the spacecraft is right now. *"Anomaly" is old astronomy-speak for "angle along the orbit." Nothing is wrong.*

Circular / elliptical / hyperbolic — orbit shapes by e: circle (e = 0), ellipse (0 < e < 1), hyperbola (e > 1, an open swing-past like a flyby).

Projection — looking at a 3-D orbit flattened onto a 2-D page, like a shadow. Why the slide pictures can be hard to read.

Geometry (of an orbit) — its size, shape and orientation: a, e, i, Ω, ω. Everything except where the spacecraft currently is.

SEC 3 · LEC 1 · SAY IT OUT LOUD

Page 4: prove it by talking

Do this AFTER the spread is written and the 3-line summary is done, with the notebook and this PDF closed. Send me what you said (typed or dictated) and I fill the box at the bottom.

🗣 The friend's opening question

"You're taking a spacecraft course? What did you learn about orbits?"

✅ Three sentences you should be able to say

- An orbit is falling and missing: you go sideways so fast the ground curves away under you.
- You can describe any orbit with six numbers: how big, how squished, how tilted, which way the tilt faces, which way the squished end points, and where the spacecraft is right now.
- The closest point is perigee, the farthest is apogee, and the size of the orbit is basically the average of the two.

No Greek letters required. If a friend asks for the names, that's a bonus, not the test.

❓ Follow-ups a friend might ask

- "Wait, why doesn't it just fall down?"
- "So is the ISS going in a circle or an oval?" (nearly circular, e ≈ 0)
- "What's the difference between tilt and which-way-the-tilt-faces?" (inclination vs Ω: tilt the hoop, then swivel it)
- "Why six numbers and not three?" (three for the shape and size won't tell you how it's turned in space)
- "If you only changed one number, which one changes every second?" (where the spacecraft is: true anomaly)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Rule from here on: a lecture isn't "done" until this page has something in both boxes.

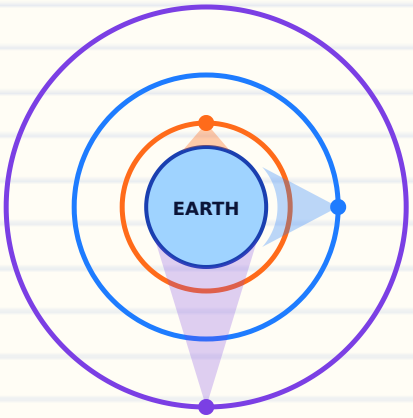
SEC 3 · LEC 2 · EARTH ORBITS + CONSTELLATIONS

Higher = sees more, moves slower, sees less detail

Left = your left notebook page. Right = your right page. You already have LEO/MEO/GEO altitudes from Section 1; the new part is periods, the trade-off, and why constellations exist.

LEFT PAGE · DRAW

S3 / L2 — Earth orbits **date · pg __**



LEO 200–2,000 km
period 1.5–2 h · ISS 400 km, 92 min
tiny footprint, sharp detail, fast

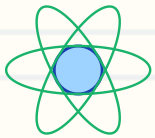
MEO 2,000–35,786 km
period 2–24 h · GPS 20,200 km, 12 h
navigation + comms

GEO 35,786 km
period 24 h = Earth's spin
hangs over one spot · over EQUATOR
sees 1/3 of Earth at once

THE TRADE-OFF (higher ↑)
↑ coverage (bigger footprint)
↓ speed · ↓ detail · ↑ radio delay

LEO sees one spot for minutes → need MANY sats = a CONSTELLATION (Iridium 66 · GPS 24+ · Starlink 1000s)

GEO: 3 sats ≈ whole Earth (not the poles)



constellation: many rings, someone always overhead

⭐ **3 TAKEAWAYS** 3 lines

1. Higher orbit = sees more, moves slower, sees less detail, longer radio delay
2. 35,786 km is the one altitude where one lap = one Earth spin, so GEO hangs still
3. LEO can't watch one place for long, so you fly a swarm: a constellation

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions) **NOTES (keywords)**

3 Earth-orbit bands: altitude + period of each?

Why exactly 35,786 km and not a round number?

Go higher: what gets better, what gets worse?

Why does LEO need a constellation and GEO doesn't?

ISS period? GPS period?

GEO has to sit over what? Why?

Name two constellations and what they're for.

? how many GEO sats to cover the whole planet

Earth orbit = goes around Earth, not the Sun (like the Moon does)
LEO 200–2,000 km · 1.5–2 h · ISS 400 km / 92 min · close, fast, small view, sharp detail
MEO 2,000–35,786 km · 2–24 h · GPS 20,200 km / 12 h · navigation + comms
GEO 35,786 km · 24 h (really 23 h 56 m) · matches Earth's spin → hangs over one spot · must be over the equator
why 35,786: the one height where one lap = one Earth turn (higher = slower, Kepler 3 later)
trade-off: ↑ altitude → bigger view, slower, blurrier, more radio delay · ↓ altitude → sharp, fast, tiny view, drag
LEO sees one spot for minutes → fly many = constellation (Iridium 66 @ 780 km · Starlink 1000s @ 550)
GPS: 24+ sats, 6 rings, 12 h → ≥ 4 always overhead → your phone finds itself · Russia GLONASS, EU Galileo, China BeiDou
3 GEO sats ≈ whole Earth (poles excluded)
no "best" orbit: pick by mission (detail vs coverage vs delay vs cost) · instructor's "36,000 km from center" is wrong: 35,786 is ALTITUDE; from center ≈ 42,164

SEC 3 · LEC 2 · TEACH + FACT CHECK

Three cameras over a football field

Start with the box. Then the detail, the caption errors, and where the instructor was loose.

🧒 Like you're 5

Imagine filming a football game three ways. A **drone ten feet up** sees one player in crisp detail, but it has to zip around fast and it only ever sees one little patch. That's LEO. A camera on a **tall pole** sees half the field, a bit blurrier, and moves slower. That's MEO. A **blimp so high** it sees the whole stadium at once, tiny and blurry, and it just hangs there. That's GEO.

Now: what if you want crisp detail *and* the whole field *all the time*? One drone can't do it. So you fly a **swarm of drones** taking turns, so there's always one over every part of the field. That swarm is a **constellation**. GPS is a swarm. Starlink is a swarm. Iridium is a swarm.

🔭 The three bands, and the one rule behind them

- **The rule:** the higher you orbit, the slower you go and the longer one lap takes. LEO laps in 90 minutes. GPS laps in 12 hours. GEO laps in exactly one day. (Kepler's third law, which the next lecture will name.)
- **LEO, 200–2,000 km, 1.5–2 h.** Close means sharp pictures and short radio hops, but a tiny view and only a few minutes over any one place. The ISS at 400 km passes over Florida and is back 92 minutes later. Most satellites ever launched live here.
- **MEO, 2,000–35,786 km, 2–24 h.** The middle ground. GPS sits at 20,200 km with a 12-hour lap: high enough that a dozen satellites see most of the planet, low enough that the signal isn't too weak or too late. Russia (GLONASS), Europe (Galileo), and China (BeiDou) run their own.
- **GEO, 35,786 km, 24 h.** One lap takes exactly one Earth rotation, so from the ground the satellite never moves. TV dishes bolt to a wall and never adjust. It must sit over the equator; anywhere else, the ground would slide underneath it as Earth turns. One GEO satellite sees about a third of the planet; three cover nearly all of it except the poles.
- **Why 35,786 exactly?** It's not chosen, it's calculated. It's the only altitude where the lap time equals one Earth spin. Lower is too fast, higher is too slow. Callback to your Section 1 page: you wrote this number down before you knew why.

🛰️ Constellations

- **Iridium:** 66 satellites at 780 km in 6 rings over the poles. Satellite phones that work anywhere on Earth, including the oceans.
- **GPS:** 24 required, ~31 flying, 6 rings tilted 55°. Arranged so at least 4 are above you at all times, which is the minimum your phone needs to compute a position.
- **Starlink:** thousands at ~550 km for internet. Same idea at a much bigger scale.

🔎 Caption errors

Captions said	Instructor meant
"meal"	**MEO**
"The Sims or your entity"	"it **seems arbitrary**" (200, 2,000 look round; why 35,786?)
"Leo's"	LEOs
"the audio of the GEO orbit"	the **idea** of the GEO orbit
"lower level bit"	low Earth orbit
"care a compromise"	**make** a compromise

▲ Where the instructor was loose

- **"Thirty-four, thirty-six thousand km from the center of Earth."** No. 35,786 km is the *altitude* above the surface. From the center it's about 42,164 km (add Earth's 6,378 km radius). He mixed the two, which is exactly the mistake your Section 1 notes warned about.
- **"GEO period 24 hours."** Close enough for conversation; the exact value is 23 h 56 min, one true rotation relative to the stars (the sidereal day from your Section 1 vocab). Four minutes matters to the people who park the satellite.
- **"At 1,000–2,000 km, 5, 6, 10-hour period."** Muddled. At 2,000 km the lap is about 2 hours. The 5–10-hour laps are up around 8,000–15,000 km. The slide's "2 to 24 h" range is right; the spoken example wasn't.
- **"Always remain at the same altitude, so the same relative position."** Constant altitude is true of any circular orbit. What makes GEO special is constant position *over the ground*, which comes from matching Earth's spin, not from altitude staying fixed.
- **Correct and worth keeping:** the slide picture isn't to scale, and

SEC 3 · LEC 2 · VOCABULARY

Every word used, in plain English

Several of these are repeats from Section 1. That's deliberate: say them again out loud. Copy only the new ones into the word bank.

Earth orbit — any path that circles Earth rather than the Sun. The Moon is in one; so is every satellite. *Opposite of an interplanetary trajectory.*
Period — the time for one complete lap. 92 minutes for the ISS, 12 hours for GPS, one day for GEO. *The orbit's "day."*
LEO / MEO / GEO — low, medium, geostationary Earth orbit: the three altitude bands (200–2,000 km, 2,000–35,786 km, 35,786 km). *Downstairs, middle floor, penthouse.*
Geostationary — a GEO orbit that is also over the equator, so the satellite seems frozen in the sky. *Repeat from Section 1: geosynchronous = one-day lap; geostationary = one-day lap AND over the equator.*
Equatorial plane — the flat sheet through Earth's middle. GEO satellites must lie in it. *The table the hula hoop sits flat on.*
Rotation period / "Earth rotation velocity" — how long Earth takes to spin once: 23 h 56 min against the stars, 24 h against the Sun. A GEO satellite's lap matches it.
Altitude vs distance from center — altitude is height above the ground; distance from center adds Earth's radius (6,378 km). GEO: 35,786 km altitude, 42,164 km from center. *The instructor mixed these up.*
Coverage / footprint — the patch of Earth a satellite can see or talk to at one moment. Tiny for LEO, a third of the planet for GEO. *The circle of light from a flashlight: closer = smaller circle.*
Global coverage — being able to see or serve every point on Earth, all the time. One satellite can't; a constellation can.
Resolution — how much detail a camera can make out. Closer = better. A LEO satellite can see a car; a GEO weather satellite sees whole storm systems. *Reading a sign from 10 feet vs 10 miles.*
Constellation — a group of satellites in planned orbits that work as one system, so at least one is always where it's needed. *A swarm of drones taking turns.*
Orbital plane / "ring" — the flat sheet one orbit lies in. Constellations spread satellites across several planes (GPS: 6) so they're not all bunched on one side of Earth.
Navigation satellite — broadcasts a precise time signal; your receiver hears several and triangulates where it is. Needs at least 4 overhead. *Repeat from Section 1.*
GPS / GLONASS / Galileo / BeiDou — the American, Russian, European, and Chinese navigation constellations. All in MEO around 20,000 km. Your phone uses several at once.
Iridium — a company and its constellation: 66 satellites at 780 km giving satellite-phone coverage everywhere on Earth. Named after element 77 because the original plan was 77 satellites.
Starlink — SpaceX's internet constellation, thousands of satellites at ~550 km. *Repeat from Section 1.*
ISS — International Space Station, 400 km, 92-minute lap. *Repeat.*
Data relay / transfer time — passing information from satellite to ground (or satellite to satellite). Higher orbit = longer radio trip = more delay. GEO adds about a quarter second each way.
Trade-off / compromise — you can't have every good thing at once; improving one (coverage) costs another (detail, delay). Orbit choice is a trade-off. *Big truck vs sports car.*
Optimize — tune the design to get the best balance for what the mission actually needs. There's no universally "best" orbit, only the best for this job.
Give or take — spoken shorthand for "approximately." He said LEO periods are "1.5 to 2 hours, give or take."

SEC 3 · LEC 2 · SAY IT OUT LOUD

Page 4: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I fill the box.

🗣 The friend's opening question

"Why are some satellites really low and some really high? Does it matter?"

✅ Three sentences you should be able to say

- The higher you go, the more of Earth you can see at once, but you move slower, you see less detail, and your radio signal takes longer.
- There's one height, about 36,000 km, where one lap takes exactly one day, so the satellite just hangs over the same spot. That's where TV and weather satellites sit, always over the equator.
- Low satellites only see a tiny patch for a few minutes, so if you want to cover the whole planet you fly a whole swarm of them. That's what GPS and Starlink are.

❓ Follow-ups a friend might ask

- "So why doesn't the ISS just hang over one spot too?" (it's too low; one lap is 92 min, so it's gone in minutes)
- "Why does the GPS one have to be in the middle?" (high enough that a few dozen see the whole planet, low enough the signal's strong and fresh)
- "How many satellites does my phone need to see?" (at least 4)
- "Why over the equator? Couldn't it hang over New York?" (Earth's spin would drag New York away underneath; only the equator moves in a matching circle)
- "Is 36,000 km from the ground or from the middle of the Earth?" (ground; from the middle add ~6,400)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Review link: this lecture repeats your Section 1 satellite page. Flip back to it first and see what you can still answer cold.

SEC 3 · LEC 3 · INTERPLANETARY TRANSFERS + FLYBYS

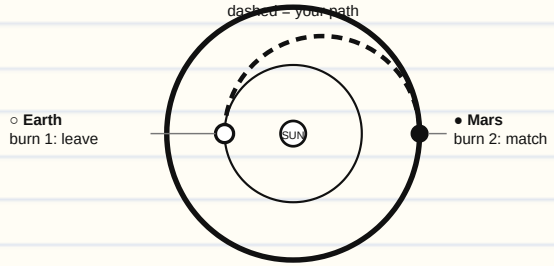
Lazy throws and bouncing off moving planets

Left = your left notebook page. Right = your right page. Draw both panels before you look closely at mine.

LEFT PAGE · DRAW

S3 / L3 — Transfers & flybys **date · pg __**

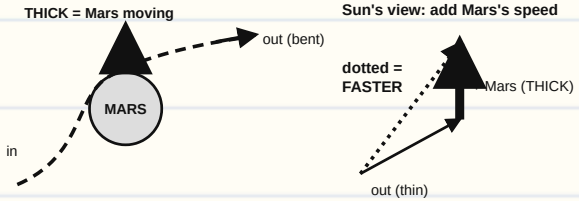
1 · HOHMANN = the lazy throw



Half an ellipse, 180° around the Sun. Cheapest Δv.
Earth → Mars ≈ 8.5 months = half the ellipse's period (your S1 quiz).

LAMBERT'S PROBLEM = the "can I get there by then?" calculator
give: 2 positions + a travel time → get: the one orbit linking them → read off the Δv
Mars in 20 days? The orbit exists. The Δv is absurd. The rocket can't be built. → NO
fast = expensive · slow = cheap · more speed at the same spot → bigger orbit → reach farther

2 · FLYBY = bounce off a moving planet
Mars's view: speed in = speed out, only the direction bends.



✏️ **PENCIL KEY** · thin solid = Earth's orbit · thick solid = Mars's orbit · dashed = your path · ○ hollow dot = start · ● filled dot = target · THICK arrow = the planet moving · dotted arrow = the result

⭐ **3 TAKEAWAYS** 3 lines

1. Getting there fast costs fuel; the Hohmann half-ellipse is the cheapest, slowest sensible path
2. A flyby bends your path around a planet and, because the planet is moving, hands you some of its speed
3. More speed at the same spot = bigger orbit = farther reach; the price of flybys is years of travel

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions) **NOTES (keywords)**

Lambert's problem: what do you give it, what does it give back?

Why can't we fly to Mars in 20 days?

Hohmann: what shape, what angle, how many burns, why cheapest?

How long is a Hohmann to Mars? What's it got to do with the S1 quiz?

Flyby: in Mars's eyes does the speed change? In the Sun's eyes? Where does the extra come from?

More speed at the same spot → what happens to the orbit?

Why did Rosetta take 10 years?

? can a flyby SLOW you down

transfer = move from one orbit around the Sun (Earth's) to another (Mars's)
Lambert's problem: give 2 positions + a travel time → math returns the ONE orbit linking them → tells you the Δv
fast = expensive: Mars in 20 days needs absurd Δv → rocket too big → not doable
Hohmann = half-ellipse touching both orbits · depart/arrive 180° apart · 2 burns · cheapest Δv (= your perigee/apogee burn line, around the Sun)
Earth→Mars Hohmann ≈ 8.5 months · TOF = half the ellipse's period (S1 quiz) · window every 26 months
Hohmann isn't required, it's the cheapest · real choice = trade: time vs fuel
far targets (Jupiter+): Hohmann Δv still too big → flyby / gravity assist instead
flyby: Mars's view → speed in = speed out, direction bends · Sun's view → add Mars's own speed → faster (pass the other side → slower)
more speed at same spot → bigger a → reach farther · energy stolen from the planet (it slows by nothing you could measure)
Rosetta (ESA): launch 2004 → Earth '05, Mars '07, Earth '07, Earth '09 → comet 2014 · cost of flybys = TIME

SEC 3 · LEC 3 · TEACH + FACT CHECK

You don't aim at Mars. You aim at where Mars will be.

Start with the box. Then the three ideas, the caption errors, and where the instructor was loose.

🧒 Like you're 5

You're on a merry-go-round and your friend is on a bigger one around the same pole, going slower. You want to throw them a ball. You could hurl it straight at them, hard, but you'd have to throw *really* hard and it'd arrive going way too fast to catch. Or you could lob it gently so it drifts outward on a long curve and arrives just as your friend comes around. The gentle lob takes longer but barely any effort. That lob is a **Hohmann transfer**, and it's how we send things to Mars.

Now the second trick. If a ball bounces off a wall, it comes back at the same speed. But if it bounces off the **front of a moving truck**, it comes back faster, because the truck gave it a push. Planets are moving trucks. Swing a spacecraft close past one and it comes out faster than it went in. That's a **flyby**, and it's how we reach Jupiter and beyond without a rocket the size of a city.

🎯 The three ideas

- **Lambert's problem** is the "can I get there by then?" calculator. You hand it where you are, where you want to be, and how long you're willing to take. It hands back the one orbit that connects those two points in that time, and from the orbit you read off how much speed change (Δv) you'd need. Ask it for Mars in 20 days and it happily gives you an orbit, but the Δv is tens of km/s, which no rocket can carry. The math works; the fuel doesn't exist. Fast is expensive. Slow is cheap.
- **The Hohmann transfer** is the cheapest answer: half an ellipse that just kisses Earth's orbit on the inside and Mars's orbit on the outside. You leave Earth at one end and arrive at Mars at the other, 180° around the Sun. Two burns: one to leave, one to match Mars. This is exactly your line from last lecture, scaled up to the Sun: burn at the low point to raise the high point. For Mars it takes about 8.5 months, which is half the period of that ellipse, which is the time-of-flight formula from your very first quiz. And because Mars has to be in the right place when you arrive, you can only launch every 26 months.
- **The flyby / gravity assist.** Seen from Mars, a passing spacecraft comes in at some speed and leaves at the same speed, just bent in a new direction. Seen from the Sun, though, Mars itself is moving at 24 km/s, and the bend lines the spacecraft up to carry some of that along. More speed at the same distance from the Sun means a bigger orbit (your semi-major axis grows), which means you can reach farther out. The energy comes from Mars, which slows down by an amount nobody could ever measure. Pass behind the planet to speed up; pass in front to slow down (that's how you brake at Mercury).

⚡ Say it as: fast costs fuel, Hohmann is the lazy throw, flybys

🔎 Caption errors

Captions said	Instructor meant
"human / home and / common / whole and transfer"	**Hohmann** transfer (every time)
"their limits problem"	**Lambert's** problem
"a connex approximation"	a **conic** approximation (ellipses, hyperbolas)
"velocity of rotation of mass"	velocity of **Mars**
"destroying"	"this drawing"
"the opposite"	the **orbit**
"flew to a comet in 2010"	launched 2004, arrived at the comet 2014 (2010 was an asteroid flyby on the way)
"get to an asteroid"	get to a **comet**

▲ Where the instructor was loose

- **"Lambert's problem takes into account the Δv and the time."** Backwards. Time (and the two positions) go *in*; the orbit and its Δv come *out*. Small, but it's the difference between understanding the tool and just naming it.
- **"Hohmann is the most efficient transfer."** True for almost every real case. (For a target more than ~12× farther out, a three-burn path can beat it. You'll never need this; just know "almost always.")
- **"Velocity and semi-major axis are related directly."** Right idea, right conclusion. The precise statement is that orbital *energy* sets the size, and more speed at a given spot means more energy. Same thing in practice.

SEC 3 · LEC 3 · VOCABULARY

Every word used, in plain English

Two people's names are in here (Hohmann, Lambert). They're just labels on ideas. Copy only the new words into the word bank.

Transfer / transfer orbit — the path you fly to get from one orbit to another, like from Earth's orbit around the Sun to Mars's. *The highway between two parking lots.*
Interplanetary — between planets, orbiting the Sun rather than Earth. *Repeat.*
Lambert's problem — the classic "find the orbit" puzzle: given a start point, an end point, and a travel time, what orbit connects them? Named after Johann Lambert, an 18th-century mathematician. *The "can I get there by Tuesday?" calculator.*
Δv (delta-v) — the total speed change a trip requires, which is really the fuel bill. *Repeat. Still the currency of spaceflight.*
Propulsion / propellant — the engine, and the fuel it burns. "Not enough propulsion" means "can't carry enough fuel to make that Δv."
Input / constraint — what you feed into a calculation / a limit you can't break (rocket size, fuel mass, time). *Repeat.*
Conic / conic approximation — treating every orbit as one of the shapes you get by slicing a cone: circle, ellipse, parabola, hyperbola. "Patched conics" = stitching an Earth-centered ellipse to a Sun-centered one to a Mars-centered hyperbola. *Lego bricks of orbit shapes.*
Hohmann transfer — the half-ellipse path that touches both the starting and ending orbit, with departure and arrival 180° apart and two burns. Cheapest in fuel. Worked out by Walter Hohmann, a German engineer, in 1925. *The lazy throw.*
Tangent — just touching, not crossing. The Hohmann ellipse is tangent to both circles, which is why the burns are small: you're already moving along the right line.
Optimal / efficient / most efficient — in this lecture, always means "least Δv," i.e. least fuel. Not fastest.
Time of flight (TOF) — how long the trip takes. For a Hohmann it's half the ellipse's period. *Your Section 1 quiz question.*
Launch window / synodic period — the stretch of days when the planets are lined up right for a Hohmann. Earth–Mars: about every 26 months. *Repeat from S1 primer.*
Phase angle — how far ahead the target planet must be when you launch so it's there when you arrive. Mars: ~44° ahead. Not on the slide; it's why the window exists.
Flyby / gravity assist / slingshot — three names for one trick: swing close past a planet so its gravity bends your path and, because the planet is moving, changes your speed relative to the Sun. *Bouncing off the front of a moving truck.*
Gravity field — the region around a planet where its pull matters. It's what bends the flyby path.
Relative velocity — your speed as seen from something else (Mars, say) rather than from the Sun. The flyby doesn't change your speed relative to Mars, only relative to the Sun. *Repeat.*
Vector — an arrow: a quantity with both a size and a direction. Velocity is a vector. The slide's arrow pictures are vector addition: spacecraft-relative-to-Mars plus Mars-relative-to-Sun equals spacecraft-relative-to-Sun. *Two pushes added tip to tail.*
Magnitude / direction — the length of the arrow / which way it points. A flyby changes the direction in Mars's frame and the magnitude in the Sun's frame.
Hyperbola — the open, swing-past shape of a flyby path near a planet ($e > 1$). *Repeat.*
Semi-major axis (a) / "mars energetic orbit" — size of the orbit. Mars speed at the same spot — more energy

SEC 3 · LEC 3 · SAY IT OUT LOUD

Page 4: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I fill the box.

🗣 The friend's opening question

"How do we even get to Mars? Don't you just point the rocket at it and go?"

✅ Three sentences you should be able to say

- No, you aim at where Mars is going to be. The cheapest way is a long lazy curve that drifts outward and meets Mars halfway around the Sun, about eight and a half months later.
- You could go faster, but every bit faster costs a huge amount more fuel, and past a point the rocket just can't be built. Fast is expensive, slow is cheap.
- For really far places like Jupiter, you cheat: you swing close past a planet and it slings you out faster, like bouncing a ball off the front of a moving truck. It costs no fuel, just years of travel.

❓ Follow-ups a friend might ask

- "Why can't you just go whenever?" (Mars has to be in the right spot when you arrive; the lineup comes every 26 months)
- "Where does the extra speed from a flyby come from? Isn't that free energy?" (from the planet; it slows by an unmeasurable amount)
- "Can a flyby slow you down?" (yes, pass in front of the planet instead of behind)
- "So why did it take Rosetta ten years?" (four flybys to build up speed for free instead of carrying fuel; time was the price)
- "What's Lambert's problem?" (a calculator: two points plus a time, gives you the orbit and the fuel)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Section 3 is done after this. Next: the Section 3 conversation test, five questions from me, party-style.

SPACE X · SECTION 4 OF 7

Section 4 · Space propulsion

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

Every engine throws mass backward. Chemical pushes hard and is thirsty; electric is gentle and efficient; nuclear never flew.

The lectures

#	Lecture	Page	Length
1	**Propulsion: definitions and types**	34	6 pages
2	**Chemical engines**	40	7 pages
3	**Electric propulsion and ion thrusters**	47	7 pages
4	**Nuclear thermal rockets**	54	7 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Number to know
$F = \dot{m} \times v_e$ · Isp: chemical 450 s, electric 3,000 s, nuclear 900 s

Don't repeat
"nuclear rockets are efficient because they run hotter" (they run cooler; the exhaust is lighter)

Talk card · 4 min

He opens with: "How does a rocket even work in space? There's nothing to push against."

- 1 · THROW IT BACK — skateboard + bowling ball · $F = \dot{m} \times v_e$ · no air needed
- 2 · CHEMICAL — fuel + oxidizer · throat Mach 1 · bell · 450 s max · tonnes per second
- 3 · ELECTRIC — ionize · grids 1,100 V → 40 km/s · neutralizer · 9 grams · power-limited
- 4 · NUCLEAR — reactor heats H_2 · light = fast · ~900 s · never flown · not Orion
- LAND — "push vs mileage; energy from propellant, panels or reactor"

Land on: Every engine is a way of throwing mass backward. Chemical gives push, electric gives mileage, nuclear would give some of both, and the mission decides which you need.

In the app: scenes 8, 9, 10 in Learn, and the "Section 4" talk in Teach.

SEC 4 · LEC 1 · PROPULSION: WHAT + TYPES

Throw something backward, go forward

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L1 — Propulsion: what + types **date · pg __**

1 · THRUST = throw mass backward



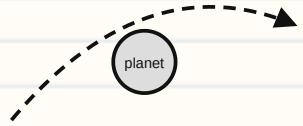
ACTION
gas thrown backward

REACTION
ship pushed forward

It pushes on its OWN exhaust. No air needed.

thrust $F = \dot{m} \times v_e$ (kg thrown per second × exhaust speed)

2 · TWO WAYS TO GET A Δv

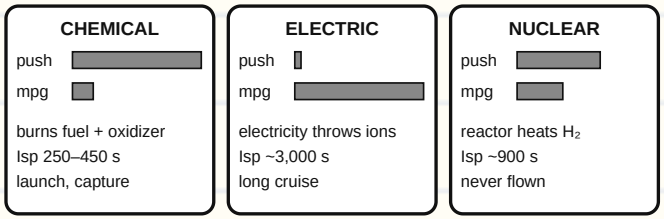


SLINGSHOT (gravity)
free, but the planet must be there



THRUST (engine)
any time, costs propellant

3 · THREE ENGINE FAMILIES



✏️ **PENCIL KEY** · thin arrow = exhaust · thick arrow = ship's motion · dashed = flight path · gray bar = how much · mpg = Isp

⭐ **3 TAKEAWAYS** 3 lines

1. Orbit change = Δv: from a planet's gravity (slingshot) or from thrust.
2. Thrust = Newton's 3rd law: throw mass back, ship goes forward. No air.
3. Chemical strong + thirsty · electric weak + efficient · nuclear between.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Why does a spacecraft need propulsion at all?	space propulsion = making thrust to change velocity → change orbit (a Δv)
Two ways to get a Δv?	Δv two ways: slingshot (planet's gravity, free, planet must be in place) · thrust (engine, any time, costs propellant)
How is thrust made? Which law?	thrust = Newton's 3rd law: throw mass back → ship goes forward · pushes on its OWN exhaust, not on air
"There's nothing to push against in space." What's wrong with that?	propellant = the stuff you throw out the back
The three engine families?	chemical (rockets): burn fuel + oxidizer · big thrust · low Isp 250–450 s
Which one pushes hardest? Which gets the best mileage?	electric / low-thrust: electricity throws ions very fast · tiny thrust (grams) · high Isp ~3,000 s · runs for years
What decides which engine a mission uses?	nuclear: reactor heats hydrogen · Isp ~900 s · political · ground-tested, never flown
? is electric propulsion really "new"	requirements: the mission picks the engine · launch / landing / capture → chemical · long slow cruise → electric
	the trade: thrust vs efficiency, you can't max both

SEC 4 · LEC 1 · TEACH

A rocket pushes on its own exhaust

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Sit on a skateboard holding a bowling ball. Throw the ball **backward** as hard as you can. You roll **forward**. You didn't push on the ground or the air. You pushed on the ball, and the ball pushed back on you. That is every rocket ever built: the ball is the gas it throws out the back. Throw it faster, or throw more of it each second, and you get a bigger push. That push is called **thrust**.

There are three kinds of thrower. **Chemical** is a fire hose: a huge amount of stuff, pretty fast, empty in minutes. **Electric** is a pea-shooter that fires each pea ten times faster than the hose: the push is tiny, about the weight of a few sheets of paper, but one small bag of peas lasts for years. **Nuclear** is a kettle heated by a reactor instead of a fire: it throws its steam about twice as fast as the fire hose can.

🏷 The official names (name first, then what it means)

- **Space propulsion** (spacecraft propulsion, in-space propulsion): any system that changes a spacecraft's velocity.
- **Thrust (F)**: the force an engine makes, in newtons (N). 1 N is about the weight of an apple.
- **Newton's third law of motion** (action–reaction, the reaction principle): forces come in equal and opposite pairs. The deeper rule behind it is **conservation of momentum**.
- **Propellant** (reaction mass, working fluid): the stuff thrown out. A **reaction engine** is any engine that works by throwing it.
- **Chemical propulsion** (chemical rocket; the instructor's "chemical internal heat engine"): the energy comes from burning the propellant itself.
- **Electric propulsion, EP** (low-thrust propulsion): electricity accelerates the propellant. Main kinds: the **ion thruster** (gridded ion engine) and the **Hall-effect thruster**.
- **Nuclear thermal propulsion, NTP** (nuclear thermal rocket): a fission reactor heats hydrogen. Its cousin **nuclear electric propulsion, NEP** uses a reactor to power electric thrusters.
- **Gravity assist** (slingshot, flyby, swing-by): the no-propellant way to change velocity. Repeat from Section 3.
- **Requirements**: what the mission needs the engine to do. Choosing against them is a **trade study**.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Momentum is kept	Momentum = mass × velocity, and the total can't change. A 1,000 kg ship at rest throws 1 kg backward at 3,000 m/s. 1 × 3,000 = 999 × v, so the ship moves forward at **3 m/s**. Air appears nowhere in that sum.
2 · Thrust	Thrust is how fast you throw momentum away: **$F = \dot{m} \times v_e$**. Throw 0.1 kg every second at 3,000 m/s and F = **300 N**.
3 · Δv	a = F ÷ m = 300 ÷ 1,000 = 0.3 m/s². Burn 10 s: Δv = 3 m/s, using 1 kg. Same answer as step 1. Thrust and time build the Δv your orbit change needs.
4 · Why electric is weak	Same 300 N from 0.01 kg/s at 30,000 m/s uses ten times less propellant. But the jet's power is $\frac{1}{2}\dot{m}v_e^2$: 0.45 MW for the first, **4.5 MW** for the second. A chemical engine gets that from the fire. An electric one needs solar panels, and the whole ISS makes about 0.1–0.2 MW. So real electric thrusters run at a few kW and push with a fraction of a newton.
5 · Real electric	NASA's NEXT ion thruster (the slide photo): 6.9 kW in, **0.236 N** out (the weight of 5 sheets of paper), exhaust at 41 km/s, Isp 4,190 s. The Dawn probe took 4 days to go 0–100 km/h, then thrust for 5.9 years: total Δv **11.5 km/s** from 425 kg of xenon.
6 · Why nuclear wins on speed	Exhaust speed goes as √(temperature ÷ molecule mass). Burning hydrogen and oxygen makes steam, mass 18. A reactor heats plain hydrogen, mass 2, to about 2,700 K. That is cooler than the flame (3,500 K), but far lighter molecules fly faster: Isp ≈ **900 s** against 450 s.
7 · What mileage buys	For a 5 km/s Δv the rocket equation says: Isp 320 s needs **80%** of the ship to be propellant; Isp 3,000 s needs **16%**.

SEC 4 · LEC 1 · COMPARE + FACT CHECK

Strong or efficient: pick one

The comparison to know cold, the caption errors, and where the instructor was loose or wrong.

⚖️ The three families side by side

Family	Thrust	Isp	Burns for	Energy comes from	Used for	Flown?
Chemical	newtons up to millions of newtons	200–450 s	seconds to minutes	the propellant's own chemical bonds	launch, landing, capture burns, anything that must happen fast	Yes, on every mission
Electric	thousandths of a newton up to about 1 N	1,500–4,000+ s	months to years	solar panels (or a reactor)	station-keeping, orbit raising, deep-space cruise: Dawn, DART, BepiColombo, Psyche, Starlink	Yes, since 1964
Nuclear thermal	tens to hundreds of thousands of newtons	about 850–900 s	minutes to hours	a fission reactor heating hydrogen	proposed for fast crewed Mars trips	Ground tests only (NERVA, 1960s–1972)

🔎 Caption errors

Captions said	Instructor meant
"tweeting definitions"	**treating** definitions
"space robots"	space **propulsion**
"space proportioned / reapportion / abortion"	space **propulsion** (every time)
"accelerating the spacecraft B or thrust"	accelerating the spacecraft **via thrust**
"So how stressed up"	"So how is **thrust obtained**?"
"intersecting some kind of Propulsid propeller"	**ejecting** some kind of **propellant**
"with Sky interventions"	**which kind of engines**
"we want to feel"	we want to **fulfil**

📒 Question bank cards

- A rocket in space has no air behind it. What does it push against?
- Thrust formula, and the two ways to make thrust bigger?
- Three engine families: rank them by thrust, then by Isp.
- Why is electric thrust so small? (Think power.)
- Why does a nuclear thermal rocket beat a chemical one on Isp?
- You need to brake into Mars orbit in 30 minutes. Chemical or electric, and why?

▲ Where the instructor was loose

- **"We need propulsion to increase velocity."** Half of it. You also need it to slow down (braking into orbit, landing) and to change direction. Δv is a change in velocity, any direction.
- **"Electric propulsion… I think it was already implemented in the 1980s… kind of new."** Wrong by two decades. Ion engines were first fired in space in **1964** (NASA's SERT-1; the Soviet Zond 2 used plasma thrusters the same year). Soviet weather satellites used Hall thrusters from 1971. Deep Space 1 (1998) was the first probe to cruise on an ion engine. Today most new communications satellites and every Starlink carry electric thrusters. It is mainstream.
- **"Nuclear… being developed especially by NASA right now."** The US built and ground-tested nuclear rockets (NERVA) until 1972. The recent NASA and DARPA flight demo, DRACO, was cancelled in 2025 as far as I know. No nuclear rocket has ever flown.
- **Not said: an RTG is not nuclear propulsion.** Voyager and New Horizons carry plutonium power packs (radioisotope thermoelectric generators). Those make electricity. They don't push anything.
- **Not said: the real sorting rule.** The three families differ in *where the energy comes from*: inside the propellant (chemical), from solar panels (electric), from a reactor (nuclear). That one idea explains the whole table above.
- **Not said: there are engines with no propellant at all.** A solar sail is pushed by sunlight. Rare, but it exists.
- **"Chemical internal heat engines"** is his own phrase. The term you will meet everywhere else is **chemical propulsion** or **chemical rocket**.

SEC 4 · LEC 1 · VOCABULARY

Every word used, in plain English

40 entries: the lecture's words plus every term I used on the other pages. "Repeat" = you have met it before. Names marked "Add to Who's Who" are missions or programs.

Space propulsion — any system that changes a spacecraft's velocity. *The spacecraft's legs.*

Velocity — speed plus direction. Changing either one counts as a velocity change. *Repeat.*

Delta-v (Δv) — the size of a velocity change; the currency every maneuver is priced in. *Repeat.*

Orbit change / maneuver — a planned burn (or flyby) that moves you from one orbit to another.

Gravitational field — the region around a body where its gravity pulls on you. *Repeat.*

Slingshot = gravity assist — swinging past a moving planet to change your speed and direction without propellant. *Repeat, scene 7.*

Thrust (F) — the force an engine produces, in newtons. *The push.*

Newton (N) — the unit of force. About the weight of one apple. 1 kN = 1,000 N, 1 MN = a million.

Accelerate / acceleration — to change velocity / how fast velocity changes (m/s^2). a = F ÷ m.

Newton's third law — every force has an equal and opposite partner. Push the gas back, the gas pushes you forward. *Skateboard and bowling ball.*

Action / reaction — the two halves of that pair: gas thrown back / ship pushed forward.

Momentum — mass × velocity. The total never changes unless something outside pushes. *The bookkeeping behind thrust.*

Eject / expel — throw out. A rocket ejects propellant.

Propellant = reaction mass — the material thrown out to make thrust. Not the same word as fuel: fuel is only the part that burns.

Reaction engine — any engine that works by throwing mass backward: rockets, ion thrusters, jets.

Exhaust — the propellant after it has left the engine. Its speed is the exhaust velocity, v_e.

Mass flow rate ($\dot{m}$, 'm-dot') — kilograms of propellant thrown out each second.

Chemical propulsion — engines whose energy comes from burning propellant. The instructor's 'chemical internal heat engines'. *Fire hose.*

Heat engine — any machine that turns heat into motion. A rocket nozzle does exactly that.

Rocket — a reaction engine that carries everything it throws, including its own oxygen.

Liquid propellant — propellant stored as liquid in tanks, as opposed to a solid block. *Next lecture.*

Electric propulsion (EP) — engines that use electricity to accelerate propellant. Very fast exhaust, very small thrust. *Pea-shooter.*

Low-thrust propulsion — the family name for engines that push gently for a very long time. Mostly means electric.

Ion / ionize — an atom with an electron knocked off, so electricity can grab and fling it / to make ions.

Ion thruster (gridded ion engine) — ionizes xenon gas and fires the ions out through charged metal grids at 30–40 km/s. *The blue glow in the slide.*

Hall-effect thruster — the other common electric engine; uses a magnetic field instead of grids. More thrust, a bit less Isp.

Xenon — a heavy, non-reactive gas. The usual propellant for ion engines because it is easy to ionize and store.

Nuclear propulsion — using a fission reactor as the energy source for an engine.

Nuclear thermal propulsion (NTP) — reactor heats hydrogen, hydrogen shoots out a nozzle. Isp about 900 s. *Reactor-heated kettle.*

Nuclear electric propulsion (NEP) — reactor makes electricity, electricity runs ion or Hall thrusters.

Fission / reactor — splitting heavy atoms to release heat / the machine that does it under control.

RTG — radioisotope thermoelectric generator: a plutonium power pack that makes electricity. Not an engine. *Voyager, New Horizons.*

Specific impulse (Isp) — engine efficiency in seconds: how much push you get per kilogram of propellant. *Miles per gallon. Full treatment next lecture.*

Requirements — what the mission needs: how much Δv, how fast, how heavy, how much power.

Trade-off / trade study — giving up one good thing to get another / the formal comparison of options. *Thrust against mileage.*

Controversial / political — here: people worry about launching a reactor, so nuclear engines need government approval.

NEXT — NASA's Evolutionary Xenon Thruster, the ion engine in the slide photo. A version flew on DART in 2021. *Add to Who's Who.*

Dawn / Deep Space 1 / DART — NASA probes that flew on ion engines: asteroid belt (2007), technology test (1998), asteroid impact test (2021). *Add to Who's Who.*

NERVA / DRACO — US nuclear rocket programs: NERVA ground-tested until 1972; DRACO, a planned flight demo, cancelled in 2025. *Add to Who's Who.*

SERT-1 / Zond 2 — the first electric thrusters fired in space, both in 1964 (US / Soviet). *Add to Who's Who.*

SEC 4 · LEC 1 · COPY INTO NOTES

Where each piece goes in the notebook

The spread on page 1 is the lecture page. Everything deeper is routed to a reserved page, same five routes as before.

A · Equations & key numbers page (p3–4)

```
F = ṁ × vₑ          thrust = kg per second × exhaust speed
                    0.1 kg/s × 3,000 m/s = 300 N
m₁v₁ = m₂v₂         momentum kept: 1 kg at 3,000 m/s back
                    → 1,000 kg ship gains 3 m/s
Isp: chemical 250–450 s · electric ~3,000 s · nuclear ~900 s
```

B · Section 4 map page (one sentence, written after the last S4 lecture)

```
propulsion: throw mass back → thrust F = ṁ × vₑ;
chemical = big ṁ, electric = big vₑ, nuclear = hot light gas
```

C · One-line word bank (new lines)

```
p__ · thrust (F) · engine's force, newtons
p__ · Newton's 3rd law = action–reaction · push gas back, go forward
p__ · propellant = reaction mass · the stuff thrown out
p__ · momentum · mass × velocity; total is kept
p__ · mass flow rate (ṁ) · kg thrown per second
p__ · chemical propulsion · burn it: big thrust, low Isp
p__ · electric propulsion (EP) · electricity flings ions: tiny thrust, high Isp
p__ · ion thruster / Hall thruster · the two main electric engines
p__ · xenon · heavy gas ion engines throw
p__ · nuclear thermal (NTP) · reactor heats H₂, Isp ~900 s, never flown
p__ · RTG · plutonium power pack; electricity, NOT thrust
p__ · trade study · comparing options against requirements
```

D · Question bank ("but how?" cards)

```
Q: What does a rocket push against in space?
A: Its own exhaust. Momentum is kept: gas goes back, ship goes forward.
Q: Why is electric thrust tiny if it is so efficient?
A: Jet power = ½ṁvₑ². Fast exhaust needs megawatts; panels give kilowatts.
Q: Why does nuclear thermal get ~900 s?
A: Cooler than a flame, but far lighter exhaust (H₂, mass 2,
   vs steam, mass 18) → about twice as fast.
```

E · Concept cards: next page, C17

C1–C16 filled the 16 pages in this notebook. C17 onward goes in the second notebook, or wherever you decided the cards live. 10 + 10 lines of 22.

WORD BANK · CONCEPT CARDS · scene 8 — **C17**

THRUST (F)

= the force an engine makes, in newtons

WHY orbits only change if a force acts.

Thrust is that force: momentum thrown away each second

\# 0.1 kg/s × 3,000 m/s = 300 N · ion engine 0.24 N · F-1 6.8 million N

≠ Δv (thrust = the push, Δv = the result)

→ F = ṁ × vₑ · a = F ÷ m

NEWTON'S THIRD LAW = action-reaction

= every force has an equal, opposite partner

WHY it is why a rocket works in vacuum: the ship pushes gas back, the gas pushes the ship forward

\# 1,000 kg ship throws 1 kg at 3,000 m/s → ship gains 3 m/s

≠ 'pushing on the air' (no air needed)

→ momentum · thrust · reaction engine

SEC 4 · LEC 1 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"There's no air in space. So what does a rocket even push against?"

✅ Three sentences you should be able to say

- It pushes against its own exhaust. That's Newton's third law: throw mass out the back and you get shoved forward, air or no air.
- The push is called thrust, and it is how much stuff you throw each second times how fast you throw it.
- There are three kinds of engine. Chemical rockets push hard but burn through propellant fast, electric engines push very gently but sip it for years, and nuclear sits in between and has never flown.

❓ Follow-ups a friend might ask

- "If electric is so efficient, why not use it for everything?" (the push is a few sheets of paper; you can't launch or brake into orbit with that)
- "Why is it so weak?" (fast exhaust takes huge power, and solar panels only give kilowatts)
- "Is the nuclear stuff on Voyager an engine?" (no, an RTG makes electricity)
- "Can you change orbit without an engine?" (yes, a gravity assist, but only when a planet is in the right place)
- "How do you choose?" (the mission's requirements: how much Δv and how fast you need it)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Next: Lecture 2, chemical engines. It turns "thrust" and "mileage" into two formulas.

SEC 4 · LEC 2 · CHEMICAL ENGINES (ROCKETS)

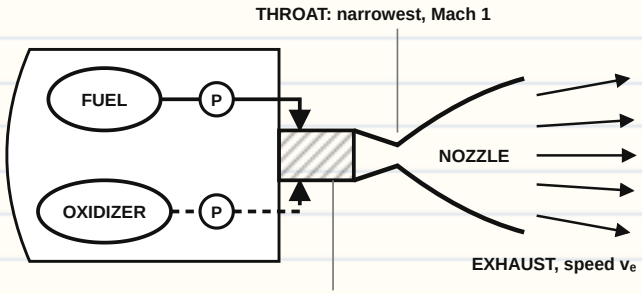
Fire, a squeeze, and a bell

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L2 — Chemical engines **date · pg __**

1 · LIQUID-PROPELLANT CHEMICAL ENGINE



P = pump
solid line = fuel · dashed = oxidizer

COMBUSTION CHAMBER
burns: ~3,300 K, 100+ bar

2 · THE NOZZLE TURNS PRESSURE INTO SPEED

CHAMBER	→	THROAT	→	BELL EXIT
pressure HIGH speed LOW		squeezed to the speed of sound		pressure LOW speed 3–4 km/s

hot + squeezed gas IN → fast gas OUT

3 · THE TWO NUMBERS

SPECIFIC IMPULSE

$$I_{sp} = v_e \div g_0$$

seconds · the mileage
g_0 = 9.81 m/s²
3,414 m/s → 348 s

THRUST

$$F = \dot{m} \times v_e$$

newtons · the push
$\dot{m}$ = kg thrown per second
287 × 3,414 ≈ 981,000 N

chemical: Isp LOW · $\dot{m}$ HUGE → thrust HUGE

✏️ **PENCIL KEY** · solid pipe = fuel · dashed pipe = oxidizer · hatched box = where it burns · thin arrows = exhaust · Ⓟ = pump

⭐ **3 TAKEAWAYS** 3 lines

1. Burn fuel + oxidizer, squeeze the gas through a throat, the bell makes it fast.
2. Isp = $v_e \div g_0$ is the mileage (s). $F = \dot{m} \times v_e$ is the push (N).
3. Chemical: low Isp, but tonnes thrown per second, so huge thrust.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What is a chemical rocket in one line?	chemical engine = reaction engine: stored propellant → high-speed jet (Newton 3)
Why carry an oxidizer?	propellant = fuel + oxidizer · no air in space → bring your own oxygen · e.g. liquid hydrogen + liquid oxygen
Path of the propellant, tank to exhaust?	tanks → pumps → combustion chamber (burns, ~3,300 K) → throat → nozzle
What does the nozzle do? Where is the gas fastest?	nozzle: pressure + heat → speed · bell shape lets the gas expand
Isp: formula, unit, what it measures?	Isp = $v_e \div g_0$ · g_0 = 9.81 m/s² · unit: seconds · = efficiency · higher v_e → less propellant for the same push
Thrust: formula? Two ways to raise it?	thrust $F = \dot{m} \times v_e$ · $\dot{m}$ = mass flow, kg per second · unit: newtons
"Low Isp but high thrust." How can both be true?	chemical: Isp low (250–450 s) but $\dot{m}$ huge (tonnes/s) → thrust huge
Solid vs liquid?	best thrust for the engine's weight · the propellant is the heavy part (~90% of a rocket)
? what do the two engines in the photos burn	solid rocket: fuel + oxidizer premixed in one block · can't shut off · often boosters
	e.g. SpaceX 2nd stage (1 engine) · Saturn V (5 engines) · both kerosene + liquid oxygen

SEC 4 · LEC 2 · TEACH

A fire with only one way out

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Blow up a balloon and let it go. Air squirts out the neck one way and the balloon shoots off the other way. A rocket engine is a balloon that keeps refilling itself by lighting a fire inside. A fire needs two things: something to burn (**fuel**) and oxygen. There is no air in space, so the rocket carries its own oxygen in a second tank (**oxidizer**).

The fire makes gas that is very hot and very squeezed, and it has only one way out: a small hole called the **throat**, then a bell called the **nozzle**. The bell is where the magic happens. It lets the squeezed gas spread out, and as it spreads it speeds up, to about four times the speed of a rifle bullet. Two numbers describe any engine: how hard it pushes (**thrust**) and how far it gets on each kilo of propellant (**specific impulse**). A chemical rocket is a drag racer: enormous push, terrible mileage.

🏷 The official names (name first, then what it means)

- **Chemical rocket engine** (chemical propulsion): makes thrust by burning propellant and expelling the gas.
- **Bipropellant** (biprop): an engine with two propellants, a **fuel** and an **oxidizer**. The slide's diagram is one. A **monopropellant** engine uses a single liquid that breaks down on a catalyst (hydrazine).
- **Cryogenic propellants**: liquids kept very cold. **LH_2** (liquid hydrogen, −253 °C) and **LOX** (liquid oxygen, −183 °C). **RP-1** is refined kerosene, stored at room temperature.
- **Turbopump** (the slide's "pressure pump"): a turbine-driven pump that forces propellant into the chamber. Small engines skip it and push with tank pressure: **pressure-fed**.
- **Combustion chamber** (thrust chamber): where the propellants mix and burn.
- **de Laval nozzle** (converging–diverging nozzle, bell nozzle): narrows to a **throat**, then widens. Turns pressure and heat into exhaust speed.
- **Exhaust velocity (v_e)** (the instructor's "velocity of ejection"): how fast the gas leaves.
- **Mass flow rate ($\dot{m}$)**, said "m-dot" (his "flux of propellant"): kilograms leaving per second.
- **Specific impulse (Isp)** = $v_e \div g_0$, in seconds. **g_0** is **standard gravity**, 9.80665 m/s².
- **Solid rocket motor** (SRM): fuel and oxidizer cast together as one solid block, the **grain**.
- **Thrust-to-weight ratio**: an engine's thrust divided by its own weight.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Two tanks	Burning is fuel joining with oxygen. Hydrogen + oxygen gives steam and 13 million joules per kg of mix. A jet plane takes its oxygen from the air; a rocket must carry it, and the oxidizer is most of the load (about 6 kg of oxygen per kg of hydrogen).
2 · Feed	The chamber is at 100+ times air pressure, so propellant has to be forced in. Each Saturn V F-1 engine had a 41 MW turbopump moving **2.6 tonnes every second**.
3 · Burn	In the chamber: about 3,300 K (hotter than steel's melting point of 1,700 K) and 100–300 bar. The gas is hot and squeezed but slow.
4 · Throat	The narrowing speeds the gas up until, at the narrowest point, it hits exactly the speed of sound (Mach 1).
5 · Bell	Past Mach 1 the rule flips: a *widening* pipe speeds the gas up. Pressure and temperature fall, speed climbs. SpaceX's Merlin Vacuum exits at **3.4 km/s**.
6 · Thrust	$F = \dot{m} \times v_e$ = 287 kg/s × 3,414 m/s ≈ **981,000 N**, enough to hold up 100 tonnes.
7 · Isp	Isp = 3,414 ÷ 9.81 = **348 s**. In words: 1 kg of propellant can give 9.81 N of push (1 kg of weight) for 348 seconds.
8 · The ceiling	The energy in a kilo of propellant is fixed by chemistry. Even at 100% conversion, 13 MJ/kg gives $v_e = \sqrt{(2 \times 13{,}000{,}000)} \approx 5.2$ km/s. Real best: 4.5 km/s, Isp about 460 s. No chemical engine will ever reach 1,000 s.
9 · Not melting	The cold fuel is piped through channels in the chamber and nozzle walls before it burns: **regenerative cooling**.

SEC 4 · LEC 2 · NUMBERS + FACT CHECK

Real engines, real numbers

Check $F = \dot{m} \times v_e$ and $Isp = v_e \div g_0$ against real engines, then the caption errors and where the instructor was loose or wrong.

🚀 Engines you can check the formulas on

Engine	Where	Propellants	Thrust	Isp	$v_e = Isp \times 9.81$	$\dot{m} = F \div v_e$
F-1	Saturn V stage 1, ×5 (right photo)	kerosene (RP-1) + LOX	6,770,000 N	263 s	2.6 km/s	2,600 kg/s each
Merlin Vacuum	Falcon 9 stage 2, ×1 (left photo)	kerosene (RP-1) + LOX	981,000 N	348 s	3.4 km/s	287 kg/s
RS-25	Space Shuttle, SLS	LH_2 + LOX	2,279,000 N	452 s	4.4 km/s	514 kg/s
Solid booster	Space Shuttle, ×2	solid grain	12,500,000 N	268 s	2.6 km/s	4,700 kg/s
Hydrazine thruster	most spacecraft	hydrazine	1–22 N	225 s	2.2 km/s	grams per second

Hydrogen gets the best mileage; kerosene and solids the worst, yet they give the biggest push. All figures rounded; F-1 and booster thrust at liftoff, the rest in vacuum.

🔎 Caption errors

Captions said	Instructor meant
"chemical IntoNow heat engines"	chemical **internal** heat engines
"they have no specific impulse"	**low** specific impulse
"velocity of a section / rejection"	velocity of **ejection** (exhaust velocity)
"divided by zero"	divided by **g-zero (g_0)**
"nine point ninety nine point eighty eight to one"	**9.81** m/s^2
"specific forces / specific examples"	specific **impulse**
"threat / frost / thirst"	**thrust**
"expecting a lot of mass"	**expelling** a lot of mass
"thermal specific properties… higher food"	**thermodynamic** properties… **fluid**
"in this cost"	in this **course**

📒 Question bank cards

- Trace a kilo of propellant from tank to exhaust: five stops.
- Where in the engine is pressure highest? Where is speed highest?
- Isp 300 s. What is the exhaust velocity? What does "300 seconds" mean?
- An engine throws 500 kg/s at 4,400 m/s. Thrust?
- Why can't a chemical engine reach Isp 1,000 s?
- Chamber gas is hotter than steel melts. Why doesn't the engine melt?

▲ Where the instructor was loose

- **"They are the lightest and have the highest thrust."** The engines have the best thrust for their own weight (a Merlin lifts about 180 times its weight). The rocket as a whole is the heaviest option, because about 90% of it is propellant.
- **"The nozzle shape maximizes the pressure we get out."** Backwards. The nozzle *spends* pressure to buy speed. Pressure is highest in the chamber and lowest at the exit.
- **"It's quite common that fuel is liquid hydrogen."** Common on upper stages. But both engines in his photos burn **kerosene** with liquid oxygen.
- **"In the United States you may use different units."** That is the whole reason Isp is in seconds: dividing by g_0 gives the same number in metric and imperial.
- **"Solid rockets, usually known as boosters."** Solid is a propellant type; booster is a job. Solids also serve as final kick stages (New Horizons left Earth on one), and boosters can be liquid.
- **"Pressure pump" on the slide** merges two feed systems: a turbopump, or no pump and pressurized tanks.
- **Not said: what spacecraft themselves carry.** Hydrogen and oxygen boil away in days. Probes use storable propellants that can sit in tanks for 20 years: hydrazine alone, or hydrazine-family fuel with nitrogen tetroxide (Isp about 320 s). They ignite on contact, with no spark: **hypergolic**.
- **Not said: why the bell is so big on the second stage.** In vacuum the gas can keep expanding, so a longer bell extracts more speed.

SEC 4 · LEC 2 · VOCABULARY

Every word used, in plain English

46 entries: the lecture's words plus every term I used on the other pages. "Repeat" = you have met it before.

Chemical engine / chemical rocket — an engine that makes thrust by burning propellant and throwing out the hot gas. *Balloon with a fire in it.*

Internal heat engine — the instructor's phrase: a machine that burns fuel inside itself and turns the heat into motion.

Stored mass / propellant — everything the rocket carries to throw out. Fuel plus oxidizer. *Repeat.*

Propulsive jet — the stream of fast gas leaving the nozzle.

Reaction engine — any engine that works by throwing mass backward (Newton's third law). *Repeat, L1.*

Thrust (F) — the engine's push, in newtons. $F = \dot{m} \times v_e$. *Repeat, L1.*

Overall thrust — just 'total thrust'. Nothing new.

Specific impulse (Isp) — exhaust velocity divided by g_0, in seconds. Push per kilo of propellant. *Miles per gallon.*

Exhaust velocity (v_e) = velocity of ejection — how fast the gas leaves the nozzle, relative to the rocket.

g_0 = standard gravity — 9.80665 m/s², the pull of gravity at Earth's surface, used as a fixed constant. *A unit converter, nothing more.*

Constant — a number that never changes. g_0 is one.

Efficiency — how much useful output you get per unit of input. For engines: Δv per kilo of propellant.

Mass flow rate ($\dot{m}$) = flux of propellant — kilograms of propellant leaving per second. Said 'm-dot'. *Repeat, L1.*

Force / newton (N) — a push or pull / its unit. kN = thousand, MN = million. *Repeat, L1.*

Fuel — the part of the propellant that burns: hydrogen, kerosene, hydrazine.

Oxidizer (oxidiser) — the part that supplies the oxygen: liquid oxygen, nitrogen tetroxide. *The air you brought with you.*

Bipropellant / monopropellant — two propellants that burn together / one liquid that breaks down by itself on a catalyst.

Liquid hydrogen (LH_2) — hydrogen chilled to −253 °C. Best mileage, very bulky, boils away.

Liquid oxygen (LOX) — oxygen chilled to −183 °C. The usual oxidizer for launch rockets.

Cryogenic — stored extremely cold to stay liquid.

Kerosene / RP-1 — refined jet-fuel-like liquid. Dense, room temperature, lower Isp than hydrogen.

Hypergolic / storable — propellants that ignite on contact, and keep for years in a tank. *What probes carry.*

Hydrazine — the standard storable spacecraft propellant. Toxic. Isp about 225 s alone.

Pump / turbopump — forces propellant into the high-pressure chamber. Driven by a small turbine.

Pressure-fed — no pump: high-pressure gas in the tanks pushes the propellant out. Simple, for small engines.

Combustion / to burn — fuel joining with oxygen fast, releasing heat.

Combustion chamber — the strong can where the burning happens.

Reaction (chemical) — here: the burning itself. Not the same 'reaction' as in action–reaction.

Thermodynamic properties — the state of a gas: its temperature, pressure and density.

Fluid — anything that flows: a liquid or a gas.

Pressure / bar — how hard a gas pushes on its walls. 1 bar ≈ normal air pressure. *Repeat.*

Throat — the narrowest point of the nozzle. Gas reaches the speed of sound there.

Nozzle — the shaped outlet that turns pressure and heat into speed. *The bell.*

de Laval nozzle = converging-diverging nozzle = bell nozzle — narrows, then widens. Named after Gustaf de Laval, 1888. *Add to Who's Who.*

Expand / expansion — gas spreading into more space. Its pressure and temperature drop.

Parabolic — the curved bell shape; a practical approximation of the ideal contour.

Speed of sound / Mach 1 / supersonic — how fast pressure waves travel in the gas / moving at that speed / faster than it.

Kinetic energy — energy of motion. The nozzle turns heat energy into it. *Repeat.*

Regenerative cooling — running cold fuel through the engine walls to keep them from melting.

Solid rocket motor / grain — fuel and oxidizer cast as one rubbery block / that block. Lights once, can't be shut off.

Booster — a rocket strapped on to help at liftoff. A job, not a propellant type.

Stage / second stage / staging — a rocket section with its own engines and tanks, dropped when empty. *Repeat idea: rocket equation.*

Thrust-to-weight ratio — thrust divided by weight. Above 1 means it can lift itself.

SpaceX / Falcon 9 / Merlin Vacuum — US rocket company / its workhorse rocket / that rocket's second-stage engine. *Add to Who's Who.*

Saturn V / F-1 — the Apollo Moon rocket (1967–1973) / its first-stage engine, the most powerful single-chamber liquid engine flown. *Add to Who's Who.*

RS-25 — the Space Shuttle main engine, hydrogen and oxygen. *Add to Who's Who.*

SEC 4 · LEC 2 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next page.

A · Equations & key numbers page (p3–4)

$Isp = v_e \div g_0$	$g_0 = 9.81$ m/s² · unit: seconds
	3,414 m/s ÷ 9.81 = 348 s (Merlin Vacuum)
$v_e = Isp \times g_0$	452 s × 9.81 = 4.4 km/s (hydrogen engine)
$F = \dot{m} \times v_e$	287 kg/s × 3,414 m/s ≈ 981,000 N
chemical ceiling	$v_e \approx 4.5$ km/s · Isp ≈ 460 s
chamber	~3,300 K · 100–300 bar · throat = Mach 1

B · Section 4 map page (add to the sentence from L1)

chemical: fuel + oxidizer burn → throat (Mach 1) → bell turns pressure into speed; low Isp, huge $\dot{m}$ → huge thrust

D · Question bank ("but how?" cards)

Q: How does a nozzle make gas go faster?
A: It spends pressure and heat to buy speed. Throat = Mach 1, then the widening bell speeds supersonic gas up further.
Q: What does "Isp = 348 s" physically mean?
A: 1 kg of propellant can give 9.81 N of thrust for 348 s. Same as v_e = 348 × 9.81 = 3.4 km/s.
Q: Low Isp but high thrust. How?
A: $F = \dot{m} \times v_e$. v_e is modest but $\dot{m}$ is tonnes per second.
Q: Why doesn't the chamber melt at 3,300 K?
A: Regenerative cooling: cold fuel runs through the walls first.

C · One-line word bank (new lines)

p__ · chemical rocket · burn propellant, throw the gas
p__ · fuel / oxidizer · what burns / the oxygen you bring
p__ · bipropellant / monopropellant · two liquids burn / one breaks down
p__ · LH_2 / LOX = liquid hydrogen / oxygen · cryogenic, best Isp
p__ · RP-1 = kerosene · dense, room temp, lower Isp
p__ · hypergolic / storable · ignite on contact, keep for years
p__ · hydrazine · spacecraft's standard propellant, ~225 s
p__ · turbopump / pressure-fed · pump forces it in / tank gas does
p__ · combustion chamber · where it burns, ~3,300 K
p__ · throat · narrowest point, Mach 1
p__ · nozzle = de Laval = bell · pressure + heat → speed
p__ · exhaust velocity (v_e) · how fast the gas leaves
p__ · g_0 = standard gravity · 9.81 m/s², a constant
p__ · specific impulse (Isp) · $v_e \div g_0$, seconds, mileage → C13
p__ · regenerative cooling · fuel cools the walls first
p__ · solid rocket motor / grain · premixed block, can't shut off
p__ · booster · strap-on helper at liftoff (a job)
p__ · stage / staging · drop empty sections
p__ · thrust-to-weight ratio · thrust ÷ weight; > 1 lifts off

Also: on card C13 (specific impulse) add in the margin: **$Isp = v_e \div g_0$ · 1 kg gives 9.81 N for Isp seconds**.

SEC 4 · LEC 2 · CONCEPT CARDS

Route E: pages C18 and C19

Two notebook pages, two cards each, same six-line shorthand as C1–C17. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 8 — **C18**

EXHAUST VELOCITY (v_e) = ejection speed

= how fast the gas leaves the nozzle

WHY it is the engine's quality in one number: faster exhaust = more push from every kilo

\# kerosene 3.4 km/s · hydrogen 4.4 · ion engine 41 · chemical max ~4.5

≠ the ship's speed (v_e is the GAS)

→ Isp = $v_e \div g_0$ · $F = \dot{m} \times v_e$ · rocket equation

MASS FLOW RATE ($\dot{m}$) = m-dot

= kg of propellant thrown out per second

WHY thrust needs speed AND amount. $\dot{m}$ is the amount; it is also how fast the tanks empty

\# Merlin Vacuum 287 kg/s · F-1 2,600 kg/s · ion engine mg/s

≠ total propellant mass ($\dot{m}$ is per second)

→ $F = \dot{m} \times v_e$ · burn time = mass ÷ $\dot{m}$

WORD BANK · CONCEPT CARDS · scene 8 — **C19**

DE LAVAL NOZZLE = bell nozzle

= narrows to a throat, then widens into a bell

WHY chamber gas is hot and squeezed but slow. The nozzle trades pressure + heat for speed

\# Mach 1 at the throat → 3–4 km/s at the exit

≠ a garden-hose nozzle (subsonic only)

→ throat · expansion · exhaust velocity

BIPROPELLANT = fuel + oxidizer

= engine that burns two propellants together

WHY fire needs oxygen and space has none, so the rocket carries its own in a second tank

\# LH_2 + LOX 450 s · kerosene + LOX 350 s · hydrazine alone 225 s

≠ monopropellant (one liquid, no burn)

→ combustion chamber · Isp · hypergolic

C18: 10 + 11 lines of 22 · C19: 10 + 10 lines of 22

SEC 4 · LEC 2 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"So how does a rocket engine actually work? Isn't it just a big fire?"

✅ Three sentences you should be able to say

- It is a fire with only one way out. You pump fuel and oxygen into a chamber, they burn, and the hot squeezed gas has to leave through a narrow throat and a bell.
- The bell is the clever part: it turns pressure into speed, so the gas leaves at three or four kilometres a second.
- Two numbers describe it. Thrust is how much you throw per second times how fast, and specific impulse is the mileage. Chemical rockets push enormously hard and get poor mileage.

❓ Follow-ups a friend might ask

- "Why two tanks?" (no air in space, so it brings its own oxygen)
- "Why is the nozzle bell-shaped?" (the gas has to expand to speed up; past the speed of sound a widening pipe makes it faster)
- "Why is specific impulse in seconds?" (exhaust speed divided by 9.81; it is how long a kilo can hold up a kilo)
- "Why doesn't it melt?" (the cold fuel runs through the walls first)
- "Can't we just build a better chemical engine?" (no: the energy per kilo is fixed by chemistry, ceiling about 460 s)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Next: the rest of Section 4 (electric and nuclear engines). Send the transcript and slides when you have watched them.

SEC 4 · LEC 3 · ELECTRIC PROPULSION + ION THRUSTERS

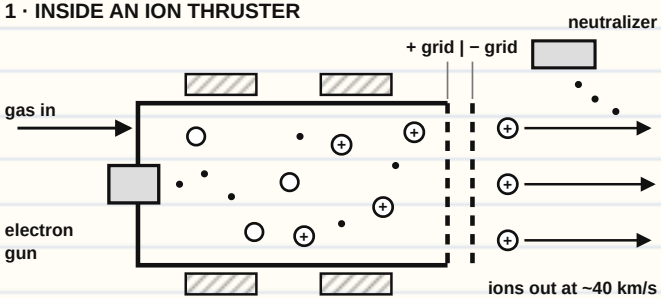
The tortoise engine

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L3 — Electric propulsion **date · pg __**

1 · INSIDE AN ION THRUSTER

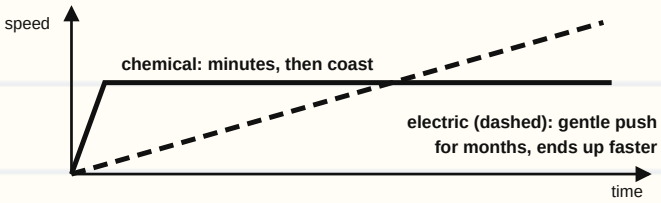


○ neutral atom ⊕ positive ion • electron
hatched = magnet · dashed = grid with holes

2 · THE FOUR STEPS

1 FEED xenon gas, a few mg per second	**2 IONIZE** electrons knock electrons off
3 ACCELERATE grids pull the ions out fast	**4 NEUTRALIZE** add electrons back to the beam

3 · TORTOISE AND HARE



✏️ **PENCIL KEY** · ○ atom · ⊕ ion · • electron · hatched = magnet · dashed line = grid · dashed curve = electric

⭐ **3 TAKEAWAYS** 3 lines

1. Electric = very fast exhaust, very little propellant, very small push.
2. Thrust is capped by electric power; it wins by pushing for months.
3. Ion thruster: ionize the gas, grids fling the ions out, neutralizer adds electrons.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What do electric and chemical engines have in common?	electric propulsion : stored propellant → high-speed jet (same idea as chemical)
Why does electric use so little propellant?	Isp very high (3,000+ s) → exhaust very fast → very little propellant
Why is the thrust so low?	thrust low : limited electric power → tiny mass flow
How does a tiny push reach a high speed?	low thrust × very long time = very high final speed
What is it used for? What can't it do?	mature tech · most new satellites carry it uses: orbit changes (primary propulsion) · orbit maintenance · NOT launch or landing (no big thrust)
Ion thruster: the four steps?	ion thruster = accelerates ions with electricity
What do the two grids do?	ionize : electron gun fires electrons at neutral atoms → knocks an electron off → positive ion · plasma = ionized gas · magnets help
Why the neutralizer?	accelerate: voltage between + grid and − grid shoots ions out at ~40 km/s
Two example missions?	neutralizer : fires electrons into the beam → ship stays uncharged Deep Space 1 (NASA, 1998) · BepiColombo (ESA + JAXA, 2018 , to Mercury)
? what exactly is plasma	

SEC 4 · LEC 3 · TEACH

Static electricity, pointed backward

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Rub a balloon on your hair and it pulls your hair toward it without touching. That pull is electricity grabbing things that are **charged**. An ion engine is that trick, aimed out the back. Step one: take a gas and knock one tiny piece (an electron) off each atom. The atom is now charged, and it is called an **ion**. Step two: put two metal screens full of holes at the back, one charged plus and one charged minus. They yank the ions through the holes and fling them out at 40 km a second, forty times faster than a rifle bullet. Step three: squirt the knocked-off electrons out as well, or the ship would end up charged and pull its own exhaust straight back.

The push is about the weight of two sheets of paper. But it never stops. A chemical rocket is the hare: one huge sprint, then it is out of breath. The ion engine is the **tortoise**: a tiny push, every second, for years, and it ends up going faster on a fifth of the propellant.

🏷 The official names (name first, then what it means)

- **Electric propulsion, EP** (**solar electric propulsion, SEP**, when solar panels supply the power): electricity, not fire, accelerates the propellant.
- **Gridded ion thruster** (electrostatic ion thruster, ion engine): the type in this lecture. Its cousin the **Hall-effect thruster** is the most common electric engine today and was not covered.
- **Ionization** (here: electron-impact ionization): knocking an electron off a neutral atom. It costs the **ionization energy**, 12.1 eV for xenon.
- **Ion**: an atom with a net charge. Lose an electron and it is a positive ion.
- **Plasma**: gas hot or energetic enough that it is a mix of ions and free electrons. Overall it is still neutral. Often called the fourth state of matter.
- **Discharge chamber** (ionization chamber): the can where the gas is ionized.
- **Discharge cathode** (hollow cathode; the slide's "electron gun") and **anode**: the negative source of electrons and the positive wall that attracts them.
- **Potential difference** (voltage): the electrical "height" between two points. Charges fall through it and speed up.
- **Ion optics**: the pair of grids. **Screen grid** (the slide's "positive grid") and **accelerator grid** ("negative grid").
- **Neutralizer** (neutralizer cathode; the slide's "neutralizing electron gun"): emits electrons into the ion beam.
- **Electron backstreaming**: outside electrons rushing back into the engine. The negative grid blocks it.
- **Recombination**: an ion and an electron rejoining into a neutral atom.
- **Primary propulsion** / **station-keeping** / **attitude control**: the main Δv job / holding an orbit / pointing.

🛠 How it actually works

Step	The mechanism, with numbers (Deep Space 1's engine)
1 · Feed	Xenon gas trickles into the chamber: about **3 milligrams per second**. Xenon because it is heavy, easy to ionize and does not react with anything.
2 · Ionize	The cathode emits electrons; about 30 volts speeds them up. One that hits a xenon atom hard enough (12.1 eV) knocks an electron off: now it is a positive ion. Magnets make the electrons spiral, so they travel far farther before reaching the wall and get many more chances to hit an atom.
3 · Accelerate	Screen grid at **+1,100 V**, accelerator grid at −180 V, half a millimetre apart. An ion that drifts into a hole falls through that voltage: energy = charge × voltage = ½mv². For xenon, v = √(2qV ÷ m) ≈ **40 km/s**.
4 · Where the push lands	The grids pull the ions backward-out; the ions pull the grids forward with the same force. Newton's third law again. The thrust is felt on the grids.
5 · Neutralize	The beam carries off 1.76 amps of positive charge. Unbalanced, the ship would charge negative in microseconds and drag its own ions back: no thrust. The neutralizer emits the same 1.76 A of electrons.
6 · Thrust	$F = \dot{m} \times v_e \approx 3$ mg/s × 30 km/s = **0.092 N**, 9 grams of push. (Effective exhaust is 30, not 40 km/s, because some gas leaves un-ionized.) Isp ≈ **3,100 s**.
7 · Power is the limit	That 0.092 N took 2,300 W, a whole solar array. Thrust ≈ 2 × efficiency × power ÷ v_e. For fixed power, faster exhaust means *less* thrust. About 40 mN per kW.
8 · Add it up	0.092 N on 486 kg adds 16 m/s every day. Deep Space 1 thrust for 678 days in total: Δv **4.3 km/s** from 74 kg of xenon. A chemical engine would need about 360 kg.

SEC 4 · LEC 3 · FACT CHECK

Where it is used, and what he got loose

The uses, the caption errors (many this time) and where the instructor was loose or wrong.

🛰️ What electric propulsion does and doesn't do

Job	Electric?	Why
Launch from the ground	Never	Thrust is 10,000+ times less than the craft's weight, and the engine only works in vacuum.
Landing, capture burns	No	They need km/s within minutes.
Raising orbit to GEO	Yes	Takes about 6 months instead of days, but saves tonnes of propellant.
Station-keeping	Yes, since the 1990s	Small, steady corrections: exactly what it is good at.
Deep-space cruise	Yes	Months of thrust add up to many km/s: Deep Space 1, Dawn, BepiColombo, Psyche.
Fine pointing	Starting to	Tiny electric thrusters already do this on a few science missions.

🔎 Caption errors

Captions said	Instructor meant
"stort much propellant… propeller"	**stored mass propellant**… propellant
"specific Kimballs" / "G-0 GZA"	specific **impulse** / **g_0**
"X and velocity", "section velocity"	**ejection** velocity
"lexing propulsion"	**electric** propulsion
"healthy object"	type of **objective**
"looking for parts and systems"	**launch or landing** systems
"I am now using a new tool Gus"	**ionizing a neutral gas**
"election", "electron gamma ray"	**electron**, electron **gun**
"a cathode and that out"	a cathode and **an anode**
"the Greeks", "negative grade"	the **grids**, negative **grid**
"the plastic"	the **plasma**
"in the camps the organization efficiency"	**enhance the ionization** efficiency
"audience", "electrons annoyance"	**ions**, electrons **and ions**
"a sound yak's a collaboration… All of them be"	**ESA and JAXA** collaboration… **BepiColombo**

▲ Where the instructor was loose

- **"Electrons collide with atoms breaking their electronic bonds."** No bonds are involved; xenon is single atoms. The collision knocks an electron *off the atom*. That is ionization.
- **"Plasma is… all of these particles which are positive."** Plasma is the ions *and* the free electrons together, neutral overall. Only the ions get pulled out by the grids.
- **"The neutralizer… so the gas will freely disperse."** The real reason is charge. Fire out only positive ions and the ship goes negative and pulls them back. This is the spacecraft charging from Section 2, self-inflicted.
- **"It cannot be used for launch… because of very low mass ejection."** Closer: its thrust is thousands of times smaller than the craft's weight, and it needs vacuum to run.
- **"By 2020 half of all satellites will have electric propulsion."** An old forecast that came true and more. Every Starlink has an electric thruster, and they are most of what gets launched.
- **BepiColombo "to perform orbital maneuvers around Mercury."** Its ion engines are for the trip *to* Mercury, braking against the Sun's pull for years. That module is dropped before arrival; chemical thrusters do the capture. Arrival was planned for November 2026, which is right now. I can't confirm whether it has happened.
- **"Electron gun" and "positive / negative grid"** are the diagram's labels. The engineering names are discharge cathode, screen grid and accelerator grid.
- **Not said: why low thrust.** He says the mass flow is low. The cause is power: every milligram needs about 500 joules to reach 30 km/s, and the panels only supply so many joules per second.
- **Not said: the Hall thruster**, the most common electric engine flying.

📒 Question bank cards

- Four steps of an ion thruster, in order.
- What limits an electric engine's thrust: propellant, or power?
- What happens if the neutralizer fails?
- Where on the engine does the push actually land?
- Name three jobs electric propulsion does and two it can't.

SEC 4 · LEC 3 · VOCABULARY

Every word used, in plain English

45 entries. This lecture is full of electricity words, so they are all here. "Repeat" = you have met it before.

Electric propulsion (EP) — engines that use electricity to accelerate propellant. *Repeat, L1.*

Solar electric propulsion (SEP) — electric propulsion powered by solar panels. Nearly all of it so far.

Stored mass propellant — the propellant carried on board. Here xenon gas. *Repeat.*

Propulsive jet — the stream leaving the engine. Here a beam of ions. *Repeat.*

Specific impulse (Isp) — exhaust speed ÷ g_0, in seconds. Electric: 1,500–4,000+ s. *Repeat.*

Ejection velocity = exhaust velocity (v_e) — how fast the propellant leaves. 30–40 km/s here. *Repeat.*

g_0 — 9.81 m/s², a constant. *Repeat.*

Thrust / low thrust — the push / here, a fraction of a newton. *Repeat.*

Electric power (watts, kW) — energy per second supplied by the solar panels. 1 kW = 1,000 W.

Mature technology — proven, flown many times, no longer experimental.

Orbit change / primary propulsion — the engine that provides the mission's main Δv.

Orbital maintenance = station-keeping — small burns that hold an orbit against drift. *Repeat, scene 1.*

Attitude control — controlling which way the spacecraft points. *Repeat. Section 6.*

Launch / landing systems — getting off a planet / down onto one. Both need thrust bigger than weight.

Peak thrust — the most push an engine can give at one moment.

Ion thruster (gridded ion engine) — ionizes gas and accelerates the ions with charged grids. *Static electricity, pointed backward.*

Atom / neutral atom — the smallest piece of an element / one with equal + and − charges, so no net charge.

Electron — the tiny negative particle around an atom. Easy to knock off.

Ion / positive ion — an atom with a net charge / one that lost an electron.

Ionize / ionization — to turn atoms into ions / the process.

Ionization energy — the energy it takes to knock one electron off. Xenon: 12.1 eV. *eV is a repeat from Section 2.*

Charge (positive / negative) — the property electricity acts on. Opposites attract, likes repel.

Electric field — the region where a charge feels a push. Between the grids it is very strong.

Potential difference = voltage (V) — electrical 'height' between two points. A charge speeds up falling through it. *A hill for charges.*

Cathode / anode — the negative electrode that gives off electrons / the positive one that collects them.

Electron gun — the slide's name for the cathode that sprays electrons into the chamber.

Electron beam — a stream of electrons.

Collide / collision — hit. Here an electron hitting an atom.

Plasma — ionized gas: ions and free electrons mixed, neutral overall. Lightning, neon signs, the Sun. *The fourth state of matter.*

Discharge chamber — the can where the gas is ionized.

Magnet / magnetic field — makes electrons spiral so they hit more atoms.

Ionization efficiency — what fraction of the gas actually gets ionized.

Grid — a thin metal plate with thousands of holes.

Screen grid = positive grid — the inner grid, at high positive voltage.

Accelerator grid = negative grid — the outer grid, slightly negative. Pulls ions out and keeps outside electrons from getting in.

Repel — push away. Negative repels negative.

Neutralizer = neutralizing electron gun — a small cathode that sprays electrons into the ion beam so the ship stays uncharged.

Neutralize / charge balance — make the + and − equal again.

Recombination — an ion and an electron joining back into a neutral atom, out in the beam.

Disperse — spread out and drift away.

Xenon — the heavy, inert gas ion engines use. *Repeat, L1.*

Ampere (A) — the unit of electric current: charge flowing per second.

Hall-effect thruster — the other main electric engine; not in this lecture. *Repeat, L1.*

Deep Space 1 — NASA's 1998 test probe, first to cruise on an ion engine. *Who's Who.*

BepiColombo / ESA / JAXA — Mercury mission launched 2018 / European and Japanese space agencies. *Who's Who.*

SEC 4 · LEC 3 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next page.

A · Equations & key numbers page (p3–4)

$v = \sqrt{(2qV \div m)}$	ion speed from grid voltage
	xenon, 1,100 V → 40 km/s
$F \approx 2 \cdot \eta \cdot P \div v_e$	thrust from power (η = efficiency)
	2,300 W → 0.092 N · ~40 mN per kW
ion engine	Isp ~3,100 s · $\dot{m}$ ~3 mg/s · 16 m/s per day
Deep Space 1	678 days thrust · 74 kg Xe · Δv 4.3 km/s

B · Section 4 map page (add to the sentence)

electric: ionize xenon → grids (+1,100 V) fling ions at 40 km/s
→ neutralizer adds electrons; tiny push for years, power-limited

D · Question bank ("but how?" cards)

Q: How does an ion engine speed the gas up with no fire?
A: Knock an electron off each atom (ion). Charged grids pull the ions through 1,100 V: v = √(2qV/m) ≈ 40 km/s.
Q: Why does it need a neutralizer?
A: Positive ions leave, so the ship would charge negative and pull them back. Equal electrons out keeps it neutral.
Q: Why is the thrust so small?
A: Power. Fast exhaust costs ~500 J per milligram; panels give only a few thousand joules per second.
Q: How can 9 grams of push get anywhere?
A: It never stops: 16 m/s per day, 4.3 km/s after 678 days.

C · One-line word bank (new lines)

p__ · electric propulsion (EP) = SEP · electricity throws it → C20
p__ · primary propulsion · the engine doing the main Δv
p__ · ion thruster = gridded ion engine · ionize, then grids → C20
p__ · atom / electron · smallest bit of an element / its − particle
p__ · ion · atom with a net charge (lost an electron = +)
p__ · ionization · knocking an electron off an atom
p__ · ionization energy · cost of that: xenon 12.1 eV
p__ · plasma = ionized gas · ions + electrons, neutral overall → C21
p__ · charge · what electricity acts on; opposites attract
p__ · voltage = potential difference · a hill charges fall down
p__ · cathode / anode · gives off electrons / collects them
p__ · electron gun = discharge cathode · sprays electrons in
p__ · discharge chamber · the can where gas is ionized
p__ · screen grid (+) / accelerator grid (−) · pull the ions out
p__ · neutralizer · electrons into beam; ship stays neutral → C21
p__ · electron backstreaming · outside electrons rushing in
p__ · recombination · ion + electron → neutral atom again
p__ · Hall-effect thruster · other electric engine, magnetic
p__ · ampere (A) · current: charge per second

SEC 4 · LEC 3 · CONCEPT CARDS

Route E: pages C20 and C21

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 9 — **C20**

ELECTRIC PROPULSION (EP)

= electricity, not fire, accelerates the propellant

WHY chemistry caps exhaust at ~4.5 km/s. Electricity has no cap, so EP uses a tenth of the propellant

\# Isp 3,000+ s · thrust under 1 N · runs for years

≠ a weak = bad engine (more total Δv)

→ power limits thrust · ion + Hall thrusters

ION THRUSTER = gridded ion engine

= ionizes gas, then grids fling the ions out

WHY a charged atom can be grabbed by voltage and thrown far faster than any flame can throw gas

\# +1,100 V → xenon ions at 40 km/s · 0.092 N from 2.3 kW

≠ Hall thruster (magnetic, no grids)

→ ionize → accelerate → neutralize

WORD BANK · CONCEPT CARDS · scene 9 — **C21**

PLASMA = ionized gas

= ions + free electrons, neutral overall

WHY neutral gas ignores electricity. Knock electrons off and you can steer and accelerate it

\# xenon: 12.1 eV per atom to ionize

≠ 'only positive particles' (electrons are in it too)

→ ionization · ionosphere · spacecraft charging

NEUTRALIZER = neutralizer cathode

= sprays electrons into the ion beam

WHY shoot out only + ions and the ship goes negative and pulls them back: thrust dies

\# Deep Space 1: 1.76 A of ions out, 1.76 A of electrons out

≠ the electron gun inside (that one ionizes)

→ spacecraft charging · recombination

C20: 11 + 10 lines of 22 · C21: 11 + 10 lines of 22

SEC 4 · LEC 3 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"An engine that pushes as hard as two sheets of paper? How is that any use?"

✅ Three sentences you should be able to say

- Because it never stops. It adds a few metres per second every day for years, and ends up faster than a chemical rocket on a fifth of the propellant.
- It works like static electricity: knock an electron off each gas atom so it is charged, then two charged grids fling those ions out the back at 40 kilometres a second.
- The catch is power. Throwing gas that fast takes a lot of electricity and solar panels only give so much, so the push is tiny and you can't launch or land with it.

❓ Follow-ups a friend might ask

- "What's an ion?" (an atom that lost an electron, so electricity can grab it)
- "What's plasma?" (gas where the atoms have been split into ions and loose electrons)
- "Why spit the electrons out too?" (otherwise the ship charges up and pulls its exhaust back)
- "Has it actually flown?" (since 1964; Deep Space 1 in 1998, Dawn, BepiColombo, every Starlink)
- "Why not just add more solar panels?" (you can, but panels are mass, and far from the Sun they fade)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Two lectures left in Section 4. Send them the same way.

SEC 4 · LEC 4 · NUCLEAR THERMAL ROCKETS

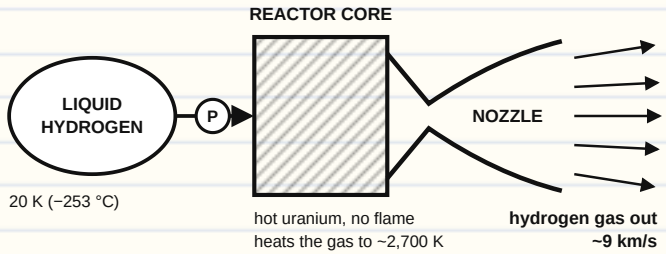
A kettle with no fire

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L4 — Nuclear thermal rockets **date · pg _**

1 · NUCLEAR THERMAL ROCKET

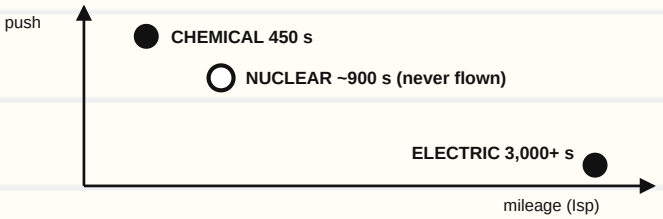


2 · WHY THE MILEAGE DOUBLES

CHEMICAL exhaust	NUCLEAR exhaust
18	2
steam: heavy	hydrogen: light
3,500 K	2,700 K (cooler!)
4.4 km/s · 450 s	8.8 km/s · 900 s

light molecules fly faster: $v_e \propto \sqrt{(T \div M)}$

3 · THE THREE FAMILIES



✏️ **PENCIL KEY** · hatched box = reactor core · Ⓟ = pump · number in circle = how heavy the exhaust molecule is · ● flown · ○ never flown

⭐ **3 TAKEAWAYS** 3 lines

1. A reactor, not a fire, heats hydrogen; the nozzle does the rest.
2. Light exhaust is fast exhaust: Isp ~900 s, twice chemical, with real thrust.
3. Ground-tested in the 1960s, never flown: cost, politics, radiation.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What heats the propellant in a nuclear thermal rocket?	nuclear thermal rocket (NTR): a reactor heats the propellant; nothing burns
Path of the hydrogen, tank to exhaust?	liquid hydrogen → pump → reactor (heated, ~2,700 K) → nozzle (expands, pressure ↓, speed ↑) → thrust
Isp compared with chemical? Why?	Isp ≈ 2× chemical (~900 s vs 450 s) · and still high thrust
What sets the temperature limit?	why: exhaust is light hydrogen, not heavy steam (not because it is hotter)
Why is it attractive for Mars?	limit = what the reactor materials can stand
What is payload?	2× Isp → about half the mass for the same trip, or 2–3× the payload
Low-enriched uranium: what and why?	payload = the useful cargo: instruments, cameras, crew
Why has it never flown?	Mars: faster trip → less radiation exposure for the crew
The two examples: what are they really?	low-enriched uranium (LEU): safer to build and test, fewer security rules, cheaper
? is it hotter than a chemical rocket	ground-tested 1960s · politics + cost → never flown
	theory: fuel mixed into the fluid (liquid / gas core) → hotter → higher Isp
	examples: bimodal NTR (drawing only) · Project Orion (really nuclear pulse: bombs)

SEC 4 · LEC 4 · TEACH

Not hotter. Lighter.

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

A normal rocket heats its gas by burning it. A nuclear rocket is a kettle with no fire. Inside is a block of special metal, uranium, that gets scorching hot all by itself when its atoms split. The engine squirts the lightest gas there is, hydrogen, through holes in that hot block. The gas comes out blazing and shoots through the same bell nozzle as before.

Why is that better? Throw a ping-pong ball and a bowling ball with the same arm: the ping-pong ball leaves much faster. The fire in a chemical rocket makes steam, which is the bowling ball. A reactor doesn't need to burn anything, so it can throw plain hydrogen, the ping-pong ball. Same kind of heat, twice the exhaust speed, twice the mileage, and it still pushes hard. The catch: the hot block is radioactive, the engine is heavy, and nobody has ever flown one. They ran on test stands in the Nevada desert in the 1960s.

🏷 The official names (name first, then what it means)

- **Nuclear thermal propulsion, NTP** (**nuclear thermal rocket, NTR**; solid-core nuclear rocket): a fission reactor heats a propellant that leaves through a nozzle.
- **Nuclear fission**: a heavy atom (uranium-235) splitting in two and releasing heat. A **reactor** keeps that going at a controlled rate. A **fuel element** is one of the rods that holds the uranium; together they are the **reactor core**.
- **Working fluid** (here the same thing as the propellant): the hydrogen that gets heated and thrown. Three different words now: **nuclear fuel** is the uranium, **propellant** is the hydrogen, and nothing is "fuel" in the burning sense.
- **Molecular mass, M** (molar mass): how heavy one molecule of the exhaust is. Hydrogen 2, steam 18.
- **Enrichment**: the share of uranium that is the splittable kind, U-235. **Low-enriched uranium, LEU**: under 20%. **Highly enriched, HEU**: 20% and up; weapons grade is about 90%.
- **Payload**: the part of the spacecraft that is the reason for the trip. **Payload fraction**: its share of the total mass.
- **Bimodal nuclear thermal rocket**: one reactor, two jobs. Thrust during burns, electricity the rest of the time.
- **Liquid-core / gas-core nuclear rocket**: the slide's "mixing nuclear fuel and the working fluid". Paper designs only.
- **Nuclear pulse propulsion** (Project Orion): riding a string of small nuclear explosions. Not a thermal rocket.
- **Radiation exposure** (dose): how much damaging radiation a crew absorbs. A **shadow shield** sits between reactor and crew.

🛠 How it actually works

Step	The mechanism, with numbers (the 1960s NERVA engine)
1 · The heat source	Splitting 1 kg of uranium-235 releases about 80 million MJ. Burning 1 kg of hydrogen and oxygen releases 13. Six million times more, so energy is never the limit.
2 · Flow path	Liquid hydrogen at 20 K → turbopump → channels in the nozzle wall (cooling it) → down through hundreds of holes in the hot core → out as gas at about **2,700 K** → throat → bell.
3 · Why it is faster	Exhaust speed goes as √(T ÷ M). Chemical: 3,500 K ÷ 13.6 (steam with spare hydrogen). Nuclear: 2,700 K ÷ 2. Ratio of the square roots ≈ 2.3. The reactor is *cooler* and still wins, because its exhaust is seven times lighter.
4 · The two numbers	$v_e \approx 8.3$ km/s, so Isp = 8,300 ÷ 9.81 ≈ **850 s** (NERVA measured 841 s). Thrust **246,000 N**, about 25 tonnes of push.
5 · Check it	$\dot{m} = F \div v_e = 246{,}000 \div 8{,}300 \approx 30$ kg of hydrogen per second. Jet power = $\frac{1}{2}\dot{m}v_e^2 \approx 1{,}000$ MW. The reactor was rated 1,100 MW. It adds up.
6 · What it buys	Leaving Earth orbit for Mars needs about 4 km/s. Rocket equation: at 450 s the ship is 60% propellant; at 900 s it is 36%. Spend the saving on payload, or on speed for a shorter trip.
7 · The limits	The core must stay solid, so about 2,700–2,900 K at most. Hot hydrogen eats the fuel rods. Liquid hydrogen is 14 times less dense than water: huge tanks. Reactor and shield weigh tonnes: thrust is only 3–4 times the engine's weight, against 70–180 for chemical.
8 · History	US Project Rover and NERVA, 1955–1973: about 20 reactors run in Nevada. In 1969 one engine was started 28 times. Cancelled when the Mars plans were. Never flown.

SEC 4 · LEC 4 · FACT CHECK

Three real errors in this lecture

This lecture has more wrong statements than any so far. The first three change what you would say to a friend.

⛔ Wrong, and worth getting right

- **"Very high specific impulse since the working temperatures are higher."** A nuclear core runs *cooler* than a chemical flame (2,700 K against 3,500 K), because the fuel rods must not melt. The Isp doubles because the exhaust is light hydrogen and not heavy steam. He half-remembers the rule ("the square root of the temperature"). The full rule is √(temperature ÷ molecular mass).
- **"NASA's ORION nuclear project… developed in 1986… nuclear thermal rockets."** Three errors. Project Orion ran **1958–1965**, led by a company (General Atomics) for the US military. It is **not a thermal rocket**: the ship drops small nuclear bombs behind itself and rides the blasts on a giant plate with shock absorbers. That is exactly what his own slide diagram labels: pusher, shock absorbers, propellant magazines. The 1963 treaty banning nuclear explosions in space killed it. And it has nothing to do with today's Orion crew capsule.
- **"It is also attractive for interstellar flight."** No. A nuclear thermal rocket gives perhaps 10–20 km/s. The nearest star is 40 trillion km away: at Voyager's 17 km/s that is **75,000 years**. It is an engine for reaching Mars, not other stars.

🔎 Caption errors

Captions said	Instructor meant
"nuclear them a rocket / rock edge"	nuclear **thermal rocket(s)**
"on the QM state of the art… react to development"	the **current** state of the art… **reactor** development
"specific import" · "lift of mass" · "tripped"	specific **impulse** · **lift-off** mass · **tripled**
"radiation explosion"	radiation **exposure** (slide: "exposition")
"hitting some kind of liquid"	**heating** some kind of liquid
"knotholes"	**nozzles**
"Nysa… by mode on nuclear rocket"	**NASA**… **bimodal** nuclear rocket
"NASA's Arrian nuclear project"	the **Orion** nuclear project
"nuclear production / proportion"	nuclear **propulsion**

▲ Where else he was loose

- **"First proposed and sketched in the 1960s… some prototypes."** Proposed in the 1940s. By the 1960s the US had built and run about 20 reactors at full power. It was far more than sketched.
- **"Right now it's 2018."** The lecture is eight years old. Since then NASA and DARPA started a flight demo (DRACO, 2023) and cancelled it in 2025, as far as I know.
- **"The required lift-off mass is half."** Nobody plans to lift off from the ground on one: too heavy for its thrust, and radioactive. It is a stage you start in orbit. The honest version: the mass you must put *into Earth orbit* for a Mars trip roughly halves.
- **"The liquid at high temperature… high velocity liquid."** It goes in as liquid at 20 K and is a gas long before it reaches 2,700 K. Liquids don't expand through nozzles.
- **"We all know what happens if we cannot withstand some nuclear reaction."** A reactor cannot explode like a bomb. The real failure is fuel rods cracking or melting.
- **"A dispute among the scientific community."** The physics isn't disputed; it was demonstrated. The obstacles are cost, testing (the exhaust is radioactive) and public acceptance.
- **"Low-enriched uranium… promoted in the European Union."** The switch to low-enriched fuel was NASA's own choice in the late 2010s. I don't know of a European programme he could mean.
- **"Mixing nuclear fuel and the working fluid."** These are liquid-core and gas-core designs: let the fuel melt or vaporize so temperature is no longer capped. Isp of 1,500 to 5,000 s on paper. None has been built.

📒 Question bank cards

- A nuclear rocket runs cooler than a chemical one. Why is its Isp double?
- Trace the hydrogen from tank to exhaust.
- Nuclear fuel, propellant, working fluid: which is which here?
- Why would a crew to Mars want one? Give two reasons.
- Project Orion: what was it really, and why did it end?

SEC 4 · LEC 4 · VOCABULARY

Every word used, in plain English

42 entries. "Repeat" = you have met it before.

Nuclear propulsion — using a nuclear reactor (or explosions) as an engine's energy source. *Repeat, L1.*

Nuclear thermal rocket (NTR) = nuclear thermal propulsion (NTP) — a reactor heats hydrogen, which leaves through a nozzle. *Kettle with no fire.*

Nuclear fission — a heavy atom splitting in two and releasing a lot of heat.

Uranium / U-235 — the heavy metal used as nuclear fuel / the kind of uranium atom that splits easily.

Nuclear reactor — a machine that keeps fission going at a steady, controlled rate.

Reactor core / fuel element — the hot heart of the reactor / one of the rods that holds the uranium.

Nuclear fuel — the uranium. It is not burned and not thrown out. *Not the same as rocket fuel.*

Working fluid — the substance that is heated and expanded to do the work. Here hydrogen, which is also the propellant.

Liquid hydrogen (LH_2) — hydrogen chilled to 20 K. Very light, very bulky. *Repeat, L2.*

Efficiency / exceptional efficiency — here: high specific impulse, a lot of Δv per kilo of propellant. *Repeat.*

Specific impulse (Isp) — exhaust speed ÷ 9.81, in seconds. NTR: about 850–900 s. *Repeat.*

Working temperature — how hot the gas gets inside the engine.

Square root (√) — the number that, multiplied by itself, gives the original. √4 = 2.

Molecular mass (M) — how heavy one molecule is, on a scale where hydrogen gas is 2 and water is 18. *Ping-pong ball against bowling ball.*

Proportional to (∝) — 'goes up and down together with'.

Theoretical — worked out on paper, not built.

Liquid-core / gas-core rocket — designs where the uranium itself is molten or gas and mixed with the propellant. Hotter, never built.

Withstand — survive without breaking or melting.

Material properties — what a material can take: melting point, strength, resistance to attack.

Interplanetary / interstellar — between planets / between stars. The second is thousands of times farther.

State of the art — the best that current technology can do.

Prototype / model — a first working version built for testing / a mock-up or a calculation.

Lift-off mass — the total mass of the rocket as it leaves the pad.

Payload — the cargo that is the reason for the mission: instruments, cameras, crew.

Payload fraction — the share of the total mass that is payload. Usually a few percent.

Radiation exposure (dose) — how much harmful radiation a person absorbs. In deep space it comes from the Sun and the galaxy.

Enrichment — raising the share of U-235 in uranium. Natural uranium has 0.7%.

Low-enriched uranium (LEU) — under 20% U-235. Cannot be made into a bomb easily, so fewer security rules.

Highly enriched uranium (HEU) — 20% U-235 and up. The 1960s engines used about 93%.

Security regulations — the legal controls on who may handle bomb-grade material.

Budget / schedule — the money / the timeline.

Turbine pump = turbopump — the pump that forces hydrogen into the reactor. *Repeat, L2.*

Nozzle / expand / pressure — the bell that turns pressure and heat into speed. *Repeat, L2.*

Fluid — anything that flows: liquid or gas. The exhaust here is a gas. *Repeat.*

Bimodal — two modes: the same reactor gives thrust and electricity.

Artist's concept — a painting of something that does not exist yet.

Nuclear pulse propulsion — propelling a ship with a series of nuclear explosions behind a pusher plate. *Project Orion.*

Pusher plate / shock absorber — the thick plate that takes each blast / the springs that smooth the kick.

Shadow shield — a slab between the reactor and the crew that blocks its radiation.

Project Rover / NERVA — the US nuclear rocket programme, 1955–1973. *Who's Who.*

Project Orion (1958–1965) — the US nuclear pulse study. Not the Orion crew capsule. *Who's Who.*

DRACO — the cancelled NASA and DARPA flight demo of the 2020s. *Who's Who.*

SEC 4 · LEC 4 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. Section 4 ends here, so route B is now the full sentence for the Section 4 map page.

A · Equations & key numbers page (p3–4)

```
vₑ ∝ √(T ÷ M)      hot + LIGHT exhaust is fast
                   chemical 3,500 K ÷ 13.6 · nuclear 2,700 K ÷ 2
                   → about 2× the exhaust speed
nuclear thermal    Isp ~850–900 s · 246,000 N · 30 kg/s H₂
                   1,100 MW reactor (NERVA, 1960s)
Mars departure     4 km/s: 60% propellant at 450 s
                           36% propellant at 900 s
```

B · Section 4 map page (write it now, from memory first)

```
S4 · propulsion: throw mass back → F = ṁ × vₑ
S4 · chemical: burn → throat → bell; big ṁ, Isp 250–450 s
S4 · electric: ionize → grids → neutralize; mg/s, Isp 3,000+ s
S4 · nuclear: reactor heats H₂ (light = fast); Isp ~900 s,
     never flown
S4 · the trade: push vs mileage; energy from propellant /
     solar panels / reactor
```

D · Question bank ("but how?" cards)

```
Q: A nuclear core is cooler than a flame. Why is Isp double?
A: vₑ ∝ √(T ÷ M). Exhaust is H₂ (mass 2), not steam (18).
Q: How does the hydrogen get hot with nothing burning?
A: It flows through holes in a core heated by uranium fission.
Q: Why not launch from the ground with it?
A: Engine too heavy for its thrust (3–4× its weight) and
   radioactive. It is started in orbit.
Q: What was Project Orion?
A: Nuclear PULSE: bombs behind a pusher plate. 1958–1965.
```

C · One-line word bank (new lines)

```
p__ · nuclear thermal rocket = NTR / NTP · reactor heats H₂  → C22
p__ · fission · heavy atom splits, releases heat
p__ · reactor / reactor core · controlled fission / its hot heart
p__ · fuel element · rod that holds the uranium
p__ · nuclear fuel vs propellant · uranium (stays) vs H₂ (thrown)
p__ · working fluid · what gets heated + expanded: H₂
p__ · molecular mass (M) · how heavy one molecule is  → C22
p__ · payload · the cargo the trip is for  → C23
p__ · payload fraction · payload ÷ total mass
p__ · lift-off mass · total mass leaving the pad
p__ · enrichment · share of U-235 in the uranium  → C23
p__ · LEU / HEU · under 20% / 20% and up (bomb-grade ~90%)
p__ · radiation exposure = dose · radiation a crew absorbs
p__ · shadow shield · slab between reactor and crew
p__ · bimodal NTR · one reactor: thrust + electricity
p__ · liquid-core / gas-core · molten or gas fuel; paper only
p__ · nuclear pulse propulsion · ride bomb blasts (Orion)
p__ · interstellar · between stars; NOT reachable this way
```

Fix one earlier card: in Lecture 1's question bank I wrote "Same heat, lighter exhaust" for nuclear. Change "Same heat" to "**Cooler than a flame, but**". Lecture 1's PDF is reissued as v2 with that corrected.

SEC 4 · LEC 4 · CONCEPT CARDS

Route E: pages C22 and C23

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 10 — **C22**

NUCLEAR THERMAL ROCKET = NTR · NTP

= a reactor heats hydrogen; a nozzle throws it

WHY fire caps exhaust at 4.5 km/s. A reactor needs no burning, so it can throw the lightest gas there is

\# ~2,700 K · 8.3 km/s · Isp ~850 s · 246,000 N (NERVA)

≠ RTG (electricity only) · nuclear pulse

→ $v_e \propto \sqrt{(T \div M)}$ · never flown

MOLECULAR MASS (M)

= how heavy one exhaust molecule is

WHY at the same heat, light molecules move faster, so light exhaust means high Isp

\# hydrogen 2 · steam 18 · exhaust speed ∝ √(T ÷ M)

≠ the propellant's total mass

→ why NTR doubles Isp while running cooler

WORD BANK · CONCEPT CARDS · scene 10 — **C23**

PAYLOAD

= the cargo that is the reason for the trip

WHY everything else (engines, tanks, propellant) exists only to move it; better Isp frees mass for it

\# Mars departure: 60% propellant at 450 s, 36% at 900 s

≠ the whole spacecraft

→ rocket equation · payload fraction

ENRICHMENT = LEU · HEU

= share of uranium that is splittable U-235

WHY more U-235 = smaller reactor, but bomb-grade material means heavy security and cost

\# natural 0.7% · LEU < 20% · 1960s engines 93%

≠ radioactivity level (different thing)

→ fission · reactor core · budget + schedule

C22: 11 + 10 lines of 22 · C23: 10 + 11 lines of 22

SEC 4 · LEC 4 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Section 4 is complete after this, so the Section 4 talk is now in the app.

🗣 The friend's opening question

"A nuclear rocket? So it's basically a bomb going off behind the ship?"

✅ Three sentences you should be able to say

- No, it's a kettle with no fire. A reactor gets hot, you blow hydrogen through it, and the hot gas goes out an ordinary rocket nozzle.
- It gets about twice the mileage of the best chemical rocket, and the reason isn't that it is hotter. It is that the exhaust is plain hydrogen, the lightest gas, and light molecules fly faster.
- It worked on test stands in the 1960s but has never flown, because it is heavy, radioactive, expensive to test and politically hard.

❓ Follow-ups a friend might ask

- "Why would anyone want one?" (half the propellant for a Mars trip, or a faster trip and less radiation for the crew)
- "Could it get us to another star?" (no: tens of thousands of years even at its best)
- "Wasn't there one that did use bombs?" (Project Orion, 1958–1965, never built)
- "Is that what powers Voyager?" (no, that is an RTG: electricity only)
- "Which engine is best, then?" (none: chemical for push, electric for mileage, nuclear would sit in between)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Section 4 done. Before Section 5: the Section 4 map page from memory, then one timed Section 4 talk from the app's Teach tab.

SPACE X · SECTION 5 OF 7

Section 5 · Communications

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

Write the message on a radio wave, aim it with a dish, send it through the radio window, and land louder than the noise.

The lectures

#	Lecture	Page	Length
1	**Antennas**	62	9 pages
2	**Atmosphere and frequency bands**	71	7 pages
3	**Satellite and spacecraft links**	78	9 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Talk card · 4 min

He opens with: "How does a probe at Saturn talk to Earth? It can't have much power out there."

- 1 · RIPPLES — c = λ × f · rod = half a wave · voice → 1,500 km · carrier, AM / FM
- 2 · FLASHLIGHT — isotropic / donut / beam · gain aims, adds nothing · 4 m dish, 0.6°, 20 W → 1 MW
- 3 · TWO WINDOWS — visible + radio · UV absorbed, long radio bounced · S, X, Ka · uplink ≠ downlink
- 4 · STADIUM — encode → modulate → up-convert → amplify · spreads ¼ per 2× · LNA · signal ÷ noise
- LAND — "write it, aim it, window, louder than the noise"

Land on: Write the message on a wave, aim it with a dish, send it through the radio window, and make sure it arrives louder than the noise.

Number to know
c = λ × f · Cassini: 20 W sent, 10^{-16} W received

Don't repeat
"a dish adds power" (it aims it) · "k is a relation for wavelength" (it is the Boltzmann constant)

In the app: scenes 11, 12, 13 in Learn, and the "Section 5" talk in Teach.

SEC 5 · LEC 1 · COMMUNICATIONS: ANTENNAS

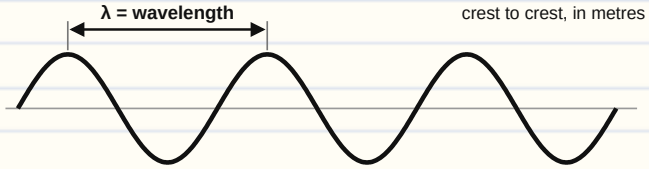
Ripples you can write on

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S5 / L1 (1/2) — Waves + modulation **date · pg __**

1 · A WAVE: wavelength and frequency

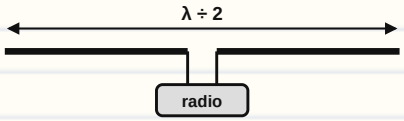


f = frequency = waves passing per second (hertz, Hz)

c = λ × f · c = 300,000 km/s, always

so: higher frequency → shorter wavelength

2 · ANTENNA = metal rod, half a wave long

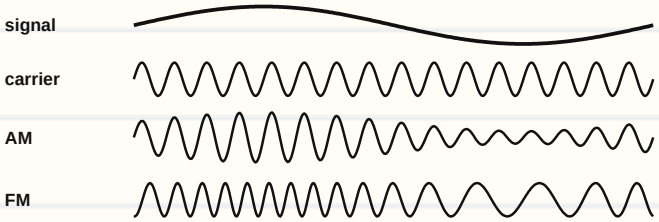


the wave pushes electrons to and fro in the rod → a voltage

FM radio 100 MHz: λ = 3 m → rod 1.5 m

voice 100 Hz: λ = 3,000 km → rod 1,500 km. Impossible.

3 · MODULATION: the voice rides a fast carrier



✏️ **PENCIL KEY** · λ arrow = one wavelength · thick bar = antenna rod · row 1 = your voice · row 2 = carrier · AM = height changes · FM = spacing changes

⭐ **3 TAKEAWAYS** 3 lines

1. Radio is an electromagnetic wave. c = λ × f, and c never changes.
2. Antenna ≈ half a wavelength, so high frequency = small antenna.
3. Modulation: a slow signal rides a fast carrier (AM height, FM frequency).

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What carries the information through space?	electromagnetic (EM) wave carries the signal · crosses empty space
The two fields: what are they, how are they arranged?	electric field ⊥ magnetic field · sinusoidal (smooth, repeating)
Wavelength? Frequency? Units?	wavelength λ = crest to crest (m) · frequency f = waves per second (Hz)
The formula that links them?	c = λ × f · c = 3×10⁸ m/s, constant → f up, λ down
How does an antenna receive? Transmit?	receive: the wave's electric field → voltage in the antenna at the same f → receiver reads it
	transmit: the reverse · the same antenna does both
What sets the size of an antenna?	antenna size ≈ λ ÷ 2 (dipole) → higher f = smaller antenna
Why can't you transmit a voice directly?	voice ~100 Hz → λ = 3,000 km → antenna 1,500 km ×
What is modulation? What is a carrier?	modulation : put the slow information signal on a fast carrier signal
AM vs FM?	AM : carrier's height follows the signal · FM : carrier's frequency follows it
? is my voice a radio wave	

SEC 5 · LEC 1 · COMMUNICATIONS: ANTENNAS

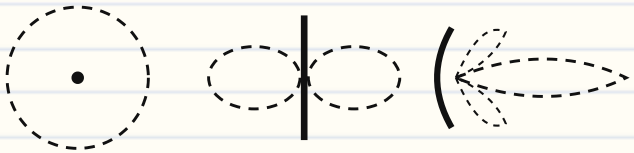
A flashlight for radio

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S5 / L1 (2/2) — Directivity + gain **date · pg __**

1 · THREE WAYS TO SPREAD THE SIGNAL

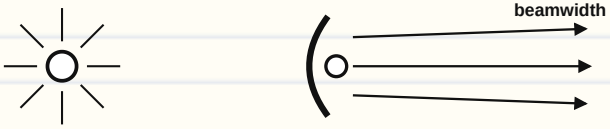


ISOTROPIC
every direction
ideal only · gain 1

OMNI (dipole)
all around the rod
none off the ends

DIRECTIONAL (dish)
main lobe + side lobes
gain 1,000s

2 · GAIN = same power, squeezed into a beam



bare bulb: dim everywhere

same bulb + reflector: bright far away

An antenna adds no power. It aims it.

3 · HIGH GAIN vs LOW GAIN

HIGH-GAIN (dish)	LOW-GAIN (rod)
narrow beam	wide beam
strong signal, fast data	weak signal, slow data
must point at Earth	works in any attitude
Cassini: 4 m dish	CubeSat: crossed dipoles
Apollo rover: TV dish	Apollo rover: voice, driving

✏️ **PENCIL KEY** · dashed outline = where the signal goes · big petal = main lobe, small = side lobes · thick line = antenna

⭐ **3 TAKEAWAYS** 3 lines

1. Isotropic = everywhere · omni = all around a rod · directional = one beam.
2. Gain concentrates power into a beam. It never creates power.
3. High gain = fast data but must be aimed. Low gain = slow but always works.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Isotropic antenna?	directivity = how much an antenna favors one direction
Omnidirectional?	isotropic: equal in ALL directions (a sphere) · ideal only · the reference
Directional? What is beamwidth?	omnidirectional: equal all around in one plane · e.g. a dipole
What is gain, really?	directional: one main lobe + small side lobes · beamwidth = angle where power is still above half (3 dB) · dish, rooftop TV antenna
Does an antenna add power?	gain = increase in signal strength due to directivity (vs isotropic) · more directional = more gain
What does high gain cost you?	an antenna adds no power: it concentrates it
Why does a CubeSat use dipoles?	high gain → better signal, more data · but must point exactly → attitude control
Why must Cassini's dish point at Earth?	CubeSat: two crossed dipoles → works whichever way it faces · weak signal
High-gain vs low-gain: when each?	Cassini: 4 m high-gain antenna, must point at Earth
? how can 20 watts reach Earth from Saturn	Apollo 17 rover: low-gain (wide, while driving) + high-gain dish (aimed, for TV)

SEC 5 · LEC 1 · TEACH

A 20-watt whisper from Saturn

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Drop a stone in a pond and ripples spread out. A radio wave is a ripple too, but in electricity and magnetism, and it needs no water and no air, so it crosses empty space at the speed of light. An **antenna** is a metal stick. When the ripple washes over it, it pushes the electricity inside the stick back and forth, like a cork bobbing, and the radio reads the bobbing. To send, do the reverse: shake the electricity in the stick and ripples fly off.

The stick works best when it is half as long as one ripple. Voice ripples are slow and would be 3,000 km long, so the stick would be longer than Texas. So we cheat. We take a fast, short ripple, the **carrier**, and make it carry the voice by changing its height (**AM**) or squeezing its spacing (**FM**) in step with the voice. Last trick: a bare bulb lights a room dimly, but the same bulb in a flashlight throws a bright spot far away. A dish antenna is a flashlight for radio. That is **gain**. The Cassini probe talked to Earth from Saturn with 20 watts, a fridge bulb, because its 4-metre dish squeezed everything into one thin beam aimed exactly at us.

🏷 The official names (name first, then what it means)

- **Electromagnetic wave** (EM wave, electromagnetic radiation): a travelling ripple of electric and magnetic field. Radio, microwaves, light, UV and X-rays are all the same thing at different frequencies: the **electromagnetic spectrum**.
- **Wavelength, λ** ("lambda"): distance from one crest to the next. **Frequency, f**: waves per second, in **hertz** (Hz; kHz, MHz, GHz = thousand, million, billion). **Period**: seconds per wave, 1 ÷ f.
- **Speed of light, c**: 299,792 km/s. **c = λ × f**.
- **Antenna**: a conductor that turns an electrical signal into a radio wave and back. **Transmitter** (TX), **receiver** (RX).
- **Half-wave dipole**: two rods end to end, λ ÷ 2 in total. The basic antenna. A **monopole** is half of one (λ ÷ 4), like a car aerial.
- **Modulation**: changing a carrier wave so that it carries information. **Carrier signal**: the fast wave. **Information signal** (baseband, message): the slow one.
- **Amplitude modulation, AM** / **frequency modulation, FM**. Spacecraft actually use **phase modulation** (phase-shift keying, PSK) to send digital bits.
- **Radiation pattern**: the shape of where an antenna sends its power. **Isotropic radiator**, **omnidirectional antenna**, **directional antenna**.
- **Directivity** and **antenna gain**: how many times stronger the signal is in the best direction than from an isotropic antenna. Written in **dBi**: 10 dBi = 10×, 20 = 100×, 30 = 1,000×.
- **Main lobe** and **side lobes**: the big beam and the small leaks beside it. **Half-power beamwidth** (3 dB beamwidth): the angle across the main lobe between the points where power has fallen to half. **Aperture**: the antenna's collecting area.
- **High-gain antenna, HGA** / **low-gain antenna, LGA**. **Parabolic reflector** (dish).

🛠 How it actually works

Step	The mechanism, with numbers
1 · The wave	A changing electric field makes a changing magnetic field, which makes a changing electric field… The pair keeps itself going, at right angles to each other and to the direction of travel, at 299,792 km/s. No medium needed.
2 · c = λ × f	FM radio 100 MHz: 300,000,000 ÷ 100,000,000 = **3 m**. Wi-Fi 2.4 GHz: **12.5 cm**. Deep-space link 8.4 GHz: **3.6 cm**.
3 · Receive	Metal is full of loose electrons. The wave's electric field shoves them one way, then the other, so a tiny alternating voltage appears along the rod at the wave's frequency: millionths of a volt. A rod half a wave long fits the wave exactly and rings loudest, like a guitar string.
4 · Transmit	The reverse. Drive an alternating voltage into the rod; electrons shaken back and forth give off a wave at that frequency.
5 · Size	Rod = λ ÷ 2. 100 MHz → 1.5 m. 2.4 GHz → 6 cm. 100 Hz → 1,500 km. That is why nothing is transmitted at voice frequency.
6 · Modulate	AM radio: a 1 MHz carrier whose height follows the voice. FM radio: a 100 MHz carrier whose frequency swings ±75 kHz with it. Digital links flip the carrier's phase to mean 1 or 0.
7 · Gain	A dish D wide squeezes the beam: gain ≈ ½ × (π × D ÷ λ)². Cassini: 4 m at 3.6 cm → about **50,000×** (47 dBi). Beamwidth ≈ 70 × λ ÷ D ≈ **0.6°**.
8 · What it buys	Cassini's transmitter: **20 W**. Through the dish, toward Earth it looked like 1,000,000 W. After 1.4 billion km, a 70 m dish on Earth caught about 0.0000000000000001 W (10^{-16}). Still readable.

SEC 5 · LEC 1 · BANDS + FACT CHECK

Same wave, different sizes

The frequency bands (including ones you already use), the caption errors, and where the instructor was loose.

📻 Check c = λ × f on things you know

Band	Frequency	Wavelength λ	Half-wave rod	Who uses it
AM radio	1 MHz	300 m	150 m	AM stations. Their towers are a quarter-wave, about 75 m.
FM radio	100 MHz	3 m	1.5 m	FM stations. A car aerial is a quarter-wave, 75 cm.
UHF	437 MHz	69 cm	34 cm	CubeSats: the tape-measure antennas in the lecture's picture.
L-band	1.575 GHz	19 cm	9.5 cm	GPS.
S-band	2.4 GHz	12.5 cm	6 cm	Wi-Fi and Bluetooth. Spacecraft command links sit nearby at 2.1–2.3 GHz.
X-band	8.4 GHz	3.6 cm	1.8 cm	Deep-space downlinks: Voyager, Cassini, the Mars orbiters.
Ka-band	32 GHz	9 mm	5 mm	The newest high-rate deep-space links.
Visible light	600,000 GHz	0.0005 mm	—	Your eyes. Ultraviolet from Section 2 is just past it.

You already own the proof: the stubby antennas on a Wi-Fi router are about 3 cm, a quarter of 12.5 cm.

🔎 Caption errors (the worst transcript so far)

Captions said	Instructor meant
"electric in a negative way", "it two magnetic waves"	**electromagnetic wave(s)**
"this to Bush tends to be seen as soil"	the **distribution** tends to be **sinusoidal**
"Nalanda" · "units of Kurds"	**lambda** (λ) · units of **hertz**
"describing a scene disorder"	describing a **sinusoid**
"a hundred foot" · "ten to the light"	a hundred **hertz** · ten to the **eight**
"deep hole antenna"	**dipole** antenna
"I am an F Am" · "amplitude in moderation"	**AM and FM** · amplitude **modulation**
"the main canvasing"	the main **carrier signal**
"indoor activity" · "beam with"	**antenna directivity** · **beamwidth**
"ID or isotropic" · "higher gangs"	**ideal** isotropic · higher **gains**
"Phi game in tennis" · "parabolic tunnels"	**high-gain antennas** · parabolic **antennas**
"cube set"	**CubeSat**

▲ Where the instructor was loose

- **"The wavelength of my voice."** Your voice is a sound wave: air pressure, 343 m/s, and it cannot cross a vacuum at all. A microphone turns it into a voltage wiggling at the same frequency. His 3,000 km is what a *radio* wave at 100 Hz would measure. His arithmetic is right.
- **Modulation "so antennas can be small."** True, and only one reason of three. It also lets thousands of users share the air, each on its own carrier frequency (that is what tuning to a station is), and a faster carrier can carry more data per second.
- **AM and FM are not what spacecraft use.** Both are analog. Space links and Wi-Fi send bits by shifting the carrier's phase.
- **"The higher the frequency, the smaller the antenna"** is for rods. A dish is different: it must be many wavelengths wide, and for the same dish a higher frequency gives a tighter beam and more gain.
- **"With a directional antenna we can get more information."** Precisely: gain compares the signal in the best direction with an isotropic antenna fed the same power. The antenna creates nothing. It moves power from where you don't need it to where you do.
- **Apollo 17.** His explanation tangles. The Moon rover carried two antennas: a low-gain one with a wide beam for voice while driving, and an umbrella-shaped high-gain dish the astronauts aimed at Earth by hand for television when parked.
- **Not said: polarization.** The receiving rod has to line up with the wave's electric field. It is the real reason the CubeSat has two dipoles at right angles.
- **Not said: an isotropic antenna cannot be built.** It is a yardstick only.

SEC 5 · LEC 1 · VOCABULARY

Every word used, in plain English

48 entries. "Repeat" = you have met it before.

Telecommunications — sending information over a distance. Here: by radio between spacecraft and ground.

Signal / information — a quantity that changes in a pattern that means something: a voice, a picture, data.

Electromagnetic (EM) wave — a travelling ripple of electric and magnetic field. Radio and light are both this. *A ripple that needs no pond.*

Electric field / magnetic field — the region where a charge feels a push / where a magnet does. In a wave the two rise and fall together. *Electric field is a repeat.*

Perpendicular (⊥) — at right angles, like the corner of a page.

Sinusoidal / sinusoid / sine wave — the smooth, endlessly repeating up-and-down shape.

Variation / oscillating — changing / swinging back and forth regularly.

Voltage / difference in voltage — electrical 'height' between two points. It makes current flow. *Repeat, ion engines.*

Induce — to cause at a distance. The wave induces a voltage in the antenna.

Speed of light (c) — 299,792 km/s, about 3×10^8 m/s. The speed of every electromagnetic wave in vacuum.

Wavelength (λ, 'lambda') — distance from one crest to the next, in metres. *Repeat, UV in Section 2.*

Frequency (f) — how many waves pass per second.

Hertz (Hz) / kHz / MHz / GHz — one per second / thousand / million / billion per second.

Period — how long one wave takes: 1 ÷ frequency.

Constant — a value that never changes. c is one. *Repeat.*

Inversely related — when one goes up the other goes down.

Antenna — a metal conductor that turns electrical signals into radio waves and back. *A stick that bobs on the ripples.*

Transmitter / receiver — the electronics that drive the antenna / that read what it picks up.

Circuit — a loop of electronic parts that current flows through.

Dipole (half-wave dipole) — two rods end to end, half a wavelength long in total.

Dimension — a size: length, width, diameter.

Modulation — changing a carrier wave in step with a signal so the carrier carries it. *Writing on the ripples.*

Carrier signal — the fast, steady wave that does the travelling.

Information signal (baseband) — the slow signal you actually want to send.

Amplitude — the height of a wave.

Amplitude modulation (AM) — the carrier's height follows the signal.

Frequency modulation (FM) — the carrier's frequency follows the signal. Resists static better.

Phase modulation / PSK — shifting the carrier's timing to send bits. What spacecraft use.

Directivity — how strongly an antenna favors one direction.

Radiation pattern — the 3-D shape of where an antenna's power goes.

Isotropic antenna — an imaginary antenna that radiates equally in every direction. *Bare bulb.*

Homogeneous — the same everywhere.

Omnidirectional antenna — equal all around in one plane: a donut shape.

Directional antenna — sends most of its power one way. *Flashlight.*

Beam / beamwidth — the cone where the signal is strong / its angle in degrees.

Main lobe / side lobes — the big petal where most power goes / the small unwanted petals beside it.

3 dB / decibel (dB) — 3 dB down means half the power. Decibels count in multiplications: 10 dB = ×10.

Aperture / large aperture — the opening or collecting area of an antenna. Bigger aperture, narrower beam.

Gain / dBi — how many times stronger than isotropic in the best direction / the unit it is quoted in.

Parabolic antenna (dish) — a curved reflector that focuses radio waves into a beam.

High-gain antenna (HGA) — a narrow-beam dish for fast data. Has to be aimed.

Low-gain antenna (LGA) — a wide-beam antenna for slow, always-available contact.

Rooftop TV antenna (Yagi) — the fishbone-shaped directional antenna on houses.

Polarization — the direction the wave's electric field swings in.

CubeSat — a tiny satellite built from 10 cm cubes. *Who's Who.*

Cassini — NASA's Saturn orbiter, 2004–2017, with a 4 m dish. *Repeat, Who's Who.*

Apollo 17 / lunar rover — the last crewed Moon landing, 1972 / the car the astronauts drove. *Who's Who.*

Positioning / rotation = attitude — which way the spacecraft points. *Section 6.*

SEC 5 · LEC 1 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next two pages.

A · Equations & key numbers page (p3–4)

$c = \lambda \times f$	$c = 3\times10^8$ m/s (299,792 km/s)
	100 MHz → 3 m · 2.4 GHz → 12.5 cm
$dipole = \lambda \div 2$	100 MHz → 1.5 m · 100 Hz → 1,500 km
$gain \approx \frac{1}{2}(\pi D \div \lambda)^2$	4 m dish at 3.6 cm → ~50,000× (47 dBi)
$beam \approx 70 \cdot \lambda \div D$	→ 0.6°
Cassini	20 W → looks like 1 MW → 10^{-16} W at Earth

B · Section 5 map page (first sentence; more after the next lectures)

S5 · antennas: radio = EM wave, c = λ × f; rod = λ ÷ 2 so high f = small; modulation = signal on a fast carrier; gain = aim the power (dish), costs pointing

C · One-line word bank (new lines)

p__ · electromagnetic wave · ripple of E + B field → C24
p__ · electromagnetic spectrum · radio … light … X-rays
p__ · wavelength (λ) · crest to crest, metres → C24
p__ · frequency (f) / hertz · waves per second → C24
p__ · period · seconds per wave = 1 ÷ f
p__ · speed of light (c) · 3×10^8 m/s; c = λ × f
p__ · sinusoid · the smooth repeating wave shape
p__ · amplitude · height of the wave
p__ · antenna · metal that turns signal ↔ radio wave
p__ · transmitter / receiver · sends / listens
p__ · dipole · rod λ ÷ 2 long; the basic antenna → C25
p__ · modulation · signal written onto a carrier → C25
p__ · carrier / information signal · fast wave / slow message
p__ · AM / FM · height follows signal / frequency follows it
p__ · phase modulation (PSK) · bits by shifting timing
p__ · isotropic · equal every direction; ideal only
p__ · omnidirectional · equal all around one plane
p__ · directional / beamwidth · one beam / its angle at half power
p__ · main lobe / side lobes · the beam / small leaks beside it
p__ · decibel (dB) · 3 dB = half or double; 10 dB = ×10
p__ · gain (dBi) · × stronger than isotropic → C26
p__ · HGA / LGA · narrow fast dish / wide slow rod → C26
p__ · polarization · direction the E field swings

D · Question bank ("but how?" cards)

Q: How does a metal rod pick up a radio wave?
A: The wave's electric field pushes the rod's electrons to and fro → an alternating voltage at the wave's frequency.
Q: Why does the rod have to be about half a wavelength?
A: It resonates: it fits the wave, like a guitar string.
Q: How does a dish make a signal stronger with no extra power?
A: It takes the power going everywhere and aims it one way.
Q: How did 20 W reach Earth from Saturn?
A: A 4 m dish gave ~50,000× gain, and a 70 m dish listened.

SEC 5 · LEC 1 · CONCEPT CARDS

Route E: pages C24 and C25

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 11 — C24

ELECTROMAGNETIC WAVE = EM wave

= travelling ripple of electric + magnetic field

WHY it needs no air or wire, so it is the only way to carry information across space

\# c = 299,792 km/s · radio, light, UV all the same thing

≠ sound (air pressure, 343 m/s, no vacuum)

→ c = λ × f · spectrum · antenna

WAVELENGTH + FREQUENCY = λ and f

= crest-to-crest distance / waves per second

WHY speed is fixed, so the two are tied: pick one and the other is set. λ sets antenna size

\# 100 MHz → 3 m · 2.4 GHz → 12.5 cm · 8.4 GHz → 3.6 cm

≠ amplitude (the wave's height)

→ c = λ × f · dipole = λ ÷ 2

WORD BANK · CONCEPT CARDS · scene 11 — C25

HALF-WAVE DIPOLE = dipole antenna

= two rods end to end, λ ÷ 2 long in total

WHY a rod that fits half a wave resonates, so it gives the biggest voltage for that frequency

\# FM 100 MHz → 1.5 m · Wi-Fi → 6 cm · voice 100 Hz → 1,500 km

≠ a dish (a reflector, many λ wide)

→ omnidirectional · monopole = λ ÷ 4

MODULATION = AM · FM · PSK

= changing a carrier wave so it carries a signal

WHY voice-speed waves need 1,500 km antennas. A fast carrier fits a small antenna and its own channel

\# AM: 1 MHz carrier · FM: 100 MHz ± 75 kHz

≠ amplification (just makes it louder)

→ carrier · AM = height · FM = spacing

C24: 11 + 10 lines of 22 · C25: 10 + 10 lines of 22

SEC 5 · LEC 1 · CONCEPT CARDS

Route E: page C26

The third card page for this lecture.

WORD BANK · CONCEPT CARDS · scene 11 — C26

ANTENNA GAIN = dBi

= strength in the best direction vs isotropic

WHY power spread over a whole sphere is wasted. Squeeze it into a beam and the far end hears more

\# dipole 1.6× · Cassini dish ~50,000× (47 dBi)

≠ added power (it only aims)

→ ≈ ½(πD ÷ λ)² · beamwidth ≈ 70λ ÷ D

HIGH-GAIN vs LOW-GAIN = HGA · LGA

= narrow-beam dish / wide-beam rod

WHY you want both: speed when you can aim, and a link that still works when you can't

\# Cassini HGA 4 m, 0.6° beam · 20 W looks like 1 MW

≠ transmitter power (same either way)

→ attitude control · safe mode · CubeSat

Why this lecture got two spreads

It is the longest lecture so far and it holds two separate ideas: how a wave carries a signal (waves, antenna size, modulation), and how you aim it (directivity, gain). Your rule is one spread per video unless it doesn't fit, and never smaller writing. This one doesn't fit.

If you are short on pages, copy spread 1 in full and add only the drawing from spread 2 under it. The gain idea is the one thing in spread 2 you can't skip.

You already know half of this. This is the physical layer under everything in networking: 2.4 GHz and 5 GHz Wi-Fi, antenna gain in dBi, why a directional antenna reaches farther. Use that. When Network+ resumes, this page is a head start on the wireless objectives.

C26: 10 + 10 lines of 22.

SEC 5 · LEC 1 · SAY IT OUT LOUD

Last page: prove it by talking

After the spreads and the 3-line summaries, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"How does a probe at Saturn talk to Earth? It can't have much power out there."

✅ Three sentences you should be able to say

- By radio, which is a ripple of electricity and magnetism that crosses empty space at the speed of light. An antenna is just a metal rod the ripple shakes.
- You can't send a voice or data directly, because slow signals would need an antenna thousands of kilometres long. So you write the signal onto a fast carrier wave. That is modulation.
- And it has very little power: Cassini had 20 watts. But a big dish squeezes it into a beam half a degree wide, which makes it tens of thousands of times stronger in our direction, as long as the dish points exactly at Earth.

❓ Follow-ups a friend might ask

- "Why does frequency matter?" (frequency times wavelength is always the speed of light, and the antenna has to be about half a wavelength)
- "AM and FM, what's the difference?" (AM changes the carrier's height, FM changes its spacing)
- "Does a dish add power?" (no, it only aims what is there, like a flashlight reflector)
- "What if the dish isn't pointing at Earth?" (you fall back to a low-gain antenna: wide beam, very slow)
- "Is that the same as Wi-Fi?" (same physics: 2.4 GHz is a 12.5 cm wave, and router antennas are a quarter of that)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ **Two more lectures in Section 5. Send them the same way.**

SEC 5 · LEC 2 · ATMOSPHERE + FREQUENCY BANDS

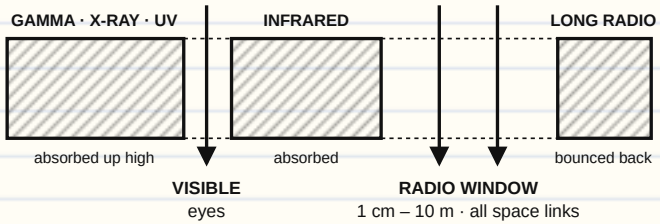
A wall with two windows

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S5 / L2 — Atmosphere + bands **date · pg __**

1 · THE AIR IS A WALL WITH TWO WINDOWS



← short wavelength · high frequency · high energy

long wavelength · low frequency · low energy →

2 · WHY: energy per packet = h × f

gamma, X, UV hit hard: rip electrons off air atoms → absorbed

visible too weak to break air, too strong to shake it → passes

infrared just right to shake water + CO_2 molecules → absorbed

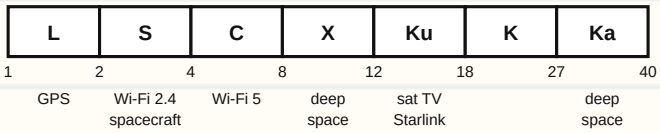
radio far too gentle to do anything → passes

long radio the ionosphere's electrons bounce it back

higher frequency = more energy in each packet (photon)

3 · BAND LETTERS (edges in GHz)

frequency rises →



UPLINK = Earth → space · DOWNLINK = space → Earth

different frequencies, so they don't drown each other

✏️ **PENCIL KEY** · hatched block = the air blocks it · gap + arrow = it gets through · boxes = band letters, numbers = GHz at each edge

⭐ **3 TAKEAWAYS** 3 lines

1. The air is a wall with two windows: visible light and radio.
2. Higher frequency = more energy per photon. Radio is too gentle to absorb.
3. Spacecraft talk in the radio window: S, X, Ka. Uplink and downlink differ.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What is atmospheric opacity?	atmospheric opacity = how much the air blocks at each wavelength (0% clear, 100% blocked)
Which waves does the air block?	short wavelength = high frequency = high energy
Which two get through?	gamma, X-ray, UV: blocked by the upper atmosphere · protects us
Why are gamma, X and UV stopped?	visible light: gets through (a gap) · why eyes use it
Why can our eyes use visible light?	infrared: mostly absorbed by gases (water, CO_2) → observe from space
What stops infrared? Long radio?	radio waves ~1 cm – 10 m: get through → telecommunications
Where do spacecraft links sit?	long-wavelength radio: blocked (bounced by the ionosphere)
The frequency-band names, low to high?	bands: VLF · LF · MF · HF · VHF · UHF · SHF · EHF (each 10× the last)
Satellite letter bands in order?	satellites: SHF 3–30 GHz · letter bands L S C X Ku K Ka (1–40 GHz)
Uplink vs downlink?	deep space: S ~2.1 / 2.3 · X 7.2 / 8.4 · Ka 34 / 32 GHz (up / down)
? why are space telescopes in space	uplink = Earth → space · downlink = space → Earth

SEC 5 · LEC 2 · TEACH

Hard hitters get stopped. Radio slips through.

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

The air around the Earth is a wall with two windows in it. One window lets in ordinary light, the kind your eyes use. The other lets in radio. Everything else gets stopped. The dangerous rays (ultraviolet, X-rays, gamma rays) are soaked up high in the sky, which is what keeps us alive. Heat rays (infrared) are soaked up by the water and carbon dioxide in the air. And very long radio waves bounce off an electric layer near the top, like a ball off a ceiling.

Why? Light of every kind arrives in tiny packets, and the faster it wiggles, the harder each packet hits. Hard-hitting packets smash into air atoms and are used up. Gentle radio packets slip between them untouched. So if you want to talk to a spacecraft from the ground you **must** use the radio window. And if you want to see X-rays or infrared from the stars, you have to lift the telescope above the wall. That is why Hubble and Webb are in space.

🏷 The official names (name first, then what it means)

- **Atmospheric opacity** (its opposite is **transmittance**): the share of radiation the air stops, as a function of wavelength.
- **Atmospheric window**: a wavelength range the air lets through. The **optical window** (visible light) and the **radio window** (about 1 cm to 10 m, or 30 GHz down to 30 MHz).
- **Photon**: one packet of electromagnetic energy. **Photon energy** $E = h \times f$, where h is the **Planck constant**.
- **Absorption**: a photon giving its energy to an atom or molecule and vanishing. **Ionization** when it knocks an electron off; **molecular vibration** when it sets a molecule shaking.
- **Ionospheric reflection**: long radio waves being turned back by the ionosphere's free electrons. The cut-off is the **critical frequency**, around 10 MHz.
- **Radio-frequency bands** (ITU bands): **VLF, LF, MF, HF, VHF, UHF, SHF, EHF**. Each covers ten times the frequency of the one before. **Microwaves** are roughly 1–100 GHz.
- **Letter bands** (IEEE radar bands): **L, S, C, X, Ku, K, Ka**. Wartime radar code letters, deliberately meaningless. Ku and Ka mean "K-under" and "K-above".
- **Uplink**: Earth to spacecraft. **Downlink**: spacecraft to Earth.
- **Deep space**, in radio rules: farther than 2 million km from Earth. Its bands are protected because the signals are so faint.
- **ITU**, the International Telecommunication Union: the UN body that decides who may use which frequency. **Frequency allocation**.
- **Rain fade**: rain and water vapour weakening signals above about 10 GHz.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Photon energy	$E = h \times f$. Handy form: energy in eV = 1,240 ÷ wavelength in nanometres. UV at 250 nm: **5 eV**. Green light, 550 nm: 2.3 eV. Infrared, 10,000 nm: 0.12 eV. X-band radio, 3.6 cm: **0.000035 eV**.
2 · Gamma, X, UV	Splitting an oxygen molecule takes about 5 eV; tearing an electron off nitrogen takes 15. These photons have that, so they are used up 50–100 km overhead. That is what makes the ozone layer and the ionosphere (Section 2).
3 · Visible	1.8 to 3.1 eV: not enough to break air molecules, far too much to match their vibrations. Nothing in clean air can take it, so it passes.
4 · Infrared	About 0.1 eV is exactly what sets water and CO_2 molecules bending and stretching. Absorbed. This is the greenhouse effect, and why infrared telescopes fly.
5 · Radio window	Millionths of an eV: too little to excite anything. Transparent from about 30 MHz to 30 GHz. Every spacecraft link lives here. Edges: water vapour absorbs near 22 GHz, oxygen near 60 GHz, and rain hurts above 10 GHz.
6 · Long radio	Below a critical frequency $\approx 9 \times \sqrt{(\text{electrons per m}^3)}$ hertz, the ionosphere's electrons move fast enough to send the wave back. With 10^{12} electrons per m^3 that is **9 MHz**. So shortwave radio skips round the world, and can't reach a spacecraft.
7 · Up ≠ down	Deep-space X-band: uplink 7.2 GHz, downlink 8.4 GHz. If both were the same, the craft's 20 W transmitter would deafen its own receiver, which is listening for a signal a billion billion times weaker.
8 · Why go higher	Same dish, Ka (32 GHz) instead of X (8.4): the wavelength is 3.8 times shorter, so the gain is $3.8^2 \approx$ **14 times** higher. Price: rain fade and a beam 3.8 times narrower to point.

SEC 5 · LEC 2 · BANDS + FACT CHECK

The names on the dial

Both naming systems from the slides with what lives in each band, then the caption errors and where the instructor was loose.

📻 The decade bands (each ×10)

Band	Frequency	Wavelength	What lives there
VLF very low	3–30 kHz	100–10 km	submarine and maritime signals
LF low	30–300 kHz	10–1 km	old navigation beacons (Loran-C)
MF medium	0.3–3 MHz	1 km–100 m	AM radio
HF high	3–30 MHz	100–10 m	shortwave: bounces off the ionosphere
VHF very high	30–300 MHz	10–1 m	FM radio, TV, aircraft; small satellites
UHF ultra high	0.3–3 GHz	1 m–10 cm	phones, GPS, Wi-Fi 2.4, CubeSats
SHF super high	3–30 GHz	10–1 cm	satellites, deep space, Wi-Fi 5, radar
EHF extremely high	30–300 GHz	1 cm–1 mm	radio astronomy, newest links, 5G mmWave

🛰️ The letter bands, and deep space's slices

Letter	GHz	You know it as	Deep-space uplink / downlink
L	1–2	GPS (1.575)	—
S	2–4	Wi-Fi 2.4, Bluetooth, microwave ovens	2.110–2.120 / 2.290–2.300
C	4–8	Wi-Fi 5 and 6 GHz, weather radar	—
X	8–12	the deep-space workhorse	7.145–7.190 / 8.400–8.450
Ku	12–18	satellite TV, Starlink to your dish	—
K	18–27	water vapour absorbs here (22 GHz)	—
Ka	27–40	Starlink gateways, newest probes	34.200–34.700 / 31.800–32.300

The X-band uplink sits just under 8 GHz, in C-band by the letters. Nobody minds: the pair is still called X-band. Edges differ a little between sources; the slide draws K/Ka at 26.

🔎 Caption errors

Captions said	Instructor meant
"Katis graphic"	**this** graphic
"an much Paik opacity", "atmospheric capacity"	**atmospheric opacity**
"a lead to a wavelength"	a **little** (short) wavelength
"very fine and energy"	very **high in** energy
"ultraviolet night"	ultraviolet **light**
"merits in navigation"	**maritime** navigation
"satellite micro telecommunications"	satellite **microwave** telecommunications
"3G Gerhard's… 30 Jihad's… ten to nine cuts"	3 **gigahertz**… 30 **gigahertz**… 10^9 **hertz**
"L as see x. Ok Kasan"	**L, S, C, X, Ku, K, Ka**
"nomenklatura"	**nomenclature** (naming system)

▲ Where the instructor was loose

- **"High-energy radiation interacts with any particle, basically always."** Not quite. A photon is absorbed when its energy matches something a molecule can do with it. Gamma, X and UV have enough to rip atoms apart. Infrared matches molecular vibrations. Visible and radio match nothing, so they pass. It is about matching, not just "more energy".
- **"Long-wavelength radio waves blocked."** By a different mechanism from the rest: the ionosphere *reflects* them. That is why shortwave radio reaches round the planet.
- **Not said: why uplink and downlink differ.** So the spacecraft can transmit and listen at once without deafening itself.
- **Not said: why anyone wants the higher bands.** The same dish gives more gain and there is more room for data. The cost is weather and tighter pointing.
- **Not said: who hands out frequencies.** The ITU, by international treaty.
- **The slide's table** lumps Ku, K and Ka together as "K, 12–40 GHz", and gives X-band's short end as 2.4 cm where 2.5 cm is correct.
- **He was right** that it is no accident we see in visible light: it is the window where the Sun is brightest and the air is clear.

SEC 5 · LEC 2 · VOCABULARY

Every word used, in plain English

39 entries. "Repeat" = you have met it before.

Atmospheric absorption — the air taking in radiation and stopping it.

Atmospheric opacity — how much of the incoming radiation the air blocks. 100% = nothing gets through. *The wall.*

Atmospheric window — a range of wavelengths the air lets through. *The window in the wall.*

Wavelength / frequency — crest-to-crest distance / waves per second. Short wavelength = high frequency. *Repeat, L1.*

Energy (of radiation) / photon — light arrives in packets; each packet is a photon and carries energy = h × f.

Planck constant (h) — the tiny fixed number that turns frequency into energy.

Electron-volt (eV) — a unit of energy the size of one chemical bond. *Repeat, Section 2.*

Radiation — energy travelling as waves or particles. Here: electromagnetic waves. *Repeat.*

Gamma rays / X-rays — the highest-energy electromagnetic waves. Stopped high in the atmosphere.

Ultraviolet (UV) — just beyond violet light. Mostly stopped by oxygen and ozone. *Repeat, Section 2.*

Visible light — the narrow band our eyes detect: 400–700 nanometres.

Infrared (IR) — just beyond red light; what warm things glow in. Mostly absorbed by water vapour and CO_2.

Radio waves / microwaves — the longest electromagnetic waves / the short end of radio, about 1 mm to 30 cm.

Spectrum — the whole range laid out in order. *Repeat, L1.*

Atom / molecule / electron configuration — what air is made of, and how the electrons are arranged in it. That arrangement decides which energies can be absorbed.

React / interact / collide — here: a photon meeting an air particle and handing over its energy.

Ionization — knocking an electron off an atom. *Repeat, ion engines.*

Atmospheric gases — here: mainly water vapour, carbon dioxide and ozone. *Repeat.*

Distortion — the twinkling and blurring that moving air adds to starlight.

Observable — able to be seen or measured.

Ionosphere — the electrically charged upper atmosphere. Reflects long radio waves. *Repeat, Section 2.*

Reflect / critical frequency — bounce back / the frequency below which the ionosphere reflects.

VLF, LF, MF, HF, VHF, UHF, SHF, EHF — very low, low, medium, high, very high, ultra high, super high, extremely high frequency.

Kilohertz, megahertz, gigahertz — thousand, million, billion waves per second. *Repeat, L1.*

Shortwave radio — HF radio, 3–30 MHz, which skips off the ionosphere.

Maritime / navigation aid — to do with ships / a radio beacon for finding position.

Radio astronomy — studying the sky by its radio waves.

Band — a named slice of the spectrum.

L, S, C, X, Ku, K, Ka bands — the letter names for slices between 1 and 40 GHz.

Nomenclature — a naming system.

Uplink / downlink — Earth to spacecraft / spacecraft to Earth.

Deep space / near-Earth — in radio rules: farther / closer than 2 million km.

Telecommunications link — one radio connection between two ends. *Next lecture.*

ITU — International Telecommunication Union: the body that allocates frequencies worldwide. *Who's Who.*

Frequency allocation — the agreed assignment of a band to a use.

Rain fade — signal loss in rain, worse at higher frequencies.

Greenhouse effect — infrared from the ground being absorbed by water vapour and CO_2. *Same physics as the infrared wall.*

Hubble / Webb (JWST) — space telescopes: Hubble for visible and UV, Webb for infrared. *Repeat, Who's Who.*

Loran-C — a retired long-range radio navigation system that used low frequencies. *Who's Who.*

SEC 5 · LEC 2 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. The concept cards (route E) are on the next page.

A · Equations & key numbers page (p3–4)

```
E = h × f           photon energy · eV = 1,240 ÷ λ in nm
                    UV 250 nm → 5 eV · radio 3.6 cm → 0.000035 eV
radio window        ~30 MHz – 30 GHz (10 m – 1 cm)
ionosphere cut-off  f ≈ 9·√(electrons per m³) → ~9 MHz
bands (GHz)         L 1–2 · S 2–4 · C 4–8 · X 8–12 · Ku 12–18
                    K 18–27 · Ka 27–40
deep space X        uplink 7.2 · downlink 8.4 GHz
```

B · Section 5 map page (second sentence)

```
S5 · atmosphere: wall with 2 windows (visible, radio);
     hard photons absorbed, long radio bounced; space links
     use S / X / Ka; uplink ≠ downlink frequency
```

D · Question bank ("but how?" cards)

```
Q: Why does the air stop UV but not radio?
A: E = h × f. A UV photon (5 eV) can break air molecules, so it
   is absorbed. A radio photon is a million times too weak.
Q: Why does infrared get absorbed if it is weaker than light?
A: Its energy matches how water and CO₂ molecules vibrate.
Q: Why can't you talk to a spacecraft on shortwave?
A: Below ~10 MHz the ionosphere reflects the wave back down.
Q: Why are uplink and downlink on different frequencies?
A: So the craft's transmitter doesn't deafen its own receiver.
```

C · One-line word bank (new lines)

```
p__ · atmospheric opacity · how much the air blocks, per λ
p__ · atmospheric window · λ range that gets through  → C27
p__ · optical / radio window · visible / ~1 cm – 10 m
p__ · photon · one packet of light
p__ · photon energy · E = h × f; more f, more energy  → C27
p__ · Planck constant (h) · turns frequency into energy
p__ · absorption · photon gives its energy to a molecule
p__ · gamma / X-ray / UV · high energy; stopped up high
p__ · infrared (IR) · heat rays; absorbed by H₂O + CO₂
p__ · microwave · short radio, ~1–100 GHz
p__ · ionospheric reflection · long radio bounced back
p__ · critical frequency · below it the ionosphere reflects
p__ · VLF … EHF · decade bands, each 10× the last
p__ · SHF · 3–30 GHz: where satellites talk
p__ · letter bands · L S C X Ku K Ka (1–40 GHz)  → C28
p__ · S / X / Ka-band · the deep-space trio
p__ · uplink / downlink · Earth → s/c / s/c → Earth  → C28
p__ · deep space (radio) · beyond 2 million km
p__ · ITU · body that allocates frequencies
p__ · rain fade · rain weakens signals above ~10 GHz
```

SEC 5 · LEC 2 · CONCEPT CARDS

Route E: pages C27 and C28

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 12 — C27

ATMOSPHERIC WINDOW = optical + radio

window

= wavelengths the air lets through

WHY the air blocks most of the spectrum.

Ground links and ground telescopes

must use the gaps

visible 400–700 nm · radio ~1 cm – 10

m (30 GHz – 30 MHz)

≠ the ozone hole (a different thing)

→ opacity · radio bands · space

telescopes

PHOTON ENERGY = E = h × f

= energy of one packet of light

WHY it decides what a wave can do to

matter: break it, shake it, or pass

through

UV 5 eV · visible 2 eV · infrared 0.1

eV · X-band 0.000035 eV

≠ total power (many weak photons)

→ eV = 1,240 ÷ nm · ionization · UV

damage

WORD BANK · CONCEPT CARDS · scene 12 — C28

FREQUENCY BANDS = L S C X Ku K Ka

= named slices of the radio spectrum

WHY everyone shares one spectrum. Named,

allocated slices keep users from

jamming each other

S 2–4 · X 8–12 · Ka 27–40 GHz · deep

space X: 8.4 down

≠ channels (finer slices inside a band)

→ ITU · SHF 3–30 GHz · Wi-Fi is S + C

UPLINK / DOWNLINK

= Earth → spacecraft / spacecraft →

Earth

WHY the craft sends and listens at once.

Separate frequencies stop its

transmitter deafening its receiver

X-band deep space: up 7.2 GHz, down

8.4 GHz

≠ upload / download speed

→ transponder · bands · link budget

C27: 11 + 10 lines of 22 · C28: 9 + 10 lines of 22

SEC 5 · LEC 2 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"Why do they put telescopes in space? And why does everything in space use radio?"

✅ Three sentences you should be able to say

- Because the air is a wall with two windows. Ordinary light gets through one and radio gets through the other, and nearly everything else is stopped.
- It comes down to how hard each packet of light hits. X-rays and ultraviolet hit hard enough to break air molecules, so they are absorbed high up, and infrared gets soaked up by water and CO_2. Radio is too gentle to do anything, so it sails through.
- So telescopes for X-rays or infrared have to sit above the air, and anything you want to say to a spacecraft has to go through the radio window, in agreed bands like S, X and Ka.

❓ Follow-ups a friend might ask

- "Why can we see, then?" (our eyes use exactly the window where the air is clear and the Sun is brightest)
- "Does all radio get through?" (no: very long waves bounce off the ionosphere, and rain hurts the highest bands)
- "What do the band letters mean?" (nothing: old radar code names; X is around 8 GHz)
- "Is Wi-Fi one of those?" (yes: 2.4 GHz is S-band, 5 GHz is C-band)
- "What's uplink and downlink?" (ground to spacecraft, spacecraft to ground, on different frequencies so they don't clash)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ One lecture left in Section 5: satellite links. Send it the same way.

SEC 5 · LEC 3 · SATELLITE + SPACECRAFT LINKS

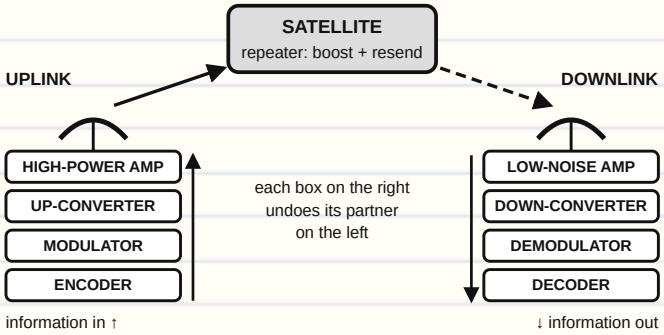
A chain, a loss, and a ratio

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

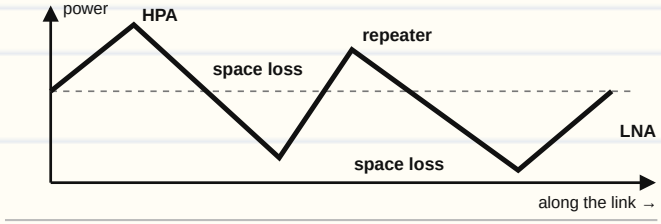
LEFT PAGE · DRAW

S5 / L3 — Satellite links **date · pg __**

1 · THE LINK IS A CHAIN



2 · POWER ALONG THE LINK



3 · THE LINK EQUATION

$$S \div N_0 = P \times G_t \times (G_r \div T) \div (k \times L_s \times L_a)$$

helps: power P · transmit gain G_t · receive gain G_r
hurts: space loss L_s · air loss L_a · noise k × T
$S \div N_0$ = signal compared with noise

bigger $S \div N_0$ = more bits per second

✏️ **PENCIL KEY** · solid arrow = uplink · dashed arrow = downlink · curved line = dish · zigzag = signal power · dotted line = starting level

⭐ **3 TAKEAWAYS** 3 lines

1. A link is a chain: every receive box undoes one transmit box.
2. Space loss is spreading, not soaking up: twice as far = a quarter the power.
3. What counts is signal ÷ noise: more power, bigger dishes, colder receiver.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions) **NOTES (keywords)**

What do encoders and decoders do?

Up-converter: what, and why after modulation?

HPA: where, and what for?

What two losses hit the signal in flight?

What does a satellite repeater do?

LNA: where, and why "low noise"?

The whole chain, in order?

$S \div N_0$: what is it? High or low is good?

Link equation: what helps, what hurts?

Why does the power graph zigzag?

? what is k really

encoder / decoder: convert information from one format or code to another (compress it, protect it) and back

modulator / demodulator: put the signal on a carrier / take it off

up-converter: raises the frequency (MHz or a few GHz → up to 30 GHz) · down-converter reverses it

HPA (high-power amplifier): boosts the signal just before the antenna

in flight: free-space loss L_s (distance) + atmospheric loss L_a

satellite repeater: boosts the signal on board and sends it back down

LNA (low-noise amplifier): boosts the weak received signal while adding as little noise as possible

chain: encode → modulate → up-convert → HPA → antenna ⇒ LNA → down-convert → demodulate → decode

$S \div N_0$ = signal over noise · high = lots of information

$S \div N_0 = P \times G_t \times (G_r \div T) \div (k \times L_s \times L_a)$

P power · G_t G_r antenna gains · T noise temperature · k Boltzmann constant

SEC 5 · LEC 3 · TEACH

Shouting across a stadium

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

You want to tell a friend a secret across a huge, noisy stadium. First you pack the message small and add a few spare words so a missed word can be guessed (**encode**). You put it on a whistle note that carries (**modulate**), move that note up to the pitch you were told to use (**up-convert**), and blow as hard as you can (**high-power amplifier**) through a megaphone (the **antenna**).

On the way, your shout does not get eaten. It **spreads out**. Twice as far away, it covers four times the area, so each ear gets a quarter of it. A helper halfway hears you faintly and shouts it on again (the **repeater**). Your friend cups a big ear (a dish) and uses a hearing aid that adds almost no hiss of its own (the **low-noise amplifier**), then undoes every step in reverse. Whether the secret arrives depends on one thing only: how loud your shout is **compared with the crowd noise** at your friend's ear. That comparison is the whole link equation.

🏷 The official names (name first, then what it means)

- **Communications link**: one complete radio path, transmitter to receiver. **Uplink**, **downlink**. The **space segment** is the satellite; the **ground segment** is the Earth stations.
- **Source coding** (compression) and **channel coding** (forward error correction, FEC): the two jobs the lecture's "encoder" does. One removes waste; the other adds protective spare bits. **Codec** = coder + decoder.
- **Modem**: modulator + demodulator.
- **Up-converter / down-converter** (frequency converter, mixer): shifts a modulated signal to a different frequency without changing what it carries. The low starting frequency is the **intermediate frequency, IF**; the on-air one is the **radio frequency, RF**.
- **High-power amplifier, HPA** (power amplifier): the last stage before the transmit antenna.
- **Free-space path loss, L_s** (spreading loss; the slide's "free space loss"). **Atmospheric loss, L_a** (atmospheric attenuation).
- **Transponder** (the lecture's "satellite repeater"): receives, shifts frequency, amplifies, retransmits. A satellite that only does this is a **bent-pipe** relay.
- **Low-noise amplifier, LNA**: the first stage after the receive antenna.
- **Thermal noise**; **noise power density, N_0** = k × T. **Boltzmann constant, k**. **System noise temperature, T**.
- **EIRP**, effective isotropic radiated power: P × G_t. **G/T** ("G over T"): receive gain ÷ noise temperature, a station's figure of merit.
- **Signal-to-noise ratio**; here **S/N_0**. **Link budget**: the gains-and-losses ledger. **Decibel, dB**. **Link margin**.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Encode	Squeeze out the waste (compression), then add structured spare bits so the far end can repair errors. Good codes let you send about 5 times more data over the same link.
2 · Modulate, then move	Write the bits onto a carrier at an easy fixed frequency, say 70 MHz. Then mix it with a 7,100 MHz tone: out comes the same signal at **7,170 MHz**, in the X-band uplink slot. Two steps, because precise modulators are easy to build low and one design then serves any band.
3 · Amplify out	The signal is strong and clean here, so noise is no worry. Only watts matter: 20 W on Cassini, 20,000 W on a Deep Space Network dish.
4 · Space loss	Nothing absorbs the wave. It spreads over a sphere of area $4\pi d^2$. **Twice the distance, a quarter of the power.** In the standard form $L_s = (4\pi d \div \lambda)^2$. To a GEO satellite: 10^{20}. Saturn to Earth: 2.5×10^{29}.
5 · Repeat	A relay satellite hears perhaps 10^{-10} W, moves it to the downlink frequency so it cannot jam its own receiver, amplifies it about 10^{11} times and sends it down.
6 · Amplify in	Everything warm hisses with radio noise: $N_0 = k \times T$ watts in every hertz, with $k = 1.38 \times 10^{-23}$. Whatever noise the *first* amplifier adds is amplified by all the stages after it, so that one stage must be the quietest thing in the station. NASA chills its to about 15 K. Then down-convert, demodulate, decode.
7 · The ratio	An amplifier cannot tell signal from noise; it lifts both. So the ratio fixed at the first amplifier is all you ever get. Data rate ≈ $S/N_0 \div$ (a factor of about 2 to 4 that the code needs).

SEC 5 · LEC 3 · THE LINK BUDGET

Cassini to Earth, line by line

The link equation with real numbers. Multiply the middle column, or add the right-hand column: decibels turn the whole thing into addition.

🧾 The ledger: Saturn to a 70 m dish, 8.4 GHz

Item	As a number	In decibels
P · transmitter power	20 W	+13 dBW
$\mathbf{G_t}$ · Cassini's 4 m dish	× 50,000	+47 dB
EIRP = P × G_t	*1,000,000 W*	*+60 dBW*
$\mathbf{L_s}$ · space loss, 1.4 billion km	÷ 2.5×10^{29}	−294 dB
$\mathbf{L_a}$ · air, clear sky	÷ 1.05	−0.2 dB
$\mathbf{G_r}$ · 70 m dish on Earth	× 25,000,000	+74 dB
S · signal received	**10^{-16} W**	**−160 dBW**
$\mathbf{N_0}$ = k × T, with T = 20 K	2.8×10^{-22} W per Hz	−216 dBW/Hz
S ÷ N_0	**about 400,000**	**56 dB-Hz**

Reading the answer. The signal is as strong as the noise in a 400,000 Hz slice. Divide by about 4 for the code's needs and a safety margin: roughly **100,000 bits per second**. Cassini's real top rate was 166,000.

Why rates collapse with distance. Voyager 1 is 17 times farther than Saturn, so its space loss is $17^2 \approx 300$ times worse. It sends 160 bits per second. New Horizons needed 15 months to return its Pluto data.

📐 Decibels in one minute

A decibel counts multiplications. **+3 dB = ×2 · +10 dB = ×10 · +20 dB = ×100 · −30 dB = ÷1,000.** Multiplying numbers means adding their decibels, which is why the right-hand column simply adds up: 13 + 47 − 294 − 0.2 + 74 = −160.

dBW means "compared with 1 watt". **dBm** means "compared with 1 milliwatt". You have seen this already: a Wi-Fi router sends about +20 dBm (0.1 W), your laptop reports the signal as about −60 dBm, and its noise floor is near −95 dBm. That is a link budget. Cassini's is the same sum with bigger numbers.

🎚 The only knobs there are

- **More power** (P): costs solar panels and heat.
- **Bigger or higher-frequency dish** (G_t): costs mass and pointing accuracy.
- **Bigger ground dish, or several combined** (G_r): 70 m is about the limit.
- **Colder receiver** (T): halve the noise temperature, double the data.
- **Better codes**: free decibels from mathematics.
- **Distance** (L_s): the one you cannot touch, and by far the largest term.

A decibel gained anywhere is worth the same as a decibel gained anywhere else.

SEC 5 · LEC 3 · FACT CHECK

What the captions mangled, and what he blurred

The caption errors, then where the instructor was loose or wrong.

🔎 Caption errors

Captions said	Instructor meant
"a few competence"	a few **components**
"upconvert as… not that process to modulation"	**up-converters**… an **added** process to modulation
"few geophones… 30 g of hurts"	few **gigahertz**… 30 **gigahertz**
"de-code us" · "compress as"	**decoders** · **compressors**
"analogous signal"	**analog** signal
"a much fake a twin ration", "a fake translation"	**atmospheric attenuation**
"low noise and profiles", "high power for fires"	low-noise **amplifiers**, high-power **amplifiers**
"and blink" · "turns meets" · "gametes"	an **uplink** · **transmits** · **emits**
"inverse Procida" · "Mody elaters"	inverse **process** · **modulators**
"mathematically rented"	mathematically **oriented**
"divided by zero and zero"	divided by **N-zero (N_0)**
"the balsam and constant"	the **Boltzmann** constant
"PTZ t" · "gang transmitted"	**P_T G_T** · **gain** transmitted
"less and y divided by L.A."	**L_S** and 1 divided by **L_A**
"space for the losses gave their names"	**free-space** losses

▲ Where the instructor was loose

- **"1/k… is just a relation for wavelength and other parameters."** Wrong. k is the **Boltzmann constant**, 1.38×10^{-23} joules per kelvin. It turns the receiver's temperature into noise power: $N_0 = k \times T$. It has nothing to do with wavelength.
- **The low-noise amplifier must "avoid mistaking the signal for noise".** No amplifier can tell them apart. It lifts both. The point is that it adds very little noise *of its own*, because anything the first stage adds is magnified by every stage after it.
- **"High-power amplifiers know the signal, so it's safe and they can amplify more."** The real reason is simpler: at the transmitter the signal is enormous compared with any noise, so only output power matters.
- **"Free-space losses."** Right that they grow with distance. Not said: nothing is lost to anything. The wave spreads, and the receiver catches a smaller share.
- **"Encoders… convert my voice into an electrical signal."** That is a microphone. Encoding means compressing the data and adding error-correcting bits. The second job is what makes deep-space links work at all.
- **"Up-conversion is similar to modulation."** Different jobs. Modulation writes the message on a carrier. Up-conversion moves the finished signal to another frequency and changes nothing in it.
- **"Gain is related to the high power and the directivity."** Antenna gain is directivity only. Power is its own term.
- **His diagram is a relay link:** ground to satellite to ground. That is a communications satellite, the "relay" type from Section 1. A science probe is one end of the link, not the middle.
- **Not said: what the ratio buys.** Bits per second. That is the reason anyone computes it.

SEC 5 · LEC 3 · VOCABULARY

Every word used, in plain English

40 entries. "Repeat" = you have met it before.

Link — one radio connection from a transmitter to a receiver.

Component — one part of a system.

Modulator / demodulator (modem) — writes the signal onto a carrier / takes it off again. *Repeat, L1.*

Up-converter / down-converter — moves a signal to a higher / lower frequency without changing what it carries. *Changing lanes, not changing the cargo.*

Encoder / decoder (codec) — turns information into a compact, protected code / turns it back.

Format / code — an agreed way of writing information as symbols.

Compressor / compression — squeezing data into fewer bits by removing what is repeated or unneeded.

Error-correcting code — spare bits added in a pattern so the receiver can find and fix mistakes.

Analog signal — a smoothly varying signal, like a voice.

Electric pulses / digital signal — information as a string of 1s and 0s.

Bit / bits per second — one 1-or-0 / how many are sent each second: the data rate.

Satellite repeater = transponder — equipment on a satellite that receives a signal, boosts it and sends it on. *The helper halfway across the stadium.*

Attenuation — weakening.

Atmospheric attenuation = atmospheric loss (L_a) — the weakening caused by the air and rain. *Repeat, L2.*

Free-space loss (L_s) — the weakening caused by the wave spreading out over distance. *Twice as far, a quarter as strong.*

Amplifier / amplify — a device that makes a signal stronger / to do so.

High-power amplifier (HPA) — the amplifier that feeds the transmitting antenna. *Blowing hard.*

Low-noise amplifier (LNA) — the first amplifier after the receiving antenna, built to add almost no noise. *A hearing aid with no hiss.*

Noise — random electrical hiss that is always present and hides weak signals.

Thermal noise — noise caused by heat: every warm object and circuit makes it.

Signal (S) — the power of the wanted signal at the receiver.

N_0 (N-zero) — noise power in each hertz of bandwidth. $N_0 = k \times T$.

Signal-to-noise ratio (S/N_0) — how strong the signal is compared with the noise. Sets the data rate.

Boltzmann constant (k) — 1.38×10^{-23} joules per kelvin. Converts temperature into noise power.

Kelvin (K) — temperature counted from absolute zero. 20 K is −253 °C. *Repeat.*

System temperature (T) — a single number for all the noise in a receiving station, as if it came from an object that warm.

Transmitted power (P) — watts fed to the transmitting antenna.

Gain (G_t, G_r) — how much the transmit / receive antenna concentrates the signal. *Repeat, L1.*

EIRP — power × transmit gain: how strong the transmitter looks from far away.

G/T — receive gain ÷ noise temperature: how good a listening station is.

Link budget — the list of every gain and loss between transmitter and receiver, added up. *A bank statement for the signal.*

Decibel (dB) / dBW / dBm — a scale that counts multiplications / relative to 1 watt / to 1 milliwatt. *Repeat, L1.*

Margin — the safety allowance left over after the sum.

Transmitting / receiving Earth station — the ground end that sends / listens.

Space segment / ground segment — the satellite part / the Earth part of a system.

Uplink / downlink — ground to satellite / satellite to ground. *Repeat, L2.*

Input / output information — what goes into the chain / what comes out the far end.

Distorted — changed in shape on the way.

Relay / bent pipe — a satellite that only passes signals on. *Repeat, Section 1.*

Voyager 1 / New Horizons / Deep Space Network — the farthest probe / the Pluto probe / NASA's 70 m dishes. *Who's Who.*

SEC 5 · LEC 3 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D. Section 5 ends here, so route B is the full set of lines for the Section 5 map page.

A · Equations & key numbers page (p3–4)

```
S/N₀ = P·Gₜ·(Gᵣ/T) ÷ (k·Lₛ·Lₐ)   the link equation
Lₛ = (4πd ÷ λ)²                  space loss · 2× distance = ¼ power
N₀ = k × T                       k = 1.38×10⁻²³ J/K · 20 K → 2.8×10⁻²² W/Hz
dB                               +3 = ×2 · +10 = ×10 · add, don't multiply
Cassini                          13 + 47 − 294 + 74 = −160 dBW = 10⁻¹⁶ W
                                 S/N₀ ≈ 56 dB-Hz → ~100,000 bits/s
Voyager 1                        160 bits per second
```

B · Section 5 map page (write it now, from memory first)

```
S5 · antennas: radio = EM wave, c = λ × f; rod = λ ÷ 2;
     modulation = signal on a fast carrier; dish gain = aim
S5 · atmosphere: wall with 2 windows (visible, radio);
     links use S / X / Ka; uplink ≠ downlink
S5 · links: encode → modulate → up-convert → HPA ⇒ space
     loss (spreading) ⇒ LNA → down-convert → demod → decode
S5 · the point: signal ÷ noise sets bits per second
```

D · Question bank ("but how?" cards)

```
Q: Where does the signal's power go on the way?
A: Nowhere. It spreads over a sphere 4πd². 2× far = ¼ power.
Q: Why must the FIRST receive amplifier be low-noise?
A: Its added noise is amplified by every later stage.
Q: Why can't you just amplify a weak signal more?
A: Amplifiers lift signal and noise together. The ratio stays.
Q: What is k in the link equation?
A: Boltzmann constant: noise per hertz = k × temperature.
```

C · One-line word bank (new lines)

```
p__ · link · one radio path, transmitter to receiver
p__ · encoder / decoder = codec · code the data / decode it
p__ · compression · fewer bits, same message
p__ · error-correcting code = FEC · spare bits fix errors → C31
p__ · up- / down-converter · shift frequency up / down
p__ · IF / RF · easy low frequency / the on-air one
p__ · HPA · high-power amp, before transmit antenna
p__ · LNA · low-noise amp, first after receive antenna → C30
p__ · free-space loss (Lₛ) · spreading; ∝ distance² → C29
p__ · atmospheric loss (Lₐ) · air + rain weakening
p__ · transponder = repeater · receive, shift, boost, resend → C31
p__ · bent pipe · satellite that only relays
p__ · thermal noise · hiss from anything warm
p__ · N₀ · noise per hertz = k × T
p__ · Boltzmann constant (k) · 1.38×10⁻²³ J/K
p__ · system noise temperature (T) · all noise as one temp
p__ · S/N₀ = signal-to-noise · sets data rate → C29
p__ · EIRP · power × transmit gain
p__ · G/T · receive gain ÷ noise temp
p__ · link budget · ledger of gains and losses → C30
p__ · dBW / dBm · dB vs 1 watt / vs 1 milliwatt
p__ · bit rate · bits per second
```

SEC 5 · LEC 3 · CONCEPT CARDS

Route E: pages C29 and C30

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 13 — **C29**

FREE-SPACE LOSS (L_s) = spreading loss

= weakening from the wave spreading out

WHY nothing eats the signal. It covers a bigger sphere, so each dish catches a smaller share

\# 2× distance = ¼ power · GEO 10^{20} · Saturn 2.5×10^{29} (294 dB)

≠ absorption (that is L_a, the air)

→ $L_s = (4\pi d \div \lambda)^2$ · inverse square · link budget

SIGNAL-TO-NOISE RATIO = S ÷ N_0

= signal power compared with noise

WHY a receiver can amplify forever, but it lifts noise too. Only the ratio decides what can be read

\# Cassini: 10^{-16} W ÷ 2.8×10^{-22} W/Hz ≈ 400,000 (56 dB-Hz)

≠ signal strength alone

→ $N_0 = k \times T$ · bits per second · coding

WORD BANK · CONCEPT CARDS · scene 13 — **C30**

LINK BUDGET

= ledger of every gain and loss on a link

WHY it shows before launch whether the far end will hear you, and how fast you can talk

\# 13 + 47 − 294 + 74 = −160 dBW (Cassini)

≠ data budget (how much to send)

→ decibels add · EIRP · G/T · margin

LOW-NOISE AMPLIFIER = LNA

= first amplifier after the receive antenna

WHY noise added by the first stage is magnified by all the rest, so it must be the quietest

\# DSN: cooled to ~15 K · system noise ~20 K

≠ HPA (transmit side: watts matter)

→ thermal noise · $k \times T$ · $S \div N_0$

C29: 10 + 9 lines of 22 · C30: 10 + 10 lines of 22

SEC 5 · LEC 3 · CONCEPT CARDS

Route E: page C31

The third card page for this lecture.

WORD BANK · CONCEPT CARDS · scene 13 — **C31**

TRANSPONDER = satellite repeater

=	receives, shifts frequency,
	amplifies, resends
WHY	after 36,000 km the signal is too
	weak to bounce. Boost it, and move it
	off the uplink frequency
#	hears ~10^{-10} W · gain ~10^{11} (110 dB)
≠	a science probe (an END of the link)
→	relay satellite · uplink / downlink ·
	bent pipe

ERROR-CORRECTING CODE = FEC

=	spare bits that let the receiver fix
	errors
WHY	weak signals arrive with mistakes.
	Patterned spare bits repair them, no
	resend needed
#	good codes ≈ 5× more data on the same
	link
≠	compression (removes bits)
→	encoder / decoder · S ÷ N_0 · data
	rate

Section 5 is complete

Three lectures, four spreads, eight card pages (C24–C31).

Before Section 6

1. Write the Section 5 map page from memory, then check it against route B on the previous page.
2. Do one timed Section 5 talk from the app's Teach tab. It is four minutes.

The thread through all three lectures

Lecture 1: how a wave carries a message and how a dish aims it. Lecture 2: which waves can get through the air. Lecture 3: adding it all up to see whether the far end can hear you. Every number in today's ledger came from one of the first two lectures.

For Network+ later

The decibel ledger on page 3 is the same arithmetic as a Wi-Fi link: transmit power in dBm, antenna gain in dBi, path loss, receiver sensitivity. You have now done it once with Saturn.

C31: 10 + 11 lines of 22.

SEC 5 · LEC 3 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Send me what you said and I will grade it like a friend would.

🗣 The friend's opening question

"Why does it take so long to get pictures back from far-away probes? Radio moves at the speed of light."

✅ Three sentences you should be able to say

- The radio is fast, but the signal arrives unbelievably faint. Nothing absorbs it; it just spreads out, so twice as far away you catch a quarter as much.
- At the receiver it has to be picked out of the electrical hiss that everything warm makes. What matters is the signal compared with that noise, and that ratio sets how many bits per second you can send.
- So engineers add up a budget: transmitter power, both dishes, the distance, the receiver's noise. From Saturn that gives about a hundred thousand bits a second. From Voyager it is 160.

❓ Follow-ups a friend might ask

- "Can't they just amplify it more?" (an amplifier boosts the noise by the same amount; the ratio doesn't change)
- "So how do you improve it?" (more power, bigger dishes, a colder receiver, cleverer codes)
- "What's a low-noise amplifier?" (the first amplifier at the receiver, chilled so it adds almost no hiss of its own)
- "What does a relay satellite do?" (hears the faint uplink, moves it to another frequency, boosts it and sends it back down)
- "What are decibels for?" (they turn all those multiplications into one column you can add)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Section 5 done. Next is Section 6: attitude determination and control.

SPACE X · SECTION 6 OF 7

Section 6 · Attitude determination and control

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

Sense, compare, twist, forever. Gyros drift and stars fix them. Wheels swap spin; jets and magnets remove it.

The lectures

#	Lecture	Page	Length
1	**Attitude: definitions and physics**	88	5 pages
2	**Sensors**	93	4 pages
3	**Actuators**	97	4 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Talk card · 4¼ min

He opens with: "How does a spacecraft point at anything? There's nothing up there to hold on to."

- 1 · LOOP — attitude = pointing · roll pitch yaw · sensors → computer → actuators · jitter
- 2 · PENCIL / COIN — moment of inertia = mass far from axis · fat way only · Explorer 1 tumbled, Sputnik a ball
- 3 · SENSORS — gyro drifts · Sun, stars (0.001°), horizon, compass · one arrow = 2 angles
- 4 · ACTUATORS — wheel one way, ship the other · saturation · thrusters + magnets empty them · CMG tilts
- LAND — "sense, compare, twist, forever"

Land on: A spacecraft keeps its pointing by sensing which way it faces, comparing that with where it should face, and twisting itself to close the gap, forever.

Number to know
star tracker 0.001° · Hubble holds 0.007 arcseconds

Don't repeat
"Explorer 1 failed" (it found the radiation belts) · "moment of inertia = mass in a direction" (distance from the axis)

In the app: scenes 14, 15, 16 in Learn, and the "Section 6" talk in Teach.

SEC 6 · LEC 1 · ATTITUDE: DEFINITIONS + PHYSICS

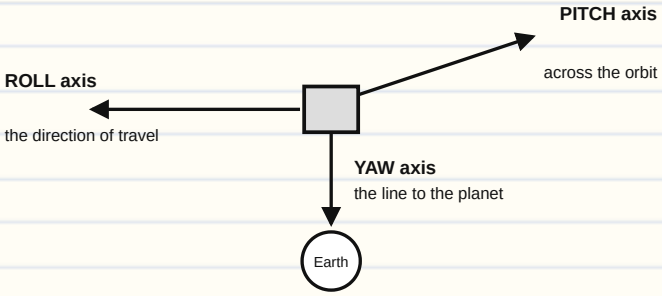
Point, check, nudge

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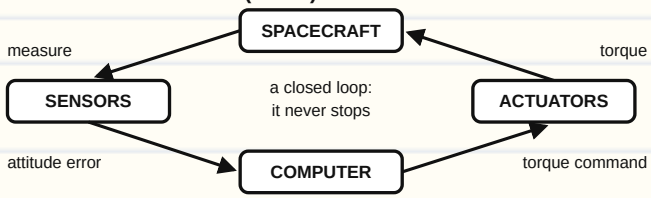
S6 / L1 (1/2) — Attitude + the loop **date · pg __**

1 · ATTITUDE = which way it points

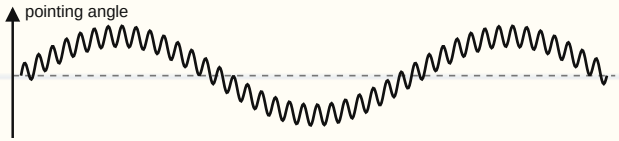


like a plane: roll = tilt the wings · pitch = nose up or down
yaw = nose left or right

2 · THE CONTROL LOOP (ADCS)



3 · JITTER: it never sits perfectly still



slow swing = the loop correcting itself
fast fuzz = jitter: vibration from wheels and pumps

✏️ **PENCIL KEY** · box = spacecraft · three arrows = the three axes · four boxes = the loop · wavy line = pointing angle · dotted = target

⭐ **3 TAKEAWAYS** 3 lines

1. Attitude = which way it points. Roll, pitch, yaw are the three ways to turn.
2. ADCS is a loop: measure, compare, command, twist, repeat.
3. Pointing always shakes a little (jitter). Keep it inside the requirement.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What is attitude? How is it different from orbit?	attitude = orientation of the spacecraft: which way it points (orbit = where it is)
Roll, pitch, yaw: which axis is each?	roll: turning about the direction of travel · pitch: about the axis across the orbit · yaw: about the line to the central body
Attitude determination vs attitude control?	attitude determination = measuring the orientation
What does ADCS stand for?	attitude control = returning the spacecraft to the desired attitude
The four boxes of the loop?	ADCS = attitude determination and control system
What travels along each arrow?	closed feedback loop: sensors → attitude error → on-board computer → torque commands → actuators → torque → spacecraft → sensors…
Why is it a "closed" loop?	attitude error = where it points − where it should point
What is attitude jitter?	torque = a twist: the turning version of a force
Can jitter be zero?	attitude jitter: small shaking of the pointing angle that never fully goes away
? measured against what	jitter must stay inside the mission's requirement

SEC 6 · LEC 1 · ATTITUDE: DEFINITIONS + PHYSICS

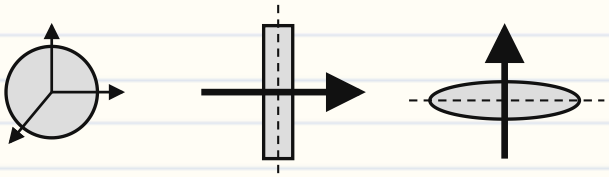
Pencils tumble, coins spin

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LEFT PAGE · DRAW

S6 / L1 (2/2) — Spin stability **date · pg __**

1 · MOMENT OF INERTIA: three shapes



SPHERE	ROD (pencil)	DISK (coin)
same about every axis	biggest: across it smallest: along it	biggest: through the face

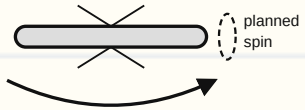
thick arrow = axis of MAXIMUM moment of inertia · dashed = minimum

2 · THE RULE

> no outside torque, and a little energy leaking away
> → the body ends up spinning about its axis of
> **MAXIMUM moment of inertia**

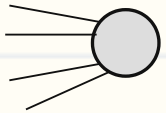
why: its spin can't change, but its energy drains away, and spinning about the biggest axis takes the least energy

3 · EXPLORER 1 (1958) vs SPUTNIK 1 (1957)



ended up: end over end

a pencil, spun on its SMALLEST axis
whip antennas flexed, energy leaked
→ tumbled within hours
still found the radiation belts



any way up is fine

a ball: no favorite axis
antennas radiate all round
so no pointing was needed

✏️ **PENCIL KEY** · thick arrow = biggest moment of inertia · dashed = smallest · dashed loop = planned spin · curved arrow = what happened

⭐ **3 TAKEAWAYS** 3 lines

1. Moment of inertia = how far the mass sits from the axis.
2. Left alone, a real body ends up spinning about its biggest-inertia axis.
3. Explorer 1 was a pencil spun the wrong way. Sputnik 1 was a ball.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What does torque-free mean?	torque-free = no outside twist acting on the body
What is moment of inertia?	moment of inertia = how hard a body is to spin about an axis · big when the mass is far from that axis
Biggest axis for a sphere? A rod? A disk?	sphere: the same about every axis · no favorite cylinder (rod): maximum about any axis across it · minimum about its own long axis disk: maximum about the axis through its face
The stable-spin rule?	rule: with no outside torque, a body spins stably only about its axis of maximum moment of inertia
How was Explorer 1 meant to spin?	Explorer 1 (USA, 1958): pencil shape, meant to spin about its long axis → tumbled
What happened instead, and why?	Sputnik 1 (USSR, 1957): sphere → no preferred axis, nothing to go wrong
Why didn't Sputnik 1 care?	
So what shape should a spinning satellite be?	lesson: spinning satellites are built like drums, not pencils
? why only the maximum axis	

SEC 6 · LEC 1 · TEACH

A thermostat for pointing

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

A spacecraft has two different problems: where it *is* (its orbit) and which way it *faces* (its **attitude**). The dish has to face Earth and the panels have to face the Sun, and nothing in space holds it steady. So it works like the thermostat in your house. A thermostat feels the temperature, compares it with what you asked for and switches the heater. The spacecraft **looks** at the stars to see which way it is facing, **compares** that with where it should face, and **nudges** itself with a small twist. Then it looks again. Forever. That loop is the whole system.

The second half is about spinning. Toss a coin spinning flat like a frisbee and it stays steady. Toss a pencil spinning around its length and it soon starts to wobble and ends up tumbling end over end. Things like to spin the "fat" way, with their weight as far from the spin line as possible. The first American satellite was a pencil spun around its length. Within hours it was tumbling. The first Soviet one was a ball, which has no fat way, so nothing could go wrong.

🏷 The official names (name first, then what it means)

- **Attitude**: the orientation of a spacecraft relative to a reference frame. **ADCS**, attitude determination and control system (also **AOCS** when orbit control is included).
- **Roll, pitch, yaw**: rotations about the three body axes. Together they are a set of **Euler angles**.
- **Reference frame**: what you measure against. An **inertial frame** is fixed to the distant stars. The **orbital frame** (LVLH, local vertical / local horizontal) turns with the orbit: one axis to the planet, one along the track, one across. The slide's picture is the orbital frame. **Nadir**: straight down to the planet.
- **Attitude determination** (measuring) and **attitude control** (correcting). **Attitude error**: measured minus wanted.
- **Closed-loop control** (feedback control). The rule the computer uses is the **control law**.
- **Sensor** and **actuator**: the parts that measure and the parts that push. Details next lecture.
- **Torque** (moment): force × lever arm, in newton-metres. The turning version of force.
- **Pointing accuracy** (how close to the target) and **pointing stability** (how steady). **Jitter**: fast pointing vibration. **Arcsecond**: 1/3600 of a degree.
- **Moment of inertia, I**: the turning version of mass. **Principal axes**: the **major axis** (maximum I), **minor axis** (minimum), and the **intermediate axis**.
- **Angular momentum, L** = I × spin rate. Conserved when no torque acts.
- **Torque-free motion**. The **major-axis rule**. **Flat spin**: tumbling about the major axis. **Spin stabilization** versus **three-axis stabilization**.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Torque	Torque = force × lever arm. Angular speed-up = torque ÷ moment of inertia, the twin of a = F ÷ m.
2 · One turn of the loop	A spacecraft with I = 100 kg·m² is 1° off target. Its wheel can twist with 0.05 N·m. Speed-up = 0.05 ÷ 100 = 0.0005 rad/s². Twist one way for 6 s, the other way for 6 s: 1° corrected in about **12 seconds**. Measure again.
3 · Jitter	Wheels are never perfectly balanced, pumps hum, panels creak as they warm. The result is a fast tremble on top of the slow correction. Hubble holds **0.007 arcseconds**: the width of a hair seen from 3 km.
4 · Moment of inertia	I = sum of (mass × distance from the axis²). A 14 kg pencil 2 m long and 16 cm wide: about its long axis ½MR² = **0.044**; about an axis across its middle **4.9** kg·m². 110 times bigger, because the ends are a metre from that axis.
5 · What is kept	With no outside torque, angular momentum L = I × spin rate cannot change. The spin *energy* is L² ÷ (2 × I): for the same L, a bigger I means less energy.
6 · The leak	Real spacecraft are not rigid. Antennas flex, fuel sloshes, and each wobble turns a little spin energy into heat. L stays, energy falls, so the body slides toward the lowest-energy spin: about the **biggest** I.
7 · Explorer 1	Spun at 750 turns a minute about its long axis: 135 joules. Its four whip antennas flexed. After the slide: 750 ÷ 110 ≈ **7 turns a minute**, end over end, with about 1 joule. The other 99% went into bending wire.
8 · The fix	Make the spin axis the fat one: build spinning satellites as squat drums. Or don't spin at all and run the loop on three axes.

SEC 6 · LEC 1 · FACT CHECK

Right conclusions, shaky reasons

Where the instructor was wrong or loose, the caption errors, and the question a friend would ask.

⛔ Wrong, and worth getting right

- **"Moment of inertia is the amount of mass in one direction."** It is about how far the mass is from the spin axis. Along a pencil's length the mass hugs the axis, so that moment is small. Across the middle the ends are far away, so it is large. His conclusions for all three shapes are right; the reason he gives is not.
- **"Explorer 1 was completely unstable… difficult to recover data."** It did tumble, and nobody expected it. But it kept working for four months and made the first great discovery of the space age: the radiation belts around the Earth.
- **"The Russians understood this… the US goes for marketing style."** Nobody understood it yet. Explorer 1's tumble is how engineers *learned* the rule. Its pencil shape came from the rocket: the satellite was the top stage's casing with instruments in the nose.
- **"The sphere would move completely random."** A sphere keeps whatever spin it starts with. It has no preferred axis, so it is neither stable nor unstable. Sputnik 1 did not care because its four antennas radiated in every direction, like the omnidirectional antennas in Section 5.

▲ Looser points

- **The rule needs a leak.** For a perfectly rigid body, spin about the smallest axis is stable too. Only the in-between axis is unstable. Try it: flip your phone over a bed about each of its three axes. One of them always wobbles.
- **"The sensors determine the difference."** Sensors only measure. The computer compares with the target.
- **Jitter "cannot really be corrected".** It can be reduced a great deal with balanced wheels and vibration isolators. It can't be made zero.
- **"An inertial reference frame."** The slide's axes (to the planet, along the velocity) turn with the orbit. That is the orbital frame. Both are used.

🔎 Caption errors

Captions said	Instructor meant
"A.D.C. s"	**ADCS**
"odorant itself"	**orient** itself
"attitude areas", "area of altitude"	attitude **errors**
"commands some talks", "actuating talks"	commands some **torques**
"de-activate is able"	**the actuator** is able
"the attitude cheater", "altitude data"	the attitude **jitter**
"talk free rotational dynamics"	**torque-free** rotational dynamics
"spin Stavely", "staple all spin"	spin **stably**, **stable** spin
"momentum of inertia", "moment to finish"	**moment of inertia**
"Selinda" · "the bass playing"	**cylinder** · the **base plane**
"this poor Nick", "this public"	**Sputnik**
"Ferric spacecraft"	**spherical** spacecraft
"the exposure from 1958"	the **Explorer** from 1958

🗣 Say it out loud

Friend: "How does a spacecraft know which way it's facing, and how does it turn with nothing to push on?"
You: It runs a loop, like a thermostat. Sensors look at the stars or the Sun to measure which way it points. The computer compares that with where it should point. Then it twists itself a little, usually by spinning a wheel inside, and checks again. It never stops, and the pointing always trembles a tiny bit.
Friend: "Why not just spin it like a top?"
You: You can, but only the fat way. Things end up spinning with their weight as far from the axis as possible. America's first satellite was a pencil spun lengthwise and it was tumbling within hours.

SEC 6 · LEC 1 · VOCABULARY

Every word used, in plain English

40 entries. "Repeat" = you have met it before.

Attitude — which way a spacecraft is pointing. Not its height: that is altitude. *Repeat.*

Orientation — the direction something faces.

ADCS — attitude determination and control system: the parts that measure pointing and correct it.

Axis / axes — an imaginary line something turns around. Three of them, at right angles.

Roll / pitch / yaw — turning about the direction of travel / about the across-orbit axis / about the line to the planet. *Wings tilt / nose up-down / nose left-right.*

Central body — the planet being orbited. *Repeat.*

Velocity direction — the way the spacecraft is travelling.

Angle of attack — aircraft term: how far the nose points above the airflow. A pitch angle.

Reference frame / inertial frame — the set of directions you measure against / one fixed to the distant stars.

Attitude determination — measuring the orientation.

Attitude control — turning the spacecraft back to the wanted orientation.

Closed feedback loop — a system that measures its own result and keeps correcting it. *A thermostat.*

Dynamics — how something moves under the pushes and twists on it.

Sensor — a device that measures: a star camera, a Sun detector, a spin meter. *Repeat; next lecture.*

On-board computer — the spacecraft's processor. Compares measured with wanted.

Attitude error — the difference between where it points and where it should.

Command — an instruction sent to a device.

Actuator — a device that pushes or twists: a wheel, a thruster, a magnetic coil. *Next lecture.*

Torque — a twisting push: force × lever arm. Captions write 'talk'. *A wrench on a bolt.*

Jitter — fast, small trembling of the pointing direction.

Phenomenon / phenomena — something that happens and can be observed.

Imprecision — not being exact.

Requirement / constraint — what the mission needs / a limit that must not be broken. *Repeat.*

Torque-free — with no outside twist acting.

Rotational dynamics — the physics of spinning things.

Spin — to turn about an axis through yourself.

Stable / unstable — returns after a nudge / runs away after a nudge.

Moment of inertia — how hard a body is to spin about a given axis. Depends on how far its mass is from that axis. *The turning version of mass.*

Maximum moment of inertia axis — the axis the mass is, on average, farthest from. *The 'fat way' to spin.*

Homogeneous — the same all the way through. *Repeat.*

Rigid — does not bend or flex.

Sphere / cylinder / disk — ball / tube or rod / flat round plate.

Center of mass — the balance point of a body.

Mass distribution — where the mass is placed inside a body.

Plane — a flat sheet. *Repeat.*

Angular momentum — amount of spin: moment of inertia × spin rate. Kept unless a torque acts. *A skater pulling in her arms.*

Flat spin — a tumble about the biggest-inertia axis.

Explorer 1 — the first US satellite, 1958. A 2 m pencil that found the radiation belts. *Who's Who.*

Sputnik 1 — the first satellite of all, USSR, 1957. A 58 cm ball that beeped for three weeks. *Who's Who.*

Radiation belts (Van Allen belts) — rings of trapped high-energy particles around the Earth. *Who's Who.*

SEC 6 · LEC 2 · ATTITUDE SENSORS

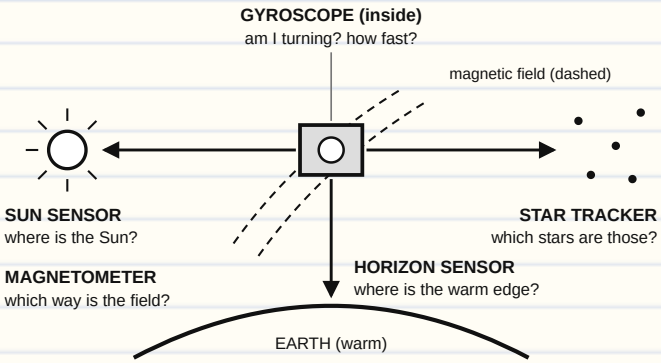
Inner ear, eyes and a compass

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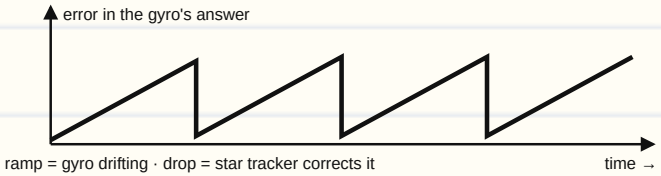
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S6 / L2 — Attitude sensors **date · pg __**

1 · WHAT EACH SENSOR LOOKS AT



2 · THE GYRO DRIFTS, THE STARS RESET IT



3 · HOW SHARP, AND THE CATCH

magnetometer	about 1°	low orbits only
Sun sensor	about 0.1°	blind in Earth's shadow
horizon sensor	about 0.1°	only tells you 'down'
star tracker	about 0.001°	blinded by the Sun; costly
gyroscope	feels turning	fast, but drifts

one direction pins down only 2 of the 3 angles
→ always combine at least two sensors

✏️ **PENCIL KEY** · box = spacecraft · arrows = what a sensor looks at · dashed = magnetic field · zigzag = gyro error over time

⭐ **3 TAKEAWAYS** 3 lines

1. A gyro feels turning. The rest look at something outside.
2. One direction fixes only two angles, so sensors are combined.
3. Gyro: fast but drifts. Star tracker: exact but slow. Use both.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
The five kinds of attitude sensor?	sensors: gyroscopes · optical (Sun sensor, star tracker, horizon scanner) · magnetometers
Which three are optical?	gyroscope: a spinning rotor keeps its direction · the frame and gimbals turn with the spacecraft → measure the turn
Gyroscope: how does it work?	Sun sensor: solar cells detect sunlight from a certain direction → angles to the Sun · light, cheap
Sun sensor: how? good and bad?	star tracker: camera matches the star field to a sky map → orientation · very accurate · complex, expensive · good far from the Sun
Star tracker: how? good and bad?	horizon scanner: infrared · warm Earth vs cold space (2.7 K) → finds the horizon → 2 axes
Horizon scanner: what does it see? Why infrared?	magnetometer: direction + strength of the magnetic field, compared with a 3D model of Earth's field · low Earth orbit only: the field fades with altitude
Magnetometer: what does it compare? Where does it work?	others: fibre-optic gyros · accelerometers + gyros · several GPS antennas
Three other options?	one direction = only 2 angles → combine sensors
Why is one sensor never enough?	gyro drifts → the star tracker resets it
? why does a gyro need a star tracker	

SEC 6 · LEC 2 · TEACH

How do you know which way you're facing?

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Sit on a spinning office chair with your eyes shut. Your **inner ear** feels the turning, so for a few seconds you know roughly which way you face. Keep going and you lose track. That is a **gyroscope**: it feels turning instantly, but small mistakes pile up. Now open your eyes. Find the **Sun**: one bright thing, easy, but it only tells you one direction, and it vanishes in the Earth's shadow. Recognize the **constellations**: now you know exactly how you are turned, though it takes a careful look. Find the **horizon**: that tells you which way is down. Check a **compass**: rough, but it works in the dark.

A spacecraft has all of these. And here is the catch that explains why: pointing at one thing is never enough. Point your arm straight at the Sun. You can still roll your wrist without your arm leaving the Sun. One direction pins down two of your three turns; you need a **second, different** direction for the third.

🏷 The official names (name first, then what it means)

- **Gyroscope** (gyro; rate gyro): measures how fast the spacecraft is rotating. A set of three, usually with accelerometers, is an **inertial measurement unit, IMU**.
- **Rotor**, **spin axis**, **gimbal**, **frame**: the spinning wheel, the line it spins about, the pivoting rings that let it stay put, and the housing.
- **Gyro drift** (bias): the slow build-up of error in a gyro's answer.
- **Fibre-optic gyroscope, FOG** and **ring laser gyro** (the lecture's "optic gyroscopes"): no spinning parts; they time light going both ways round a loop.
- **Sun sensor**: measures the direction to the Sun. **Coarse** ones are bare solar cells; **fine** (digital) ones use a slit or pinhole over a detector.
- **Star tracker** (star sensor, star camera): photographs stars and matches them to a **star catalogue**.
- **Horizon sensor** (horizon scanner, Earth sensor): finds the edge of the Earth's warm disc in infrared.
- **Magnetometer**: measures the direction and strength of the local magnetic field. The map it is compared with is a **geomagnetic field model** (IGRF).
- **GPS attitude determination**: several GPS antennas on one spacecraft, comparing signal arrival.
- **Absolute sensor** (sees an outside reference) versus **relative sensor** (feels change only: the gyro).
- **Sensor fusion**: blending several sensors into one best answer, usually with a **Kalman filter**.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Two arrows	One measured direction fixes two angles and leaves you free to spin about that direction. Two different directions (Sun + magnetic field, or two stars) fix all three.
2 · Gyroscope	A spinning rotor keeps its axis fixed in space (angular momentum, last lecture), so the turning of the frame around it can be read off. It reports turn *rate*, 100 times a second. Add the rates up to get the angle, and the errors add up too: a drift of 0.01° per hour is 0.24° after a day.
3 · Sun sensor	A solar cell's current falls as the Sun moves off straight-on (as the cosine of the angle). Put four cells under one pinhole and compare their currents: two angles to the Sun. From about 1° for bare cells to 0.01° for fine ones. Tens of grams.
4 · Star tracker	A small camera photographs 10 to 50 stars, measures the angles between them and finds the same pattern in a catalogue of a few thousand stars. One picture gives all three angles to about **3 arcseconds** (0.001°), several times a second.
5 · Horizon sensor	Earth radiates like a body at 255 K, peaking near 11 µm (2,898 ÷ T). Space is 2.7 K. At 15 µm the planet is a sharp, even, warm disc day and night. From 400 km it is 140° wide. Find its centre and you know "down": roll and pitch, not yaw.
6 · Magnetometer	A three-axis compass. In low orbit the field is 25–65 microtesla. Measure its direction, look up what the field model says it should be at your position, and compare: about 1°. The field falls with distance cubed, so at GEO it is about 0.1 microtesla and little use.
7 · Fusion	The gyros carry the answer from moment to moment; every star-tracker picture pulls it back to the truth. That sawtooth is on your drawing.

SEC 6 · LEC 2 · COMPARE + FACT CHECK

Six sensors, one table

The comparison to know, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📋 The sensors side by side

Sensor	What it senses	Accuracy	Gives	Weakness
Gyroscope	its own rotation	drifts 0.001° to 10° per hour	turn rate, all 3 axes	relative only: needs resetting
Sun sensor	direction to the Sun	1° coarse, 0.01° fine	2 angles	nothing in eclipse
Star tracker	star patterns	about 0.001° (3 arcseconds)	all 3 angles	blinded by Sun, Earth, Moon; fails when tumbling; costly
Horizon sensor	edge of the warm planet in infrared	about 0.1°	roll + pitch	near a planet only; no yaw
Magnetometer	direction of the magnetic field	about 1°	2 angles	low orbit only; needs a field map; upset by the craft's own magnetism
GPS, several antennas	differences in signal arrival	about 0.3°	all 3 angles	only below the GPS satellites

▲ Where the instructor was loose

- **"A gyroscope determines the position and orientation."** It measures rotation only: how fast you are turning. It knows nothing about position, and it cannot tell you where you point unless something else told it the starting direction. Then it drifts.
- **"The Cubli, a cubic representation of a gyroscope."** The Cubli is a small cube from ETH Zurich that jumps up and balances on a corner by spinning wheels inside it. Those are *reaction wheels*: actuators, the subject of the next lecture. It is not a gyroscope.
- **Star trackers are for "interplanetary… the Sun's intensity is very low".** The Sun is still by far the brightest thing at Jupiter. Star trackers are chosen for accuracy, and because one picture gives all three angles. Nearly every modern satellite carries one, in Earth orbit too.
- **"Earth emits around 100–1,000 K."** About 255 K. Space is 2.7 K, as he says.
- **Horizon scanner "determines orientation on 2 axes".** Right, and it matters: it finds "down", so roll and pitch. It cannot see yaw.
- **Several GPS sensors each "receive an accurate position".** GPS positions are good to a few metres, and the antennas sit a metre apart, so positions alone cannot show the tilt. The method compares the exact moment the same radio wave reaches each antenna.
- **Magnetometers "not used for upper orbits".** Correct. The reason: the field falls with the cube of distance, so at GEO it is about 300 times weaker than in low orbit.
- **Not said at all: one sensor is never enough**, and gyros drift. Those two facts are why a spacecraft carries several kinds.

🔎 Caption errors

Captions said	Instructor meant
"basically telescopes", "a charge scope"	**gyroscopes**, a gyroscope
"a story of some sensors"	such as **Sun sensors**
"spin X is" · "Montane constant"	**spin axis** · **maintained** constant
"Kimberlin… Joe's frame… Jones Cup"	**gimbal and gyroscope frame**
"the Cobley"	the **Cubli**
"Startrek", "Star-Trek us", "stockbroker's"	**star tracker(s)**
"sunsets" · "Sun Center"	**Sun sensors** · Sun sensor
"an iPhone 6 keeps out"	a **NanoAvionics CubeSat** (Sun sensor)
"Corelli's on the scanner"	**horizon** scanner
"thermal pot"	thermal **part**
"optic Jungers scopes"	**optical gyroscopes**
"CPS sensors… GPL"	**GPS** sensors
"for Paul BNZ cannot be used"	for **upper orbits** (per the slide)

🗣 Say it out loud

Friend: "How does a spacecraft know which way it's pointing?"
You: The same ways you would. It has gyroscopes, like an inner ear, that feel every turn but slowly lose track. And it has eyes: a sensor that finds the Sun, a camera that recognizes star patterns, one that finds the warm edge of the Earth, and a compass for the magnetic field. No single one is enough, because one direction only pins down two of your three turns. So it blends them: the gyros second by second, and the star camera to pull them back to the truth.

SEC 6 · LEC 2 · VOCABULARY Every word used, in plain English

37 entries. "Repeat" = you have met it before.

Sensor — a device that measures something. *Repeat.*

Gyroscope (gyro) — a sensor that feels rotation. Captions write 'telescope'. *Your inner ear.*

Rotor — the spinning wheel inside a mechanical gyroscope.

Spin axis — the line the rotor spins about. It stays pointing the same way.

Gimbal — a pivoting ring that lets the rotor keep its direction while the frame turns.

Gyroscope frame — the housing, fixed to the spacecraft.

Drift — slow build-up of error.

Rotation rate — how fast something is turning, in degrees per second.

Optical sensor — a sensor that works by detecting light.

Sun sensor — measures the direction to the Sun. *Find the brightest thing.*

Solar cell — a device that turns light into electric current. *Section 7.*

Body angles — angles measured relative to the spacecraft's own body.

Accuracy — how close a measurement is to the truth.

Star tracker — a camera that recognizes star patterns to find the orientation. *Reading the constellations.*

Star field / star catalogue — the stars in view / the stored map they are matched against.

Interplanetary trajectory — a path between planets. *Repeat.*

Horizon scanner (horizon sensor, Earth sensor) — finds the edge of the planet's disc. *Finding 'down'.*

Horizon — the line where the planet ends and space begins.

Infrared — heat radiation: light just beyond red, given off by anything warm. *Repeat, Section 5.*

Kelvin (K) — temperature counted from absolute zero. *Repeat.*

Deep space (2.7 K) — the background of space, almost absolute zero.

Emit / radiate — give off radiation. *Repeat.*

Magnetometer — measures the direction and strength of a magnetic field. *A three-axis compass.*

Magnetic field — the region where a compass needle feels a pull. Earth has one. *Repeat.*

Magnitude / direction — size / which way. *Repeat.*

Field model — a mathematical map of Earth's magnetic field at every point.

Low Earth orbit (LEO) — orbits below about 2,000 km. *Repeat.*

Fibre-optic gyroscope — a gyro with no moving parts that times light going round a coil of glass fibre.

Accelerometer — a sensor that feels speeding up and slowing down in a straight line.

IMU — inertial measurement unit: gyros and accelerometers in one box.

GPS sensor / antenna — a receiver for the navigation satellites' signals. *Repeat.*

Absolute / relative sensor — sees an outside reference / only feels change.

Sensor fusion — blending several sensors into one best answer.

Eclipse — being in the planet's shadow. *Repeat.*

Arcsecond — 1/3600 of a degree. *Repeat.*

The Cubli — a balancing cube from ETH Zurich that uses reaction wheels. Not a gyroscope.

NanoAvionics / Vectronic Aerospace — companies that sell the Sun sensor and star tracker in the slide photos.

SEC 6 · LEC 3 · ATTITUDE ACTUATORS

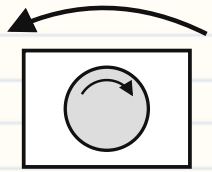
Spin a wheel, turn the ship

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S6 / L3 — Attitude actuators **date · pg __**

1 · INSIDE: wheels swap spin with the craft

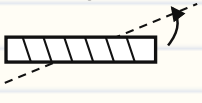


REACTION WHEEL
change the wheel's SPEED
→ the craft turns the other way



CONTROL MOMENT GYRO
speed stays the same
TILT its axis → a big twist

2 · OUTSIDE: push on something else

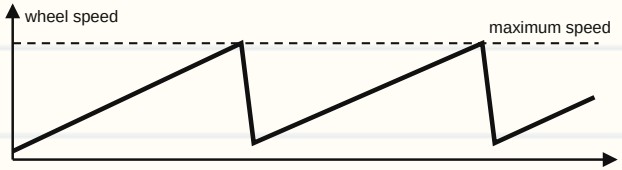


MAGNETIC TORQUER
a coil = an electromagnet
Earth's field twists it, like a compass
weak · no fuel · low orbit only



THRUSTERS
small jets, off-centre, in pairs
strong · but they use propellant

3 · WHEELS FILL UP (saturation)



ramp = soaking up a steady outside nudge · drop = dumping it

wheels and CMGs only SWAP spin with the craft
thrusters and magnetic torquers REMOVE it

✏️ **PENCIL KEY** · thin arrow = wheel or exhaust · thick arrow = the craft's turn · dashed arrow = tilting · dashed line = magnetic field or limit

⭐ **3 TAKEAWAYS** 3 lines

1. Spin a wheel one way and the spacecraft turns the other way.
2. Wheels and CMGs only swap spin, so they fill up (saturate).
3. Thrusters and magnetic torquers push on the outside and empty them.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

? RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
The five actuators?	actuators twist the spacecraft: momentum wheels · reaction wheels · magnetic torquers · control moment gyros · thrusters
Why does spinning a wheel turn the spacecraft?	conservation of angular momentum: spin a wheel one way → the craft turns the other (helicopter + tail rotor)
Momentum wheel vs reaction wheel?	momentum wheel: always spinning, 5,000+ rpm · change its speed to turn the craft · fast, so the wheel can be light
What is saturation? What fixes it?	saturation: a wheel at top speed can do no more → thrusters or other help
Why 3 or 4 wheels?	DC motor → flywheel → spacecraft rotation · too fast → the wheel can break
CMG: what changes? Why use one?	reaction wheel: same idea, starts at rest · change the SPEED · 3–4 wheels for 3 axes
Magnetic torquer: how? Good and bad?	CMG: wheel at constant speed · change the DIRECTION of its axis → large torque · high slew rate · the ISS uses them
Thrusters: how do they make a twist?	magnetic torquer: coil (solenoid) makes a magnet that the field of the Earth twists · weak · no moving parts, reliable, cheap
What is slew rate?	thrusters: small cold-gas or hot-gas jets placed off-centre · torque = force × arm · use propellant
? which ones can empty a wheel	

SEC 6 · LEC 3 · TEACH

Turning with nothing to push on

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Sit on a swivel chair with your feet up and swing your arms hard to the left. The chair turns to the right. You pushed on nothing; you and your arms just traded spin. That is a **reaction wheel**: a motor spins a heavy wheel one way and the whole spacecraft slowly turns the other way. Stop the wheel and the spacecraft stops, pointing somewhere new. No fuel at all.

The catch: sunlight and thin air keep nudging the spacecraft the same way, day after day. To hold still, the wheel must keep spinning faster to soak that up, until it hits top speed. It is full: **saturated**. To empty it you must push on something *outside*: a puff from a small jet (**thruster**), or an electromagnet that leans on the Earth's magnetic field like a compass needle (**magnetic torquer**). For very big spacecraft there is a stronger trick. Hold a spinning bicycle wheel by its axle and try to tilt it: it twists hard in your hands. A **control moment gyro** is that wheel, tilted on purpose. Four of them steer the Space Station.

🏷 The official names (name first, then what it means)

- **Actuator**: any device that applies a torque. **Momentum-exchange devices**: the ones that work by trading spin (wheels and CMGs).
- **Reaction wheel** (RW; the slide's "reactor wheel" is a typo): a wheel that normally sits near zero speed and is spun either way.
- **Momentum wheel** (bias momentum wheel): a wheel kept spinning fast all the time. Its spin also stiffens the spacecraft against tipping: **gyroscopic stiffness**. **Flywheel**: the heavy rotor itself.
- **Angular momentum** (h or L): moment of inertia × spin rate, in newton-metre-seconds. **Conservation of angular momentum**.
- **Saturation**: a wheel at its top speed. **Momentum dumping** (desaturation, momentum unloading): emptying it with an outside torque.
- **Control moment gyroscope, CMG**: a wheel at constant speed on a motor-driven **gimbal**. Output torque = angular momentum × **gimbal rate**.
- **Slew**: a deliberate turn to a new target. **Slew rate**: how fast it can be done, in degrees per second.
- **Magnetorquer** (magnetic torquer, torque rod, torque coil): a coil that makes a **magnetic dipole moment**; the planet's field twists it.
- **Reaction control system, RCS**: the set of small attitude thrusters. **Cold-gas thruster** (just releases stored gas) and **monopropellant thruster** (hot gas). A **couple**: two equal thrusters on opposite sides, giving a pure twist. **Moment arm**: distance from the centre of mass.
- **Internal torque** (wheels, CMGs) versus **external torque** (thrusters, magnetorquers, and the disturbances).

🛠 How it actually works

Step	The mechanism, with numbers
1 · The swap	Wheel spin + craft spin = constant. Wheel inertia 0.01, craft 100 kg·m²: a ratio of 10,000. Spin the wheel up to 1,000 rpm and the craft turns at 0.1 rpm (0.6° per second) the other way. Brake the wheel and the craft stops.
2 · Three axes	One wheel per axis, plus a fourth mounted at a slant as a spare. The Kepler telescope lost two of its four in 2013 and could no longer hold its target.
3 · Filling up	A steady outside twist of 0.00001 N·m for a day (86,400 s) adds 0.86 N·m·s. A wheel that holds 4 N·m·s is full in under **5 days**.
4 · Dumping	Switch on a magnetorquer or fire a thruster to twist the craft one way while the wheel slows down to cancel it. The extra spin leaves the spacecraft for good.
5 · CMG	The Space Station's: a 98 kg wheel at 6,600 rpm stores 4,760 N·m·s. Tilt its axis at 3.1° per second (0.054 rad/s) and out comes 4,760 × 0.054 ≈ **258 N·m**. A reaction wheel gives about 0.1 N·m.
6 · Magnetorquer	Torque = coil strength × field: 10 A·m² × 0.00003 tesla = **0.0003 N·m**. Tiny but free. It cannot twist about the field line itself, and it needs a strong field: low orbit only.
7 · Thrusters	Torque = force × arm. Two 1 N jets, each 1 m from the centre, firing opposite ways: **2 N·m** and no sideways push. Every puff is propellant you never get back.
8 · Why wheels burst	The rim of a 20 cm wheel at 6,000 rpm moves at 63 m/s. At 65,000 rpm it would be 680 m/s, and the stress (density × speed²) about 3.6 GPa: more than steel can take.

SEC 6 · LEC 3 · COMPARE + FACT CHECK

Five actuators, one table

The comparison to know, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📋 The actuators side by side

Actuator	How it twists	Kind	Typical torque	Uses up	The catch
Reaction wheel	changes its speed, either way	inside: swaps	0.01–0.5 N·m	electricity	saturates; bearings wear out
Momentum wheel	spins fast always; small speed changes	inside: swaps	0.01–0.5 N·m	electricity	controls one axis; saturates
Control moment gyro	tilts a constant-speed wheel	inside: swaps	100s of N·m	electricity	heavy, complex; saturates too
Magnetic torquer	electromagnet in the planet's field	outside: removes	about 0.0003 N·m	electricity	low orbit only; nothing about the field line
Thrusters	small jets off-centre, in pairs	outside: removes	0.1 to 100s of N·m	propellant	runs out; coarse; exhaust can coat optics

▲ Where the instructor was loose

- **Not said: the inside / outside split.** Wheels and CMGs only trade spin with the spacecraft; the total never changes. Only thrusters and magnetic torquers can change the total. That is why every wheel system also carries one of those. It is the most important idea in this lecture.
- **Saturation "occurs commonly due to instabilities".** It comes from steady outside nudges: sunlight pressure, thin air, the pull of gravity on a long body. They push the same way for days, and the wheel has to keep absorbing them.
- **Momentum wheel versus reaction wheel.** He blurs them. A momentum wheel keeps a large spin all the time. A reaction wheel rests near zero and goes either way.
- **"Speeds from 3,500 to 65,000 rpm."** Flight wheels run at a few thousand rpm. As far as I know, 65,000 belongs to experimental energy-storage flywheels.
- **"CMGs are a complete set of momentum wheels in all directions."** One CMG is one wheel on a gimbal. Several are needed; the Station has four.
- **"High slew rate… very time-responsive, not much lag."** Slew rate is how fast the spacecraft can turn. CMGs give big torque, so heavy craft turn quickly.
- **Magnetic torquers "align the sensor" and "work similarly to CMGs".** They are actuators, and nothing like a CMG: they push on the Earth's field, which is outside the spacecraft.
- **Cold gas "usually carbon dioxide".** Usually nitrogen. Hot-gas ones burn or decompose hydrazine.
- **Slide typos:** "Reactor wheels" = reaction wheels. "3 axis determination" = 3-axis *control*.
- **He was right** about the helicopter: the tail rotor exists to cancel the body's reaction to the main rotor.

🔎 Caption errors

Captions said	Instructor meant
"attrited determination"	**attitude** determination
"momentum Boyle's", "moments of will"	**momentum wheels**
"magnetic talks / talkers", "restoring talk"	magnetic **torquers**, restoring **torque**
"thrust us"	**thrusters**
"helicopters need to tail"	need **a tail rotor**
"the amount of the wind can blow"	the **mass of the wheel can be low**
"separation limit"	**saturation** limit
"Maybrick"	**may break**
"reactor Wilt's"	**reaction wheels**
"Control Moment Hiros", "momand Jews"	**control moment gyros**
"solar noise"	**solenoids** (coils)
"calls of hot thrusters"	**cold or hot gas** thrusters
"froggeries ontology axis"	probably the **horizontal** axis (angle θ on the slide)

🗣 Say it out loud

Friend: "How does a spacecraft turn with nothing to push against?"
You: Mostly by spinning a wheel inside. Spin the wheel one way and the spacecraft turns the other, like swinging your arms on a swivel chair. No fuel. The catch is that outside nudges make the wheels spin faster and faster until they are full. Then it has to push on something outside to empty them: a little jet, or an electromagnet leaning on the Earth's magnetic field. Big things like the Space Station use a spinning wheel they tilt, which twists much harder.

SEC 6 · LEC 3 · VOCABULARY

Every word used, in plain English

37 entries. "Repeat" = you have met it before.

Actuator — a device that pushes or twists the spacecraft. *Repeat.*

Torque — a twist: force × lever arm. Captions write 'talk'. *Repeat.*

Angular momentum — amount of spin: moment of inertia × spin rate. *Repeat.*

Conservation of angular momentum — the total spin of a closed system never changes. *Swing your arms on a swivel chair.*

Momentum wheel — a wheel kept spinning fast all the time; small speed changes turn the craft.

Reaction wheel — a wheel that rests near zero and is spun either way to turn the craft.

Flywheel — a heavy wheel built to store spin.

rpm — revolutions per minute.

DC motor — an electric motor run from steady (direct) current. It drives the wheel.

Saturation / saturation limit — the wheel has reached top speed and can absorb no more. *A full bucket.*

Momentum dumping — emptying a wheel by pushing on something outside. *Repeat.*

Instability — a tendency to drift or tip away.

Centrifugal force — the outward pull felt by anything going round. It tries to tear a fast wheel apart.

Stress — force inside a material per unit of area.

Three-axis control — controlling roll, pitch and yaw separately.

Control moment gyro (CMG) — a constant-speed wheel whose axis is tilted to make a large torque. *A tilted bicycle wheel.*

Gimbal — the pivoting mount that tilts the wheel. *Repeat.*

Slew / slew rate — a turn to a new target / how fast the craft can turn.

Tracking — keeping pointed at a moving target.

Lag — delay between command and response.

International Space Station (ISS) — the crewed station; it steers with four CMGs. *Repeat.*

Magnetic torquer (magnetorquer, torque rod) — a coil that becomes a magnet when current flows; the planet's field twists it. *A compass needle.*

Solenoid — a coil of wire.

Magnetic dipole (moment) — how strong a magnet the coil is.

Restoring torque — a twist that pulls something back into line.

Electrical current — the flow of electric charge, in amperes.

Static (no moving parts) — nothing in it moves, so nothing wears out.

Reliable — unlikely to fail.

Thruster — a small rocket engine used for pointing or small manoeuvres. *Repeat, Section 4.*

Cold-gas thruster — a valve that lets stored gas (usually nitrogen) squirt out.

Hot-gas thruster — one that heats or burns its propellant, usually hydrazine.

Moment arm — the distance from the centre of mass to where a force acts.

Couple — two equal, opposite forces on opposite sides: a pure twist.

Reaction control system (RCS) — the full set of small attitude thrusters.

Internal / external torque — a twist from inside (wheels) / from pushing on something outside.

Disturbance torque — small unwanted outside twists that never stop. *Repeat, Section 2.*

Kepler — NASA's planet-hunting telescope; it lost two of its four reaction wheels. *Who's Who.*

SPACE X · SECTION 7 OF 7

Section 7 · Space power systems

Start here. The sentence, the talk card, and the page of every lecture in this section.

Say this

A sunlight sandwich near the Sun, a bucket for the dark, and a warm rock where the Sun doesn't reach.

The lectures

#	Lecture	Page	Length
1	**Solar arrays**	102	4 pages
2	**Batteries**	106	4 pages
3	**Radioisotope generators (RTG)**	110	4 pages

In each one: page 1 is the spread to copy, the next page teaches it, the one after is what the instructor got wrong.

Number to know
1,361 W/m² at Earth · cells 30% · plutonium-238 halves in 87.7 years

Don't repeat
"three layers triple the efficiency" (20% becomes 30%) · "RTGs last 500 years" (decades)

Talk card · 3½ min

He opens with: "Where does a spacecraft get its electricity?"

- 1 · SANDWICH — n + p layers · photon frees electron · junction pushes · 3 layers ≈ 30% · hot = less
- 2 · DISTANCE — 1,361 W/m² here · ÷ distance² · Jupiter 50 · Juno 60 m² for 500 W
- 3 · BUCKET — eclipse 36 of 92 min · 16 cycles a day · depth of discharge · life vs weight
- 4 · WARM ROCK — Pu-238 decays · Seebeck: hot + cold → volts · 6% · half in 88 years · not a reactor
- LAND — "sunlight, a bucket, a warm rock"

Land on: Sunlight when you have it, a battery for the dark, and plutonium where the Sun doesn't reach.

In the app: scenes 17, 18, 19 in Learn, and the "Section 7" talk in Teach.

SEC 7 · LEC 1 · SOLAR ARRAYS

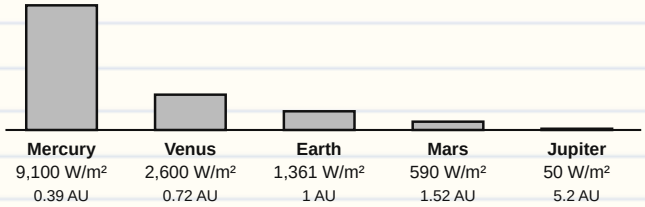
A sandwich that turns light into current

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LEFT PAGE · DRAW

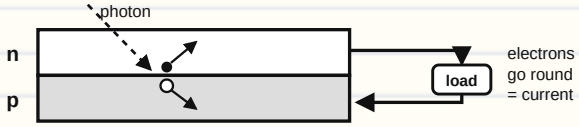
S7 / L1 — Solar arrays **date · pg __**

1 · SUNLIGHT FADES WITH DISTANCE (solar flux)



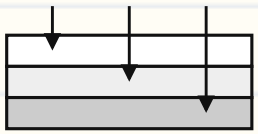
flux = 1,361 ÷ distance² · twice as far = a quarter

2 · INSIDE A SOLAR CELL



• electron · ○ hole · the junction pushes electrons up, holes down
one cell: 0.6 volts (silicon) · 2.7 volts (triple-junction)

3 · THREE LAYERS CATCH MORE COLOURS



top cell: short waves (blue)
middle cell: medium (red)
bottom cell: long (infrared)

one layer ≈ 20% · three layers ≈ 30% · the rest is heat

4 · HOW MUCH POWER

power = flux × area × efficiency

1 m² at Earth: 1,361 × 0.30 ≈ 400 W · hot cells give less

✏️ **PENCIL KEY** · bar = how much sunlight · dashed arrow = photon · • electron · ○ hole · arrows down = light reaching each layer

⭐ **3 TAKEAWAYS** 3 lines

1. Sunlight fades with distance squared: 1,361 W/m² at Earth, 50 at Jupiter.
2. A photon frees an electron; the junction pushes it round the wire.
3. Three stacked layers catch about 30%. Hot cells make less.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What is solar flux? Units?	solar array : turns sunlight into electricity
How does it change with distance?	solar flux = solar radiation per unit of surface (W/m²) · more flux = more power · farther from the Sun = less flux
Deployable vs body-mounted arrays?	Mercury: huge · Mars: needs big arrays · Jupiter: very big, efficient arrays · beyond: not enough
What are n-type and p-type?	deployable arrays : folded for launch, opened in space → lots of area (Juno, Rosetta)
How does a photon become current?	body-mounted arrays : cells on the body → less area (CubeSat, Philae)
Why stack three cells? Which catches what?	photovoltaics : photon energy → electrical energy · two semiconductor layers: n-type (spare electrons) + p-type (holes)
Typical efficiency?	photon knocks an electron free → it travels round the circuit = current
What does heat do to a cell?	each material uses certain wavelengths → stack three: triple-junction gallium arsenide cell ≈ 30%
How much power from an array?	top: short waves · middle: medium · bottom: long
? where does the other 70% go	power = volts × amps · hotter cell = less power · coldest (best) just out of eclipse

SEC 7 · LEC 1 · TEACH

Light in, electricity out

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Sunlight is a rain of tiny packets of energy (the photons from Section 5). A solar cell is a sandwich made of two slightly different slices of crystal. Where the slices touch there is a built-in one-way slope. A packet of light knocks an electron loose, like a ball knocked off a shelf. The slope rolls it to one side, and the only way back is the long way round, through a wire, and through your equipment on the way. That flow is electricity. Nothing moves, nothing burns.

Two catches. First, each crystal only uses packets of the right size: too small and they pass straight through, too big and the extra is wasted as heat. So the best cells stack **three** crystals, each tuned to a different colour, and catch about 30% of the light. Second, sunlight spreads out exactly like the radio signal did: twice as far from the Sun, a quarter as much. At Jupiter it is 27 times weaker than here, so the Juno probe needs three panels, each as long as a bus, to make what a hair dryer uses.

🏷 The official names (name first, then what it means)

- **Solar array** (solar panel, photovoltaic array): many **solar cells** (photovoltaic cells, PV cells) wired together.
- **Solar flux** (solar irradiance): sunlight power per square metre. At Earth's distance it is the **solar constant**, 1,361 W/m². **Astronomical unit, AU**: the Earth–Sun distance, 150 million km. **Inverse-square law**.
- **Deployable array** (solar wings) and **body-mounted array**. A **solar array drive** turns the wings to face the Sun. **Hold-down and release mechanism**: what lets the folded wings go, often a small pyrotechnic cutter.
- **Photovoltaic effect**: light producing a voltage in a material.
- **Semiconductor**: a material such as silicon that conducts a little. **Doping**: adding a trace of another element. **n-type**: extra electrons. **p-type**: missing electrons, called **holes**.
- **p–n junction**: where the two meet. Its **built-in electric field** (in the **depletion region**) is the one-way slope.
- **Band gap**: the energy a photon needs to free an electron in that material.
- **Multi-junction cell** (**triple-junction**, 3J): several junctions stacked. Usual stack: gallium indium phosphide / **gallium arsenide** / germanium.
- **Efficiency**: electrical power out ÷ sunlight power in.
- **I–V curve**: current against voltage for a cell. **Maximum power point**: the spot on it where volts × amps is largest.
- **Thermal cycling**: repeated heating and cooling, once per orbit. **Beginning of life / end of life** power: arrays fade with age.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Flux	Flux = 1,361 ÷ (distance in AU)². Mercury (0.39): 9,100. Venus (0.72): 2,600. Mars (1.52): 590. Jupiter (5.2): **50**. Saturn (9.5): 15 W/m².
2 · Two slices	Silicon with a trace of phosphorus has spare electrons (n). With a trace of boron it has gaps where electrons are missing (p). Join them: electrons near the boundary drop into gaps, leaving a thin zone with a permanent electric field.
3 · Photon in	A photon with at least the band-gap energy frees an electron and leaves a hole. Silicon's gap is 1.1 eV, so any light shorter than 1,240 ÷ 1.1 ≈ 1,100 nm works (the eV rule from Section 5).
4 · Sorted	The built-in field sweeps the electron to the n side and the hole to the p side. That piles up charge: about **0.6 V** for silicon. Connect a wire and electrons flow from n, through the load, to p.
5 · The losses	Photons below the gap pass through unused. Photons above it lose their extra as heat. One material can never beat about 33%; real silicon reaches about 20%.
6 · Three layers	Top, gap 1.9 eV: takes light up to 650 nm. Middle, 1.4 eV: up to 890 nm. Bottom, 0.67 eV: up to 1,850 nm. Each uses its colours well and lets the rest through: about **30%**, 2.7 V per cell.
7 · Power	P = flux × area × efficiency × cos(tilt). 1 m² at Earth: 1,361 × 0.30 ≈ **410 W**. Tilt the panel 60° away from the Sun and you get half.
8 · Heat	On the slide's graph the current stays near 3.5 A while the top voltage falls from 0.63 V at 25 °C to 0.55 V at 60 °C. Power is volts × amps, so a hot cell gives less: about 0.2–0.4% per degree. A panel is at its best cold, just out of Earth's shadow.
9 · Age	Radiation knocks atoms out of place: roughly 1–3% of power lost per year. Arrays are sized for the last day of the mission.

SEC 7 · LEC 1 · SIZING + FACT CHECK

How big for one kilowatt?

A sizing table for your quiz on designing a solar array, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📐 Array area needed for 1,000 W

Where	Distance	Flux, W/m²	Area for 1,000 W
Mercury	0.39 AU	9,100	0.46 m²
Venus	0.72 AU	2,600	1.6 m²
Earth	1 AU	1,361	3.1 m²
Mars	1.52 AU	590	7.1 m²
Jupiter	5.2 AU	50	83 m²
Saturn	9.5 AU	15	280 m²

🧮 The recipe (for the quiz)

1. Flux = 1,361 ÷ (distance in AU)².
2. Power per square metre = flux × efficiency × losses. Here: 30% cells, and 0.8 for wiring, gaps between cells and ageing.
3. Area = power needed ÷ power per square metre.
Example, Mars: 590 × 0.30 × 0.8 = 142 W per m². 1,000 ÷ 142 = **7.1 m²**.
Check on a real one: Juno carries about 60 m² of cells. Near Earth they would give about 14,000 W. At Jupiter, 27 times less sunlight: about **500 W**.
If the quiz gives different numbers for efficiency or losses, use theirs. The three steps stay the same.

▲ Where the instructor was loose

- **"We multiply by three the efficiency."** No. One layer gets about 20%, three get about 30%. The layers share the same sunlight; each takes the colours it uses best. It is 1.5 times better, not 3.
- **The photon gives the electron "inertia to move through the circuit".** The photon only frees the electron. What pushes it one way is the electric field built into the junction. Without that field the electron would just fall back.
- **His account of which way electrons and holes go is tangled.** The clean version: freed electrons collect on the n side, holes on the p side, and electrons travel through the outside wire from n to p.
- **"Low Earth orbit… around two hours… 12 cycles a day."** Low orbits take 90 to 100 minutes (Section 1), so about 15 or 16 a day: nearly 6,000 hot-cold cycles a year.
- **Slide: "Thermal cycles: low efficiencies".** Two different things. Being *hot* lowers efficiency. *Cycling* between hot and cold cracks joints and ages the array.
- **"I think this image is from Rosetta."** It looks like the Juno array in a clean room, the same photo as in Section 2. I can't be certain.
- **"The maximum was around 40%."** Flight cells are 30–32%. Laboratory records with concentrated light are about 47%.
- **He was right** that no solar-powered craft has yet worked beyond Jupiter's distance, and that a cold panel fresh from the shadow gives the most power.
- **Not said: arrays make nothing in the shadow.** That is the reason for the next lecture, batteries.

🔎 Caption errors

Captions said	Instructor meant
"several of our race", "efficiency race"	**solar arrays**
"sort of flux"	**solar flux**
"3-D used" · "culpably"	**reduced** · **probably**
"the playable arrays"	**deployable** arrays
"body mounts at the race"	**body-mounted arrays**
"keeps out"	**CubeSat**
"fil-A… the land of founders at"	**Philae**, the lander of **Rosetta**
"full of all types theory"	**photovoltaics** theory
"solar foton", "fortune"	solar **photon**
"pea type" · "election"	**p-type** · **electron**
"kind of whole… they name it all"	a **hole**
"clean room terms"	**equilibrium** terms
"unperson" · "Curran's times the Volt"	**amperes** · **current × voltage**

🗣 Say it out loud

Friend: "How do solar panels actually make electricity?"
You: A solar cell is two slices of crystal with a built-in one-way slope where they meet. A packet of light knocks an electron loose, the slope pushes it to one side, and it has to go round through the wire to get back. That flow is the current. The best space cells stack three layers for three colours and turn about 30% of the light into power. And sunlight thins out with distance squared, so at Jupiter you need panels the size of buses.

SEC 7 · LEC 1 · VOCABULARY

Every word used, in plain English

37 entries. "Repeat" = you have met it before.

Space power system — everything that makes, stores and hands out electricity on a spacecraft.

Solar array / solar panel — many solar cells wired together.

Solar cell (photovoltaic cell) — one unit that turns light into electricity.

Solar flux — sunlight power landing on each square metre, in W/m². *How hard it is raining light.*

Solar radiation — the light and other electromagnetic waves from the Sun. *Repeat.*

Watt (W) / W/m² — unit of power / power per square metre.

Astronomical unit (AU) — the Earth–Sun distance: 150 million km.

Inverse-square law — twice as far, a quarter as much. *Repeat, Section 5.*

Deployable array — panels folded for launch and opened in space.

Body-mounted array — cells fixed to the spacecraft's own skin.

Deploy — to unfold or release.

Interplanetary — between planets. *Repeat.*

Photovoltaics — making electricity directly from light.

Photon — one packet of light. *Repeat, Section 5.*

Electron — the tiny negative particle that carries electric current. *Repeat.*

Semiconductor — a material, such as silicon, that conducts electricity a little.

n-type — semiconductor with spare electrons (negative carriers).

p-type — semiconductor with missing electrons (positive 'holes').

Hole — a gap where an electron is missing. It behaves like a positive charge.

Junction (p–n junction) — the boundary between the n and p layers. *The one-way slope.*

Bond / escape the bond — an electron held in place by its atom / being knocked free.

Circuit — the closed loop that current flows round. *Repeat.*

Electrical current — the flow of electrons, in amperes. *Repeat.*

Voltage (volts) — electrical push. *Repeat.*

Power = voltage × current — watts = volts × amperes.

Efficiency — useful power out ÷ power in, as a percentage. *Repeat.*

Wavelength — the size of a light wave; it sets the colour and the photon's energy. *Repeat.*

Multi-junction / triple-junction cell — two or more / three cells stacked, each for a different colour.

Gallium arsenide (GaAs) — the semiconductor family used in the best space cells.

Upper / intermediate / lower cell — the three layers: short, medium and long wavelengths.

Band gap — the energy needed to free an electron in a given material.

Temperature effect — hot cells give lower voltage and so less power.

Thermal cycle — one round of heating in sunlight and cooling in shadow.

Eclipse / shadow — the part of the orbit where the planet blocks the Sun. *Repeat.*

Degradation — slow loss of performance with age and radiation. *Repeat, Section 2.*

Juno / Rosetta / Philae — solar-powered Jupiter orbiter / comet orbiter / its small lander. *Repeat, Who's Who.*

CubeSat — a tiny satellite with cells on its faces. *Repeat.*

SEC 7 · LEC 2 · BATTERIES

The bucket for the dark

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

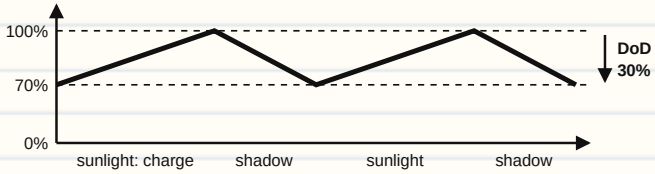
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S7 / L2 — Batteries **date · pg __**

1 · TWO KINDS OF BATTERY

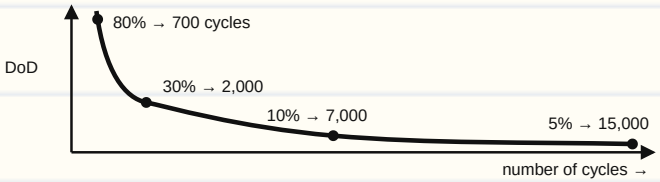
PRIMARY	SECONDARY
one shot · NO recharge most energy per kilo one-time jobs: pyro, stage separation, opening arrays	rechargeable from the array less energy per kilo runs everything while the craft is in shadow

2 · ONE CYCLE EVERY ORBIT



depth of discharge (DoD) = how much you drain each time

3 · DRAIN IT LESS, IT LASTS LONGER



shallow cycles = long life, but a bigger, heavier battery

4 · HOW BIG A BATTERY

battery energy = load × shadow time ÷ DoD

1,000 W × 0.6 h ÷ 0.30 = 2,000 Wh (about 13 kg)
watt-hours = ampere-hours × volts
2,000 Wh ÷ 28 V = 71 Ah

✏️ **PENCIL KEY** · zigzag = battery charge over two orbits · dotted lines = full and lowest level · dots = points on the life curve

⭐ **3 TAKEAWAYS** 3 lines

1. Primary = one shot, most energy per kilo. Secondary = rechargeable.
2. Depth of discharge = how much you drain each orbit.
3. Shallow drains last far longer but need a heavier battery.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
Primary vs secondary battery?	two kinds: primary and secondary batteries
What are primary batteries used for?	primary: one shot, no recharge · high energy density · missiles, rockets · one-time jobs: pyro, stage separation, deploying arrays
When does a spacecraft run on its battery?	secondary: rechargeable from the solar array · lower energy density · runs the systems in the shadow
Capacity vs stored energy: units?	capacity in ampere-hours (Ah) · stored energy in watt-hours (Wh) = Ah × volts
What is depth of discharge?	depth of discharge (DoD) = % of capacity used · 80% left → DoD 20%
Three questions for designing a battery?	one charge + one discharge per orbit
How does DoD affect battery life?	design: 1) time in shadow · 2) rate of discharge (the load) · 3) rate of charge → capacity + power
What does a small DoD cost?	must survive a very high number of cycles
How many cycles a day in low orbit?	small DoD → long life · but a bigger, heavier battery → trade mass against life
? why does deep draining wear it out	

SEC 7 · LEC 2 · TEACH

Saving sunlight for the shadow

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Solar panels are like rain: wonderful while it lasts, but in low orbit the spacecraft flies into the Earth's shadow sixteen times a day. So it carries a bucket. In sunlight the panels run everything and fill the bucket. In the dark the bucket runs everything. That bucket is the **secondary battery**, the rechargeable kind, like the one in your phone. The other kind, the **primary battery**, is a firework: it holds more punch for its weight but works once. It is used for one-time jobs like firing the little charges that release the solar panels.

The whole lecture is about one choice: how far down do you drain the bucket each time? That is the **depth of discharge**. Your phone battery lasts years longer if you keep it between 80% and 100% than if you run it flat every day. A spacecraft battery is the same, and it gets 6,000 cycles a year. Drain it gently and it lives. But to drain it gently you need a much bigger bucket, and every kilo costs. Life against weight: that is the trade.

🏷 The official names (name first, then what it means)

- **Cell**: one electrochemical unit, a few volts. A **battery** is several cells wired together.
- **Primary battery** (non-rechargeable). Space types: lithium thionyl chloride, silver-zinc.
- **Secondary battery** (rechargeable; storage battery). Space types, oldest to newest: **nickel-cadmium** (NiCd), **nickel-hydrogen** (NiH_2), **lithium-ion** (Li-ion).
- **Energy density** (properly **specific energy**): watt-hours stored per kilogram.
- **Capacity**: charge a battery can deliver, in **ampere-hours** (Ah). **Stored energy**: in **watt-hours** (Wh) = Ah × volts.
- **Depth of discharge, DoD**: percentage of the capacity taken out. Its mirror is **state of charge, SoC** = 100% − DoD.
- **Charge rate / discharge rate** (C-rate): how fast. 1C empties or fills the battery in one hour.
- **Cycle**: one discharge plus one recharge. **Cycle life**: how many it survives.
- **Eclipse** (the lecture's "penumbra time"): the part of the orbit in the planet's shadow. **Umbra** is the full shadow; **penumbra** is only its fuzzy edge.
- **Anode, cathode, electrolyte**: the two electrodes and the ion-carrying material between them.
- **Pyrotechnic device** (pyro): a small explosive charge that cuts a bolt or opens a valve, once.
- **Bus voltage**: the spacecraft's supply voltage, commonly 28 V.

🛠 How it actually works

Step	The mechanism, with numbers
1 · Inside a cell	Two electrodes hold lithium with different strength. The material between them lets charged lithium atoms (ions) through but not electrons. So when lithium moves across inside, its electrons must go the long way round through the wire: the same trick as the solar cell. About 3.6 V per lithium-ion cell; eight in a row make 28 V.
2 · One shot or many	In a primary cell the reaction cannot be run backwards, which allows more energetic chemistry: about 400–500 Wh per kg. Lithium-ion: 150–250. Older nickel-hydrogen: about 60.
3 · Units	Energy (Wh) = capacity (Ah) × volts. 71 Ah × 28 V ≈ 2,000 Wh.
4 · The shadow	At 400 km an orbit takes 92 minutes and up to 36 of them (0.6 h) are dark. About 16 orbits a day: 5,800 cycles a year, 29,000 in five years.
5 · Sizing	A 1,000 W spacecraft uses 1,000 × 0.6 = 600 Wh each shadow. Allow only a 30% drain: battery = 600 ÷ 0.30 = **2,000 Wh**, about 13 kg. Allow 80%: only 750 Wh and 5 kg, but it wears out several times sooner.
6 · Why depth hurts	The electrodes swell and shrink as lithium moves in and out. Deeper swings strain them more and feed side reactions that lock lithium away for good. The slide's curve: 80% → 700 cycles, 30% → 2,000, 10% → 7,000, 5% → 15,000. Today's lithium-ion space cells manage tens of thousands at 30%, with the same shape of curve.
7 · High orbits	A GEO satellite is in shadow only around the two equinoxes: about 90 eclipses a year, up to 72 minutes. That is 1,350 cycles in 15 years, so it can drain 70–80% each time.
8 · Refilling	The 600 Wh must go back in during 56 minutes of sunlight: 600 ÷ 0.93 h ÷ 0.9 (losses) ≈ 710 W. So the array for a 1,000 W spacecraft must make about **1,700 W**.

SEC 7 · LEC 2 · SIZING + FACT CHECK

How big a bucket, and for how long

A sizing recipe with low orbit and high orbit side by side, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

🧮 The recipe

1. Shadow time in hours.
2. Energy used = load (W) × shadow time (h).
3. Battery energy = energy used ÷ depth of discharge.
4. Capacity (Ah) = battery energy ÷ bus voltage.
5. Mass = battery energy ÷ energy per kilo.
Worked: 1,000 W × 0.6 h = 600 Wh. ÷ 0.30 = 2,000 Wh. ÷ 28 V = 71 Ah. ÷ 150 Wh/kg = **13 kg**.
His formula is step 3 turned round: DoD (%) = energy taken out ÷ (capacity × voltage) × 100.

🛰️ Low orbit against GEO

	Low orbit, 400 km	GEO
One orbit	92 minutes	24 hours
Longest shadow	36 minutes	72 minutes
Cycles per year	about 5,800	about 90
Usual depth of discharge	20–40%	60–80%
What wears the battery	the number of cycles	mostly age

GEO satellites only pass through the Earth's shadow for a few weeks around each equinox.

▲ Where the instructor was loose

- **"Auxiliary power, low power."** In the shadow the battery is the *only* power on board. It has to carry the whole spacecraft.
- **"Penumbra time."** The word is eclipse. The penumbra is just the thin fuzzy edge of the shadow where the Sun is partly covered.
- **"Two-hour cycle… 12 cycles a day."** Again: low orbits take about 92 minutes, so about 16 cycles a day. Earlier in this same lecture he says 1.5 hours.
- **"Secondary batteries are low energy density."** Lower than primary ones, not low: 150–250 Wh per kg against 400–500.
- **The charge graph from 0% to 100%.** He says himself it is the limit case. No real design goes anywhere near 0%.
- **The life curve's numbers.** The shape is right for every battery. The values look like an older battery type; modern lithium-ion cells survive several times more cycles at the same depth, as far as I know.
- **Not said: how a battery works, or which types fly.** Both are on the teach page.
- **Not said: the array has to be bigger because of the battery.** It must run the spacecraft and refill the battery in the same sunlit stretch. That links this lecture to your solar array quiz.
- **He was right** about the phone: keeping it between 80% and 100% makes it last longer than running it flat.

🔎 Caption errors

Captions said	Instructor meant
"mistrials rockets"	**missiles**, rockets
"look cool and just one one time"	**occur** just one time
"the solar Appollo"	the solar **power**
"measured in per hour"	measured in **ampere-hours**
"Watchtower's", "vat's house"	**watt-hours**
"the Andy Awards… they empower hours"	the **ampere-hours**
"depth of this charge", "deficit of this charge"	**depth of discharge**
"d o d", "diod D", "deodorize", "deity", "the O.D."	**DoD**
"multi-byte bytes are found"	**multiplied by a hundred**
"one charge and these talks Bob be it"	one charge and **discharge per orbit**
"on the shuttle"	in the **shadow**
"rate of chach" · "eyeball"	rate of **charge** · **are able**
"lineal", "Linnea" · "Vardi"	**linear** · **vary**

🗣️ Say it out loud

Friend: "What happens to a satellite's solar power when it goes behind the Earth?"
You: It runs on a rechargeable battery. In sunlight the panels power everything and fill the battery; in the shadow the battery takes over, sixteen times a day in a low orbit. The design question is how far to drain it each time. Drain it only 30% and it lasts many times longer than if you run it flat, but then the battery has to be more than three times bigger than the energy you actually use. It is a trade between weight and lifetime.

SEC 7 · LEC 2 · VOCABULARY

Every word used, in plain English

34 entries. "Repeat" = you have met it before.

Battery — a device that stores energy chemically and gives it back as electricity.

Cell — one unit of a battery, a few volts.

Primary battery — used once; cannot be recharged. *A firework.*

Secondary battery — rechargeable. *A bucket you refill.*

One shot — works a single time.

Recharge — push energy back in.

Energy density — energy stored per kilogram (or per litre).

Missile / rocket — vehicles whose flights last minutes, so one-shot batteries suit them.

Pyrotechnic (pyro) — a small explosive device that cuts or releases something.

Stage separation — dropping an empty rocket stage. *Repeat, Section 4.*

Deploy — unfold or release. *Repeat.*

Auxiliary power — back-up or secondary power.

Penumbra / umbra — the partly shaded edge of a shadow / the fully dark part.

Eclipse — the time the spacecraft spends in the planet's shadow. *Repeat.*

Capacity — how much charge a battery holds, in ampere-hours. *The size of the bucket.*

Ampere-hour (Ah) — one ampere flowing for one hour. *Your phone lists it as mAh.*

Stored energy — the energy in the battery, in watt-hours.

Watt-hour (Wh) — one watt for one hour. Wh = Ah × volts.

Voltage / average discharge voltage — electrical push / its average value while the battery empties.

Depth of discharge (DoD) — the percentage of the capacity that has been used. *How far down the bucket is.*

State of charge — the percentage left. 100% − DoD.

Charge rate / discharge rate — how fast energy goes in / comes out.

Cycle — one discharge followed by one recharge.

Battery life / life expectancy — how many cycles, or years, it lasts.

Linear — changing at a steady rate: a straight line on a graph.

Empirical — found by measurement, not derived from theory.

Iterative — repeated, improving the answer each round.

Mass — how much matter; what the launcher has to lift. *Repeat.*

Lithium-ion (Li-ion) — today's standard rechargeable battery, in phones and spacecraft.

Nickel-cadmium / nickel-hydrogen — the older rechargeable types used on spacecraft.

Electrode / electrolyte — the two plates of a cell / the material between that carries ions.

Ion — an atom with an electric charge. *Repeat, Section 4.*

Bus voltage — the spacecraft's main supply voltage, often 28 V.

Equinox — the two times a year when the Sun is over the equator; GEO eclipse seasons.

SEC 7 · LEC 3 · RADIOISOTOPE GENERATORS (RTG)

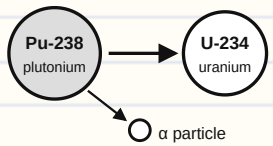
The rock that stays warm

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LEFT PAGE · DRAW

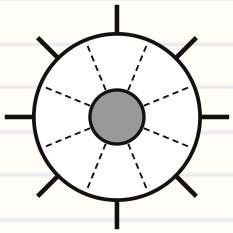
S7 / L3 — RTGs **date · pg __**

1 · A LUMP THAT HEATS ITSELF (decay)

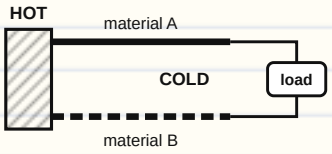


α particle = 2 protons + 2 neutrons
it stops inside the fuel → **HEAT**
1 gram gives 0.57 W of heat
half the atoms gone in 87.7 years
nothing to switch on or off

2 · HEAT → ELECTRICITY (Seebeck effect)



hot fuel in the middle
cold shell with fins



hot end + cold end → a voltage
only about 6% becomes electricity
the rest leaves through the fins
no moving parts

3 · IT FADES SLOWLY

Galileo: 290 W → 250 W in 14 years



4 · WHICH POWER WHERE

SOLAR ARRAY near the Sun, out to Jupiter
BATTERY through the shadow
RTG far from the Sun, or dark for weeks (Pluto, Mars night)

an RTG is not a reactor and not an engine

✏️ **PENCIL KEY** · big circle = atom · small circle = α particle · grey = hot · solid bar = material A · dashed bar = material B · curve = power left

⭐ **3 TAKEAWAYS** 3 lines

1. Plutonium-238 decays by itself and the decay makes heat.
2. Hot inside, cold outside: thermocouples turn about 6% into electricity.
3. Works anywhere, for decades. Used where sunlight is too weak.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What does RTG stand for?	RTG = radioisotope thermal (thermoelectric) generator
What do Voyager 1 and New Horizons have in common?	Voyager 1 + New Horizons: no solar panels → very far from the Sun → RTGs
Where does the heat come from?	heat from the natural decay of plutonium-238 → uranium-234 + an alpha particle (2 protons + 2 neutrons)
What does Pu-238 decay into?	
How does heat become electricity?	Seebeck effect: two different materials, one end hot, one end cold → a voltage
What are the fins for?	hot centre, cold outside · fins keep the outside cold
How long does it last?	power does not depend on the Sun
Why is it controversial?	lasts decades (slide: up to 100 years) · fades as the plutonium decays
Is it a nuclear reactor?	expensive · plutonium is controversial · NASA uses it, ESA rarely
Galileo's RTG: the numbers?	NOT a nuclear reactor: natural decay, no fission chain
? why plutonium-238 and not another	Galileo's: 290 W new → 250 W at end · 55 kg · 7.5 kg Pu-238 · 114 cm long

SEC 7 · LEC 3 · TEACH

Power for where the Sun doesn't reach

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Imagine a lump of metal that is hot to the touch and stays hot for a hundred years with nothing plugged in. Plutonium-238 is that lump. Its atoms are unstable: one by one, each spits out a tiny piece of itself, and every spit is a little kick of heat. A fresh pellet glows red by itself. Nobody starts it and nobody can stop it.

Now the trick for getting electricity. Join two different materials, heat one end and keep the other end cold, and a small voltage appears between the cold ends. So you wrap hundreds of these little pairs around the hot lump and put cooling fins on the outside. That is the whole machine: no moving parts, nothing to wear out. It works in the dark, in a dust storm, at Pluto. The catches: only about 6% of the heat becomes electricity, the plutonium is rare and radioactive so launching it is a political fight, and the power slowly fades as the lump uses itself up. Voyager has been running on three of them since 1977.

🏷 The official names (name first, then what it means)

- **Radioisotope thermoelectric generator, RTG** (the slide's "thermal generator"; nuclear battery): makes electricity from the heat of radioactive decay.
- **Radioisotope** (radioactive isotope): a version of an element whose atoms are unstable. **Radioactive decay**: an unstable nucleus changing by itself and releasing energy.
- **Alpha decay**: decay by throwing out an **alpha particle** (α; a helium nucleus: 2 protons + 2 neutrons).
- **Half-life**: the time for half the atoms to decay. Plutonium-238: 87.7 years.
- **Plutonium-238, Pu-238**: the usual fuel, used as a ceramic, plutonium dioxide. It decays to **uranium-234**.
- **Seebeck effect** (thermoelectric effect): a temperature difference across two joined materials produces a voltage.
- **Thermocouple** (in an RTG, a **unicouple**): one such pair. **Hot junction**, **cold junction**. Material: silicon-germanium.
- **General purpose heat source, GPHS**: the protective brick that holds the fuel pellets.
- **Radiator fins**: carry the waste heat away so the outside stays cold.
- **Thermal power** (heat, watts-thermal) versus **electrical power** (watts-electric). **BOL / EOL**: beginning of life / end of life.
- **Nuclear reactor**: a machine that runs a controlled **fission chain reaction**. An RTG is not one.
- **Radioisotope heater unit, RHU**: a small pellet used only to keep equipment warm.

🛠 How it actually works

Step	The mechanism, with numbers (Galileo's RTG, from the slide)
1 · Decay	A plutonium-238 nucleus (94 protons, 144 neutrons) throws out an alpha particle and becomes uranium-234 (92 and 142). Count them: 94 − 2, 144 − 2.
2 · Heat	The alpha particle is stopped inside the fuel within a hair's width, and its energy turns into heat. One gram gives 0.57 W. Galileo's 7.5 kg: about **4,300 W** of heat. The fuel sits near 1,000 °C.
3 · Seebeck	Heat makes the charge carriers in a material drift from the hot end to the cold end. Two different materials drift by different amounts, so joining them at the hot end leaves a voltage between the cold ends. 572 pairs in series give about 30 V.
4 · Efficiency	Hot side about 1,000 °C, cold side about 300 °C. 290 W of electricity from 4,300 W of heat: **6.7%**. The other 4,000 W leaves through the fins (and helps keep the spacecraft warm).
5 · Fading	Power = starting power × 0.5^(years ÷ 87.7). That is 0.8% a year. Galileo flew 14 years: 0.5^(14 ÷ 87.7) = 0.895, so 290 W → 260 W from decay alone. The thermocouples age too, which gives the slide's **250 W**.
6 · Voyager	470 W at launch in 1977. About 220 W today, as far as I know, so instruments are being switched off one at a time.
7 · Why this isotope	Alpha particles are stopped by a sheet of paper, so almost no shielding is needed. And 88 years is long enough for a mission but short enough to be hot. Strontium-90 (29 years) needs heavy shielding. Americium-241 (432 years) gives a fifth of the heat per gram.
8 · Safety	The fuel is ceramic pellets clad in iridium inside graphite shells built to survive a launch accident and re-entry. Apollo 13's is lying intact on the floor of the Pacific. Plutonium-238 is not bomb material; that is plutonium-239.

SEC 7 · LEC 3 · COMPARE + FACT CHECK

Three ways to keep the lights on

Section 7 in one table, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📋 The power sources side by side

Source	Energy comes from	Typical power	Lasts	Use it when	The catch
Solar array	sunlight	100s of W to 100,000+ W	15+ years, fading 1–3% a year	anywhere from Mercury to Jupiter	nothing in shadow; huge beyond Mars
Secondary battery	stored from the array	the whole spacecraft, for an hour	thousands of cycles	every eclipse	stores, does not make; heavy
Primary battery	its own chemicals, once	W to kW for hours	one use	pyro, short-lived probes	cannot be recharged
RTG	decay heat of plutonium-238	100–300 W each	decades, fading 0.8% a year	beyond Jupiter; long nights; dust	6% efficient; costly; scarce fuel; politics

RTG users: Voyager 1 and 2, Galileo, Cassini, New Horizons, and the Mars rovers Curiosity and Perseverance (for the dust and the nights, not the distance).

▲ Where the instructor was loose

- **The name.** His slides say radioisotope *thermal* generator. The proper name is radioisotope **thermoelectric** generator.
- **"They can last up to 500 years."** His own slide says 100. The honest answer: the heat halves every 87.7 years and the thermocouples age as well, so useful life is decades. Voyager is at 49 years and running short.
- **"One point fifty metres long."** His slide says 114 cm.
- **"Temperature is ejected from the nucleus."** A particle is ejected. It is stopped inside the fuel and its energy of motion becomes heat.
- **"Really only applicable for space."** The Soviet Union powered hundreds of remote lighthouses with strontium-90 units, and early heart pacemakers ran on plutonium-238.
- **"ESA is developing strontium."** As far as I know, the European programme is built on americium-241. Strontium-90 was the Soviet ground choice.
- **Only for "very far from the Sun".** Also for long darkness and dust: the Mars rovers, and the Apollo instruments that had to survive two-week lunar nights.
- **"New Horizons took a few poor photos of Pluto."** Probably the captions. It returned sharp close-ups of Pluto in 2015.
- **Not said: the efficiency (about 6%)**, and that an RTG cannot be turned off. It is hot on the launch pad and has to be cooled there.
- **He was right** that it is not a reactor: no chain reaction, no fission, nothing to control. And this is the "RTG is not an engine" point from Section 4.

🔎 Caption errors

Captions said	Instructor meant
"radio isotope with more generators"	radioisotope **thermal** generators
"apart from the intent"	apart from the **antenna**
"far apart from soccer"	far from the **Sun**
"the five to one"	**Voyager 1**
"plutonium to 3:8"	plutonium-**238**
"the case time to uranium… not a particle"	**decays into** uranium… **an alpha** particle
"to newsrooms and to printers"	**two neutrons and two protons**
"X thermic"	**exothermic** (gives off heat)
"Seabeck", "seabag", "C-PAC effect"	**Seebeck** effect
"occult material", "Hunter called Hellmann's"	**a cold** material, **hot and cold elements**
"a strong team"	**strontium**
"loan to date on RTG needs"	**long debate** on RTGs
"regarding Cephalopoda"	regarding **RTGs**

🗣 Say it out loud

Friend: "How does Voyager still have power after fifty years with no solar panels?"
You: It carries lumps of plutonium-238, which is naturally hot because its atoms keep breaking down. Around the hot lump are hundreds of little pairs of two materials: hot on one end, cold on the other, and that difference makes a voltage. No moving parts, works in total darkness. It is not a reactor, just natural decay. It only turns about 6% of the heat into electricity and it loses a bit under 1% a year, which is why Voyager is now switching instruments off.

SEC 7 · LEC 3 · VOCABULARY

Every word used, in plain English

32 entries. "Repeat" = you have met it before.

Radioisotope thermoelectric generator (RTG) — a device that turns the heat of radioactive decay into electricity. *A rock that stays warm, wrapped in heat-to-electricity strips.*

Radioisotope — a form of an element whose atoms are unstable and break down by themselves.

Isotope — one version of an element; versions differ in their number of neutrons.

Radioactive decay / natural decay — an unstable atom changing into another by itself, releasing energy.

Spontaneous — happening by itself, with no trigger.

Nucleus — the tiny heavy centre of an atom, made of protons and neutrons.

Proton / neutron — the positive / neutral particles in the nucleus.

Plutonium-238 (Pu-238) — the radioisotope used as RTG fuel. Half-life 87.7 years.

Uranium-234 (U-234) — what plutonium-238 turns into.

Alpha particle (α) — two protons and two neutrons thrown out together: a helium nucleus. *Stopped by a sheet of paper.*

Half-life — the time it takes for half the atoms to decay.

Exothermic — giving off heat.

Seebeck effect (thermoelectric effect) — a voltage produced by a temperature difference across two joined materials.

Thermocouple / junction — a pair of different materials joined together / the place they join.

Hot / cold side — the end against the fuel / the end against the finned shell.

Fins (radiator fins) — thin plates that throw away waste heat so the outside stays cold. *Repeat idea, Section 2.*

Voltage — electrical push. *Repeat.*

Watt (W) — unit of power. *Repeat.*

BOL / EOL — beginning of life / end of life: power when new and when the mission ends.

Interplanetary / interstellar — between planets / between stars. *Repeat.*

Controversial — argued over. *Repeat.*

Strontium-90 / americium-241 — other radioisotopes that can fuel such generators.

Nuclear reactor — a machine that splits atoms in a controlled chain reaction. *Repeat, Section 4.*

Fission / fusion — splitting heavy atoms / joining light ones. An RTG does neither.

Chain reaction — each split atom triggering more splits.

Shielding — material placed to stop radiation.

Heat source (GPHS) — the protective brick holding the plutonium pellets.

Multi-foil insulation — layers of thin foil that keep the heat going through the thermocouples. *Like MLI, Section 2.*

Voyager 1 — launched 1977; now beyond the Sun's bubble, in interstellar space. *Repeat, Who's Who.*

New Horizons — the Pluto probe; one RTG. *Repeat, Who's Who.*

Galileo — NASA's Jupiter orbiter of the 1990s; two RTGs. *Repeat, Who's Who.*

NASA / ESA — the US and European space agencies. *Repeat.*

SPACE X · BACK OF THE NOTEBOOK

The back pages: words, names and cards

Reference material. You look things up here; you don't read it through.

What	Page	Length
Teach-it-back kit, Sections 1–3 (scenes, talks, quiz)	115	16 pages
Concept cards in notebook layout, C1–C16	131	10 pages
Word bank, concept level, Sections 1–3	141	9 pages
Glossary, every word, Sections 1–3	150	10 pages
One-line word bank, Sections 1–3	160	3 pages
Copy-into-notes sheet (equations, map sentences), Sections 1–3	163	2 pages
Who's Who: every mission, engine and person named	165	4 pages
Map of the paper notebook	169	1 pages

Where to look

- **A word you don't know:** the lecture's own vocabulary page first, then the glossary here. For every word in the course, the app's Terms tab.
- **A name** (a probe, an engine, a person): Who's Who.
- **Sections 1–3 as a talk:** the teach-it-back kit.
- **What to write in the paper notebook:** the concept cards and the copy-into-notes sheet.

The word banks and glossary here cover Sections 1–3. For Sections 4–7 the words are on each lecture's vocabulary page.

TEACH-IT-BACK KIT · SECTIONS 1-3 · PAGE 1 OF 16

One story, seven minutes

You've done three sections. They're not seven separate topics; they're one story with seven scenes. Learn the story, and the facts hang off it.

THE STORY (memorize this paragraph; everything else is detail):
Space gives you nothing for free. **Getting there** is falling and missing: go sideways fast enough and the ground curves away under you. **Where you end up** depends on how high you go: low is fast and sharp but sees little, high is slow and sees everything, and one exact height hangs still. **Describing** any orbit takes six numbers, which are really six ways to describe a hula hoop. **Staying alive** up there means doing everything the air used to do for you, because the vacuum does none of it, and the thin air that's left eats your plastic. **Going somewhere else** is a fuel bill: fast is expensive, the lazy half-ellipse is cheap, and the only free lunch is bouncing off a moving planet.



1 FALLING & MISSING	2 THREE SHELVES	3 HULA HOOP	4 FIVE FREE SERVICES	5 THE AIR BITES BACK	6 THE FUEL BILL	7 THE FREE LUNCH
orbit = fall + miss	LEO · MEO · GEO	a e i Ω ω ν	vacuum, outgassing	layers, atomic O	Lambert, Hohmann	flyby, Rosetta
S1 · S3	S1 · S3	S3	S2	S2	S1 · S3	S1 · S3

The 7 minutes at a glance

0:00–0:45	0:45–1:45	1:45–2:45	2:45–3:45	3:45–4:30	4:30–5:45	5:45–6:30	6:30–7:00
HOOK	SHELVES	HULA HOOP	FREE SERVICES	AIR BITES	FUEL BILL	FREE LUNCH	LANDING
falling & missing	LEO MEO GEO	6 numbers	vacuum	layers, atomic O	Hohmann	flyby	"nothing's free"

Pages 2–3: each scene like you're in 5th grade. Pages 4–10: the same scenes at engineer level, how it actually works, with numbers you can check, one scene per page. Page 11: the full 7-minute script. Pages 12–13: the three section talks with cue cards. Pages 14–15: quiz and answers. Page 16: how to rehearse.

Scenes 1–4: getting there and staying alive

🏀 Scene 1 · Falling and missing

Throw a ball sideways. It falls and lands. Throw it harder, it lands farther. Now throw it so hard that by the time it falls, the ground has curved away underneath it. It never lands. It just keeps falling *around* the Earth. That's an orbit. The space station isn't floating; it's falling, and missing, 16 times a day.

> **SAY: "An orbit is falling and missing. Go sideways fast enough and you never hit the ground."**

Draw: a mountain, three throws, the third one wrapping around.
Friend asks "so why do astronauts float?" → Because they're falling too, together with the station. Same as the drop on a roller coaster, but forever.

🏢 Scene 2 · Three shelves

Satellites live on three shelves. **Low shelf (LEO):** 200–2,000 km up, laps Earth every 90 minutes, sees a tiny patch in sharp detail. Space station, Starlink. **Middle shelf (MEO):** up to 36,000 km, 12-hour laps, GPS lives here. **Top shelf (GEO):** exactly 35,786 km, one lap takes exactly one day, so from the ground it hangs still. TV and weather satellites.
Higher = sees more, moves slower, sees less detail, and the radio takes longer. Low satellites only see one spot for minutes, so to cover the planet you fly a swarm: a constellation. GPS is a swarm of 24+; your phone needs 4 overhead.

> **SAY: "Higher means you see more but move slower. One exact height hangs still, and it has to be over the equator."**

Draw: Earth, three rings, a little flashlight cone from each to show footprint size.
Friend asks "why over the equator?" → Every orbit is centered on the middle of the Earth, and the equator is the only ground circle that matches. Park it over New York and New York slides away underneath.

🌀 Scene 3 · The hula hoop

Earth is a basketball; the orbit is a hula hoop around it. To tell someone exactly how the hoop sits you need six things: **how big** (a), **how squished** (e), **how tilted** (i), **which way the tilt faces** (Ω, swivel), **which way the squished end points** (ω, spin), and **where the ball is right now** (ν). Five freeze the hoop; one moves.
Most orbits are ovals with Earth off to one side, so there's a closest point (perigee) and a farthest point (apogee). Like a swing: fastest at the bottom (perigee), slowest at the top (apogee). And those are the two spots where you fire the engine: push at perigee and the far side rises; push at apogee and the near side rises.

> **SAY: "Six numbers describe any orbit: size, squash, tilt, swivel, spin, and where you are. Fastest at the low point, slowest at the high point."**

Draw: an egg with Earth off-center, two dots labeled fast and slow.
Friend asks "why six, not three?" → Size and shape don't tell you how the hoop is turned in space. That takes three more.

🧯 Scene 4 · Five free services

Down here, the air does five jobs you never notice: it's **sunscreen** (ozone blocks harsh UV), it's **pressure** (holds gas inside materials), it's a **damper** (a buzzing ruler stops because air steals its energy), it's a **fan** (hot air rises and carries heat away), and it's a **ground wire** (static leaks away). Vacuum cancels all five. So in space: UV snaps plastics, trapped gas pushes out and fogs your lenses (outgassing), vibrations never die, hot spots stay hot (only radiators work), and charge builds until it sparks. And it all starts on the ground: clean rooms, because one speck can kill a sensor.

> **SAY: "Earth's air is doing five jobs for you right now. Space does none of them, so the spacecraft has to do all five itself."**

Draw: a box labeled spacecraft, five arrows pointing in, each labeled with what's missing.
Friend asks "so why does the ISS have those big white panels?" → Radiators. It makes its own heat and there's no air to blow it away, so it glows it off into space.

Scenes 5–7: the air bites, the fuel bill, the free lunch

🌡 Scene 5 · The air bites back

The air is a layer cake. Bottom layer (troposphere, 0–11 km): weather, planes, gets colder going up. Next (stratosphere, to 50 km): gets *warmer*, because the ozone is here soaking up UV and turning it into heat. Next (mesosphere, to 90 km): coldest, meteors burn. Then the thermosphere (to 500 km): the thermometer says 1000 °C but there's almost no air, so nothing can hand that heat to you. The space station flies through it and doesn't melt. Temperature is how fast molecules move; heat is how many there are to hit you.

Up there UV also snaps oxygen pairs into single oxygen atoms, which are desperate to grab onto anything. The spacecraft plows into them at 7.5 km/s and they eat its plastics like sandpaper. Metal shrugs; Kapton film gets shredded. Fix: coatings.

What the air is made of, bottom to top: below 100 km it's a stirred mix, the same everywhere: nitrogen pairs 78%, oxygen pairs 21%, argon 0.9%, carbon dioxide 0.04%, a little water vapor, and ozone (three oxygen atoms) as a few parts per million in the stratosphere. Above 100 km the stirring stops and gases sort by weight: single oxygen atoms rule 200–600 km (LEO), helium 600–1,500 km, and hydrogen on top, so thin it leaks into space. The ionosphere (60–1,000 km) overlaps all of it: sunlight knocks electrons off atoms, and the charged soup bends radio and delays GPS.

> **SAY: "The top of the atmosphere is 1000 degrees but can't burn you, because there's nothing there. What is there is loose oxygen atoms that eat plastic."**

Draw: a stack of five bands with a zigzag temperature line beside it.

Friend asks "1000 degrees and it doesn't melt?" → One sparkler is 1000 degrees and doesn't warm a room.

💸 Scene 6 · The fuel bill

First, the ladder of ways to visit a planet: fly past it, orbit it, drop a probe into its air, land on it, drive on it, or crash into it on purpose. Each rung costs more fuel to slow down and brings back more science.

Now, getting there. You don't aim at Mars; you aim at where Mars will be. The cheapest route is a lazy half-ellipse that drifts outward and meets Mars halfway around the Sun: the **Hohmann transfer**, about 8.5 months, two engine burns. You could go faster, but every bit faster costs wildly more fuel, and past a point the rocket can't be built. **Lambert's problem** is the calculator: two points plus a time, out comes the orbit and the fuel bill. And you can only launch when Earth and Mars line up, every 26

🚚 Scene 7 · The free lunch

For really far places the lazy throw is still too expensive. So you cheat. Swing your spacecraft close past a planet. In the planet's eyes you come in and go out at the same speed, just bent. But the planet is moving around the Sun, and the bend lines you up to carry some of that motion along. You leave faster than you arrived, for free. That's a **flyby** or **gravity assist**: a ball bouncing off the front of a moving truck. The planet pays for it by slowing down an amount nobody could ever measure.

More speed means a bigger orbit, which means you can reach farther. Pass in front of the planet instead of behind and you slow down, which is how you brake at Mercury. The price is time: Rosetta took 10 years and four flybys to reach a comet. New Horizons took 9.5 years to Pluto with one Jupiter boost, and arrived too fast to ever stop, which is why it was a flyby.

> **SAY: "Bounce off a moving planet and you leave faster than you came. It's free, but it takes years."**

Draw: a planet with a thick arrow showing it moving, a dashed path bending around it.

Friend asks "isn't that free energy?" → No, it comes from the planet. Jupiter slowed down by about a trillionth of nothing.

🎯 The landing line (end on this)

"So that's the whole thing. Space gives you nothing for free. Getting there is falling and missing. Staying alive means doing the air's five jobs yourself. And going anywhere is a trade between time and fuel, unless you can borrow a planet. Every spacecraft ever built is just a different answer to those trades."

> **Then stop talking. Let them ask.**

🔁 Where this came from

Scenes 1, 2, 6, 7 = Section 1 + Section 3. Scenes 4, 5 = Section 2. Scene 3 = Section 3. Your notebook already has every fact on this page; what's new here is the order and the connecting sentences.

PAGE 4 OF 16 · HOW IT ACTUALLY WORKS · 🛠 ENGINEER LEVEL

Scene 1 · Falling & missing: How it actually stays up (no engines involved)

A spacecraft in orbit does nothing to stay in orbit. It doesn't fire sideways; it doesn't fire at all. Gravity is the only force on it, and gravity is what keeps it there. Here is the arithmetic you can check yourself, for the ISS at 400 km.

Step	In short	How it actually works	Official name / why it matters
1 · The rocket's job	give it sideways speed, once	Launch looks vertical for a minute, then tips over. Most of the burn is spent going sideways. At engine cutoff the spacecraft is moving horizontally at ~7.7 km/s. Height is cheap; sideways speed is the expensive part.	Official name: orbital velocity (circular velocity), $v = \sqrt{(\mu/r)}$.
2 · The fall	gravity pulls it down ~4.3 m every second	Gravity at 400 km is 8.7 m/s², about 90% of surface gravity. In one second the spacecraft drops $\frac{1}{2} \cdot 8.7 \cdot 1^2 \approx 4.3$ m toward Earth.	There is plenty of gravity up there. 'No gravity in space' is false.
3 · The miss	the ground curves away ~4.3 m every second	In that same second it moves 7.7 km sideways. Over 7.7 km, Earth's surface curves away by $d^2/(2R) = (7{,}670\text{ m})^2 / (2 \times 6{,}771{,}000\text{ m}) \approx 4.3$ m. Drop equals curve. Distance to the ground never changes.	That cancellation IS the orbit. Falling and missing is arithmetic, not a metaphor.
4 · Why gravity is the string	gravity = centripetal force	Swing a ball on a string: the string's inward pull is what bends a straight path into a circle. Gravity is Earth's string. A circular orbit is exactly the speed where gravity's pull equals the inward pull that circle needs.	Official name: centripetal force. For a satellite, gravity provides it.
5 · Wrong speed?	no engine needed for any of these	Too slow for the height: drops more than the ground curves, path bends into the atmosphere. Too fast: climbs, slows, comes back round as an ellipse. Faster than escape speed ($\sqrt{2} \times$ circular): never comes back.	Different starting speeds, same physics. See vis-viva.
6 · What engines DO run for	station-keeping and attitude, not 'staying up'	Station-keeping: undoing tiny losses. ISS loses ~2 km/month to thin air and reboosts for a few minutes about monthly. GEO sats do small burns every few weeks against the Moon's and Sun's tug to hold their slot. Attitude control: pointing, via reaction wheels; it never changes the orbit. A tumbling spacecraft is still in a perfect orbit.	Two separate jobs, two separate names: orbit maintenance (station-keeping, reboost) vs attitude control (ADCS).

- The sentence to own: it doesn't stay up by pushing. It stays up by moving sideways so fast it keeps missing the ground. Engines only run to fix pointing or to undo tiny drag losses.
- Why 'weightless': you feel weight only when something pushes back on you (the floor). In free fall nothing does, so you float, even though gravity is 90% of normal.
- A GEO satellite's lifetime is set by its station-keeping fuel. When that runs out it still orbits fine; it just drifts out of its parking slot and is moved to a graveyard orbit.

PAGE 5 OF 16 · HOW IT ACTUALLY WORKS · 🛠 ENGINEER LEVEL

Scene 2 · Three shelves: Why higher is slower, how a satellite gets to GEO, and how GPS actually finds you

Three mechanisms hide behind the three shelves. Each one has numbers you can check.

Step	In short	How it actually works	Official name / why it matters
1 · Why higher = slower	gravity weakens with distance (inverse square)	Gravity falls off as $1/r^2$. At LEO (r = 6,771 km) you need 7.67 km/s sideways to balance it; at GPS height (26,560 km) only 3.87 km/s; at GEO (42,164 km) only 3.07 km/s. Slower speed AND a bigger circle, so the lap time balloons: 92 min → 12 h → 24 h.	$v = \sqrt{(\mu/r)}$ and $T = 2\pi\sqrt{(r^3/\mu)}$. Period grows with r^1.5 (Kepler's third law).
2 · Why 35,786 km exactly	solve T = one sidereal day for r	Set T = 86,164 s (23 h 56 m 4 s) in $T = 2\pi\sqrt{(r^3/\mu)}$ with $\mu = 398{,}600$ km³/s² and solve: r = 42,164 km from the center. Subtract Earth's 6,378 km equatorial radius: 35,786 km altitude. Nobody chose it; the equation did.	Any other altitude gives a different period, so the satellite drifts east or west across the sky.
3 · How a satellite reaches GEO	parking orbit → GTO → apogee kick	Rocket drops it in a ~200–300 km parking orbit (7.8 km/s). Burn 1 at perigee, +2.4 km/s, stretches the orbit into a Geostationary Transfer Orbit with apogee at 35,786 km. Coast 5.3 hours to apogee (now moving only 1.6 km/s). Burn 2 there, +1.5 km/s, rounds it off at 3.07 km/s. Total ~3.9 km/s after launch.	It's your perigee/apogee burn rule, applied. A launch from 28.5° latitude (Florida) also needs a plane change to reach i = 0°, which is why GEO launches from near the equator (Kourou, 5°) are cheaper.
4 · How GPS actually finds you	time stamps → distances → spheres	Each GPS satellite broadcasts the exact time and its own orbit. Your phone compares the time it hears with its own clock: the delay × the speed of light = distance to that satellite. One distance puts you on a sphere; three spheres cross at a point. The fourth satellite corrects your phone's cheap clock, which is wrong by far more than a few meters' worth of light-time.	Light travels 30 cm per nanosecond, so GPS needs clocks good to nanoseconds. The satellites carry atomic clocks; your phone doesn't, hence the fourth satellite.
5 · Why MEO for GPS	high enough to see a lot, low enough to be strong and fast	At 20,200 km each satellite sees ~38% of Earth, so 24 satellites in 6 planes guarantee 4+ overhead anywhere. GEO would need fewer but the signal would be weaker and the poles uncovered; LEO would need hundreds.	Orbit choice is a trade-off between coverage per satellite and signal strength/delay.
6 · Holding a GEO slot	station-keeping box ±0.05°	The Moon and Sun tug a GEO satellite north-south about 0.85° per year; Earth's slightly lumpy equator drifts it east-west. Small burns every few weeks, ~50 m/s of Δv per year, keep it in its box. When the fuel's gone, a last burn lifts it ~300 km into a graveyard orbit.	A typical GEO satellite carries ~15 years of station-keeping propellant; that, not the electronics, ends its life.

- The one rule behind the three shelves: gravity weakens with distance, so higher orbits are slower, longer, and hang still only at the single height where one lap equals one day.
- Why launch sites matter: Earth's spin gives a free 0.46 km/s eastward at the equator, zero at the poles. Launching east, near the equator, is the cheapest way up.
- GPS gives position; it does not track you. Your phone only listens. Nothing goes up.

PAGE 6 OF 16 · HOW IT ACTUALLY WORKS · 🛠 ENGINEER LEVEL

Scene 3 · The hula hoop: How a burn changes the six numbers, and where the numbers come from

The Classical Orbital Elements aren't just labels; each engine burn moves specific ones and leaves the rest alone. That's how you steer. And every set of elements you'll ever see online was measured, not assumed.

Step	In short	How it actually works	Official name / why it matters
1 · Prograde burn at perigee	raises apogee → a ↑, e ↑	Speed up at the low point and the far side of the orbit rises. Size (a) grows, the ellipse stretches (e grows). Tilt (i), swivel (Ω) and spin (ω) don't change, because you pushed along the direction you were already moving, inside the plane, at the perigee itself.	This is burn 1 of a Hohmann / GTO.
2 · Prograde burn at apogee	raises perigee → a ↑, e ↓	Speed up at the high point and the near side rises; the ellipse rounds out toward a circle. Same three untouched: i, Ω, ω.	This is burn 2, the circularization or apogee kick.
3 · Retrograde burns	the mirror image	Slow down at apogee and perigee drops (how you de-orbit: a retrograde burn at the high point dips the low point into the atmosphere). Slow down at perigee and apogee drops.	De-orbit burns for returning capsules are retrograde, roughly 100 m/s from the ISS.
4 · Sideways (out-of-plane) burn at a node	changes i	Push perpendicular to the orbital plane while crossing the equator and the tilt changes. Cost: $\Delta v = 2 \cdot v \cdot \sin(\Delta i/2)$. At LEO speed, 1° of tilt costs ~134 m/s; 28.5° to 0° costs ~3.8 km/s, more than reaching orbit. So you change planes at apogee, where v is smallest, or launch into the right plane to begin with.	Why inclination is the expensive element and GEO launches happen near the equator.
5 · Out-of-plane burn away from a node	changes Ω (and i)	The same sideways push done elsewhere on the orbit swivels the plane around Earth's axis instead of (or as well as) tilting it. Rarely done on purpose; Ω is usually left to drift.	Nature changes Ω for free via Earth's bulge, see row 7.
6 · Radial burn	rotates ω	Push straight toward or away from Earth and the line of apsides rotates: perigee moves around the orbit without the size changing much. Used to re-point an elliptical orbit's low point over a target.	Rare in practice; expensive for what it does.
7 · Where the numbers come from	state vector → elements, at an epoch	Radar, laser ranging or onboard GPS measure the satellite's position and velocity (six numbers: x, y, z, vx, vy, vz, called the state vector) at one instant (the epoch). Pure geometry converts that into a, e, i, Ω, ω, ν. The public format is the Two-Line Element set (TLE), which is why every element comes with an epoch date.	Elements are a different way of writing position-and-velocity, chosen because five of them barely change.
8 · Why elements drift	J_2 and friends	Earth isn't a perfect sphere; its equatorial bulge (the J_2 term) makes an inclined orbit's Ω swivel slowly westward (~5° per day for the ISS) and ω rotate within the plane. Engineers use it: pick i ≈ 98° and Ω swivels exactly once a year, giving a sun-synchronous orbit.	Perturbations aren't just noise; the right one is a free engine.

- Steering summary: speed along your motion changes size and shape; pushing sideways changes tilt and swivel; pushing toward Earth rotates the ellipse. Six numbers, three kinds of push.
- Why 'burn at the low point': the same Δv gives a bigger energy change where you're already moving fastest (the Oberth effect). Perigee burns are the most efficient place to spend fuel.
- Every orbit-change plan you'll ever read is a list of which element each burn moves.

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Scene 4 · Five free services: How heat, gas and charge actually behave in vacuum, with numbers

The five free services are the picture. These are the mechanisms underneath, the ones the engineers design to.

Step	In short	How it actually works	Official name / why it matters
1 · How a radiator works	power radiated ∝ T^4 (Stefan–Boltzmann)	Every surface glows heat as infrared light at a rate $\sigma \cdot \varepsilon \cdot T^4$ per square meter. A 1 m² panel at 300 K (27 °C) with emissivity $\varepsilon \approx 0.9$ sheds about 410 W into cold space. Double the absolute temperature and you shed 16× as much. So radiators are kept hot and big, and painted with coatings that have high emissivity (radiate well) but low solar absorptivity (don't soak up sunlight): white paint, silvered Teflon.	The ISS radiators total ~1,700 m² to dump ~75 kW. No fan on Earth comes close for that area.
2 · How heat gets to the radiator	conduction, heat pipes, pumped loops	Metal straps conduct a little. Heat pipes do most of it: a sealed tube with a liquid (ammonia) that boils at the hot end, flows as vapor to the cold end, condenses, and wicks back. No pump, no power, moves kilowatts. Big stations use pumped ammonia loops.	Convection replaced by a plumbing trick that still works in free fall.
3 · Why hot and cold at once	sun side +120 °C, shade side −150 °C	Sunlight delivers 1,361 W/m². A surface facing the Sun with nothing cooling it settles where radiated = absorbed, which for a black surface is ~120 °C; a shaded surface radiating to 3 K space drops toward −150 °C. Spinning the spacecraft, MLI blankets and heaters keep electronics in their −20 to +50 °C window.	Hence 'barbecue roll' on Apollo: rotate slowly so no side bakes.
4 · How outgassing contaminates	hot source → cold sink	Trapped water, solvents and plasticizers diffuse out faster when warm. The vapor travels in straight lines (no air to scatter it) and sticks to whatever cold surface it hits. Optics and detectors are deliberately the coldest things on board, so they're the sink. A monolayer of film on a star tracker lens costs real pointing accuracy.	Standard test (ASTM E595): 125 °C in vacuum for 24 h; pass = TML < 1% and CVCM < 0.1%. Bake-out beforehand gets the loss out on the ground.
5 · How charging builds and arcs	photoelectric vs plasma collection	Sunlit surfaces lose electrons to UV (photoelectric effect) and go positive; shaded insulating surfaces collect plasma electrons and go negative. Two surfaces a few cm apart can end up kilovolts apart. When the gap breaks down, an arc jumps: a current spike that resets computers or punches through solar-array insulation.	Fix: every surface conductive (indium tin oxide over glass, carbon-loaded paints) and bonded to the structure with straps, so nothing can be more than a volt from anything else. GEO, in Earth's magnetotail during storms, is the worst place for it.
6 · Why vibration persists	damping ratio ~0.1–1%	A metal structure's own internal damping is tiny: a strut rung like a bell in vacuum loses only a fraction of a percent of its energy per cycle. Launch shakes everything hard (that's what vibration testing checks). In orbit the worries are reaction-wheel jitter and slewing: a big antenna swung quickly keeps wobbling for minutes, blurring images.	Hence dampers, slow slews, and 'settle time' built into every observation plan.

- Design rule: in vacuum you must create every heat path and every charge path on purpose. Nothing happens for free, and nothing happens by accident either, which is also why thermal and charging behavior is so predictable once modeled.
- The T^4 law is why a small temperature rise makes a radiator much more effective, and why cold things in space are very hard to cool further.
- A cheap sanity check for any spacecraft picture: the gold/silver crinkly stuff is MLI, the big white flat panels are radiators, the dark flat panels are solar arrays.

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Scene 5 · The air bites back: What the air is made of, floor by floor

Think of the atmosphere as a smoothie at the bottom and a layered drink at the top. Below about 100 km everything is stirred together, so the recipe is the same everywhere. Above about 100 km the stirring stops and the gases sort themselves by weight: heavy ones low, light ones high. Sunlight also starts breaking molecules apart up there.

Height	Zone	What's there	Why it matters
0–100 km	the stirred mix (homosphere)	N_2 78% · O_2 21% · Ar 0.9% · CO_2 0.04% · water vapor 0–4% (mostly below 11 km) · trace neon, helium, methane · ozone O_3 is a few parts per million, peaking at 15–35 km	Same recipe at every height here. This is the air you breathe and the air planes fly in.
100–200 km	the breakup zone	N_2 still the main gas, but UV is now splitting O_2 into single O atoms; atomic O climbs fast	Where the atmosphere stops being 'air' as you know it. Meteors have already burned up below this.
200–600 km	atomic oxygen country	atomic O is the number-one gas; N_2 fading; a little N and He	LEO. The ISS, Hubble and Starlink fly here. Atomic O erodes polymers; what's left of the density causes drag.
600–1,500 km	helium shell	He takes over from O (the boundary moves up and down with the Sun's activity)	Too thin to matter much for drag; GPS and GEO are far above this.
1,500 km and up	hydrogen fringe (exosphere / geocorona)	mostly single H atoms, so thin they can escape Earth entirely	A faint hydrogen halo glows in UV out to tens of thousands of km.

- Why the order: nitrogen pairs and oxygen pairs are heavy, single oxygen atoms are half an O_2, helium is lighter still, hydrogen is the lightest thing there is. Once the stirring stops, each gas settles at the height its weight allows, like oil over water.
- Where ozone fits: ozone is just three oxygen atoms stuck together. It forms in the stratosphere when UV hits O_2 and the loose atoms grab onto other O_2. It's rare (a few molecules per million) but it absorbs so well that it blocks most harmful UV. Above the stratosphere there isn't enough air for it to form, so UV comes through raw.
- The ionosphere overlaps all of this, roughly 60–1,000 km: sunlight knocks electrons off atoms, leaving charged particles. It bends and reflects radio, which is why AM radio travels far at night and why GPS signals get slightly delayed. It also grows and shrinks between day and night.
- 'Anomalous O' on the lecture's chart: a small population of extra-hot oxygen atoms the model adds above ~300 km. Still oxygen; not a separate thing to memorize.
- Why a spacecraft cares, in one line each: atomic O eats plastics (LEO); the thin air causes drag that the Sun's activity can double or triple; the ionosphere messes with radio and GPS; helium and hydrogen are too thin to matter.

Scene 6 · The fuel bill: The Hohmann and rocket-equation numbers, worked

The lazy throw isn't just cheapest in spirit; here is what each burn costs and why fast transfers are impossible, with the actual arithmetic.

Step	In short	How it actually works	Official name / why it matters
1 · LEO → GEO Hohmann, worked	$\Delta v_1 = 2.42$, $\Delta v_2 = 1.47$, total 3.89 km/s	Circular speed at 300 km: √(398600/6671) = 7.73 km/s. Transfer ellipse: a = (6671 + 42164)/2 = 24,418 km. Vis-viva at perigee: √(398600·(2/6671 − 1/24418)) = 10.15 km/s, so burn 1 = 10.15 − 7.73 = 2.42. At apogee: 10.15 × 6671/42164 = 1.61 km/s. Circular speed at GEO = 3.07, so burn 2 = 3.07 − 1.61 = 1.47. Time of flight = $\pi\sqrt{(a^3/\mu)}$ = 5.3 h.	Every number here comes from two formulas you already have: vis-viva and circular speed.
2 · Earth → Mars Hohmann, worked	~3.6 km/s to leave, ~2 km/s to capture, 259 days	Around the Sun: Earth moves at 29.8 km/s, the transfer ellipse needs 32.7 at perihelion, so you must leave Earth's neighborhood with $v_\infty \approx 2.9$ km/s. But you start in a gravity well: from LEO the burn is $\sqrt{(v_{esc}^2 + v_\infty^2)} - v_{circ} = \sqrt{(11.0^2 + 2.9^2)} - 7.7 \approx 3.6$ km/s. You arrive at Mars with $v_\infty \approx 2.6$ km/s; braking into a loose orbit costs ~2 km/s more. Travel time: half the transfer ellipse's period ≈ 259 days.	Why the departure burn is only 3.6 and not 11 + 2.9: you're already doing 7.7 km/s in LEO, and escape speed is √2 × that.
3 · Why you can only go every 26 months	phase angle 44°	Mars must be ~44° ahead of Earth at departure so it arrives at the far end of the ellipse when you do. Earth laps the Sun in 365 days, Mars in 687; the relative angle repeats every 1/(1/365 − 1/687) = 780 days.	Miss the window by a month and the Δv climbs steeply; miss by six and it's not possible.
4 · Why 20 days is impossible	the rocket equation	Tsiolkovsky: $\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$. A good hydrogen engine has Isp ≈ 450 s, so $Isp \cdot g_0 \approx 4.4$ km/s. For the 3.6 km/s Hohmann departure, $m_0/m_f = e^{(3.6/4.4)} \approx 2.3$: 56% of the departing mass is propellant, fine. A 20-day trip needs ~60 km/s: $e^{(60/4.4)} \approx 800{,}000$. The ship would need to be 99.9999% propellant. No structure can be that light.	Fuel cost grows exponentially with Δv. That single fact shapes every mission.
5 · How Lambert's problem is actually solved	one free parameter, one equation	Infinitely many ellipses with the Sun at a focus pass through two given points. They differ by their semi-major axis. Lambert proved the time of flight between the points depends only on a, the straight-line distance between them, and the sum of the two radii. So you pick the a whose time of flight matches your deadline (by iteration), and the departure and arrival velocities fall out, hence the Δv.	Mission designers run this for thousands of date pairs and plot the Δv as a contour map, the 'porkchop plot', to pick the launch day.
6 · Why burns happen at perigee	the Oberth effect	Kinetic energy is $\frac{1}{2}mv^2$, so adding a fixed Δv adds more energy when v is already large. A 1 km/s burn at 10 km/s adds 10.5 units of energy per kg; the same burn at 2 km/s adds only 2.5. Deep in a gravity well you're moving fastest, so that's where fuel buys the most.	Why interplanetary departures burn from LEO, and why flybys dip as low as they dare.

- Chain of reasoning to keep: vis-viva gives speeds at each end → differences give Δv → rocket equation turns Δv into propellant fraction → exponential growth makes fast trips impossible.
- GTO numbers (2.42 + 1.47) and Mars numbers (3.6 + 2) are worth memorizing; they anchor every other mission you'll read about.
- Isp (specific impulse, in seconds) is the engine's fuel efficiency: chemical 300–450 s, ion engines 3,000+ s but with tiny thrust. Coming in the propulsion section.

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Scene 7 · The free lunch: The flyby, worked: vectors, turning angle, and where the energy goes

A gravity assist is two frames of reference and one addition. Here is the mechanism with numbers, using a Jupiter flyby.

Step	In short	How it actually works	Official name / why it matters
1 · In Jupiter's frame	speed in = speed out, path bends by δ	Far from Jupiter you approach at some relative speed v_∞. Gravity bends you around a hyperbola and you leave at the same v_∞, rotated by the turning angle δ. Energy relative to Jupiter is conserved: nothing is gained or lost in this frame.	Hyperbolic excess velocity v_∞; turning angle δ. The closer you pass, the bigger δ.
2 · How much it bends	$\sin(\delta/2) = 1 / (1 + r_p \cdot v_\infty^2 / \mu)$	r_p = closest approach distance, μ = Jupiter's gravitational parameter (1.27×10^8 km³/s²). Example: v_∞ = 10 km/s, r_p = 500,000 km → $r_p \cdot v_\infty^2/\mu = 0.39$ → $\sin(\delta/2) = 0.72$ → δ ≈ 92°. Pass closer or slower and δ grows toward 180°.	Big planet, close pass, slow approach = big bend. Mars bends you little; Jupiter bends you a lot.
3 · In the Sun's frame	add Jupiter's own velocity	Jupiter moves around the Sun at 13 km/s. Your Sun-frame velocity is (your velocity relative to Jupiter) + (Jupiter's velocity). Before the flyby the two vectors partly cancel; after the bend they partly add. The magnitude changes even though the relative speed didn't.	Same addition as walking on a moving train: your speed relative to the ground depends on which way you walk.
4 · Worked gain	up to ~2 × v_{planet} in the ideal case	In the extreme 180° bounce, approaching head-on at v_∞ against the planet's motion and leaving along it, the Sun-frame speed goes from ($v_{planet} - v_\infty$) to ($v_{planet} + v_\infty$): a gain of $2 \cdot v_\infty$, capped by geometry at about $2 \cdot v_{planet}$. Real flybys bend 60–150°. Voyager 2 picked up roughly 10 km/s of Sun-frame speed at Jupiter.	Why Jupiter is the slingshot of choice: it's massive (big δ) and fast (13 km/s to borrow from).
5 · Where the energy comes from	momentum conservation	The spacecraft's gain is the planet's loss: $m_{spacecraft} \times \Delta v_{spacecraft} = M_{planet} \times \Delta v_{planet}$. With a 1-ton probe gaining 10 km/s, Jupiter (1.9×10^{27} kg) slows by ~5×10^{-24} m/s. Not free energy; just a ridiculously lopsided trade.	Energy and momentum are both conserved. The planet pays an amount no instrument could ever measure.
6 · Behind vs in front	gain vs brake	Pass behind the planet (on the trailing side of its motion) and the bend points you along its velocity: you gain. Pass in front and the bend points you against it: you lose. MESSENGER used six flybys of Earth, Venus and Mercury to shed speed so it could be captured at Mercury; Parker Solar Probe uses Venus the same way to fall toward the Sun.	Same trick, opposite sign. Geometry decides.
7 · Why it costs years	you must meet the planet where it is	A gravity assist only works if the planet is at the right place when you pass, so trajectories string together windows: Earth→Venus→Earth→Jupiter (Cassini, 1997–2004). Each loop around the inner solar system is a year or more of coasting.	Time is the currency you spend instead of propellant.

- In Jupiter's eyes nothing changed but your direction. In the Sun's eyes your speed changed because Jupiter was moving. Both are true at once; that's what 'frame of reference' means.
- Design levers: approach speed, closest-approach distance, and which side you pass on. Those three set the bend and the sign of the gain.
- After the flyby you're on a new, bigger (or smaller) orbit around the Sun with a new semi-major axis; that's the 'more speed → bigger a' line from Section 3.

PAGE 11 OF 16 · THE 7-MINUTE SCRIPT SKELETON

Minute by minute: key line, napkin drawing, hand-off

Not a script to memorize word for word. Memorize the bold key line for each block and the drawing. Everything else you say in your own words. Total: 420 seconds.

0:00 0:45 **HOOK.** "Did you know the space station is falling right now? It just keeps missing." Ball off a mountain, three throws. **Key line:** an orbit is falling and missing. *Hand-off:* "So how high you go changes everything…"

0:45 1:45 **THREE SHELVES.** Low shelf 90 min, sharp, tiny view. Middle shelf GPS 12 h. Top shelf one day, hangs still, 35,786 km, over the equator. Higher = sees more, slower, blurrier. Swarms for low orbit. **Key line:** one exact height hangs still. *Hand-off:* "But how do you even describe an orbit to someone?"

1:45 2:45 **HULA HOOP.** Six things: size, squish, tilt, swivel, spin, where the ball is. Egg with Earth off-center: fastest at the low point, slowest at the high point, fire the engine at those two spots. **Key line:** six numbers, five freeze the hoop, one moves. *Hand-off:* "Okay, you're up there. Now space tries to kill you."

2:45 3:45 **FIVE FREE SERVICES.** Sunscreen, pressure, damper, fan, ground wire. Vacuum cancels all five: UV snaps plastic, outgassing fogs lenses, vibrations never die, hot spots stay hot, static sparks. Clean rooms. **Key line:** the air does five jobs for you; space does none. *Hand-off:* "And the little bit of air that IS up there is worse."

3:45 4:30 **THE AIR BITES.** Layer cake, zigzag temperature. Top layer 1000 °C but empty: a sparkler. Loose oxygen atoms eat plastic at 7.5 km/s. **Key line:** hot but empty, and the oxygen is hungry. *Hand-off:* "Now let's leave Earth entirely."

4:30 5:45 **THE FUEL BILL.** Ladder: fly past, orbit, probe, land, drive, crash. Don't aim at Mars, aim where it'll be. Lazy half-ellipse, 8.5 months, two burns, every 26 months. Lambert = the "by Tuesday?" calculator. **Key line:** fast is expensive, slow is cheap. *Hand-off:* "Unless you cheat."

5:45 6:30 **THE FREE LUNCH.** Bounce off the front of a moving truck. Planet's view: same speed, bent. Sun's view: faster. Rosetta 10 years, New Horizons 9.5. Pass in front to brake. **Key line:** free speed, but it costs years. *Hand-off:* "So here's the whole thing in one breath."

6:30 7:00 **LANDING.** "Space gives you nothing for free. Getting there is falling and missing. Staying alive is doing the air's five jobs yourself. Going anywhere is time versus fuel, unless you borrow a planet." Then stop and let them ask.

⚡ Running long? Cut Scene 5 (the air) to one sentence: "and the thin air that's left eats plastic." Running short? Add the ISS-radiator question to Scene 4. Never cut the hook or the landing.

Practice in pieces: Section 1 and Section 2

Each section is its own short talk with its own opener, blocks, landing line and cue card. Get each one to a clean run 3 before attempting the full 7 minutes. These match the Teach tab in the app.

🛰️ Section 1 · Intro & mission design · 3¼ min

Friend opens with: “So what even is a satellite, and why are there different kinds?”

0:00 0:45 **HOOK.** An orbit is falling and missing. *Draw:* ball off a mountain, three throws. *Hand-off:* “So once something's up there, what is it FOR?”

0:45 1:30 **FOUR JOBS.** Every satellite looks out, looks down, relays, or carries something alive. *Draw:* four boxes: telescope, camera pointing down, radio dish, a little dog. *Hand-off:* “And where it lives depends on the job…”

1:30 2:30 **THREE SHELVES.** Low is fast and sharp, high sees everything, one exact height hangs still over the equator. *Draw:* Earth, three rings, flashlight cones. *Hand-off:* “Now leave Earth. How do you visit another planet?”

2:30 3:15 **THE LADDER.** Fly past, orbit, probe, land, drive, crash. Each rung costs more fuel and gives more science. *Draw:* a planet with six paths, numbered.

Landing line: “Every spacecraft is a different answer to one question: how close do you dare get, and what will it cost?”

1 · FALL & MISS — ball off a mountain
2 · 4 JOBS — look out / look down / relay / live
3 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
4 · LADDER — flyby → orbiter → probe → lander → rover → penetrator
LAND — “every rung costs more and gives more”

🧯 Section 2 · Space environment · 3 min

Friend opens with: “What's actually so dangerous about space? There's nothing there.”

0:00 1:15 **FIVE FREE SERVICES.** The air does five jobs for you: sunscreen, pressure, damper, fan, ground wire. Space does none. *Draw:* box with five arrows in, each labeled. *Hand-off:* “And the little bit of air that IS up there is worse.”

1:15 2:15 **AIR BITES BACK.** The top of the atmosphere is 1000 degrees but empty, and the loose oxygen atoms eat plastic. *Draw:* layer cake with a zigzag temperature line. *Hand-off:* “So what's actually floating around up there?”

2:15 3:00 **WHAT'S IN THE AIR.** Stirred nitrogen and oxygen near the ground, then single oxygen, then helium, then hydrogen on top. *Draw:* five bands labeled with the gas that rules each.

Landing line: “Space isn't dangerous because something's there. It's dangerous because nothing is, and you have to bring all five of the air's jobs with you.”

1 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
2 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
3 · WHAT'S IN IT — N_2/O_2 → atomic O (200–600) → He → H · ozone · ionosphere
LAND — “space does nothing for you; build all five”

PAGE 13 OF 16 · SECTION TALKS

Practice in pieces: Section 3

The longest of the three, because it carries the two ideas people actually ask about: how do you steer, and how do you get to Mars.

🌀 Section 3 · Orbital mechanics · 4¼ min

Friend opens with: “How do you even steer something in space? Isn't it just floating?”

0:00 1:00 **HULA HOOP.** Six numbers describe any orbit: size, squash, tilt, swivel, spin, where you are. Fastest low, slowest high. *Draw:* egg with Earth off-center, then a tilted hoop over a flat disk. *Hand-off:* “Pick the right hoop and you get the three shelves…”

1:00 1:45 **EARTH ORBITS.** Higher sees more but moves slower; 35,786 km hangs still; low orbits need swarms. *Draw:* three rings with periods written on them. *Hand-off:* “Now leave Earth entirely.”

1:45 3:00 **THE FUEL BILL.** You aim at where Mars will be. Fast is expensive, the lazy half-ellipse is cheap, every 26 months. *Draw:* two circles, dashed half-oval, a burn at each end. *Hand-off:* “Unless you cheat.”

3:00 3:45 **THE FREE LUNCH.** Bounce off a moving planet and you leave faster than you came. Free, but it costs years. *Draw:* planet with a thick arrow, dashed path bending around it. *Hand-off:* “So here's the whole thing in one breath.”

3:45 4:15 **LANDING.** Going anywhere is a trade between time and fuel, unless you can borrow a planet. Then stop.

🃏 Section 3 cue card

1 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
3 · FUEL BILL — lazy throw 8.5 mo · every 26 mo · Lambert = by-Tuesday calculator · fast = $$$
4 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
LAND — “time vs fuel, unless you borrow a planet”

🧭 How the three talks become the full one

- The full 7-minute talk (page 4) is Section 1's hook, Section 3's hula hoop and shelves, Section 2's two blocks, then Section 3's fuel bill and free lunch, with one landing line for all of it.
- Section 1's "four jobs" and "the ladder" blocks are cut from the full talk to make time. Bring them back if a friend asks "what kinds are there?"
- Order of work: Section 1 to a clean run 3 → Section 2 → Section 3 → full talk.

⚡ On the call: keep this page and page 5 open. Read the key line, say it in your own words, draw the napkin picture, use the hand-off to move on.

PAGE 14 OF 16 · OUR QUIZ · 24 QUESTIONS

Do it cold. Then check page 6.

Three kinds of question, mixed: facts (F), explain-why (W), and "a friend asks" (A). The W and A ones are the ones that matter for the 7 minutes. Score yourself honestly.

1. (W) In one sentence, why doesn't an orbiting satellite fall down?
2. (F) LEO, MEO, GEO: altitude band and lap time for each.
3. (W) Why is GEO at 35,786 km and not a round number?
4. (A) "Why can't a TV satellite hang over New York?"
5. (F) Geosynchronous vs geostationary: which one hangs over one spot?
6. (W) Why does GPS need 24+ satellites when a TV satellite needs 1?
7. (F) Name the six orbital parameters and what each one answers in one word.
8. (W) Where is a satellite fastest, and which two spots do you fire the engine at? What does each burn do?
9. (F) a = ? (formula, and measured from where?)
10. (F) e = 0 means? e near 1? e over 1?
11. (A) "What does the air do for us that space doesn't?" Name all five.
12. (W) Why is a vibration a bigger problem on a spacecraft than on a car?
13. (F) What is outgassing, and why keep outgassing materials away from sensors?
14. (A) "The thermosphere is 1000 °C. Why doesn't the ISS cook?"
15. (W) Why does the stratosphere get warmer as you go up?
16. (F) What is atomic oxygen, where does it come from, and where is it the main gas?
17. (W) Who's moving at 7.5 km/s in an atomic-oxygen hit, and why does it matter?
18. (F) Rank the six "visit a planet" spacecraft types from least to most committed.
19. (W) Why can't we fly to Mars in 20 days even though an orbit for it exists?
20. (F) Describe a Hohmann transfer: shape, angle, number of burns, time to Mars.
21. (F) Lambert's problem: what goes in, what comes out?
22. (A) "Isn't a gravity assist free energy?"
23. (W) In the planet's frame, what changes during a flyby? In the Sun's frame?
24. (A) "Why did Rosetta take 10 years?" and "Why couldn't New Horizons stop at Pluto?"

Scoring: 20+ solid → you're ready to rehearse. 15–19 → rehearse, but redo the missed ones first. Under 15 → one more review pass on the weak scene before you rehearse. Every miss becomes a question-bank card.

PAGE 15 OF 16 · ANSWERS

Answers, and which scene to review if you missed it

1. It is falling; it's just moving sideways so fast the ground curves away as fast as it falls. *(Scene 1)*
2. LEO 200–2,000 km, 1.5–2 h. MEO 2,000–35,786 km, 2–24 h (GPS 12 h). GEO 35,786 km, 24 h. *(Scene 2)*
3. It's the one altitude where one lap takes exactly one Earth rotation. Calculated, not chosen. *(Scene 2)*
4. Every orbit is centered on Earth's center; only the equator moves in a circle an orbit can match. Over New York it would drift north and south daily. *(Scene 2)*
5. Geostationary (one-day lap AND over the equator). Geosynchronous = one-day lap at any tilt. *(Scene 2)*
6. GPS is low (MEO), so each satellite sees a limited patch and moves; you need many so 4 are always overhead. A GEO satellite sees a third of Earth and never moves. *(Scene 2)*
7. a size, e shape, i tilt, Ω swivel, ω spin, ν position. *(Scene 3)*
8. Fastest at perigee. Burn at perigee → apogee rises; burn at apogee → perigee rises. *(Scene 3)*
9. a = (r_perigee + r_apogee) ÷ 2, both measured from Earth's center. *(Scene 3)*
10. 0 = circle. Near 1 = long cigar. Over 1 = open path that never closes (a flyby). *(Scene 3)*
11. Sunscreen (ozone vs UV), pressure, damper, fan (convection), ground wire (discharge). *(Scene 4)*
12. On Earth the air absorbs the vibration and it dies out. In vacuum nothing does; the spacecraft must damp it itself. *(Scene 4)*
13. Trapped gas leaking out of materials in vacuum. It condenses on cold surfaces, usually lenses and sensors, and fogs them. *(Scene 4)*
14. Almost no molecules up there. Temperature is molecule speed; heat needs numbers. A sparkler is 1000 °C and won't warm a room. *(Scene 5)*
15. The ozone layer lives there and absorbs UV; absorbed light becomes heat. *(Scene 5)*
16. Single O atoms made when UV splits O_2. Main gas from ~200–600 km, exactly where LEO satellites fly. *(Scene 5)*
17. The spacecraft. The atoms are nearly still; the spacecraft slams into them, so each hit carries enough energy to break a chemical bond. *(Scene 5)*
18. Flyby → orbiter → atmospheric probe → lander → rover → penetrator. Each step costs more fuel/risk, returns more science. *(Scene 6)*
19. The Δv needed is absurd; no rocket could carry the fuel. Fast is expensive. *(Scene 6)*
20. Half an ellipse touching both orbits; depart and arrive 180° apart; two burns; ~8.5 months to Mars. *(Scene 6)*
21. In: two positions + a travel time. Out: the one orbit connecting them, and therefore the Δv. *(Scene 6)*
22. No. The energy comes from the planet, which slows down by an unmeasurably tiny amount. *(Scene 7)*
23. Planet's frame: speed unchanged, direction bends. Sun's frame: speed changes because you add the planet's own motion. *(Scene 7)*
24. Rosetta used four flybys to build speed for free instead of carrying fuel; time was the price. New Horizons left Earth so fast (plus a Jupiter boost) that no fuel could brake it into Pluto orbit. *(Scene 7)*

PAGE 16 OF 16 · REHEARSAL PLAN + CUE CARD

Three runs, then do it for real

🎬 Three rehearsal runs

- **Run 1, with page 4 open.** Talk through all seven blocks, reading the key lines. Don't time it. Goal: feel the order.
- **Run 2, with only the cue card** (right). Phone timer running. Expect to run long. Note which block ate the time.
- **Run 3, nothing in front of you, phone recording.** Timer running. Then listen back and mark: every "um," every sentence that ended in a word you didn't explain, every place you reached for a fact and it wasn't there. Those three things are your fix list.
- **Then send me the recording or a transcript.** I'll grade it the way the friend would hear it, not the way a textbook would.

🧯 When you get stuck mid-sentence

- Pivot to the picture: "The way I think about it is…" then the analogy for that block (ball off a mountain, three shelves, hula hoop, five jobs, sparkler, lazy throw, moving truck).
- Forgot a number? Say the idea instead. "About a day" beats a frozen pause over "23 hours 56 minutes."
- Lost your place? Go to the hand-off line of the block you're in. They all start with "So…" or "But…"

📏 What "holding your own" sounds like

- You use the analogy first and the real word second: "it's like a hula hoop around a basketball, the six orbital parameters…"
- You can answer one "why" past each fact. Not two. One is enough for a friend.
- You say "I'm not sure, but I think…" for the things you're not sure about. That's what people who actually know things sound like.

🃏 Cue card (copy onto one index card)

1 · FALL & MISS — ball off a mountain
2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
3 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
4 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
5 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
6 · FUEL BILL — ladder · lazy throw 8.5 mo · every 26 mo · fast = $$$
7 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
LAND — "space gives you nothing for free"

Eight lines. If you can talk for 7 minutes from these eight lines, you own it.

⚡ **Order of work today: read pages 2–3 once → do the quiz cold (page 5) → check (page 6) → fix the misses → Run 1 → Run 2 → Run 3 → send me the recording.**

SPACE X · COPY INTO NOTEBOOK

Concept cards for the word bank · S1–S3 (v1)

32 core terms · two cards per notebook page · **16 pages (8 spreads), numbered C1–C16** · 309 handwritten lines. Every page that follows is one notebook spread at full size: copy it line for line.

The five symbols (write this key at the top of C1's facing page or on your abbreviation page)

Write	Means	What that line does for you
[NAME] = …	official name = also called	Box the name in pencil. Say it out loud before you write the rest.
=	meaning	The dictionary line. What the word points at.
WHY	why it exists	The problem it solves. This is the line that makes the word stick.
#	real numbers	One example you can check. No number = you can't test yourself.
≠	don't confuse with	The look-alike, and in brackets how to tell them apart.
→	connects to	Formulas and terms it plugs into. Your next sentence in a conversation.

Pencil only. Nothing here needs color. The tiny gray numbers 1–22 down the left edge are your ruled lines; don't copy them. The dashed line is one skipped line between the two cards.

How to copy so it actually sticks

1. **One spread per sitting** (4 cards, about 40 lines, 20–25 minutes). Eight sittings. Don't do all 16 pages in one go; by page 5 you would be copying shapes, not ideas.
2. **Order:** C3–C7 first (scene 3, the orbital elements: your biggest gap and the heart of the 7-minute talk). Then C11–C15 (fuel bill + flybys). Then C1–C2, C8–C10, C16.
3. **Per card:** read the whole card. Say the WHY line out loud. Then write the card. Writing first and reading later is the wrong way round.
4. **Close the loop:** when a spread is done, cover everything except the four boxed names and say each WHY from memory. Miss one twice → it becomes a question-bank card.
5. **Link the two banks:** in the one-line word bank, write "→ C5" (the card's page) at the end of that term's line. Index of all 32 is on the last page of this PDF.

Where the 16 pages go, and what it costs the rest of the notebook, is on the last page. Short version: they start 44 pages from the end, right in front of the talk cards. Nothing you have already written moves.

SPREAD 1 OF 8 **Left page C1 · right page C2**

19 lines · 20 lines of 22

WORD BANK · CONCEPT CARDS · scene 1 — C1

CENTRIPETAL FORCE = center-seeking force — scene 1

= inward pull that bends a straight path into a circle

WHY nothing circles on its own. For a sat the pull IS gravity, so no engine is needed

\# ISS: 8.7 m/s² inward at 7.67 km/s = what a 6,771 km circle needs

≠ centrifugal 'force' (your inertia)

→ v = √(μ/r): gravity = needed pull

STATION-KEEPING = orbit maintenance — scene 1

= small burns that cancel drift

WHY drag, Moon, Sun, lumpy Earth nudge every orbit. Fixing it burns fuel = what ends a sat's life

\# GEO ~50 m/s per yr · ISS reboost a few m/s monthly

≠ 'holding it up' (nothing does)

→ reboost · perturbation · graveyard

WORD BANK · CONCEPT CARDS · scene 2 — C2

GEOSTATIONARY ORBIT (GEO) = Clarke orbit — scene 2

= 1-day lap, circular, over equator: stays put in the sky

WHY a fixed dish can point at it forever. Needs T = 1 Earth spin: true at one height only

\# alt 35,786 km · r 42,164 km · 3.07 km/s · 23 h 56 m

≠ geosynchronous (tilted: figure-8)

→ T = 2π√(a³/μ) · GTO · station-keeping

CONSTELLATION — scene 2

= many sats in planned orbits working as one system

WHY one sat = one place, and low sats move fast. Need enough planes so one is always overhead

\# GPS 24+ in 6 planes · Iridium 66 · Starlink 1000s

≠ sats that just share an orbit

→ footprint · revisit time · planes

SPREAD 2 OF 8 **Left page C3 · right page C4**

19 lines · 15 lines of 22

WORD BANK · CONCEPT CARDS · scene 3 **C3**

CLASSICAL ORBITAL ELEMENTS = Keplerian scene 3

elements (COEs)

= a e i Ω ω ν: six numbers that fully describe an orbit

WHY position + velocity change every second. Describe the PATH instead and five barely change

\# ISS: a 6,790 km · e 0.0005 · i 51.6° · ν laps in 92 min

≠ lat/long (a ground point, not a path)

→ vis-viva and T use a · burn rules

SEMI-MAJOR AXIS (a) = size of the orbit scene 3

= half the long width of the ellipse

WHY one number for 'how big'. Lap time depends ONLY on a

\# ISS 6,790 km → 92 min · GPS 26,560 → 12 h · GEO 42,164 → 24 h

≠ altitude (height above ground)

→ $T = 2\pi\sqrt{(a^3/\mu)}$ · $a = (r_1+r_2)/2$

WORD BANK · CONCEPT CARDS · scene 3 **C4**

ECCENTRICITY (e) = shape · squash scene 3

= how far the ellipse is from a circle

WHY an ellipse needs one 'how squashed' number: 0 = circle, 1 = barely open

\# ISS 0.0005 · GTO 0.73 · escape > 1

≠ inclination (e squash, i tilt)

→ $r_p = a(1-e)$ · $r_a = a(1+e)$

INCLINATION (i) = tilt scene 3

= angle between orbit plane and equator

WHY sets what you fly over: never past latitude = i. Costliest element to change later

\# GEO 0° · ISS 51.6° · sun-sync ~98°

≠ Ω (i = how much, Ω = which way)

→ $\Delta v = 2v\cdot\sin(\Delta i/2)$ · launch latitude

SPREAD 3 OF 8 **Left page C5 · right page C6**

19 lines · 20 lines of 22

WORD BANK · CONCEPT CARDS · scene 3 **C5**

RAAN (Ω) = right ascension of the scene 3

ascending node

= angle from ♈ to where the orbit

crosses the equator going north

WHY a tilted plane can face any way round

the axis. Ω pins which way

GPS: 6 planes with Ω 60° apart

≠ ω (Ω in equator, ω in orbit plane)

→ J_2 drift → sun-synchronous

ARGUMENT OF PERIAPSIS (ω) = arg. of scene 3

perigee

= angle in the orbit plane from the

node to perigee

WHY says where the near end points: over

the target, or away so apogee lingers

Molniya ω ≈ 270° → apogee hangs over

the north

≠ perigee distance (ω is an angle)

→ radial burn rotates ω · J_2 drift

WORD BANK · CONCEPT CARDS · scene 3 **C6**

TRUE ANOMALY (ν) = θ scene 3

= angle from perigee to the sat, right

now

WHY five elements freeze the orbit. ν

says WHERE on it = the clock

0° perigee · 180° apogee · ISS 360°

per 92 min

≠ mean anomaly M (even with time)

→ Kepler 2: faster near perigee

PERIAPSIS / APOAPSIS = perigee / apogee scene 3

at Earth

= nearest + fastest point / farthest +

slowest point

WHY speed is purely sideways there, so

burns go there. The two distances

define a and e

GTO: 300 km at 10.15 km/s · 35,786 km

at 1.61 km/s

≠ new orbits (-gee Earth, -helion Sun)

→ prograde @ peri raises apo

SPREAD 4 OF 8 **Left page C7 · right page C8**

19 lines · 20 lines of 22

WORD BANK · CONCEPT CARDS · scene 3 — C7

VIS-VIVA EQUATION = orbital energy eqn — scene 3

= $v^2 = \mu(2/r - 1/a)$: speed anywhere on any orbit

WHY orbit energy is fixed, so speed trades with distance. One line replaces every special case

GTO: r 6,671, a 24,418 → 10.15 km/s · apogee 1.61

≠ $v = \sqrt{(\mu/r)}$ (vis-viva with a = r)

→ Δv = speed after − speed before

GRAVITATIONAL PARAMETER (μ) = GM — scene 3

= G × the body's mass, as one constant

WHY every orbit formula has G×M in it, and GM is known far better than G or M alone

Earth 398,600 km^3/s^2 · Sun 1.327×10^{11} · Mars 42,828

≠ $g = 9.8\ m/s^2$ ($g = \mu/r^2$ at surface)

→ vis-viva · period · escape speed

WORD BANK · CONCEPT CARDS · scene 4 — C8

OUTGASSING = offgassing — scene 4

= gas leaking out of solid materials in vacuum

WHY materials made in air hold trapped gas. Vacuum frees it and it lands on the coldest surface = your lens

TML < 1% · CVCM < 0.1% · bake-out days at 100 °C+

≠ a leak from a tank or cabin

→ contamination · clean room · MLI

SPACECRAFT CHARGING = static build-up — scene 4

= surfaces collect charge; no air to bleed it off

WHY UV knocks e^- off the sunlit side (+). Plasma e^- land on the shaded side (−). The gap arcs

kV differences measured at GEO in storms

≠ radiation damage (particles in chips)

→ ESD · conductive coating · bonding

SPREAD 5 OF 8 **Left page C9 · right page C10**

21 lines · 19 lines of 22

WORD BANK · CONCEPT CARDS · scene 4 + scene 5 — C9

STEFAN–BOLTZMANN LAW = P = σεT⁴ — scene 4

= power radiated ∝ temperature to the 4th power

WHY radiation is the ONLY exit for heat in vacuum. T⁴: hot radiators work, cold ones barely

\# 1 m², ε 0.9: 300 K ≈ 410 W · 350 K ≈ 760 W · 200 K ≈ 80 W

≠ conduction (needs a temp difference)

→ radiator size · emissivity · hot spot

ATOMIC OXYGEN (AO) = O · monatomic — scene 5

oxygen

= single O atoms: the main gas at 200–600 km

WHY no ozone up there, so UV splits O_2. Too thin to re-pair. Sat rams the atoms at 7.5 km/s

\# eats bare Kapton tens of µm per yr · LDEF: shredded in 5.8 yr

≠ ozone O_3 (protects) · O_2 (breathe)

→ ram / wake · coatings · drag

WORD BANK · CONCEPT CARDS · scene 5 — C10

IONOSPHERE — scene 5

= charged layer of the upper air, ~60–1,000 km

WHY UV + X-rays knock e^- off atoms. Charged gas bends and delays radio

\# GPS error up to tens of m if uncorrected

≠ ozone layer · magnetosphere

→ space weather · radio · GPS

HOMOSPHERE / HETEROSPHERE = mixed / — scene 5

sorted

= below ~100 km a stirred mix; above, sorted by weight

WHY weather stirs low air into one recipe. Above the turbopause each gas settles by weight

\# N_2 + O_2 → O (200–600 km) → He → H

≠ temperature layers (tropo-, strato-)

→ turbopause · AO · geocorona

WORD BANK · CONCEPT CARDS · scene 6 C11

DELTA-V (Δv) = the fuel bill scene 6

= total speed change a mission needs

WHY reach is limited by speed changes, not distance. Each costs propellant → budget in km/s

surface→LEO ~9.4 · LEO→GEO 3.9 · LEO→Mars orbit ~5.6

≠ speed (Δv = sum of changes)

→ rocket equation · Hohmann = min Δv

HOHMANN TRANSFER = min-energy transfer scene 6

= half-ellipse touching both orbits; 2 burns

WHY both burns are tangent, so nothing is wasted sideways. Cheapest, NOT fastest

LEO→GEO 2.42 + 1.47 km/s, 5.3 h · Mars ~3.6 + ~2, 259 d

≠ the fastest route (it's the slowest)

→ vis-viva · TOF = $\pi\sqrt{(a^3/\mu)}$ · synodic

WORD BANK · CONCEPT CARDS · scene 6 C12

LAMBERT'S PROBLEM scene 6

= 2 places + the time between → the orbit that fits

WHY planners start from 'be at Mars on this date'. Lambert turns that into an orbit + fuel bill

Earth Nov 2026 → Mars Aug 2027 → Δv ≈ 3.7 km/s

≠ Hohmann (one special case: 180°)

→ porkchop plot = 1000s of solutions

SYNODIC PERIOD scene 6

= time for two planets to line up the same way again

WHY planets lap the Sun at different rates. This is why launch windows exist

Earth–Mars 780 d (26 months) · Venus 584 d

≠ orbital period (Mars yr = 687 d)

→ phase angle 44° · launch window

SPREAD 7 OF 8 **Left page C13 · right page C14**

19 lines · 20 lines of 22

WORD BANK · CONCEPT CARDS · scene 6 — C13

ROCKET EQUATION = Tsiolkovsky — scene 6

= Δv = Isp·g₀·ln(m₀/m_f): Δv ↔ propellant fraction

WHY a rocket carries all it will throw, so fuel grows EXPONENTIALLY with Δv

\# Isp 450 s: 3.6 km/s → 56% fuel · 9 → 87% · 60 → 99.9999%

≠ '2× fuel = 2× speed' (ratio SQUARES)

→ Isp · staging · no 20-day Mars

SPECIFIC IMPULSE (Isp) = engine mpg — scene 6

= push per kg of propellant, in seconds; × 9.81 = exhaust speed

WHY engines differ in what they get from each kg. Isp is the 'miles per gallon'

\# solid ~250 s · kerosene ~320 · hydrogen 450 · ion 3,000+

≠ thrust (Isp = mpg, thrust = hp)

→ rocket equation · Section 4

WORD BANK · CONCEPT CARDS · scene 6 + scene 7 — C14

SPHERE OF INFLUENCE (SOI) — scene 6

= region where a planet's gravity beats the Sun's

WHY lets you switch 'planet rules' ↔ 'Sun rules', so each leg is a simple 2-body problem

\# Earth ≈ 925,000 km · Jupiter ≈ 48 million km

≠ a wall (gravity doesn't stop there)

→ patched conics · v∞ at the edge

GRAVITY ASSIST = flyby · slingshot — scene 7

= swing past a moving planet to change speed + direction

WHY planet frame: same speed, bent path. Sun frame: you take some of the planet's motion

\# Voyager 2 +10 km/s at Jupiter · New Horizons saved 3 yr

≠ free energy (planet pays 10^{-24} m/s)

→ $\sin(\delta/2) = 1/(1 + r_p \cdot v_\infty^2/\mu)$

SPREAD 8 OF 8 **Left page C15 · right page C16**

21 lines · 19 lines of 22

WORD BANK · CONCEPT CARDS · scene 7 — C15

HYPERBOLIC EXCESS VELOCITY = v∞ — scene 7

= speed left over once you are far from the planet

WHY near a planet speed depends on depth in the well. Far away it settles to one number

\# at Mars (Hohmann) ≈ 2.6 km/s · leaving Earth ≈ 2.9

≠ escape velocity (gives v∞ = 0)

→ C3 = v∞² · capture burn · δ

CHARACTERISTIC ENERGY (C3) = launch energy — scene 7

= v∞² at Earth departure: how hard you were thrown

WHY the one number a launch provider needs. Rocket performance charts are drawn against it

\# Mars ~10–15 km²/s² · Jupiter ~80 · New Horizons ~158

≠ Δv (C3 = rocket's job)

→ v∞ = √C3 · departure burn

WORD BANK · CONCEPT CARDS · scene 7 — C16

REACTION WHEEL = momentum wheel · CMG — scene 7

= spinning wheel inside; the s/c turns the other way

WHY nothing to push on in vacuum, so push against yourself. No fuel, very precise

\# Hubble: 4 wheels, 0.007 arcsec · ISS: 4 CMGs

≠ thrusters (expel gas, use fuel)

→ momentum dumping · disturbances

DISTURBANCE TORQUE — scene 7

= tiny unwanted twist that never stops

WHY gravity gradient, sunlight, thin air, magnetism each twist a little. Nothing damps it

\# small sat ~10^{-5} N·m · ISS several N·m. Tiny but nonstop

≠ perturbation (path, not pointing)

→ reaction wheels · momentum dumping

NOTEBOOK MAP · v3

Where the concept cards live, and what it costs

Count from the LAST page so the rule works whatever your notebook's real page count is. The page numbers in brackets assume ~95 pages; check your last page number once and adjust.

p1–5 index · equations · abbrev key	p6 → p51 **LECTURE SPREADS + section maps** S1–S3 already used ≈ p6–24 · room left for Sections 4, 5, 6	**16 pages [p52–67]** **CONCEPT CARDS C1–C16** NEW · tab at 44 from end	2 talk cards	12 pages **ONE-LINE WORD BANK** unchanged	last 14 **QUESTION BANK** unchanged

The page count, honestly

Block	Pages	Why
Concept cards, S1–S3	**16**	32 terms × 9–11 lines = 309 lines. Two cards fit a 22-line page; three don't.
Concept cards, S4–S7 (estimate)	**~14**	About 25–30 more core terms at the same rate (propulsion, comms, attitude, power).
One-line word bank, whole course	12 (have) · ~15 (need)	S1–S3 is 161 lines ≈ 7½ pages; four more sections won't fit in the 4½ left.
Word bank total by end of course	**~45**	The old map gave it 12.

What that means: one ~95-page notebook cannot hold 25 lectures, a 45-page word bank and a 14-page question bank. That is ~120 pages of content. The old 2-page buffer is gone and the lecture block now ends at p51, which is the end of **Section 6**.

The plan

1. **Now:** count 44 pages back from the last page, put a sticky tab there, label the next 16 pages C1–C16 in pencil (top outer corner). Copy the eight spreads into them.
2. **Nothing else moves.** Talk cards, the 12 one-line pages and the question bank stay exactly where they are. Old tabs at 30 / 28 / 14 from the end stay.
3. **Before Section 7:** get a second small notebook. It takes Section 7's lecture spreads, the S4–S7 concept cards (C17 onward) and the one-line overflow. Number its concept pages on from C17 so the "→ C__" pointers keep working.
4. **If you would rather keep every lecture in this one book:** buy the second notebook now and put C1–C16 in it instead. Same cards, same numbering. Then this notebook's map stays exactly as v2.

My recommendation is 1–3: you are reviewing Sections 1–3 this week and need the cards next to those notes today.

Index: which page each card is on (add "→ C#" to that term's line in the one-line word bank)

ARGUMENT OF PERIAPSIS (ω) **C5**
ATOMIC OXYGEN (AO) **C9**
CENTRIPETAL FORCE **C1**
CHARACTERISTIC ENERGY (C3) **C15**
CLASSICAL ORBITAL ELEMENTS **C3**
CONSTELLATION **C2**
DELTA-V (Δv) **C11**
DISTURBANCE TORQUE **C16**
ECCENTRICITY (e) **C4**
GEOSTATIONARY ORBIT (GEO) **C2**
GRAVITATIONAL PARAMETER (μ) **C7**
GRAVITY ASSIST **C14**
HOHMANN TRANSFER **C11**
HOMOSPHERE / HETEROSPHERE **C10**
HYPERBOLIC EXCESS VELOCITY **C15**
INCLINATION (i) **C4**
IONOSPHERE **C10**
LAMBERT'S PROBLEM **C12**
OUTGASSING **C8**
PERIAPSIS / APOAPSIS **C6**
RAAN (Ω) **C5**
REACTION WHEEL **C16**
ROCKET EQUATION **C13**
SEMI-MAJOR AXIS (a) **C3**
SPACECRAFT CHARGING **C8**
SPECIFIC IMPULSE (Isp) **C13**
SPHERE OF INFLUENCE (SOI) **C14**
STATION-KEEPING **C1**
STEFAN–BOLTZMANN LAW **C9**
SYNODIC PERIOD **C12**
TRUE ANOMALY (ν) **C6**
VIS-VIVA EQUATION **C7**

SPACE X · WORD BANK

Word bank, concept level — S1–S3 (v4)

32 core terms. Each one answers five questions, in this order, every time.

How to read an entry (and how to write it in your notebook)

- **Name** in bold first, then **Also called**. Say the official name out loud before anything else; that's the rule.
- **Meaning** = the one-line definition you'd put in a dictionary. This is what you've been copying so far. It is not enough on its own.
- **Concept** = *why the thing exists*: what problem it solves, what would go wrong without it. This is the part that makes the word stick, because now it has a job.
- **Numbers** = one real example with a real value. A term with no number attached is a term you can't check yourself.
- **Not this** = the thing people mix it up with, and the one-line test that separates them.
- **Connects** = which formulas or other terms it plugs into. This is how you walk from one idea to the next in a conversation instead of stalling.

In the notebook: the back word-bank pages only have room for Name · Also called · Meaning (one line). Put Concept and Numbers in the margin of the lecture page the term came from, or say them out loud when you review. Don't copy all five lines for all 32 terms; the app has them.

Order of the entries: by scene, in the order the app shows them (1 Falling & missing → 7 The free lunch). The number badge is the scene.

Test yourself: cover everything but the name. Say the concept line. If you can't say *why it exists*, you don't own the word yet, even if you can define it.

WORD BANK · 1/8

Concept-level word bank

Centripetal force

1

Also called: center-seeking force

MEANING The inward force that bends straight-line motion into a curve. For a satellite, gravity is the centripetal force; for a ball on a string, the string's tension. Without it, motion continues in a straight line (Newton's first law).

CONCEPT Nothing moves in a circle on its own; something must pull it inward constantly, or it goes straight. For a satellite that something is gravity. Seeing gravity as the string, not the enemy, is the key to understanding why orbits need no engine.

NUMBERS ISS: gravity provides 8.7 m/s² inward; at 7.67 km/s that's exactly the v^2/r a 6,771 km circle requires.

NOT THIS 'Centrifugal force,' the outward feeling, which isn't a real force; it's your inertia resisting the inward pull.

CONNECTS Circular velocity formula comes from setting gravity equal to the required centripetal pull.

Station-keeping

1

Also called: orbit maintenance

MEANING Small periodic burns that cancel perturbations (drag in LEO, the Moon's and Sun's gravity in GEO) so the orbit stays where the mission needs it. Sets a GEO satellite's useful life. Not what 'holds the satellite up'; nothing does.

CONCEPT Orbits aren't perfectly stable: drag, the Moon, the Sun and Earth's lumpy shape all nudge a satellite off its intended path. Station-keeping is the routine of small burns that cancel those nudges. It's maintenance, and it's what actually ends most satellites' lives, because it uses fuel.

NUMBERS GEO: ~50 m/s per year, burns every few weeks. ISS: a reboost of a few m/s roughly monthly.

NOT THIS 'Holding the satellite up.' Nothing holds it up; station-keeping only corrects drift from where it would be anyway.

CONNECTS Reboost, perturbations, graveyard orbit when fuel runs out.

Geostationary orbit (GEO)

2

Also called: Clarke orbit

MEANING The special geosynchronous orbit that is also circular and equatorial (i = 0), at 35,786 km altitude (42,164 km radius), so the satellite appears fixed. TV and weather satellites.

CONCEPT A satellite that doesn't move in the sky lets you point a fixed dish at it forever. That needs a lap time of exactly one Earth rotation, which happens at only one altitude, and it needs the orbit to lie over the equator, because the equator is the only ground circle centered on Earth's center.

NUMBERS Altitude 35,786 km, radius 42,164 km, speed 3.07 km/s, period 23 h 56 m. Three satellites cover almost all of Earth except the poles.

NOT THIS Geosynchronous: same period but tilted, so it traces a figure-8 instead of staying put.

CONNECTS Kepler's third law gives the altitude; station-keeping holds the slot; GTO gets you there.

Constellation

2

MEANING A set of satellites in coordinated orbits that function as one system (GPS, Iridium, Starlink). Arranged in several orbital planes so coverage is continuous.

CONCEPT One satellite can only be in one place, and low ones move fast. For continuous coverage of the whole planet you need enough satellites, spread over enough orbital planes, that one is always overhead wherever you are. That coordinated swarm is a constellation.

NUMBERS GPS 24+ in 6 planes. Iridium 66 in 6 polar planes. Starlink thousands in dozens of planes.

NOT THIS A group of satellites that merely share an orbit. A constellation is designed as one system with coverage guarantees.

CONNECTS Footprint per satellite, revisit time, orbital planes (Walker).

WORD BANK · 2/8

Concept-level word bank

Classical Orbital Elements (COEs)

3

Also called: Keplerian elements · orbital parameters · the six elements

MEANING The six numbers that completely describe an orbit: a, e, i, Ω, ω, ν. Five fix the shape and orientation; the sixth says where the body is. The hula hoop's six settings.

CONCEPT Position and velocity (six numbers) describe where a satellite is right now, but they change every second. Rewrite the same information as six numbers that describe the orbit itself, and five of them barely change for years. That's the whole point: a stable description of a path instead of a snapshot of a point.

NUMBERS ISS: a ≈ 6,790 km, e ≈ 0.0005 (nearly circular), i = 51.6°, Ω and ω drift slowly, ν changes 360° every 92 minutes.

NOT THIS Not the same as latitude/longitude, which describe a point on the ground. Elements describe the whole path in space.

CONNECTS Vis-viva uses a; period uses a; the burn rules say which element each burn moves.

Semi-major axis (a)

3

Also called: half the major axis · 'size' of the orbit

MEANING Half the longest width of the ellipse; the average of perigee and apogee distances from the center of the body. Sets the orbit's size and, through Kepler's third law, its period.

CONCEPT One number that captures how big the orbit is. Because period depends only on a (Kepler's third law), a tells you how long a lap takes without knowing anything else.

NUMBERS ISS a = 6,790 km → 92 min. GPS a = 26,560 km → 12 h. GEO a = 42,164 km → 24 h.

NOT THIS Altitude. a is measured from Earth's center and is the average of the near and far distances; altitude is height above the ground at one point.

CONNECTS Kepler's third law, vis-viva, Hohmann transfer (a of the transfer ellipse = $(r_1 + r_2)/2$).

Eccentricity (e)

3

Also called: 'shape' of the orbit

MEANING How far the ellipse departs from a circle. 0 = circle, 0 < e < 1 = ellipse, e = 1 = parabola (escape), e > 1 = hyperbola (flyby).

CONCEPT Orbits are ellipses, and an ellipse needs a 'how squashed' number. e is that number, built so 0 is a circle and 1 is the edge of not closing at all. Everything you need to know about shape is in one dimensionless number.

NUMBERS ISS 0.0005 (circle). Molniya 0.74 (cigar). GTO 0.73. New Horizons leaving Earth: > 1 (hyperbola, never comes back).

NOT THIS Inclination. e is squash, i is tilt. An orbit can be very squashed and flat, or circular and steeply tilted.

CONNECTS Perigee/apogee distances: r_p = a(1−e), r_a = a(1+e).

Inclination (i)

3

Also called: tilt

MEANING Angle between the orbital plane and the reference plane (Earth's equator for Earth satellites). 0° equatorial, 90° polar, > 90° retrograde.

CONCEPT Where on Earth a satellite can fly over is set entirely by tilt: a satellite never goes to a latitude higher than its inclination. So i is chosen by the mission (what must you see?) and is the most expensive element to change afterward.

NUMBERS GEO 0° (equator only). ISS 51.6° (sees up to 51.6° N/S). Polar/sun-synchronous ~98° (sees everything).

NOT THIS Ω. Both are angles describing orientation; i is how much the plane is tilted, Ω is which way the tilt faces.

CONNECTS Plane change Δv = 2v·sin(Δi/2); launch latitude sets the minimum i you can reach cheaply.

WORD BANK · 3/8

Concept-level word bank

Right ascension of the ascending node (Ω, RAAN) 3

Also called: longitude of the ascending node · 'swivel'

MEANING Angle, measured in the reference plane from the vernal equinox (♈), to the point where the orbit crosses the plane heading north.

CONCEPT A tilted plane can face any direction around Earth's axis. Ω pins that direction down by saying where the orbit crosses the equator going north, measured from a fixed star-based reference. Without it two orbits with the same tilt could be at right angles to each other.

NUMBERS GPS uses six planes with Ω spaced 60° apart so satellites are spread around the globe rather than bunched.

NOT THIS Argument of periapsis ω. Ω is where the plane crosses the equator (measured in the equator plane); ω is where perigee sits (measured inside the orbit plane).

CONNECTS J_2 makes Ω drift; sun-synchronous orbits exploit that drift.

Argument of periapsis (ω) 3

Also called: argument of perigee · 'spin in plane'

MEANING Angle, measured in the orbital plane, from the ascending node to periapsis. Says which way the long axis of the ellipse points.

CONCEPT An elliptical orbit has a near end and a far end, and it matters where the near end points: over the target you want to photograph up close, or over the region you want to linger above. ω records that.

NUMBERS Molniya orbits set ω ≈ 270° so apogee hangs over the northern hemisphere for hours.

NOT THIS Ω (see above). Also 'perigee' itself: ω is the angle to perigee, not the perigee distance.

CONNECTS Radial burns rotate ω; J_2 makes ω drift for most inclinations.

True anomaly (ν or θ) 3

MEANING Angle from periapsis to the body's current position, measured at the central body. The only element that changes quickly as the orbit proceeds.

CONCEPT Five elements freeze the orbit; one number is still needed to say where on it the satellite is at this instant. That's ν. It's the only element that changes fast, which is why the other five are the useful 'address' and ν is the 'time.'

NUMBERS ν = 0° at perigee, 180° at apogee. The ISS sweeps all 360° every 92 minutes.

NOT THIS Mean anomaly M, which grows evenly with time; ν does not, because the satellite is faster near perigee.

CONNECTS Kepler's second law explains why ν changes unevenly.

Periapsis / apoapsis 3

Also called: perigee/apogee (Earth) · perihelion/aphelion (Sun) · perilune/apolune (Moon) · periareion/apoareion (Mars)

MEANING The closest and farthest points of an orbit from the central body. Fastest at periapsis, slowest at apoapsis.

CONCEPT Any non-circular orbit has one nearest and one farthest point, and they're where the physics is simplest: speed is purely sideways there, which is why burns are planned there and why their distances define a and e.

NUMBERS GTO: perigee 300 km altitude at 10.15 km/s, apogee 35,786 km at 1.61 km/s.

NOT THIS The suffix changes with the body: -gee Earth, -helion Sun, -lune Moon, -apsis generic. Same two points.

CONNECTS Burn rules: prograde at periapsis raises apoapsis; at apoapsis raises periapsis.

WORD BANK · 4/8

Concept-level word bank

Standard gravitational parameter (μ) 3

Also called: GM · 'mu'

MEANING The body's mass times the gravitational constant, in one number: Earth μ = 398,600 km³/s². Shows up in every orbit formula.

CONCEPT Every orbit formula has G × M in it, and G × M is known far more precisely than either G or M alone. So engineers bundle it into one constant per body and never separate it.

NUMBERS Earth 398,600 km³/s². Sun 1.327 × 10^{11}. Mars 42,828. Jupiter 1.267 × 10^{8}.

NOT THIS g (9.8 m/s²), which is surface gravity, a different thing. μ/r² gives g at any distance.

CONNECTS Appears in vis-viva, period, escape speed, turning angle.

Vis-viva equation 3

Also called: orbital energy equation

MEANING $v^2 = \mu(2/r - 1/a)$. Speed anywhere on any orbit from distance and semi-major axis. Gives circular speed (a = r) and escape speed (a → ∞).

CONCEPT The energy of an orbit is fixed, so speed and distance trade off along it. Vis-viva is that trade-off written as one formula: know how far you are and how big the orbit is, and you know how fast you're going. It replaces a dozen special cases with one line.

NUMBERS GTO at perigee: $v^2 = 398{,}600 \cdot (2/6{,}671 - 1/24{,}418) \rightarrow v = 10.15$ km/s. At apogee: 1.61 km/s.

NOT THIS Circular velocity $v = \sqrt{(\mu/r)}$ is just vis-viva with a = r. Escape velocity is vis-viva with a = ∞.

CONNECTS Every Δv calculation: speed before and after a burn come from vis-viva.

Stefan–Boltzmann law 4

MEANING Radiated power per area = $\sigma \cdot \varepsilon \cdot T^4$. Doubling absolute temperature radiates 16× more. The physics behind radiator sizing.

CONCEPT Radiation is the only way heat leaves a spacecraft, and this law says how fast: proportional to the fourth power of absolute temperature. The steep T^4 means radiators work far better when hot, cold things barely radiate, and sizing a radiator is a one-line calculation.

NUMBERS 1 m² at 300 K, ε = 0.9: ~410 W. At 350 K: ~760 W. At 200 K: only ~80 W.

NOT THIS Conduction/convection, which depend on a temperature difference, not an absolute temperature to the fourth.

CONNECTS Radiator sizing, emissivity/absorptivity coatings, why hot spots persist.

Outgassing 4

MEANING Release of trapped gas and volatiles from materials in vacuum. Measured as TML (total mass loss) and CVCM (collected volatile condensable material); spacecraft materials must stay under 1% and 0.1%.

CONCEPT Every material made in air has gas dissolved and trapped inside it. In vacuum there's nothing holding it in, so it slowly leaks out, and in vacuum vapor travels in straight lines to the coldest surface it can find: your lens. So materials are chosen and baked for low outgassing before launch.

NUMBERS Limits: TML < 1%, CVCM < 0.1% (ASTM E595, 125 °C for 24 h). Bake-outs run days at 100+ °C.

NOT THIS Leaks from tanks or cabins. Outgassing is from solid materials themselves: plastics, adhesives, paints.

CONNECTS Contamination of optics, clean rooms, MLI material choice.

WORD BANK · 5/8

Concept-level word bank

Spacecraft charging

4

Also called: surface charging · electrostatic discharge (ESD)

MEANING Build-up of electric charge on surfaces with no path to bleed it off, until it arcs. Mitigated by conductive coatings and bonding (grounding straps).

CONCEPT On Earth static bleeds off into air and ground. In space it can't, so charge collects: sunlight knocks electrons off lit surfaces (positive) while plasma electrons land on shaded ones (negative). Two surfaces can end up kilovolts apart and arc, which fries electronics. Every surface is therefore made conductive and strapped to every other.

NUMBERS Differential potentials of several kV have been measured on GEO satellites during geomagnetic storms; arcs have killed solar-array strings.

NOT THIS Radiation damage. Charging is static electricity; radiation is energetic particles passing through chips.

CONNECTS Photoelectric effect, ESD, conductive coatings (ITO), bonding straps.

Disturbance torque

4

Also called: perturbing torque

MEANING A small twisting force on a spacecraft: gravity-gradient, solar radiation pressure, aerodynamic, magnetic.

CONCEPT Even in 'empty' space tiny twists never stop: gravity pulls the near end harder, sunlight pushes on big panels, leftover air drags unevenly, Earth's magnetism tugs. Each is tiny, none ever stops, nothing damps them. Attitude control exists because of them.

NUMBERS Small satellite: ~10^{-5} N·m. ISS: several N·m, enough to tumble it within hours if uncontrolled.

NOT THIS Orbital perturbations, which change the path. Disturbance torques change the pointing.

CONNECTS Reaction wheels counter them; momentum dumping resets the wheels.

Reaction wheel / momentum wheel / control moment gyro (CMG)

4

MEANING Internal spinning masses used to turn or stabilize a spacecraft without fuel; CMGs tilt a spinning wheel for much larger torque.

CONCEPT In vacuum there's nothing to push against to turn, so you push against yourself: spin a wheel inside and the spacecraft turns the other way. No fuel, very precise. The catch is that constant one-way pushes make the wheels spin ever faster until they saturate.

NUMBERS Hubble uses four reaction wheels for pointing to 0.007 arcseconds. The ISS uses four CMGs. Kepler lost two wheels and had to be reinvented.

NOT THIS Thrusters, which push on expelled gas and use fuel. Wheels only trade angular momentum internally.

CONNECTS Momentum dumping, disturbance torques, attitude sensors.

Homosphere / heterosphere

5

MEANING Below ~100 km the air is mixed to a uniform composition (homosphere). Above, gases separate by molecular weight (heterosphere).

CONCEPT Weather stirs the lower air so thoroughly that its recipe is the same everywhere. Above about 100 km the stirring stops and each gas settles at the height its weight allows: heavy low, light high. That boundary explains why the gas list changes with altitude.

NUMBERS Below 100 km: N_2 78%, O_2 21% everywhere. 200–600 km: atomic O. 600–1,500: He. Above: H.

NOT THIS The temperature layers (troposphere etc.). Homo/heterosphere is about composition; those are about temperature.

CONNECTS Turbopause, atomic oxygen, geocorona.

WORD BANK · 6/8

Concept-level word bank

Atomic oxygen (AO, O) 5

MEANING Single oxygen atoms produced by UV photodissociation of O_2 above ~100 km; the dominant species ~200–600 km. Highly reactive; erodes polymers at orbital velocity.

CONCEPT Up high there's no ozone to absorb UV, so UV splits O_2 into single atoms, and the air is too thin for them to find partners again. The result is a sea of hungry single atoms exactly where LEO satellites fly, and a satellite plowing through it at 7.5 km/s gets chemically sandblasted.

NUMBERS Dominant gas 200–600 km. Bare Kapton on the ISS ram face erodes tens of micrometers per year; the LDEF experiment showed shredded samples after 5.8 years.

NOT THIS Ozone (O_3, three atoms, stratosphere, protective) and O_2 (two atoms, what you breathe). AO is one atom, high up, destructive.

CONNECTS Ram/wake, erosion yield × fluence, coatings, drag (same thin air).

Ionosphere 5

MEANING ~60–1,000 km, overlapping the upper layers: solar UV and X-rays ionize atoms, leaving free electrons that refract and reflect radio and delay GPS signals.

CONCEPT Sunlight's UV and X-rays knock electrons off atoms in the upper atmosphere, leaving a charged layer. Charged layers bend and reflect radio waves, which is why AM radio reaches far at night, why GPS signals get delayed, and why the layer grows and shrinks with day and solar activity.

NUMBERS ~60–1,000 km. GPS delay error up to tens of meters uncorrected; receivers model it out.

NOT THIS The ozone layer (much lower, chemical not electrical) and the magnetosphere (farther out, Earth's magnetic field).

CONNECTS Space weather indices, radio comms section, GPS accuracy.

Specific impulse (Isp) 6

MEANING An engine's fuel efficiency, in seconds: exhaust speed divided by g_0. Chemical rockets 300–450 s; ion engines 3,000+ s with tiny thrust.

CONCEPT Engines differ in how much push they get per kilogram of propellant. Isp is that efficiency, expressed in seconds for historical reasons: multiply by g_0 (9.81) and you get exhaust speed in m/s, which is what the rocket equation actually uses.

NUMBERS Solid rocket ~250 s. Kerosene 300–340. Hydrogen 450. Ion engine 3,000+ (but thrust like the weight of a sheet of paper).

NOT THIS Thrust. Isp is efficiency (miles per gallon); thrust is power (horsepower). Ion engines are high Isp, low thrust.

CONNECTS Rocket equation; propulsion section of the course.

Delta-v (Δv) 6

Also called: velocity change · 'the fuel bill'

MEANING Total change in speed a maneuver or mission requires (km/s). Converts to propellant mass through the rocket equation.

CONCEPT Where you can go isn't limited by distance or time; it's limited by how much you can change your speed, because every speed change costs propellant and propellant is the heaviest thing you carry. So missions are budgeted in km/s, not kilometers.

NUMBERS LEO to GEO: 3.9 km/s. LEO to Mars orbit: ~5.6. Earth surface to LEO: ~9.4 (including losses).

NOT THIS Speed. Δv is the total of all changes, not how fast you're going. A spacecraft can have a huge speed and tiny Δv left.

CONNECTS Rocket equation converts Δv to propellant mass; Hohmann minimizes Δv.

WORD BANK · 7/8

Concept-level word bank

Hohmann transfer

6

Also called: minimum-energy two-impulse transfer

MEANING Half an ellipse tangent to both the starting and target orbit, 180° apart, two prograde burns. Cheapest two-burn transfer between coplanar circular orbits (for radius ratios under ~11.9).

CONCEPT To move between two circular orbits you must change energy twice: once to leave, once to arrive. The cheapest way is an ellipse that just touches both, because both burns are then tangent (no wasted sideways component). It's cheapest, not fastest.

NUMBERS LEO→GEO: 2.42 + 1.47 km/s, 5.3 h. Earth→Mars: ~3.6 from LEO + ~2 to capture, 259 days.

NOT THIS 'Fastest route.' A Hohmann is the slowest sensible route; fast routes cost exponentially more.

CONNECTS Vis-viva for the speeds, time of flight = half the ellipse period, synodic period for when it's possible.

Lambert's problem

6

MEANING Given two position vectors and the time of flight between them, find the orbit that connects them (and so the Δv). The core of trajectory design.

CONCEPT Mission planners don't start with an orbit; they start with 'be at Mars on this date.' Lambert's problem is the tool that turns two places and a deadline into the one orbit that works, and therefore into the fuel bill. It's how every launch date is actually chosen.

NUMBERS Earth on 2026-11-01, Mars on 2027-08-01 → the solver finds the ellipse and reports departure Δv ≈ 3.7 km/s.

NOT THIS Hohmann, which is one special solution (180°, minimum energy). Lambert handles any angle and any time.

CONNECTS Porkchop plots are thousands of Lambert solutions on a grid.

Synodic period

6

Also called: launch-window spacing

MEANING Time between successive identical Earth–target alignments: Earth–Mars ≈ 780 days (26 months).

CONCEPT Two planets circling the Sun at different rates line up the same way only periodically. That repeat time, the synodic period, is why launch windows exist and why a missed Mars window costs two years.

NUMBERS Earth–Mars 780 days (26 months). Earth–Venus 584 days. Earth–Jupiter 399 days.

NOT THIS Orbital period. Mars's year is 687 days; the Earth–Mars lineup repeats every 780 days.

CONNECTS Phase angle: the lineup needed; launch window: the dates around it.

Sphere of influence (SOI)

6

MEANING The region around a planet where its gravity dominates the Sun's for trajectory purposes. Earth's ≈ 925,000 km.

CONCEPT The Sun and a planet both pull on you everywhere, which would make trajectories impossible to compute by hand. The SOI is the practical boundary where you switch from 'planet rules' to 'Sun rules,' so each leg becomes a simple two-body problem.

NUMBERS Earth's SOI ≈ 925,000 km (2.4× the Moon's distance). Jupiter's ≈ 48 million km.

NOT THIS The planet's gravity doesn't stop at the SOI; it's a bookkeeping line, not a physical wall.

CONNECTS Patched conics stitch the legs together at SOI crossings.

WORD BANK · 8/8

Concept-level word bank

Characteristic energy (C3)

6

Also called: launch energy

MEANING C3 = v_∞^2 at Earth departure (km^2/s^2). C3 = 0 is bare escape; Mars needs ~10–15.

CONCEPT Launch providers need one number for 'how hard must I throw this.' C3 is v_∞ squared at Earth departure: zero means barely escaping, bigger means a faster interplanetary departure, and rocket performance charts are drawn against it.

NUMBERS Mars: ~10–15 km^2/s^2. Jupiter direct: ~80. New Horizons: ~158, the highest ever.

NOT THIS Δv. C3 is the energy the rocket must deliver at Earth; Δv is the spacecraft's own budget afterward.

CONNECTS $v_\infty = \sqrt{C3}$; departure burn from LEO = $\sqrt{(v_{esc}^2 + C3)} - v_{circ}$.

Tsiolkovsky rocket equation

6

Also called: ideal rocket equation

MEANING $\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$. Links fuel fraction to Δv; why fast transfers are impossible. Coming in the propulsion section.

CONCEPT A rocket pushes on its own exhaust, so it must carry everything it will ever throw. The result is that propellant need grows exponentially with Δv: each extra km/s multiplies the fuel, not adds to it. This single equation explains staging, why fast transfers are impossible, and why Δv is the budget that matters.

NUMBERS Isp 450 s → 4.4 km/s per e-fold. 3.6 km/s needs 56% propellant; 9 km/s (to orbit) needs 87%; 60 km/s needs 99.9999%.

NOT THIS 'More fuel = proportionally more speed.' No: doubling Δv squares the mass ratio.

CONNECTS Specific impulse sets the exhaust speed; Δv comes from the mission plan.

Hyperbolic excess velocity (v∞)

7

MEANING Your speed relative to a planet when far from it, before capture or after escape. The 'in' and 'out' speed of a flyby, equal in magnitude in the planet's frame.

CONCEPT Near a planet, your speed depends on how deep you are in its gravity well, so it's not a useful single number. Far away, gravity's effect has faded and your speed settles to one value: v_∞. It's the clean 'arrival speed' that every flyby and capture calculation starts from.

NUMBERS Earth→Mars Hohmann: v_∞ at Mars ≈ 2.6 km/s. New Horizons left Earth with v_∞ ≈ 12.6 km/s.

NOT THIS Escape velocity. v_∞ is what's left over after you've escaped; escape velocity is the minimum to get $v_\infty = 0$.

CONNECTS C3 = v_∞^2; capture burn size depends on v_∞.

Gravity assist

7

Also called: flyby · swing-by · slingshot

MEANING Passing close to a moving planet so its gravity bends your path; in the Sun's frame you gain (pass behind) or lose (pass in front) speed, borrowed from the planet's orbital motion.

CONCEPT A planet is a moving, massive object. Pass near it and gravity swings you around; because the planet is moving, that swing hands you some of its orbital motion in the Sun's frame. It's the only way to reach the outer planets without impossible amounts of fuel.

NUMBERS Voyager 2 gained ~10 km/s at Jupiter. New Horizons shaved ~3 years off its Pluto trip at Jupiter.

NOT THIS 'Free energy.' The planet pays, by an unmeasurable amount. Also: in the planet's own frame your speed doesn't change at all.

CONNECTS Turning angle formula, hyperbolic excess velocity, momentum conservation.

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 1 OF 10

Every word, fully defined

Same order as the word bank. Bold = the term · grey = what else it's called · orange = the short line you write in the notebook · then the full meaning, with the picture in italics.

S1 · Satellite types & orbits

satellite
anything that orbits something bigger
anything that orbits something bigger. The Moon is Earth's natural satellite; everything we launch is an artificial one. *A ball on a string, swung so fast the string is gravity.*

orbit
falling and missing, forever
the closed path a satellite follows around a body. It's falling toward Earth and missing, forever. *Throw a ball sideways fast enough and the ground curves away as fast as it falls.*

altitude
height above the ground, not from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

LEO / MEO / GEO · also: low / medium / geosynchronous
200–2000 km · 2000–35,786 · 35,786
Low, Medium, Geosynchronous Earth Orbit: the three height bands. LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km. *Downstairs, middle floor, penthouse.*

orbital period (T)
time for one lap: 90 min / 12 h / 24 h
Time for one revolution: $T = 2\pi\sqrt{(a^3/\mu)}$. Depends only on a and the central body.

geosynchronous (GSO)
1-day lap, any tilt
an orbit whose period is exactly one day, at any tilt. The satellite returns to the same spot in the sky once a day but may drift north-south in between.

geostationary (GEO)
1-day lap + over equator, stays put
geosynchronous AND directly over the equator with no tilt, so it appears frozen in one spot. TV dishes point at these and never move. *Sync = same time every day. Stationary = never leaves.*

equator / equatorial plane
flat sheet through Earth's middle
the flat sheet through Earth's middle, perpendicular to its spin axis. The reference plane for Earth orbits.

sidereal day
true spin vs stars, 23 h 56 m
one true rotation of Earth relative to the stars: 23 h 56 min, not 24 h. That's the real period a GEO satellite matches. *The 4-minute difference is Earth also moving around the Sun.*

astronomical satellite · also: space telescope
looks out: Hubble, JWST
a telescope in space pointed away from Earth (Hubble, JWST). *Looks out.*

Earth observation (EO)
looks down: Landsat
cameras and radar pointed down at Earth for mapping, farming, disasters (Landsat). *Looks down.*

weather satellite
looks down from GEO: GOES
looks down at clouds and storms, usually from GEO so it watches the same region nonstop (GOES).

reconnaissance satellite · also: spy sat
looks down, military
a military spy satellite. Same idea as Earth observation, different customer and classified detail.

navigation satellite
broadcasts time, phone finds itself: GPS
broadcasts a precise time signal; your phone hears four of them and works out where it is (GPS, Galileo). *Tells you where you are.*

communication satellite · also: comms / relay
radio up, radio down: Starlink
a radio relay in the sky: receives a signal from one place on Earth and sends it back down somewhere else (Starlink, DirecTV). *A mirror for radio.*

biosatellite
carries living things: Laika
carries living things (plants, animals, cells) to study how they cope with space. Sputnik 2 carried Laika the dog in 1957.

space station
satellite people live in: ISS
a satellite people live in (ISS, China's Tiangong).

constellation
swarm so one is always overhead
A set of satellites in coordinated orbits that function as one system (GPS, Iridium, Starlink). Arranged in several orbital planes so coverage is continuous.

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 2 OF 10

Every word, fully defined

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S1 · Satellite types & orbits

footprint · also: coverage area
patch a sat sees; bigger when higher
The patch of Earth a satellite can see or communicate with at one moment. Grows with altitude. Also called: field of view (FOV) · swath

eclipse (orbital)
Earth blocks the Sun; batteries cover it
when Earth blocks the Sun from the satellite. LEO sats get ~15 a day; batteries have to cover them.

atomic oxygen / drag
LEO-only problems: eats plastic / slows you
a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

S1 · Types of interplanetary spacecraft

interplanetary
between planets, orbiting the Sun
between planets. An interplanetary spacecraft leaves Earth's neighborhood entirely, as opposed to a satellite that stays in Earth orbit.

flyby
passes once, never captured
a spacecraft that passes a planet once at high speed and keeps going. It never slows down enough to be caught. *Driving past a house and photographing it out the window.*

orbiter
brakes into orbit, stays
a spacecraft that fires its engine to slow down at the planet so gravity captures it into orbit. *Pulling into the driveway and staying.*

atmospheric probe
dives into the air, short life
a small craft that dives into a planet's air on purpose, usually on a parachute or balloon, and reports until it's crushed or runs out of power (Galileo probe at Jupiter, Huygens at Titan).

lander
touches down, stays put
a craft that survives touchdown and sits in one place doing science (Viking, InSight, Philae).

rover
lander with wheels
a lander with wheels: it can drive to new spots (Sojourner, Spirit, Opportunity, Curiosity, Perseverance).

penetrator · also: impactor
crashes in on purpose
a craft designed to slam into the surface on purpose, either to bury sensors or to kick up material for another craft to analyze (Deep Impact hit a comet; LCROSS hit the Moon).

observatory spacecraft · also: space telescope
telescope platform, target = sky
a telescope platform whose target is the sky, not a planet (Hubble, JWST, Kepler).

communications spacecraft · also: relay
forwards messages for others
a relay whose job is passing messages between other spacecraft and Earth. Mars orbiters relay for the rovers; China's Queqiao relays for the far side of the Moon.

commitment ladder · also: mission types
flyby→orbiter→probe→lander→rover→penetrator
my phrase, not the instructor's: flyby → orbiter → atmospheric → lander → rover → penetrator, ordered by how much fuel, mass, and risk each needs, and how much science it returns.

delta-v (Δv) · also: velocity change · 'the fuel bill'
speed change = fuel bill
Total change in speed a maneuver or mission requires (km/s). Converts to propellant mass through the rocket equation. Also called: velocity change · 'the fuel bill'

braking burn · also: capture burn
orbit insertion · slow down so gravity grabs you
firing the engine against your direction of travel when you arrive, to slow down enough for the planet's gravity to hold you. A flyby skips this; an orbiter can't.

prograde / retrograde · also: posigrade / —
burn to speed up / slow down
prograde = firing so you speed up along your direction of motion; retrograde = firing to slow down. *Gas pedal vs brake.*

gravity assist · also: flyby
slingshot · steal speed from a moving planet
swinging close past a planet to steal a bit of its orbital speed, for free. New Horizons used Jupiter this way. *Grabbing a moving merry-go-round to fling yourself faster.*

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 3 OF 10

Every word, fully defined

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S1 · Types of interplanetary spacecraft

EDL · also: entry, descent, landing
7 minutes of terror
Entry, Descent, Landing: the sequence a lander or rover goes through from hitting the top of the atmosphere to touching down. On Mars it's ~7 minutes and fully automatic. *"Seven minutes of terror."*

C3 · also: characteristic / launch energy
how hard you throw it from Earth
a number for how hard you have to throw a spacecraft to leave Earth and reach a target, in km²/s². Bigger C3 = bigger rocket. Earth escape is C3 = 0; Mars needs ~10–15; Pluto direct needed far more.

Kuiper Belt
icy ring beyond Neptune, Pluto lives there
the ring of icy bodies beyond Neptune. Pluto is the biggest known member. New Horizons kept flying into it after Pluto.

comet vs asteroid
ice + dust with a tail / a rock
a chunk of ice and dust that grows a tail when it nears the Sun. Rosetta orbited one and dropped the Philae lander onto it.

S2 · Space vacuum conditions

vacuum
almost no gas: no pressure, no air
space with (almost) no air in it. Not "nothing"; just no gas pushing on things. *Like the inside of a sealed jar after you pump all the air out.*

ozone layer (O_3)
Earth's sunscreen, blocks UV < 0.3 µm
a thin band of a special oxygen (O_3) high in Earth's air that soaks up harsh sunlight before it reaches you. *Earth's sunscreen.*

ultraviolet (UV)
invisible high-energy light, sunburns
light that's a bit "bluer than blue," too high-energy for your eyes to see. It's what sunburns you. *The invisible part of sunlight that does damage.*

wavelength
size of one ripple; shorter = more energy
the size of one ripple of a light wave. Shorter wavelength = more energy = more damage. UV is shorter than visible light. *Small ripples hit harder.*

µm (micrometer) · also: micron
millionth of a meter; 0.3 µm = 300 nm
one millionth of a meter. 0.3 µm = 300 nm. A human hair is ~70 µm. *The caption said "millimeters": wrong by 1,000×.*

polymer · also: plastic
long molecule chains: Kapton, Teflon
any plastic, rubber, or similar material made of long repeating chains of molecules. Kapton, Teflon, epoxies, cable insulation. *Long molecular spaghetti.*

organic chemical bond · also: bond energy
carbon glue; UV photon beats it
the "glue" between carbon atoms that holds those chains together. Takes a certain amount of energy to break. *UV has more than that amount, so it snaps the glue.*

degradation / decomposition
material falling apart over time
the material slowly falling apart: going brittle, yellowing, cracking, flaking. *Like a garden hose left in the sun for 5 years.*

corrosion
chemical attack (rust); loose word for UV damage
chemical attack on a material, like rust on iron. The instructor used it loosely for UV damage; strictly, UV damage is bond-breaking, not corrosion. *Rust = corrosion. Sun-rotted plastic = degradation.*

UV photodegradation · also: UV degradation
photons snap polymer bonds
Breaking of chemical bonds in polymers by UV photons whose energy exceeds the bond energy (~4 eV at 300 nm vs ~3.6 eV for C–C). Also called: UV degradation

outgassing
gas leaking out of solids in vacuum
gas slowly leaking out of a solid material once it's in vacuum. *Like a new car's "new car smell": plastics releasing trapped stuff, but in space it lands on your camera lens.*

TML / CVCM
outgassing limits: < 1% / < 0.1%
Total Mass Loss and Collected Volatile Condensable Material: the two numbers a material gets from a standard outgassing test (heated in vacuum for 24 hours). TML is how much of its mass escapes as gas; CVCM is how much of that gas condenses on a cold plate nearby, which is what would fog a lens. Spacecraft materials must stay under 1% TML and 0.1% CVCM. *The 'how much will this thing sweat in space' report card.*

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 4 OF 10

Every word, fully defined

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S2 · Space vacuum conditions

bake-out
outgas it on the ground first
Heating components in a vacuum chamber before launch so they outgas on the ground instead of in orbit.

void
bubble inside a material
a tiny bubble or pocket inside a material left over from manufacturing, usually full of air. *A bubble in a cake.*

pressure difference
1 atm inside, 0 outside → cracks
one side pushing harder than the other. Air trapped in a void pushes out at 1 atmosphere; vacuum pushes back with nothing. *A balloon taken up a mountain swells for the same reason.*

clean room
filtered air, no specks on sensors
a sealed workspace with filtered air, bunny suits, and strict rules, so dust and skin flakes don't land on the spacecraft. *An operating theater for hardware.*

perturbation
small unwanted nudge
any small unwanted push, nudge, or wobble. A thruster puff, a solar panel unfolding, a wheel spinning up. *Someone bumping your desk.*

dissipation · also: damping
energy leaking away until motion stops
energy leaking away into the surroundings until the motion stops. *The reason a plucked guitar string goes quiet.*

Newton's 1st law · also: inertia
motion keeps going unless pushed
a thing keeps doing what it's doing (moving or sitting still) unless something pushes on it. *On Earth, air and friction are the "something." In space, there isn't one.*

vibration / damping
shaking / what makes it fade
a shaking back and forth / anything that makes it fade out. *Shock absorbers on a car are dampers.*

convection
heat carried by moving air; gone in vacuum
heat carried away by moving air or liquid: hot air rises, cool air replaces it. *Why a radiator heats a whole room.*

conduction
heat through touching solids
heat moving through a solid by touch, atom to atom. *A metal spoon getting hot in soup.*

radiation (heat)
heat glowing away; works in vacuum
heat glowing away as invisible light, needing no air at all. *Feeling a campfire on your face from a distance.* In vacuum, conduction + radiation are the only two left.

radiator
panel that glows heat into space
A panel that dumps the spacecraft's waste heat into space by radiation, the only heat exit in vacuum.

MLI · also: multilayer insulation
thermal blanket · crinkly gold wrap
Stacks of thin reflective film that slow radiative heat loss. The crinkly gold/silver wrapping on spacecraft. Also called: thermal blanket

equilibrium / hot spot
settled temperature / unspread heat
Thermal equilibrium is when heat has spread out so everything sits at the same temperature. A hot spot is a place where heat has piled up and not spread yet. On Earth convection evens hot spots out; in vacuum they stay hot until conduction or radiation slowly moves the heat. *A pan on the stove: the whole pan warms evenly in air; in vacuum only the spot over the flame would glow.*

electrical discharge · also: ESD
a spark
built-up charge suddenly jumping somewhere: a spark. *The zap from a doorknob after walking on carpet.*

spacecraft charging · also: surface charging · electrostatic discharge (ESD)
static builds, no air to bleed it, arcs
the spacecraft slowly collecting electric charge from space (sunlight knocks electrons off, charged particles stick). With no air to bleed it away, it builds until it sparks. *You on a carpet, forever, with no doorknob.*

passive vs active system
no power (coating) / powered (heater)
Passive: works with no power and no decisions, by its nature (a coating, a grounding strap, a radiator, a thermal blanket). Active: does something on command, using power (a heater, a reaction wheel, a thruster, a charge-bleeding device). Engineers prefer passive where possible because it can't fail by losing power. *A window shade vs. air conditioning.*

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 5 OF 10

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S2 · Earth atmosphere

atmosphere
blanket of gas held by gravity
the blanket of gas held around a planet by gravity. Thick at the bottom, fading to nothing a few hundred km up. *Earth's air, all of it.*

troposphere
0–11 km, weather, colder up
bottom floor, 0–11 km. Weather, clouds, airliners. *"Tropo" = turning: the air churns here.*

stratosphere
11–50 km, ozone, warmer up
11–50 km. Calm, layered, holds the ozone layer, warms with height. *"Strato" = layered.*

mesosphere
50–90 km, coldest, meteors burn
50–90 km. The coldest layer; meteors burn up here. *"Meso" = middle.*

thermosphere
90–500 km, hot but empty, ISS here
90–500 km. Extremely hot by thermometer, almost no air. The ISS and most LEO satellites fly inside it. *"Thermo" = heat.*

exosphere
500 km+, fades to space
500 km and beyond. So thin that atoms can drift off into space. *"Exo" = outside.*

homosphere / heterosphere
stirred mix / sorted by weight
homo = the bottom ~100 km where gases are stirred into one uniform mix. Hetero = above that, where gases separate by weight. *Smoothie vs. a layered cocktail.*

turbopause · also: homopause
~100 km, mixing stops
The ~100 km boundary where turbulent mixing stops and diffusive separation begins. Also called: homopause

temperature vs heat
molecule speed / energy delivered
temperature = how fast molecules move. Heat = how much energy they can actually give you, which also depends on how many there are. *One sparkler is 1000 °C and won't warm a room.*

density / number density
how much stuff per volume (m^{-3})
how much stuff is packed into a space. Numerical density (m^{-3}) on the chart = how many molecules per cubic meter. *People per square meter in a crowd.*

log scale
each tick = 100,000×
the chart's vertical axis where each tick (10^{10}, 10^{15}, 10^{20}) is 100,000× the one below. Used when numbers span huge ranges. *Zooming out so a mountain and a pebble fit on one page.*

constituent · also: component
one gas in the mix
one of the gases the air is made of. *Ingredients.*

N_2 / O_2 / Ar / CO_2
78% / 21% / 0.9% / 0.04%
nitrogen gas (two N atoms), oxygen gas (two O atoms), argon. The three main ingredients of normal air: 78%, 21%, 1%.

O / N / H / He · also: atomic forms
single atoms, up high
single atoms of oxygen, nitrogen, hydrogen, and helium gas. The "atomic" forms. Up high, radiation breaks the paired molecules into these singles. *Loose Lego bricks instead of snapped-together pairs.*

atomic oxygen (AO)
lone O atom, 200–600 km, eats plastic
a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

photodissociation
light splits $O_2 \rightarrow O + O$
Light breaking a molecule apart. $O_2 + UV \rightarrow O + O$.

anomalous O · also: hot oxygen
minor extra-hot O above 300 km
a tiny extra population of unusually energetic oxygen atoms that the chart's model adds above ~300 km. Minor; not something you need to remember beyond "it's still oxygen."

ionosphere
60–1000 km, charged, bends radio, delays GPS
~60–1,000 km, overlapping the upper layers: solar UV and X-rays ionize atoms, leaving free electrons that refract and reflect radio and delay GPS signals.

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 6 OF 10

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S2 · Earth atmosphere

geocorona
hydrogen halo glowing in UV
The faint hydrogen halo of the exosphere, glowing in UV, extending tens of thousands of km.

VLEO / LEO
below ~250 km heavy drag / up to 2000
Very Low Earth Orbit (below ~250 km, heavy drag, short life) vs Low Earth Orbit (up to 2,000 km). The green line on the chart marks the split.

reactive
eager to bond, sticky
eager to bond chemically with other stuff. *Sticky.*

oxidation
oxygen bonding to a material; rust
oxygen bonding to a material and changing it. Rust is iron oxidizing. Atomic oxygen oxidizes spacecraft surfaces; here "corrosion" is the right word. *Rust, but in orbit, and fast.*

erosion
worn away bit by bit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

erosion yield / fluence
loss per atom / total atoms that hit
material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

kinetic energy / relative velocity
energy of motion / closing speed
kinetic energy = energy of motion. Relative velocity = how fast two things approach each other; it doesn't matter which one is "really" moving. The s/c at 7.5 km/s hitting a still atom is the same as the atom hitting a still s/c at 7.5 km/s.

eV (electron-volt)
tiny energy unit; bond ≈ few eV
a tiny unit of energy used for single atoms and bonds. A chemical bond is a few eV; an atomic-oxygen hit at 7.5 km/s carries ~5 eV. *Enough to snap the bond.*

ram / wake
front face / sheltered back
Ram: the side facing the direction of motion, which takes the atomic-oxygen and drag hits. Wake: the sheltered trailing side.

atmospheric drag · also: aerodynamic drag
thin air slows LEO sats; ISS reboosts
Resistance from residual air in LEO that slowly lowers the orbit. Grows with solar activity; the ISS reboosts to counter it. Also called: aerodynamic drag

Kapton · also: polyimide film
gold insulation, AO's victim
a gold-colored polymer film used everywhere on spacecraft for insulation. Famous for being eaten by atomic oxygen unless coated.

F10.7 / Ap · also: space weather indices
Sun activity; puffs up the air
two numbers on the chart title describing how active the Sun and Earth's magnetic field were for the model run. The atmosphere puffs up when the Sun is active. Ignore the values.

S3 · Orbit definition & parameters

orbital mechanics
math of where a s/c goes under gravity
the math of how things move under gravity: given where a spacecraft is and how fast it's going, where will it be later? *Ballistics, but the "ground" is a planet.*

celestial body
planet, moon, Sun, comet
any big natural object in space: planet, moon, Sun, asteroid, comet. The thing you orbit.

transfer orbit
the highway between two orbits
an orbit you use to get from one place to another, like Earth to Mars, rather than to stay. *The highway, not the parking lot.*

Classical Orbital Elements · also: Keplerian elements
the six · a e i Ω ω ν = the hoop's settings
The six numbers that completely describe an orbit: a, e, i, Ω, ω, ν. Five fix the shape and orientation; the sixth says where the body is. The hula hoop's six settings. Also called: Keplerian elements · orbital parameters · the six elements

semi-major axis (a) · also: half the major axis · 'size' of the orbit
size: (r_p + r_a)/2
half the longest width of the ellipse. Sets the orbit's size and its period. *"Semi" = half, "major axis" = the long way across.*

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 7 OF 10

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S3 · Orbit definition & parameters

eccentricity (e) · also: 'shape' of the orbit
squash: 0 circle → 1 cigar → >1 open
a number from 0 up that says how squashed the ellipse is. 0 = circle, near 1 = very stretched, 1 or more = doesn't close. *"Eccentric" literally means off-center.*

perigee / apogee · also: periapsis / apoapsis
closest, fastest / farthest, slowest
closest and farthest points of an orbit around Earth, measured from Earth's center. *"Peri" = near, "apo" = far, "gee" = Earth.*

perihelion / aphelion
same, around the Sun
The closest and farthest points of an orbit around the Sun. Same idea as perigee and apogee, with the suffix changed for the body: -gee for Earth, -helion for the Sun, -lune for the Moon, -apsis for anything. Earth's own perihelion is in early January, about 147 million km from the Sun; aphelion in early July, about 152 million km. *Perigee and apogee, Sun edition.*

apsis (apsides)
either extreme point
either of those two extreme points. The line joining them is the long axis of the ellipse.

line of apsides · also: major axis
perigee↔apogee line
The straight line through periapsis and apoapsis, passing through the central body. Its half-length is a. Also called: major axis

inclination (i) · also: tilt
tilt vs equator; 0° flat, 90° polar
the tilt angle between the orbit's plane and the reference plane (the equator, for Earth). 0° flat, 90° over the poles.

ecliptic
plane of Earth's orbit; reference for Sun orbits
the flat sheet Earth's orbit around the Sun lies in. The reference plane for orbits around the Sun. *Not the same as the equator; they're tilted 23.4° apart.*

plane of reference
what you measure tilt from
the flat sheet you measure tilt from. Equator for Earth orbits, ecliptic for Sun orbits.

vernal equinox (♈) · also: First Point of Aries
reference direction · fixed zero direction in space
a fixed "12 o'clock" direction in space to measure angles from: the direction to the Sun on the first day of spring. The ♈ symbol (Aries) marks it on diagrams. *North on a compass, but for space.*

ascending node
hoop crosses equator going north
the two points where a tilted orbit pierces the reference plane. The ascending node is the one where the satellite is crossing from south to north. *Where the hoop pokes up through the table.*

line of nodes
where orbital plane cuts equator
The line where the orbital plane cuts the reference plane; it joins the ascending and descending nodes.

RAAN (Ω) · also: longitude of ascending node
swivel: ♈ → ascending node
Angle, measured in the reference plane from the vernal equinox (♈), to the point where the orbit crosses the plane heading north. Also called: longitude of the ascending node · 'swivel'

argument of periapsis (ω) · also: argument of perigee
spin in plane: node → perigee
the angle, measured in the orbit's own plane, from the ascending node to perigee. The orbit's spin-in-place. *"Argument" is old math-speak for "angle." Nobody is arguing.*

true anomaly (ν)
where the bead is now
the angle from perigee to where the spacecraft is right now. *"Anomaly" is old astronomy-speak for "angle along the orbit." Nothing is wrong.*

epoch
timestamp the elements are valid for
The timestamp the elements are valid for. Elements drift, so a set of COEs always comes with an epoch.

orbital plane
flat sheet the orbit lies in; passes through center
The flat sheet the orbit lies in. Always passes through the center of the body being orbited. Why GEO must be over the equator.

circular / elliptical / hyperbolic
e = 0 / 0–1 / > 1
orbit shapes by e: circle (e = 0), ellipse (0 1, an open swing-past like a flyby).

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 8 OF 10

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S3 · Orbit definition & parameters

polar / equatorial orbit
i = 90° / i = 0°
the imaginary line around Earth's middle. A geostationary satellite has to sit above it; a satellite above Texas can't hover there because Earth's spin would carry Texas away underneath.

geometry (of an orbit)
size + shape + orientation
The orbit's size, shape and orientation taken together: a, e, i, Ω and ω. Everything about the orbit except where the spacecraft currently is on it (that's ν). Two spacecraft can share one geometry and be at different points. *The shape and tilt of the hoop, ignoring the bead.*

vis-viva · also: orbital energy equation
$v^2 = \mu(2/r - 1/a)$
$v^2 = \mu(2/r - 1/a)$. Speed anywhere on any orbit from distance and semi-major axis. Gives circular speed (a = r) and escape speed (a → ∞). Also called: orbital energy equation

μ (gravitational parameter) · also: GM
Earth 398,600 km^3/s^2
The body's mass times the gravitational constant, in one number: Earth μ = 398,600 km^3/s^2. Shows up in every orbit formula. Also called: GM · 'mu'

S3 · Earth orbits & constellations

Earth orbit · also: geocentric
goes around Earth, not the Sun
~200–2,000 km altitude; period 1.5–2 h. ISS, Hubble, Starlink, most Earth imaging.

rotation period
Earth spins once: 23 h 56 m
how long one full trip around takes. LEO ~90 min, GPS 12 h, GEO 24 h. Higher = longer. *Bigger circle, slower runner.*

altitude vs distance from center
GEO 35,786 alt = 42,164 from center
height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

coverage / global coverage
what a sat sees / everything, always
Coverage is the patch of Earth a satellite can see or talk to at a given moment (its footprint). Global coverage means every point on Earth is covered all the time, which one satellite can never do and a constellation can. *A flashlight's circle of light vs. a stadium's floodlights.*

resolution
how much detail; closer = sharper
How much detail an instrument can make out: the smallest thing it can tell apart. Depends on the optics and on distance, so a LEO camera can see a car while a GEO weather camera sees whole storm systems. Higher orbit = bigger coverage but lower resolution. *Reading a street sign from 10 feet vs. from a mile.*

orbital plane (constellation) · also: Walker constellation
several rings so sats aren't bunched
The flat sheet the orbit lies in. Always passes through the center of the body being orbited. Why GEO must be over the equator.

revisit time
how long until same spot again
How long until a satellite (or constellation) sees the same spot again. Shorter with more satellites or higher altitude.

GPS / GLONASS / Galileo / BeiDou
US / Russia / EU / China nav constellations
Global Positioning System, the US navigation constellation at ~20,200 km in MEO.

Iridium
66 sats, 780 km, sat phones anywhere
A company and its satellite constellation: 66 active satellites at about 780 km in six near-polar orbital planes, giving satellite-phone and data coverage everywhere on Earth including the poles and oceans. Named for element 77 because the original plan had 77 satellites. *The swarm that makes a sat phone work in the middle of the ocean.*

data relay / transfer time
sat → ground; higher = more delay
Data relay is passing information from one spacecraft to another or down to Earth, often through a dedicated relay satellite. Transfer time is how long the signal takes; radio travels at light speed, so it grows with distance: ~0.12 s one way to GEO, 4–24 minutes to Mars. Higher orbits mean longer delay. *A game of telephone across the solar system, with real lag.*

trade-off · also: compromise
better coverage costs detail + delay
Giving up some of one good thing to get more of another, because you can't have all of both. Orbit choice is a trade-off: higher gets you coverage and costs you detail and delay; lower gets you detail and costs you coverage and lifetime. Engineering is mostly choosing trade-offs on purpose. *A big truck vs. a sports car: you don't get both.*

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 9 OF 10

Every word, fully defined

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S3 · Earth orbits & constellations

sun-synchronous (SSO)
polar-ish LEO, same local sun time
A near-polar LEO (~98° inclination) whose plane precesses once a year (via J_2) so it passes over any spot at the same local solar time. Consistent lighting for cameras.

J_2 / nodal precession
Earth's bulge swivels Ω; makes SSO work
Earth's equatorial bulge slowly swivels an inclined orbit's Ω. Engineers exploit it to make sun-synchronous orbits.

S3 · Transfers & flybys

Lambert's problem
2 points + a time → the orbit + Δv
the classic "find the orbit" puzzle: given a start point, an end point, and a travel time, what orbit connects them? Named after Johann Lambert, an 18th-century mathematician. *The "can I get there by Tuesday?" calculator.*

input / constraint
what you feed in / limit you can't break
what you feed into a calculation / a limit you can't break (rocket size, fuel mass, time). *Repeat.*

conic / patched conics · also: conic approximation
stitched 2-body pieces of a trajectory
Approximating an interplanetary trajectory as a chain of two-body conics: Earth-centered hyperbola → Sun-centered ellipse → target-centered hyperbola, 'patched' at the sphere-of-influence boundaries. Also called: conic approximation

sphere of influence (SOI)
where a planet's gravity wins; Earth ≈ 925,000 km
The region around a planet where its gravity dominates the Sun's for trajectory purposes. Earth's ≈ 925,000 km.

Hohmann transfer · also: minimum-energy two-impulse transfer
lazy half-ellipse, 180°, 2 burns, cheapest
the half-ellipse path that touches both the starting and ending orbit, with departure and arrival 180° apart and two burns. Cheapest in fuel. Worked out by Walter Hohmann, a German engineer, in 1925. *The lazy throw.*

tangent
just touching; why burns are small
just touching, not crossing. The Hohmann ellipse is tangent to both circles, which is why the burns are small: you're already moving along the right line.

optimal / efficient
least Δv, not fastest
in this lecture, always means "least Δv," i.e. least fuel. Not fastest.

time of flight (TOF) · also: transfer time
Hohmann = half the ellipse's period
how long the trip takes. For a Hohmann it's half the ellipse's period. *Your Section 1 quiz question.*

launch window / synodic period
lineup repeats; Mars every 26 months
the stretch of days when the planets are lined up right for a Hohmann. Earth–Mars: about every 26 months. *Repeat from S1 primer.*

phase angle · also: lead angle
how far ahead target must be: Mars ~44°
how far ahead the target planet must be when you launch so it's there when you arrive. Mars: ~44° ahead. Not on the slide; it's why the window exists.

impulsive burn · also: impulsive maneuver
engine firing treated as instant
An engine firing treated as instantaneous. Hohmann assumes two of them. Also called: impulsive maneuver

circularization burn · also: apogee kick
2nd burn rounds the ellipse
The second burn of a Hohmann: at apoapsis, speed up to round the ellipse into the target circle. Also called: apogee kick

plane change · also: inclination change
tilting the orbit; very expensive
A burn to tilt the orbital plane. Very expensive: $\Delta v = 2v \cdot \sin(\Delta i/2)$. Done at apogee where v is small. Also called: inclination change

bi-elliptic transfer
3-burn alternative, only for huge jumps
A three-burn alternative that goes far out and comes back. Beats Hohmann only for very large radius ratios. Named so you recognize it.

gravity field
region where the planet's pull matters
the region around a planet where its pull matters. It's what bends the flyby path.

GLOSSARY · COMPLETE DEFINITIONS · S1–S3 · PAGE 10 OF 10

Every word, fully defined

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S3 · Transfers & flybys

relative velocity
speed as seen from Mars vs from Sun
your speed as seen from something else (Mars, say) rather than from the Sun. The flyby doesn't change your speed relative to Mars, only relative to the Sun. *Repeat.*

vector / magnitude / direction
arrow / its length / which way
an arrow: a quantity with both a size and a direction. Velocity is a vector. The slide's arrow pictures are vector addition: spacecraft-relative-to-Mars plus Mars-relative-to-Sun equals spacecraft-relative-to-Sun. *Two pushes added tip to tail.*

v∞ (hyperbolic excess velocity)
speed relative to planet, far away
Your speed relative to a planet when far from it, before capture or after escape. The 'in' and 'out' speed of a flyby, equal in magnitude in the planet's frame.

turning angle (δ) · also: deflection angle
how much the flyby bends you
How much a flyby bends your direction. Larger for slower approaches and closer passes. Also called: deflection angle

orbital energy · also: specific orbital energy
speed + height budget; bigger = bigger a
the combined speed-plus-height "budget" of an orbit. Fixed unless you burn or do a flyby. Bigger budget = bigger orbit.

aerobraking
use atmosphere to shed speed instead of fuel
Using a planet's upper atmosphere to shed speed instead of fuel (gradually, over many passes = aerobraking; in one pass = aerocapture, not yet flown).

Tsiolkovsky rocket equation · also: ideal rocket equation
$\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$
$\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$. Links fuel fraction to Δv; why fast transfers are impossible. Coming in the propulsion section. Also called: ideal rocket equation

Rosetta / 67P / ESA
comet mission 2004→2014, 4 flybys
the European Space Agency's comet mission: launched 2004, four flybys (Earth, Mars, Earth, Earth), arrived at comet 67P/Churyumov–Gerasimenko in 2014, dropped the Philae lander. *Rosetta and ESA are repeats; 67P is new.*

deep space
beyond the Moon; outer planets
far beyond Earth and the Moon; the outer planets and beyond.

WORD BANK · COMPLETE · S1-S3 · PAGE 1 OF 3

Every word, by section and lecture

161 lines total. Shape: **page · term = also called · short meaning**. Fill the page number from your notebook. Say the meaning before you write it. One line per word; if the line's long, abbreviate the meaning, not the term.

S1 · Satellite types & orbits

p__ · satellite · anything that orbits something bigger
p__ · orbit · falling and missing, forever
p__ · altitude · height above the ground, not from center
p__ · LEO / MEO / GEO = low / medium / geosynchronous · 200–2000 km · 2000–35,786 · 35,786
p__ · orbital period (T) · time for one lap: 90 min / 12 h / 24 h
p__ · geosynchronous (GSO) · 1-day lap, any tilt
p__ · geostationary (GEO) · 1-day lap + over equator, stays put
p__ · equator / equatorial plane · flat sheet through Earth's middle
p__ · sidereal day · true spin vs stars, 23 h 56 m
p__ · astronomical satellite = space telescope · looks out: Hubble, JWST
p__ · Earth observation (EO) · looks down: Landsat
p__ · weather satellite · looks down from GEO: GOES
p__ · reconnaissance satellite = spy sat · looks down, military
p__ · navigation satellite · broadcasts time, phone finds itself: GPS
p__ · communication satellite = comms / relay · radio up, radio down: Starlink
p__ · biosatellite · carries living things: Laika
p__ · space station · satellite people live in: ISS
p__ · constellation · swarm so one is always overhead
p__ · footprint = coverage area · patch a sat sees; bigger when higher
p__ · eclipse (orbital) · Earth blocks the Sun; batteries cover it
p__ · atomic oxygen / drag · LEO-only problems: eats plastic / slows you

S1 · Types of interplanetary spacecraft

p__ · interplanetary · between planets, orbiting the Sun
p__ · flyby · passes once, never captured
p__ · orbiter · brakes into orbit, stays
p__ · atmospheric probe · dives into the air, short life
p__ · lander · touches down, stays put
p__ · rover · lander with wheels
p__ · penetrator = impactor · crashes in on purpose
p__ · observatory spacecraft · telescope platform, target = sky
p__ · communications spacecraft = relay · forwards messages for others
p__ · commitment ladder = mission types · flyby → orbiter → probe → lander → rover → penetrator
p__ · delta-v (Δv) · speed change = fuel bill
p__ · braking burn = capture burn · orbit insertion · slow down so gravity grabs you
p__ · prograde / retrograde · burn to speed up / slow down
p__ · gravity assist = flyby · slingshot · steal speed from a moving planet
p__ · EDL = entry, descent, landing · 7 minutes of terror
p__ · C3 = characteristic / launch energy · how hard you throw it from Earth
p__ · Kuiper Belt · icy ring beyond Neptune, Pluto lives there
p__ · comet vs asteroid · ice + dust with a tail / a rock

S2 · Space vacuum conditions

p__ · vacuum · almost no gas: no pressure, no air
p__ · ozone layer (O_3) · Earth's sunscreen, blocks UV < 0.3 µm
p__ · ultraviolet (UV) · invisible high-energy light, sunburns

p__ · wavelength · size of one ripple; shorter = more energy
p__ · µm (micrometer) = micron · millionth of a meter; 0.3 µm = 300 nm
p__ · polymer = plastic · long molecule chains: Kapton, Teflon
p__ · organic chemical bond = bond energy · carbon glue; UV photon beats it
p__ · degradation / decomposition · material falling apart over time
p__ · corrosion · chemical attack (rust); loose word for UV damage
p__ · UV photodegradation · photons snap polymer bonds
p__ · outgassing · gas leaking out of solids in vacuum
p__ · TML / CVCM · outgassing limits: < 1% / < 0.1%
p__ · bake-out · outgas it on the ground first
p__ · void · bubble inside a material
p__ · pressure difference · 1 atm inside, 0 outside → cracks
p__ · clean room · filtered air, no specks on sensors
p__ · perturbation · small unwanted nudge
p__ · dissipation = damping · energy leaking away until motion stops
p__ · Newton's 1st law = inertia · motion keeps going unless pushed
p__ · vibration / damping · shaking / what makes it fade
p__ · convection · heat carried by moving air; gone in vacuum
p__ · conduction · heat through touching solids
p__ · radiation (heat) · heat glowing away; works in vacuum
p__ · radiator · panel that glows heat into space
p__ · MLI = multilayer insulation · thermal blanket · crinkly gold wrap
p__ · equilibrium / hot spot · settled temperature / unspread heat
p__ · electrical discharge = ESD · a spark
p__ · spacecraft charging · static builds, no air to bleed it, arcs

WORD BANK · COMPLETE · S1-S3 · PAGE 2 OF 3

Every word, by section and lecture

161 lines total. Shape: **page · term = also called · short meaning**. Fill the page number from your notebook. Say the meaning before you write it. One line per word; if the line's long, abbreviate the meaning, not the term.

p__ · passive vs active system · no power (coating) / powered (heater)

S2 · Earth atmosphere

p__ · atmosphere · blanket of gas held by gravity
p__ · troposphere · 0–11 km, weather, colder up
p__ · stratosphere · 11–50 km, ozone, warmer up
p__ · mesosphere · 50–90 km, coldest, meteors burn
p__ · thermosphere · 90–500 km, hot but empty, ISS here
p__ · exosphere · 500 km+, fades to space
p__ · homosphere / heterosphere · stirred mix / sorted by weight
p__ · turbopause = homopause · ~100 km, mixing stops
p__ · temperature vs heat · molecule speed / energy delivered
p__ · density / number density · how much stuff per volume (m^{-3})
p__ · log scale · each tick = 100,000×
p__ · constituent = component · one gas in the mix
p__ · N_2 / O_2 / Ar / CO_2 · 78% / 21% / 0.9% / 0.04%
p__ · O / N / H / He = atomic forms · single atoms, up high
p__ · atomic oxygen (AO) · lone O atom, 200–600 km, eats plastic
p__ · photodissociation · light splits $O_2 \rightarrow O + O$
p__ · anomalous O = hot oxygen · minor extra-hot O above 300 km
p__ · ionosphere · 60–1000 km, charged, bends radio, delays GPS
p__ · geocorona · hydrogen halo glowing in UV
p__ · VLEO / LEO · below ~250 km heavy drag / up to 2000
p__ · reactive · eager to bond, sticky
p__ · oxidation · oxygen bonding to a material; rust
p__ · erosion · worn away bit by bit
p__ · erosion yield / fluence · loss per atom / total atoms that hit

p__ · kinetic energy / relative velocity · energy of motion / closing speed
p__ · eV (electron-volt) · tiny energy unit; bond ≈ few eV
p__ · ram / wake · front face / sheltered back
p__ · atmospheric drag · thin air slows LEO sats; ISS reboosts
p__ · Kapton = polyimide film · gold insulation, AO's victim
p__ · F10.7 / Ap = space weather indices · Sun activity; puffs up the air

S3 · Orbit definition & parameters

p__ · orbital mechanics · math of where a s/c goes under gravity
p__ · celestial body · planet, moon, Sun, comet
p__ · transfer orbit · the highway between two orbits
p__ · Classical Orbital Elements = Keplerian elements · the six · a e i Ω ω ν = the hoop's settings
p__ · semi-major axis (a) · size: $(r_p + r_a)/2$
p__ · eccentricity (e) · squash: 0 circle → 1 cigar → >1 open
p__ · perigee / apogee = periapsis / apoapsis · closest, fastest / farthest, slowest
p__ · perihelion / aphelion · same, around the Sun
p__ · apsis (apsides) · either extreme point
p__ · line of apsides = major axis · perigee↔apogee line
p__ · inclination (i) · tilt vs equator; 0° flat, 90° polar
p__ · ecliptic · plane of Earth's orbit; reference for Sun orbits
p__ · plane of reference · what you measure tilt from
p__ · vernal equinox (♈) = First Point of Aries · reference direction · fixed zero direction in space
p__ · ascending node · hoop crosses equator going north

p__ · line of nodes · where orbital plane cuts equator
p__ · RAAN (Ω) = longitude of ascending node · swivel: ♈ → ascending node
p__ · argument of periapsis (ω) = argument of perigee · spin in plane: node → perigee
p__ · true anomaly (ν) · where the bead is now
p__ · epoch · timestamp the elements are valid for
p__ · orbital plane · flat sheet the orbit lies in; passes through center
p__ · circular / elliptical / hyperbolic · e = 0 / 0–1 / > 1
p__ · polar / equatorial orbit · i = 90° / i = 0°
p__ · geometry (of an orbit) · size + shape + orientation
p__ · vis-viva · $v^2 = \mu(2/r - 1/a)$
p__ · μ (gravitational parameter) = GM · Earth 398,600 km^3/s^2

S3 · Earth orbits & constellations

p__ · Earth orbit = geocentric · goes around Earth, not the Sun
p__ · rotation period · Earth spins once: 23 h 56 m
p__ · altitude vs distance from center · GEO 35,786 alt = 42,164 from center
p__ · coverage / global coverage · what a sat sees / everything, always
p__ · resolution · how much detail; closer = sharper
p__ · orbital plane (constellation) = Walker constellation · several rings so sats aren't bunched
p__ · revisit time · how long until same spot again
p__ · GPS / GLONASS / Galileo / BeiDou · US / Russia / EU / China nav constellations
p__ · Iridium · 66 sats, 780 km, sat phones anywhere

WORD BANK · COMPLETE · S1-S3 · PAGE 3 OF 3

Every word, by section and lecture

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p__ · data relay / transfer time · sat → ground; higher = more delay
p__ · trade-off = compromise · better coverage costs detail + delay
p__ · sun-synchronous (SSO) · polar-ish LEO, same local sun time
p__ · J_2 / nodal precession · Earth's bulge swivels Ω; makes SSO work

S3 · Transfers & flybys

p__ · Lambert's problem · 2 points + a time → the orbit + Δv
p__ · input / constraint · what you feed in / limit you can't break
p__ · conic / patched conics = conic approximation · stitched 2-body pieces of a trajectory
p__ · sphere of influence (SOI) · where a planet's gravity wins; Earth ≈ 925,000 km
p__ · Hohmann transfer · lazy half-ellipse, 180°, 2 burns, cheapest
p__ · tangent · just touching; why burns are small
p__ · optimal / efficient · least Δv, not fastest
p__ · time of flight (TOF) · Hohmann = half the ellipse's period
p__ · launch window / synodic period · lineup repeats; Mars every 26 months
p__ · phase angle = lead angle · how far ahead target must be: Mars ~44°
p__ · impulsive burn · engine firing treated as instant
p__ · circularization burn = apogee kick · 2nd burn rounds the ellipse
p__ · plane change · tilting the orbit; very expensive
p__ · bi-elliptic transfer · 3-burn alternative, only for huge jumps
p__ · gravity field · region where the planet's pull matters
p__ · relative velocity · speed as seen from Mars vs from Sun
p__ · vector / magnitude / direction · arrow / its length / which way

p__ · v_∞ (hyperbolic excess velocity) · speed relative to planet, far away
p__ · turning angle (δ) = deflection angle · how much the flyby bends you
p__ · orbital energy = specific orbital energy · speed + height budget; bigger = bigger a
p__ · aerobraking · use atmosphere to shed speed instead of fuel
p__ · Tsiolkovsky rocket equation = ideal rocket equation · $\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$
p__ · Rosetta / 67P / ESA · comet mission 2004→2014, 4 flybys
p__ · deep space · beyond the Moon; outer planets

COPY INTO NOTES · ENGINEER LEVEL · PAGE 1 OF 2

What goes where: equations, numbers, and section lines

Nothing from the engineer pages goes on the full lecture spreads. Each line below has a home in a reserved page. Say each line before you write it.

A · Equations & key numbers page (p3–4)

Box each formula. One worked number beside it. Units always.

$v = \sqrt{(\mu/r)}$ circular speed · LEO 400 km: 7.67 km/s · GPS: 3.87 · GEO: 3.07

$v_{esc} = \sqrt{(2\mu/r)} = \sqrt{2} \times v_{circ}$ · from LEO ≈ 11 km/s

$T = 2\pi\sqrt{(r^3/\mu)}$ period · GEO: T = 86,164 s → r = 42,164 km → alt 35,786

$v^2 = \mu(2/r - 1/a)$ vis-viva · GTO perigee: 10.15 km/s

$TOF = \pi\sqrt{(a^3/\mu)}$ Hohmann · LEO→GEO 5.3 h · Earth→Mars 259 d

$\Delta v_{plane} = 2v \cdot \sin(\Delta i/2)$ · 1° at LEO ≈ 134 m/s · 28.5°→0° ≈ 3.8 km/s

$\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$ rocket eqn · Isp 450 s → 4.4 km/s per e-fold

$\sin(\delta/2) = 1/(1 + r_p \cdot v_\infty^2/\mu)$ flyby bend · Jupiter 10 km/s @ 500,000 km → 92°

$P = \sigma \cdot \varepsilon \cdot T^4$ radiator · 1 m² @ 300 K, ε 0.9 ≈ 410 W

$drop = \frac{1}{2}gt^2$ · $curve = d^2/(2R)$ · ISS 1 s: 4.3 m = 4.3 m

μ_{Earth} = 398,600 km³/s² · R_{Earth} = 6,371 (6,378 eq.) km · g at 400 km = 8.7 m/s²

GTO burns: +2.42 then +1.47 km/s = 3.89 · Mars: 3.6 leave + ~2 capture

Earth 29.8 km/s · Jupiter 13 km/s around the Sun · sunlight 1,361 W/m²

synodic Earth–Mars 780 d (26 mo) · phase angle 44° · GEO station-keeping ~50 m/s/yr

B · Section map pages (1 page after each section)

One mechanism sentence per scene, under that scene's box. These are the lines you say out loud when someone asks "but how?"

S1 · falling & missing: drops 4.3 m/s, ground curves 4.3 m/s → never lands; engines only for pointing + drag

S1 · shelves: gravity ∝ $1/r^2$ → higher = slower; GEO is where T = 1 sidereal day; GPS = time delays → spheres, 4 sats

S1 · ladder: each rung = more Δv to slow down; flyby needs no capture burn

S2 · vacuum: only exit for heat is radiation ∝ T^4 → big hot panels; outgas → cold optics; sunlit + vs shaded − → arcs

S2 · air: stirred mix to 100 km, then sorted by weight: N_2/O_2 → O → He → H; UV splits O_2 → hungry O at 200–600

S3 · hoop: prograde @ perigee raises apogee (a↑ e↑); sideways @ node tilts i (expensive); radial rotates ω

S3 · orbits: LEO → GTO (+2.42) → apogee kick (+1.47) → GEO

S3 · fuel bill: vis-viva → Δv → rocket eqn → fuel grows exponentially → 20-day Mars impossible

S3 · free lunch: planet frame: same speed, bent δ; Sun frame: + planet's 13 km/s → faster; planet pays 10^{-24} m/s

C · Word bank (back pages) — 15 new lines

Same shape as the rest: p__ · term = also called · short meaning

centripetal force · gravity = the string

station-keeping · small burns, holds the slot

reboost · ISS +2 km/month

free fall · why you float

graveyard orbit · GEO + 300 km, retired

state vector · x y z vx vy vz at an epoch

TLE · public element format

Oberth effect · burn where fastest

Isp · engine efficiency, seconds

porkchop plot · Δv vs dates map

Stefan–Boltzmann · P ∝ T^4

ε / α · radiates well / absorbs little

heat pipe · boil-condense loop, no pump

photoelectric effect · UV knocks electrons off

GTO burns · 2.42 + 1.47 km/s

COPY INTO NOTES · ENGINEER LEVEL · PAGE 2 OF 2

Question bank cards: the “but how?” questions

Back pages, Q on the left page, A on the facing right. These are the ones a friend who knows a little would ask. 20 cards.

Q (left page)

1 · How does a satellite stay up with its engines off?
2 · ISS drops 4.3 m every second. Why isn't it lower?
3 · What does the rocket actually provide: height or speed?
4 · Why do astronauts float if gravity is 90%?
5 · Why is a higher orbit slower?
6 · Where does 35,786 km come from?
7 · Two burns from LEO to GEO: where, which way, how big?
8 · How does GPS turn a radio signal into a position? Why 4?
9 · Why do GEO launches happen near the equator?
10 · Burn prograde at perigee: which elements change, which don't?
11 · Why is a plane change so expensive? Where do you do it?
12 · Where do published orbital elements come from?
13 · How does a radiator get rid of heat with no air?
14 · Why is one side of a spacecraft +120 °C and the other −150 °C?
15 · How does charge build up and what stops the arc?
16 · Walk the LEO→GEO Hohmann numbers.
17 · Why can't we reach Mars in 20 days? (use the rocket equation)
18 · How is Lambert's problem actually solved?
19 · In Jupiter's frame what changes in a flyby? In the Sun's?
20 · What three things set how much a flyby bends you?

A (facing right page)

1 · It's moving sideways at 7.7 km/s; it falls and the ground curves away equally. Gravity is the centripetal force.
2 · In that second it moved 7.7 km and Earth curved away 4.3 m. Drop = curve.
3 · Speed. 7.7 km/s sideways. Height is cheap.
4 · Free fall: nothing pushes back on you. Weight is the push, not the pull.
5 · Gravity $\propto 1/r^2$: less pull → less speed needed, and a bigger circle. $T \propto r^{1.5}$.
6 · Solve $T = 86{,}164$ s in $T = 2\pi\sqrt{(r^3/\mu)}$: r = 42,164 km, minus 6,378 = 35,786.
7 · Prograde at perigee +2.42 km/s (raises apogee), prograde at apogee +1.47 (circularizes). 5.3 h apart.
8 · Delay × c = distance → sphere; 3 spheres = a point; 4th fixes the phone's clock.
9 · Launch latitude = orbit tilt; GEO needs 0°; plane change costs $2v\cdot\sin(\Delta i/2)$, ~3.8 km/s from Florida.
10 · a↑ and e↑ (apogee rises). i, Ω, ω unchanged.
11 · $\Delta v = 2v\cdot\sin(\Delta i/2)$: 1° ≈ 134 m/s at LEO. Do it at apogee where v is small, or launch into the right plane.
12 · Tracking gives position + velocity (state vector) at an epoch; geometry converts to a e i Ω ω ν; published as a TLE.
13 · Radiation: $P = \sigma\varepsilon T^4$. Big, hot, high-ε panels; heat pipes carry heat to them.
14 · No convection to even it out: sunlit side settles where absorbed = radiated (~120 °C); shade radiates to 3 K space.
15 · UV kicks electrons off sunlit surfaces (+); shade collects plasma electrons (−); kV gap arcs. Fix: conductive coatings + bonding straps.
16 · 7.73 → 10.15 at perigee (+2.42); 1.61 at apogee → 3.07 (+1.47); a = 24,418 km; TOF 5.3 h.
17 · ~60 km/s needed; $e^{(60/4.4)} \approx 800{,}000$ → 99.9999% propellant. 3.6 km/s (Hohmann) needs 56%.
18 · Infinitely many ellipses fit 2 points; TOF depends on a; iterate a until TOF matches; velocities fall out.
19 · Jupiter frame: speed same, direction bent by δ. Sun frame: add Jupiter's 13 km/s → faster (behind) or slower (in front).
20 · Approach speed v_∞, closest approach r_p, planet's μ: $\sin(\delta/2) = 1/(1 + r_p v_\infty^2/\mu)$.

WHO'S WHO · PAGE 1 OF 4

Every name I've used so far, and what it is

Colored tag = which kind of thing it is. You don't need to memorize these. They're the "for example" behind each category, so when a name shows up you know what bucket it belongs in.

LIVE **Laika** — a Moscow stray dog, the first animal to orbit Earth, inside Sputnik 2 (1957). Proved a living body survives launch and weightlessness. Didn't survive the flight.

LIVE **ISS** — International Space Station. A football-field-sized lab in LEO (~400 km) where 6–7 astronauts live for months. Built by the US, Russia, Europe, Japan, Canada. Orbits every 90 minutes.

LOOK OUT **Hubble** — a bus-sized telescope orbiting Earth since 1990 at ~540 km. Above the air, so razor-sharp pictures. Source of most famous deep-space photos.

LOOK OUT **Kepler** — a space telescope (2009–2018) that stared at one patch of sky and found thousands of planets around other stars.

LOOK DOWN **GOES** — US weather satellites in GEO. The cloud loops on the TV weather report are GOES pictures.

RELAY **Starlink** — SpaceX's internet constellation: thousands of small satellites in LEO (~550 km) relaying internet to dishes on the ground.

FLYBY **Voyager 1 & 2** — twin NASA probes launched 1977. Flew past Jupiter and Saturn (Voyager 2 also Uranus and Neptune), then kept going. Both have now left the solar system and are still transmitting.

LIVE **Sputnik 2** — the Soviet satellite that carried Laika. Sputnik 1, a month earlier, was the very first artificial satellite of any kind: a beeping metal ball.

LIVE **Tiangong** — China's own space station, smaller than ISS, also in LEO.

LOOK OUT **JWST** — James Webb Space Telescope (2021). Hubble's bigger successor, sees infrared (heat-light). Parked 1.5 million km from Earth at a gravity balance point, not in Earth orbit.

LOOK DOWN **Landsat** — US series of Earth-mapping satellites running since 1972. Farms, forests, cities, floods, all photographed from LEO.

RELAY **GPS** — Global Positioning System: ~30 US satellites in MEO (~20,200 km) broadcasting time signals. Your phone hears several and computes where you are. Europe's version is Galileo.

RELAY **Queqiao** — a Chinese relay satellite parked beyond the Moon so a lander on the Moon's far side (Chang'e-4) could talk to Earth, since the Moon blocks direct radio.

FLYBY **New Horizons** — NASA probe launched 2006, flew past Pluto July 2015 after 9.5 years, took the famous heart-shaped-glacier photos, then continued into the Kuiper Belt.

WHO'S WHO · PAGE 2 OF 4

Orbiters, probes, landers, rovers, and the places

Second half. Places at the bottom, because a few of them (Titan, Pluto, Kuiper Belt) came up as destinations.

ORBITER **Cassini** — NASA spacecraft that orbited Saturn 2004–2017, photographing the rings and moons. Deliberately crashed into Saturn at the end so it couldn't contaminate a moon.

ATMOSPHERIC **Galileo probe** — dropped by the Galileo orbiter into Jupiter's atmosphere in 1995. Transmitted for 58 minutes before heat and pressure crushed it. No surface to land on.

ORBITER **MRO** — Mars Reconnaissance Orbiter (2006–). Maps Mars in high detail AND relays radio for the rovers on the ground. Orbiter and relay at once.

LANDER **InSight** — NASA Mars lander (2018–2022). Planted a seismometer and listened for marsquakes. Died when dust covered its solar panels.

ROVER **Sojourner** — the first Mars rover (1997), microwave-oven sized.

ROVER **Curiosity & Perseverance** — car-sized, nuclear-powered Mars rovers (2012 and 2021). Perseverance carried the Ingenuity helicopter and is collecting samples for a future return.

PENETRATOR **LCROSS** — NASA mission (2009) that crashed a spent rocket stage into a dark Moon crater and flew through the dust plume. Found water ice.

Pluto — dwarf planet at the edge of the solar system, ~33 times farther from the Sun than Earth. Visited once, by New Horizons.

L2 — a gravity balance point 1.5 million km behind Earth (away from the Sun) where a spacecraft can hover with little fuel. JWST lives there.

Walter Hohmann / Johann Lambert — not missions, people. Hohmann: German engineer who worked out the cheapest two-burn transfer in 1925. Lambert: 18th-century mathematician whose "two points plus a time" orbit problem still carries his name.

GLONASS / BeiDou — Russia's and China's navigation constellations, both in MEO like GPS. Your phone usually listens to all of them.

ATMOSPHERIC **Huygens** — a European probe Cassini carried and dropped onto Titan (Saturn's biggest moon) in 2005. Parachuted through the orange haze, landed, sent pictures for 90 minutes. One mission, two types.

ORBITER **Juno** — NASA Jupiter orbiter, arrived 2016. The huge solar arrays in your clean-room slide are Juno's; it's the farthest solar-powered spacecraft ever.

LANDER **Viking 1 & 2** — NASA's first successful Mars landers (1976). Sat in one spot, photographed the surface, tested the soil for life.

LANDER **Rosetta / Philae** — Rosetta = European spacecraft that orbited a comet (2014). Philae = the small lander it dropped onto the comet's surface. Philae bounced twice, landed in a shadow, ran out of solar power in 3 days.

ROVER **Spirit & Opportunity** — twin golf-cart-sized Mars rovers (2004). Designed for 90 days; Opportunity drove for 14 years. The rover in your slide is this design.

PENETRATOR **Deep Impact** — NASA mission (2005) that fired a washing-machine-sized copper block into comet Tempel 1 and watched what sprayed out.

Titan — Saturn's largest moon. Thick orange atmosphere, lakes of liquid methane. The only moon anyone has landed a probe on (Huygens).

Kuiper Belt — the ring of icy leftovers beyond Neptune. Pluto is its biggest known member.

Comet 67P / Churyumov–Gerasimenko — the duck-shaped comet Rosetta orbited from 2014 and Philae landed on. Named after the two astronomers who spotted it in 1969.

Galileo (two things) — the European navigation constellation (GPS rival, MEO), AND NASA's 1990s Jupiter orbiter that dropped the atmospheric probe. Context tells you which.

NASA / ESA / SpaceX — US space agency / European Space Agency / the private US rocket company (Falcon rockets, Starlink). Not affiliated with your notebook title.

WHO'S WHO · PAGE 3 OF 4 · updated 11/12

Section 4: rockets, engines, and the people behind them

Every proper name used in the Section 4 propulsion PDFs so far. Same rule as before: if I use a name, it is defined here.

ROCKET **Saturn V** — the Apollo Moon rocket (flew 1967–1973). 110 m tall, three stages. Its first stage had five F-1 engines and burned 13 tonnes of propellant every second. The rocket lying on its side in your slide.

ROCKET **Space Shuttle / SLS** — NASA's reusable spaceplane (1981–2011) and its current Moon rocket. Both use RS-25 hydrogen engines plus two big solid boosters.

ENGINE **Merlin / Merlin Vacuum** — SpaceX's kerosene + liquid oxygen engine. Nine on Falcon 9's first stage; one 'Vacuum' version with a huge bell on the second stage (981,000 N, Isp 348 s).

ENGINE **NEXT (and NSTAR)** — NASA's ion thrusters. NSTAR flew on Deep Space 1 and Dawn; NEXT, the blue glow in your slide, is the bigger successor (0.236 N, Isp 4,190 s) and flew on DART.

PROBE **Dawn** — NASA ion-engine probe (2007–2018). Orbited the asteroid Vesta, left, then orbited the dwarf planet Ceres: the first spacecraft to orbit two different bodies, only possible with electric propulsion.

PROBE **BepiColombo / Psyche** — European–Japanese Mercury mission (launched 2018, ion engines) / NASA mission to a metal asteroid (launched 2023, Hall thrusters). Today's electric-propulsion flagships.

PROGRAM **NERVA** — the US nuclear thermal rocket program. Built and ground-tested working engines in Nevada in the 1960s; ended in 1972 without ever flying.

PROGRAM **Project Rover** — the US reactor-testing effort (from 1955) that fed into NERVA. About 20 nuclear rocket reactors were run in Nevada between 1959 and 1972.

CAPSULE **Orion (crew capsule)** — NASA's current astronaut capsule for Moon missions. Shares only the name with Project Orion.

THING **RTG** — radioisotope thermoelectric generator: a plutonium power pack that turns decay heat into electricity. Powers Voyager, New Horizons, Curiosity. Not an engine.

PERSON **Konstantin Tsiolkovsky** — Russian schoolteacher who published the rocket equation in 1903, before any liquid rocket existed.

PERSON **Gustaf de Laval** — Swedish engineer who invented the narrowing-then-widening nozzle in 1888, for steam turbines. Every rocket bell is one.

ROCKET **Falcon 9** — SpaceX's two-stage workhorse rocket. The first stage flies back and lands; the second stage, the one in your slide, has a single Merlin Vacuum engine.

ENGINE **F-1** — the Saturn V first-stage engine: kerosene + liquid oxygen, 6.77 million newtons each. Still the most powerful single-chamber liquid engine ever flown.

ENGINE **RS-25** — the Space Shuttle main engine: liquid hydrogen + liquid oxygen, Isp 452 s. About the best mileage a chemical engine gets.

PROBE **Deep Space 1** — NASA technology test (1998). The first probe to cruise between planets on an ion engine.

PROBE **DART** — NASA mission that deliberately hit a small asteroid moon in 2022 to test deflecting one. Carried a NEXT ion thruster.

TEST **SERT-1 / Zond 2** — the first electric thrusters fired in space, both in 1964: a NASA ion-engine test and a Soviet Mars probe's plasma thrusters. Electric propulsion is 60 years old.

PROGRAM **DRACO** — a NASA and DARPA plan to fly a nuclear thermal rocket in orbit. Cancelled in 2025, as far as I know.

PROGRAM **Project Orion (1958–1965)** — a US study of nuclear PULSE propulsion: a ship riding small nuclear explosions on a pusher plate. Never built; ended by the 1963 treaty banning nuclear explosions in space. Not a nuclear thermal rocket, and not the Orion crew capsule.

PLACE **Proxima Centauri** — the nearest star to the Sun, 4.2 light years away: about 40 trillion km. At Voyager's speed, 75,000 years.

PERSON **Isaac Newton** — English scientist whose three laws of motion (1687) are the whole basis of rocketry. The third: every action has an equal and opposite reaction.

PERSON **Robert Goddard** — American who flew the first liquid-fuel rocket in 1926. A 1920 newspaper editorial mocked him for thinking rockets could work in vacuum; the paper corrected itself in 1969, as Apollo 11 flew to the Moon.

WHO'S WHO · PAGE 4 OF 4 · updated 11/12

Section 5: radio, antennas, and who listens

Every proper name used in the Section 5 communications PDFs so far. More will be added as the section goes on.

PERSON **James Clerk Maxwell** — Scottish physicist who showed in the 1860s that electricity and magnetism together make waves that travel at the speed of light. Every radio link rests on his equations.

THING **CubeSat** — a tiny satellite built from 10 cm cubes, cheap enough for universities. The one in your slide has four thin rod antennas that spring out after launch.

MISSION **Apollo 17** — the last crewed Moon landing, December 1972. Two astronauts spent three days on the surface.

NETWORK **Deep Space Network (DSN)** — NASA's three ground stations in California, Spain and Australia, spaced so one always faces any spacecraft. Each has a 70-metre dish.

AGENCY **ITU** — the International Telecommunication Union, the United Nations body that decides which radio frequencies may be used for what. It is why GPS, Wi-Fi and a probe at Saturn do not jam each other.

PERSON **Ludwig Boltzmann** — Austrian physicist (1844–1906) who connected temperature to the random motion of particles. The constant named after him, k, turns a receiver's temperature into its noise power.

PERSON **Heinrich Hertz** — German physicist who first made and detected radio waves in 1887, proving Maxwell right. The unit of frequency is named after him.

ORBITER **Cassini (its radio)** — the Saturn orbiter from page 2. Its 4-metre dish was its high-gain antenna; the transmitter behind it was only 20 watts.

VEHICLE **Lunar Roving Vehicle** — the battery-powered car driven on the Moon on Apollo 15, 16 and 17. It carried a wide-beam low-gain antenna for voice while driving and an umbrella-shaped high-gain dish for television when parked.

THING **Yagi antenna** — the fishbone-shaped rooftop TV antenna, named after one of its two Japanese inventors (1926). A simple directional antenna.

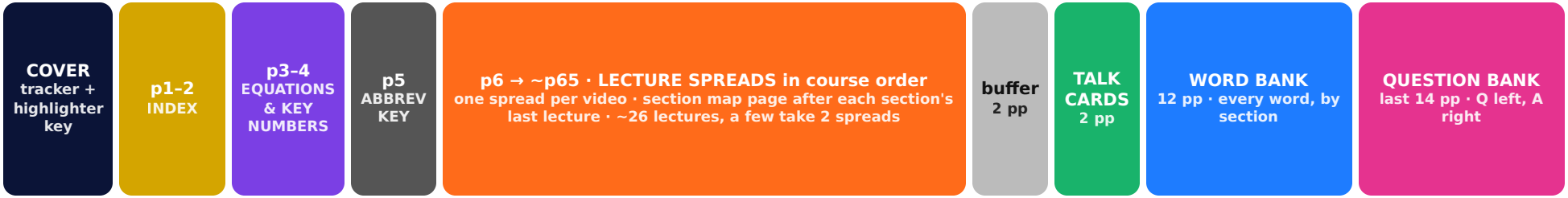
THING **Loran-C** — a long-range radio navigation system that used low frequencies (around 100 kHz) before GPS replaced it. Named on your frequency slide.

PROBE **Voyager 1 (its radio)** — the farthest spacecraft, launched 1977. Its 23-watt transmitter now reaches Earth at 160 bits per second.

NOTEBOOK MAP · TAPE THIS INSIDE THE COVER

The whole notebook, front to back

About 95 pages. Number every page in pencil first. Sticky tabs at the three back-section boundaries. Then never think about layout again.



Block	Pages	Rule
Inside front cover	—	Review tracker: 7 section rows × watched / +1 / +3 / +7 / +14 / +30. Highlighter key under it.
Index	1–2	One line per lecture, written when you START the spread: "S3 L2 Earth orbits … p19".
Equations & key numbers	3–4	Every formula boxed with one worked number; key numbers for lectures with no formula. Nothing else.
Abbreviation key	5	sat · s/c · Δv · AO · ∴ · → … plus "→ WB" = see word bank.
Lecture spreads	6 → ~65	Left: draw + 3 takeaways + 3-line summary. Right: cues + notes. Doesn't fit → next spread, never smaller writing.
Section map	1 page after each section	From memory, nothing open: the section's lectures as boxes with one line each. Compare to index after.
Buffer	2	Blank. Absorbs a lecture that runs long.
Talk cards	2	The cue card for each section's talk, ~8 lines each. Copy from the app's Teach tab.
Word bank	12	By section and lecture, EVERY word the lectures used. "p13 · RAAN (Ω) = longitude of ascending node · swivel". ~22 lines a page; S1–S3 alone is 161 lines ≈ 7½ pages.
Question bank	last 14	Q on left page, A on facing right page. Every quiz miss, every "?", every word missed twice.

Set-up, in order, today

1. **Number every page** in pencil, bottom outer corner, 1 to the end.
2. Count **30 from the last page**. Sticky tab. Everything behind it is the back section.
3. Tab at **14 from the end** (question bank starts) and **28 from the end** (talk cards start; word bank is the 12 pages between).
4. Write the block name at the top of the first page of each back block.
5. Fill the word bank from the complete word-bank sheet, guessing each meaning first.

If your notebook is a different size

- **Under 95 pages:** shrink in this order: question bank → 10, word bank → 8 (two words per line, term only + 2-word meaning), lectures take what's left. Never shrink index or equations.
- **Over 95 pages:** keep every block the same size and let the buffer grow. Don't spread the back section out.
- **Already have S1–S3 spreads in it:** fine. Number around them; they're the first lecture spreads.

⚡ Why 30 back pages: the full vocabulary runs ~160 words for 3 sections, so ~250 for the course ≈ 12 pages at 22 lines; ~150 question cards over 7 Q/A spreads; 7 talk cue cards ≈ 2 pages. Everything has a home before you need it.